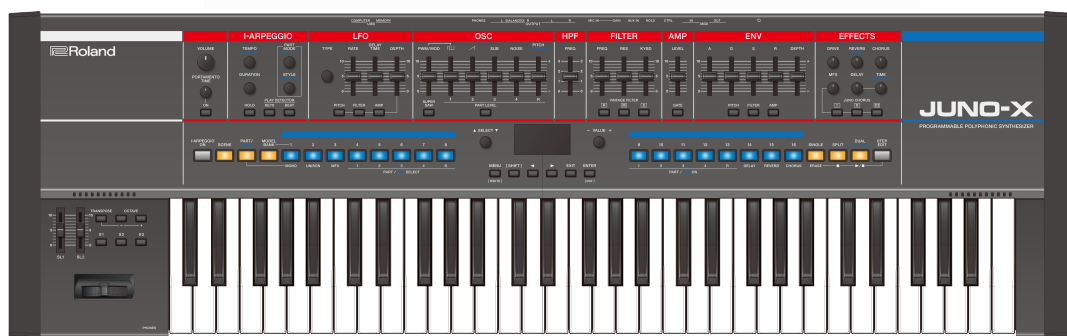




# JUNO-X

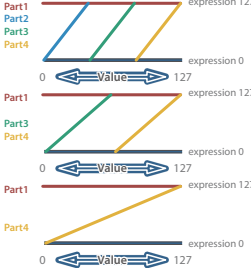
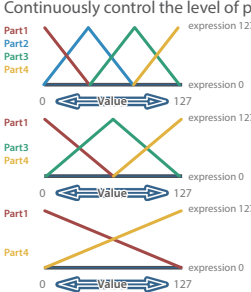
## Parameter Guide

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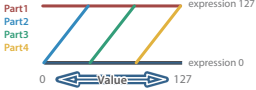
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# Scene Parameter

## SCENE COMMON

| Parameter   | Value                         | Explanation                                                                                                                                                                                                                  |
|-------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scene Level | 0–127                         | Adjusts the overall volume of the scene.                                                                                                                                                                                     |
| Tempo       | 20.00–300.00                  | Specifies the tempo of the scene (including the arpeggio).                                                                                                                                                                   |
| ASSIGN      |                               |                                                                                                                                                                                                                              |
| SL1–SL2     | SL1, SL2 function assignments |                                                                                                                                                                                                                              |
|             | OFF                           | No function is assigned.                                                                                                                                                                                                     |
|             | CC01–CC95                     | Controller number 1–95                                                                                                                                                                                                       |
|             | AFT                           | Aftertouch                                                                                                                                                                                                                   |
|             | BEND DOWN                     | Applies the same effect as when the pitch bend wheel is moved downward.                                                                                                                                                      |
|             | BEND UP                       | Applies the same effect as when the pitch bend wheel is moved upward.                                                                                                                                                        |
|             | CHO LEVEL                     | Chorus level is assigned.                                                                                                                                                                                                    |
|             | REV LEVEL                     | Reverb level is assigned.                                                                                                                                                                                                    |
|             | DLY LEVEL                     | Delay level is assigned.                                                                                                                                                                                                     |
|             | ARP SHUFFLE                   | I-ARP's G-Shuffle parameter is assigned.                                                                                                                                                                                     |
|             | ARP DURATION                  | I-ARP's G-Duration parameter is assigned.                                                                                                                                                                                    |
|             | PART FADE1                    | Continuously control the level of parts 1–4.<br>                                                                                           |
|             | PART FADE2                    | Continuously control the level of parts 1–4.<br>                                                                                          |
|             | LEVEL P-1–P-R                 | Each part's volume is assigned.                                                                                                                                                                                              |
|             | AGE                           | The SYSTEM > AGE parameter is assigned.                                                                                                                                                                                      |
|             | S1–S3 function assignments    |                                                                                                                                                                                                                              |
|             | OFF                           | No function is assigned.                                                                                                                                                                                                     |
|             | CC01–CC95                     | Controller number 1–95                                                                                                                                                                                                       |
|             | AFT                           | Aftertouch                                                                                                                                                                                                                   |
|             | MONO/POLY                     | Monophonic/polyphonic switch is assigned.                                                                                                                                                                                    |
|             | SCENE DOWN (*)                | Switch the scene to the previous number.                                                                                                                                                                                     |
|             | SCENE UP (*)                  | Switch the scene to the next number.                                                                                                                                                                                         |
|             | TONE DOWN (*)                 | Switch the tone to the previous number.                                                                                                                                                                                      |
|             | TONE UP (*)                   | Switch the tone to the next number.                                                                                                                                                                                          |
|             | PANEL DEC (*)                 | Decreases the parameter's value.                                                                                                                                                                                             |
|             | PANEL INC                     | Increases the parameter's value.                                                                                                                                                                                             |
|             | CHO SW                        | Chorus on/off is assigned.                                                                                                                                                                                                   |
|             | REV SW                        | Reverb on/off is assigned.                                                                                                                                                                                                   |
|             | DLY SW                        | Delay on/off is assigned.                                                                                                                                                                                                    |
| S1–S3 Func  | ARP SW (*)                    | Applies the same effect as when the panel's I-ARPEGGIO [ON] button is pressed.                                                                                                                                               |
|             | ARP HOLD (*)                  | Applies the same effect as when the panel's I-ARPEGGIO [HOLD] button is pressed.                                                                                                                                             |
|             | DETECT KEYS (*)               | Applies the same effect as when the panel's I-ARPEGGIO [KEYS] button is pressed.                                                                                                                                             |
|             | DETECT BEAT                   | Applies the same effect as when the panel's I-ARPEGGIO [BEAT] button is pressed.                                                                                                                                             |
|             | UNISON SW                     | Unison on/off is assigned.                                                                                                                                                                                                   |
|             | BEND MODE                     | Switches the bend mode.                                                                                                                                                                                                      |
|             | AUTO TUNING                   | Executes tuning for the voice slot of a model that simulates an analog synthesizer. The tuning will be corrected in a few seconds, but will subsequently return to the pitch discrepancies specified by the tone parameters. |
|             | TAP TEMPO                     | Sets the tap tempo function.                                                                                                                                                                                                 |
|             | START/STOP                    | Starts/stops the step editor.                                                                                                                                                                                                |
|             | DRV SW                        | Overdrive on/off is assigned.                                                                                                                                                                                                |

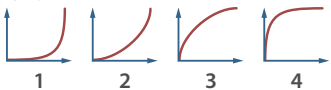
| Parameter  | Value                                                           | Explanation                                                                                                                                                                                                                  |
|------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S1–S3 Func | DETECT BEAT (*)                                                 | Applies the same effect as when the panel's I-ARPEGGIO [BEAT] button is pressed.                                                                                                                                             |
|            | UNISON SW                                                       | Unison on/off is assigned.                                                                                                                                                                                                   |
|            | BEND MODE                                                       | Switches the bend mode.                                                                                                                                                                                                      |
|            | AUTO TUNING (*)                                                 | Executes tuning for the voice slot of a model that simulates an analog synthesizer. The tuning will be corrected in a few seconds, but will subsequently return to the pitch discrepancies specified by the tone parameters. |
|            | TAP TEMPO (*)                                                   | Sets the tap tempo function.                                                                                                                                                                                                 |
|            | START/STOP (*)                                                  | Starts/stops the step editor.                                                                                                                                                                                                |
|            | DRV SW                                                          | Overdrive on/off is assigned.                                                                                                                                                                                                |
| S1–S3 Mode | VOC/MIC                                                         | When the Vocoder tone is selected, this selects whether to use the vocoder sound or the mic input sound.                                                                                                                     |
|            | LATCH                                                           | The assigned function is switched each time you press the button.                                                                                                                                                            |
|            | MOMENTARY                                                       | The assigned function is effective only while you hold down the button.                                                                                                                                                      |
| Hold       | Hold pedal function assignments                                 |                                                                                                                                                                                                                              |
|            | OFF                                                             | No function is assigned.                                                                                                                                                                                                     |
|            | CC01–CC95                                                       | Controller number 1–95                                                                                                                                                                                                       |
|            | AFT                                                             | Aftertouch                                                                                                                                                                                                                   |
|            | MONO/POLY                                                       | Monophonic/polyphonic switch is assigned.                                                                                                                                                                                    |
|            | SCENE DOWN                                                      | Switch the scene to the previous number.                                                                                                                                                                                     |
|            | SCENE UP                                                        | Switch the scene to the next number.                                                                                                                                                                                         |
|            | TONE DOWN                                                       | Switch the tone to the previous number.                                                                                                                                                                                      |
|            | TONE UP                                                         | Switch the tone to the next number.                                                                                                                                                                                          |
|            | PANEL DEC                                                       | Decreases the parameter's value.                                                                                                                                                                                             |
|            | PANEL INC                                                       | Increases the parameter's value.                                                                                                                                                                                             |
|            | CHO SW                                                          | Chorus on/off is assigned.                                                                                                                                                                                                   |
|            | REV SW                                                          | Reverb on/off is assigned.                                                                                                                                                                                                   |
|            | DLY SW                                                          | Delay on/off is assigned.                                                                                                                                                                                                    |
|            | ARP SW                                                          | Applies the same effect as when the panel's I-ARPEGGIO [ON] button is pressed.                                                                                                                                               |
|            | ARP HOLD                                                        | Applies the same effect as when the panel's I-ARPEGGIO [HOLD] button is pressed.                                                                                                                                             |
|            | DETECT KEYS                                                     | Applies the same effect as when the panel's I-ARPEGGIO [KEYS] button is pressed.                                                                                                                                             |
|            | DETECT BEAT                                                     | Applies the same effect as when the panel's I-ARPEGGIO [BEAT] button is pressed.                                                                                                                                             |
|            | UNISON SW                                                       | Unison on/off is assigned.                                                                                                                                                                                                   |
|            | BEND MODE                                                       | Switches the bend mode.                                                                                                                                                                                                      |
|            | AUTO TUNING                                                     | Executes tuning for the voice slot of a model that simulates an analog synthesizer. The tuning will be corrected in a few seconds, but will subsequently return to the pitch discrepancies specified by the tone parameters. |
|            | TAP TEMPO                                                       | Sets the tap tempo function.                                                                                                                                                                                                 |
|            | START/STOP                                                      | Starts/stops the step editor.                                                                                                                                                                                                |
|            | DRV SW                                                          | Overdrive on/off is assigned.                                                                                                                                                                                                |
|            | VOC/MIC                                                         | When the Vocoder tone is selected, this selects whether to use the vocoder sound or the mic input sound.                                                                                                                     |
| Hold Pole  | Specifies the polarity of the pedal connected to the HOLD jack. |                                                                                                                                                                                                                              |
|            | STANDARD                                                        | Specifies standard polarity.                                                                                                                                                                                                 |
|            | REVERSE                                                         | Specifies reverse polarity.                                                                                                                                                                                                  |

| Parameter      | Value                                | Explanation                                                                                                                        |
|----------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Ctrl           | Ctrl pedal function assignments      |                                                                                                                                    |
|                | OFF                                  | No function is assigned.                                                                                                           |
|                | CC01–CC95                            | Controller number 1–95                                                                                                             |
|                | AFT                                  | Aftertouch                                                                                                                         |
|                | BEND DOWN                            | Applies the same effect as when the pitch bend wheel is moved downward.                                                            |
|                | BEND UP                              | Applies the same effect as when the pitch bend wheel is moved upward.                                                              |
|                | CHO LEVEL                            | Chorus level is assigned.                                                                                                          |
|                | REV LEVEL                            | Reverb level is assigned.                                                                                                          |
|                | DLY LEVEL                            | Delay level is assigned.                                                                                                           |
|                | ARP SHUFFLE                          | I-ARP's G-Shuffle parameter is assigned.                                                                                           |
|                | ARP DURATION                         | I-ARP's G-Duration parameter is assigned.                                                                                          |
|                | PART FADE1                           | Continuously control the level of parts 1–4.<br>  |
|                | PART FADE2                           | Continuously control the level of parts 1–4.<br> |
|                | LEVEL P-1–P-R                        | Each part's volume is assigned.                                                                                                    |
|                | AGE                                  | The SYSTEM > AGE parameter is assigned.                                                                                            |
| Part XFade Pos | 0–127                                | This parameter stores the current value of PART FADE as a scene setting.                                                           |
| CTRL SOURCE    |                                      |                                                                                                                                    |
| CtrlSrc1–4     | OFF, CC01–CC31, CC33–CC95, BEND, AFT | Specify the MIDI messages used for tone control.                                                                                   |
| VOICE RSRV     |                                      |                                                                                                                                    |
| Part1–5        | 0–10                                 | Specifies the number of voices reserved for each part when the performance exceeds the maximum polyphony.                          |

## SCENE PART

| Parameter   | Value                                                              | Explanation                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part Level  | 0–127                                                              | Specifies the volume of each part.                                                                                                                                                                                                                                                   |
| Pan         | L64–63R                                                            | Specifies the pan of each part's sound when outputting in stereo.                                                                                                                                                                                                                    |
| Rev Send    | 0–127                                                              | Specifies the send level to reverb.                                                                                                                                                                                                                                                  |
| Cho Send    | 0–127                                                              | Specifies the send level to chorus.                                                                                                                                                                                                                                                  |
| Delay Send  | 0–127                                                              | Specifies the send level to delay.                                                                                                                                                                                                                                                   |
| Output      | THRU, DRIVE                                                        | Specifies whether the output of each part goes through the OVER DRIVE effect (DRIVE) or does not go through it (THRU).                                                                                                                                                               |
| Part Sw     | OFF, ON                                                            | Specifies whether the part is enabled (ON) or disabled (OFF).                                                                                                                                                                                                                        |
| Mute Sw     | OFF, MUTE                                                          | Specifies the part mute setting.                                                                                                                                                                                                                                                     |
| PITCH       |                                                                    |                                                                                                                                                                                                                                                                                      |
| Coarse Tune | –48–+48                                                            | Shifts the pitch in units of a semitone.                                                                                                                                                                                                                                             |
| Fine Tune   | –50–+50                                                            | Finely adjusts the pitch in units of one cent.                                                                                                                                                                                                                                       |
| Oct Shift   | –3–+3                                                              | Shifts the pitch of the keyboard in units of one octave.                                                                                                                                                                                                                             |
| Bend Range  | 0–24, TONE                                                         | Specifies the range of pitch change controlled by pitch bend, in semitone units. To use the setting of the tone, choose TONE.                                                                                                                                                        |
| Bend Mode   | Specifies the behavior when the pitch bend controller is operated. |                                                                                                                                                                                                                                                                                      |
|             | NORMAL                                                             | The conventional pitch bend effect occurs.                                                                                                                                                                                                                                           |
|             | C+L (CATCH + LAST)                                                 | The pitch bend effect applies only to the last-played note. If a note-on occurs while pitch bend is already applied, the new note sounds at the center pitch. The pitch starts changing only after the controller passes through the center position.                                |
| Bend Mode   | TONE                                                               | The tone's settings are used.                                                                                                                                                                                                                                                        |
| MODIFY      |                                                                    |                                                                                                                                                                                                                                                                                      |
| Cutoff      | –64–+63                                                            | Adjusts how far the filter is open. Increasing this value makes the sound brighter, and decreasing it makes the sound darker.                                                                                                                                                        |
| Resonance   | –64–+63                                                            | Emphasizes the portion of the sound in the region of the cutoff frequency, adding character to the sound. Excessively high settings can produce oscillation, causing the sound to distort. Increasing this value strengthens the character, and decreasing it weakens the character. |
| Attack      | –64–+63                                                            | Adjusts the time over which the sound reaches its maximum volume after you press the key. Higher values produce a milder attack; lower values produce a sharper attack.                                                                                                              |
| Decay       | –64–+63                                                            | Adjusts the time over which the volume decreases from its maximum value. Larger settings of this value make the decay longer, and smaller settings make the decay shorter.                                                                                                           |
| Release     | –64–+63                                                            | The time it takes after the key is released for a sound to become inaudible. Larger settings of this value make the sound linger, and smaller settings make the sound end more sharply.                                                                                              |
| Vib Rate    | –64–+63                                                            | Adjusts the vibrato speed (the rate at which the pitch is modulated). The pitch will be modulated more rapidly for higher settings, and more slowly with lower settings.                                                                                                             |
| Vib Depth   | –64–+63                                                            | Adjusts the depth of the vibrato effect (the depth at which the pitch is modulated). The pitch will be modulated more greatly for higher settings, and less with lower settings.                                                                                                     |
| Vib Delay   | –64–+63                                                            | Adjusts the time until vibrato (pitch modulation) starts to apply. Higher settings will produce a longer delay time before vibrato begins, while lower settings produce a shorter time.                                                                                              |



| Parameter                          | Value                                                                          | Explanation                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CTRL</b>                        |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Mono/Poly</b><br>Part 1–4 only  | MONO,<br>POLY,<br>TONE                                                         | Choose “MONO” if you want the tone assigned to the part to play monophonically, or “POLY” if you want to play it polyphonically.<br>To use the setting of the tone, choose “TONE.”                                                                                                                                                                                                      |
| <b>Legato Sw</b><br>Part 1–4 only  | OFF,<br>ON,<br>TONE                                                            | Legato can be applied when playing monophonically. “Legato” is a playing technique that smooths the transition between notes, minimizing the sense of a gap between them. The effect is similar to the guitar performance techniques of hammering-on and pulling-off. Choose “ON” to apply legato, or “OFF” if not. Choose “TONE” if you want to use the setting specified by the tone. |
| <b>Porta Sw</b><br>Part 1–4 only   | OFF,<br>ON,<br>TONE                                                            | Specifies whether portamento is applied. Select “ON” to apply portamento, or “OFF” if you don’t want to apply portamento. Choose “TONE” if you want to use the setting specified by the tone.                                                                                                                                                                                           |
| <b>Porta Time</b><br>Part 1–4 only | 0–127,<br>TONE                                                                 | When portamento is used, this specifies the time over which the pitch will change. Higher settings will cause the pitch change to the next note to take more time. Choose “TONE” if you want to use the setting specified by the tone.                                                                                                                                                  |
| <b>Unison Sw</b><br>Part 1–4 only  | OFF,<br>ON,<br>TONE                                                            | This layers a single sound. Choose “ON” if you want to play using unison, or “OFF” if not. Choose “TONE” if you want to use the setting specified by the tone.                                                                                                                                                                                                                          |
| <b>Velo Sens</b>                   | -63–+63                                                                        | Adjusts the velocity sensitivity. Larger settings raise the sensitivity.                                                                                                                                                                                                                                                                                                                |
| <b>Voice Assign</b>                | Sets the way sounds are played when the same key is pressed a number of times. |                                                                                                                                                                                                                                                                                                                                                                                         |
|                                    | SINGLE                                                                         | Only one sound can be played at a time when the same key is pressed. With continuous sounds where the sound plays for an extended time, the previous sound is stopped when the following sound is played.                                                                                                                                                                               |
|                                    | LIMIT                                                                          | Layer the sound of the same keys. If long-sustaining notes are played consecutively, the previous notes are turned off after a certain number of notes accumulate.                                                                                                                                                                                                                      |
|                                    | FULL                                                                           | Layer the sound of the same keys. Even with continuous sounds where the sound plays for an extended time without previously played sounds being eliminated.                                                                                                                                                                                                                             |
| <b>KBD Velo</b>                    | REAL,<br>FIXED                                                                 | Specifies whether the velocity value changes according to the actual strength of your playing (REAL) or is always a fixed velocity value regardless of how you play (FIXED).                                                                                                                                                                                                            |
| <b>KBD Fixed Velo</b>              | 1–127                                                                          | Specifies the velocity value when KBD Velo is “FIXED.”                                                                                                                                                                                                                                                                                                                                  |
| <b>Velo Curve</b>                  | OFF, 1–4                                                                       | For each part, select one of the following four velocity curves as appropriate for the playing touch of your MIDI keyboard. If you want to use the velocity curve of this unit’s keyboard, choose “OFF.”<br>                                                                                         |

| Parameter                                                               | Value         | Explanation                                                                                                                                                                                    |
|-------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MIDI</b>                                                             |               |                                                                                                                                                                                                |
| <b>Rx PC</b>                                                            | OFF, ON       | Specifies whether program change is received (ON) or not received (OFF).                                                                                                                       |
| <b>Rx Bank</b>                                                          | OFF, ON       | Specifies whether bank select is received (ON) or not received (OFF).                                                                                                                          |
| <b>Rx Bend</b>                                                          | OFF, ON       | Specifies whether pitch bend is received (ON) or not received (OFF).                                                                                                                           |
| <b>Rx Poly Pres</b>                                                     | OFF, ON       | Specifies whether polyphonic aftertouch is received (ON) or not received (OFF).                                                                                                                |
| <b>Rx Ch Pres</b>                                                       | OFF, ON       | Specifies whether channel aftertouch is received (ON) or not received (OFF).                                                                                                                   |
| <b>Rx Mod</b>                                                           | OFF, ON       | Specifies whether modulation is received (ON) or not received (OFF).                                                                                                                           |
| <b>Rx Volume</b>                                                        | OFF, ON       | Specifies whether volume is received (ON) or not received (OFF).                                                                                                                               |
| <b>Rx Pan</b>                                                           | OFF, ON       | Specifies whether pan is received (ON) or not received (OFF).                                                                                                                                  |
| <b>Rx Exp</b>                                                           | OFF, ON       | Specifies whether expression is received (ON) or not received (OFF).                                                                                                                           |
| <b>Rx Hold-1</b>                                                        | OFF, ON       | Specifies whether hold 1 is received (ON) or not received (OFF).                                                                                                                               |
| <b>Rx Ch</b>                                                            | 1–16          | Specifies the MIDI receive channel of each part.<br>* If Tx Mode is ON, this is also used as the MIDI transmit channel setting.                                                                |
| <b>SCALE</b>                                                            |               |                                                                                                                                                                                                |
| <b>Part 1–4 only</b><br><br><br><br><br><br><br><br><br><br><b>Type</b> | CUSTOM        | <b>Custom:</b> This lets you create a custom scale.                                                                                                                                            |
|                                                                         | EQUAL         | <b>Equal Temperament:</b> This tuning divides an octave into 12 equal parts. Every interval produces about the same amount of slight dissonance.                                               |
|                                                                         | JUST-MAJ      | <b>Just (Major):</b> This scale eliminates dissonance in fifths and thirds. It is unsuited to playing melodies and cannot be transposed, but is capable of beautiful sonorities.               |
|                                                                         | JUST-MIN      | <b>Just (Minor):</b> The scales of the major and minor just intonations are different. You can get the same effect with the minor scale as with the major scale.                               |
|                                                                         | PYTHAGORE     | <b>Pythagorean:</b> This scale, devised by the philosopher Pythagoras, eliminates dissonance in fourths and fifths. Dissonance is produced in thirds, but melodies are euphonious.             |
|                                                                         | KIRNBERGE     | <b>Kirnberger:</b> This scale is a modification of the meantone and just intonations that permits greater freedom in transposition to other keys. Performances are possible in all keys (III). |
|                                                                         | MEANTONE      | <b>Meantone:</b> This scale makes some compromises in just intonation, enabling transposition to other keys.                                                                                   |
|                                                                         | WERCKMEIS     | <b>Werckmeister:</b> This is a combination of the Meantone and Pythagorean scales. Performances are possible in all keys (first technique, III).                                               |
|                                                                         | ARABIC        | <b>Arabic Scale:</b> This scale is suitable for Arabic music.                                                                                                                                  |
|                                                                         | <b>Key</b>    | C, C#, D, D#, E, F, F#, G, G#, A, A#, B<br>Sets the keynote.                                                                                                                                   |
| <b>C–B</b>                                                              | -64–+63       | Finely adjusts the pitch.                                                                                                                                                                      |
| <b>EQ</b>                                                               |               |                                                                                                                                                                                                |
| <b>Switch</b>                                                           | OFF, ON       | Turns the equalizer (EQ) on/off.                                                                                                                                                               |
| <b>In Gain</b>                                                          | -24–+24 [dB]  | Specifies the amount of boost/cut for the input sound.                                                                                                                                         |
| <b>Low Gain</b>                                                         | -24–+24 [dB]  | Specifies the amount of boost/cut for the low-frequency region.                                                                                                                                |
| <b>Low Freq</b>                                                         | 20–16000 [Hz] | Specifies the frequency of the low-frequency region.                                                                                                                                           |
| <b>Mid Gain</b>                                                         | -24–+24 [dB]  | Specifies the amount of boost/cut for the mid-frequency region.                                                                                                                                |
| <b>Mid Freq</b>                                                         | 20–16000 [Hz] | Specifies the frequency of the mid-frequency region.                                                                                                                                           |
| <b>Mid Q</b>                                                            | 0.5–16.0      | Specifies the width of the mid-frequency region. Higher values produce a narrower width.                                                                                                       |

| Parameter | Value         | Explanation                                                      |
|-----------|---------------|------------------------------------------------------------------|
| High Gain | -24--+24 [dB] | Specifies the amount of boost/cut for the high-frequency region. |
| High Freq | 20–16000 [Hz] | Specifies the frequency of the high-frequency region.            |

## SCENE ZONE

| Parameter         | Value                                                                                                                                        | Explanation                                                                                                                                                               |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keyboard Sw       | OFF, ON                                                                                                                                      | Turns on/off the part played by the keyboard.                                                                                                                             |
| Key Rng Low       | C--G9                                                                                                                                        | Set the keyboard range in which each part will sound.<br>Make these settings when you want different key ranges to play different tones.                                  |
| Key Rng Upp       | C--G9                                                                                                                                        | Specify the lower limit (Key Rng Low) and upper limit (Key Rng Upp) of the key range.                                                                                     |
| Key Fade Low      | 0–127                                                                                                                                        | Specifies the degree to which the part is sounded by notes played below the Key Rng Low. If you don't want the tone to sound at all, set this parameter to "0."           |
| Key Fade Upp      | 0–127                                                                                                                                        | Specifies the degree to which the part is sounded by notes played above the Key Rng Upp. If you don't want the tone to sound at all, set this parameter to "0."           |
| Velo Rng Low      | 1–127                                                                                                                                        | Specify the lower limit (Velo Rng Low) and upper limit (Velo Rng Upp) of the velocities that will sound the tone.                                                         |
| Velo Rng Upp      | 1–127                                                                                                                                        | Make these settings when you want different tones to sound depending on keyboard playing dynamics.                                                                        |
| Velo FadeLow      | 0–127                                                                                                                                        | Specifies the degree to which the part is sounded by notes played more softly than Velo Rng Low. If you don't want the tone to sound at all, set this parameter to "0."   |
| Velo FadeUpp      | 0–127                                                                                                                                        | Specifies the degree to which the part is sounded by notes played more strongly than Velo Rng Upp. If you don't want the tone to sound at all, set this parameter to "0." |
| <b>Ctrl Rx</b>    | * If a function that is not in units of an individual part is assigned to a controller, the operation is received regardless of the setting. |                                                                                                                                                                           |
| Rx S1 -S3         | OFF, ON                                                                                                                                      | Specifies whether [S1]–[S3] button operations are received (ON) or not received (OFF).                                                                                    |
| Rx SL1 -SL2       | OFF, ON                                                                                                                                      | Specifies whether [SL1], [SL2] slider operations are received (ON) or not received (OFF).                                                                                 |
| Rx HoldPdl        | OFF, ON                                                                                                                                      | Specifies whether hold pedal operations are received (ON) or not received (OFF).                                                                                          |
| Rx CtrlPdl        | OFF, ON                                                                                                                                      | Specifies whether Ctrl pedal operations are received (ON) or not received (OFF).                                                                                          |
| <b>Master Kbd</b> |                                                                                                                                              |                                                                                                                                                                           |
| Tx Mode           | ON, OFF, MKB                                                                                                                                 | Specifies whether MIDI messages are transmitted (ON) or not transmitted (OFF).<br>If you're using this unit as a master keyboard, choose "MKB."                           |
| Mkb Ch            | 1–16                                                                                                                                         | Specifies the transmit channel for MIDI messages of the keyboard part.                                                                                                    |
| Mkb MSB           | OFF, 0–127                                                                                                                                   | Here you can enter numerical values for program number and bank select MSB/LSB to switch sounds on an external MIDI device.                                               |
| Mkb LSB           | OFF, 0–127                                                                                                                                   |                                                                                                                                                                           |
| Mkb PC            | OFF, 1–128                                                                                                                                   |                                                                                                                                                                           |
| Mkb Volume        | OFF, 0–127                                                                                                                                   | Adjusts the volume of an external MIDI device.                                                                                                                            |

## SCENE PART MFX

| Parameter      | Value                                                                                                                                                                                                                                                                                                                                                                                     | Explanation                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| FillwToneMFX   | OFF, ON                                                                                                                                                                                                                                                                                                                                                                                   | If this is "OFF," the following parameters are shown.                                       |
| Categ          | → "MFX List" (p. 36)                                                                                                                                                                                                                                                                                                                                                                      |                                                                                             |
| Type           | → "MFX List" (p. 36)                                                                                                                                                                                                                                                                                                                                                                      |                                                                                             |
| Switch         | OFF, ON                                                                                                                                                                                                                                                                                                                                                                                   | Switches the MFX on/off.                                                                    |
| MFX Parameters | → "MFX List" (p. 36)                                                                                                                                                                                                                                                                                                                                                                      |                                                                                             |
| Cho Send       | 0–127                                                                                                                                                                                                                                                                                                                                                                                     | Adjusts the amount of chorus.<br>If you don't want to add the chorus effect, set it to "0." |
| Rev Send       | 0–127                                                                                                                                                                                                                                                                                                                                                                                     | Adjusts the amount of reverb.<br>If you don't want to add the reverb effect, set it to "0." |
| Src1–4         | Specifies the MIDI message that will control the corresponding MFX CONTROL parameter.                                                                                                                                                                                                                                                                                                     |                                                                                             |
|                | OFF                                                                                                                                                                                                                                                                                                                                                                                       | MFX CONTROL will not be used.                                                               |
|                | CC01–31                                                                                                                                                                                                                                                                                                                                                                                   | Controller number 1–31                                                                      |
|                | CC33–95                                                                                                                                                                                                                                                                                                                                                                                   | Controller number 33–95                                                                     |
|                | BEND                                                                                                                                                                                                                                                                                                                                                                                      | Pitch bend                                                                                  |
|                | AFT                                                                                                                                                                                                                                                                                                                                                                                       | Aftertouch                                                                                  |
| Sens1–4        | Use the controller that is assigned by the System Control Source 1–4.                                                                                                                                                                                                                                                                                                                     |                                                                                             |
|                | Specifies the depth of MFX CONTROL.<br>Specify a positive (+) value if you want to change the value of the assigned destination in a positive direction (larger, toward the right, faster, etc.), or specify a negative (-) value if you want to change the value in a negative direction (smaller, toward the left, slower, etc.). Larger values will allow a greater amount of control. |                                                                                             |
|                | -63--+63                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                             |
|                |                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             |
|                |                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             |
| Asgn1–4        | → "MFX List" (p. 36)                                                                                                                                                                                                                                                                                                                                                                      |                                                                                             |

\* These parameters are not shown if the model assigned to the part is RD-PIANO.

## SCENE EFFECT: Dly (Delay)

If the “SYSTEM EFFECT: Dly” (p. 34) Source parameter is set to “SCENE,” the following parameters are shown, allowing you to edit the delay type and other parameters.

If the parameter is set to “SYS,” the screen indicates “Source is System,” and the following parameters are not shown.

| Parameter        | Value                                                                                                                                                              | Explanation                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Switch           | OFF, ON                                                                                                                                                            | Switches the delay on/off.                                  |
| DlyType          | → “Delay Parameters” (p. 7)                                                                                                                                        |                                                             |
| Level            | 0–127                                                                                                                                                              | Specifies the output level of the sound with delay applied. |
| Rev Send         | 0–127                                                                                                                                                              | Specifies the send level to reverb.                         |
| Delay parameters | Edit the parameters of the selected delay. The available parameters differ depending on the type of chorus you selected in DlyType.<br>→ “Delay Parameters” (p. 7) |                                                             |

## Delay Parameters

### 00 OFF

### 01 Delay

This is a stereo delay.

| Parameter | Value                         | Explanation                                                                                                              |
|-----------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly Sync  | OFF, ON                       | If this is “ON,” the delay synchronizes with the tempo.                                                                  |
| Dly Msec  | 1–1300                        | Adjusts the delay time from the direct sound until the delay sound is heard.                                             |
| Dly Note  | Note (*1)                     |                                                                                                                          |
| Feedback  | -98–+98 [%]                   | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp   | 200–8000 [Hz],<br>BYPASS (*2) | Adjusts the frequency above which the delay sound fed back to the effect is filtered out (BYPASS: no cut).               |

### 02 T-Ctrl Dly (Time Control Delay)

A stereo delay in which the delay time can be varied smoothly.

| Parameter    | Value                         | Explanation                                                                                                                                                                                      |
|--------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dly Sync     | OFF, ON                       | If this is “ON,” the delay synchronizes with the tempo.                                                                                                                                          |
| Dly Msec     | 1–1300                        | Adjusts the delay time from the direct sound until the delay sound is heard.                                                                                                                     |
| Dly Note     | Note (*1)                     |                                                                                                                                                                                                  |
| Acceleration | 0–15                          | When you change the delay time, this specifies the time over which the current delay time changes to the specified delay time. This affects the speed of pitch change as well as the delay time. |
| Feedback     | -98–+98 [%]                   | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.                                                                         |
| HF Damp      | 200–8000 [Hz],<br>BYPASS (*2) | Adjusts the proportion of the delay sound that is fed back into the effect (BYPASS: no cut).                                                                                                     |

### 03 Delay → Trem (Delay → Tremolo)

Tremolo is applied to the delay sound.

| Parameter | Value     | Explanation                                                                  |
|-----------|-----------|------------------------------------------------------------------------------|
| Input     | MONAURAL  | The input is mono-mixed.                                                     |
|           | STEREO    | The sound is input in stereo.                                                |
| Dly Sync  | OFF, ON   | If this is “ON,” the delay synchronizes with the tempo.                      |
| Dly Msec  | 1–1300    | Adjusts the delay time from the direct sound until the delay sound is heard. |
| Dly Note  | Note (*1) |                                                                              |

| Parameter   | Value                         | Explanation                                                                                                              |
|-------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Feedback    | -98–+98 [%]                   | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp     | 200–8000 [Hz],<br>BYPASS (*2) | Adjusts the frequency above which the filtered out (BYPASS: no cut).                                                     |
| Trm Switch  | OFF, ON                       | Switches the tremolo effect on/off                                                                                       |
| Trm ModWave | Modulation Wave               |                                                                                                                          |
|             | TRI                           | Triangle wave                                                                                                            |
|             | SQR                           | Square wave                                                                                                              |
|             | SIN                           | Sine wave                                                                                                                |
|             | SAW1                          | Sawtooth wave                                                                                                            |
|             | SAW2                          |                                                                                                                          |
|             | TRP                           | Trapezoidal wave                                                                                                         |
| Trm Sync    | OFF, ON                       | If this is “ON,” the tremolo synchronizes with the tempo.                                                                |
| Trm Hz      | 0.05–10.00 [Hz]               | Adjusts the tremolo rate.                                                                                                |
| Trm Note    | Note (*1)                     |                                                                                                                          |
| Trm Depth   | 0–127                         | Adjusts the tremolo depth.                                                                                               |

### 04 2Tap PanDly (2 Tap Pan Delay)

Delayed sound is heard from the two locations you specify.

| Parameter  | Value                         | Explanation                                                                                                              |
|------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly Sync   | OFF, ON                       | If this is “ON,” the delay synchronizes with the tempo.                                                                  |
| Dly Msec   | 1–1300                        | Adjusts the delay time from the direct sound until the second delay sound is heard.                                      |
| Dly Note   | Note (*1)                     |                                                                                                                          |
| Feedback   | -98–+98 [%]                   | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200–8000 [Hz],<br>BYPASS (*2) | Adjusts the frequency above which the delay sound fed back to the effect is filtered out (BYPASS: no cut).               |
| Dly1 Pan   | L64–63R                       | Adjusts the stereo location of delay 1.                                                                                  |
| Dly2 Pan   | L64–63R                       | Adjusts the stereo location of delay 2.                                                                                  |
| Dly1 Level | 0–127                         | Adjusts the volume of delay 1.                                                                                           |
| Dly2 Level | 0–127                         | Adjusts the volume of delay 2.                                                                                           |

### 05 3Tap PanDly (3 Tap Pan Delay)

Delayed sound is heard from the three locations you specify.

| Parameter  | Value                         | Explanation                                                                                                              |
|------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly Sync   | OFF, ON                       | If this is “ON,” the delay synchronizes with the tempo.                                                                  |
| Dly Msec   | 1–2600                        | Adjusts the delay time from the direct sound until the third delay sound is heard.                                       |
| Dly Note   | Note (*1)                     |                                                                                                                          |
| Feedback   | -98–+98 [%]                   | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200–8000 [Hz],<br>BYPASS (*2) | Adjusts the frequency above which the delay sound fed back to the effect is filtered out (BYPASS: no cut).               |
| Dly1 Pan   | L64–63R                       | Adjusts the stereo location of delay 1.                                                                                  |
| Dly2 Pan   | L64–63R                       | Adjusts the stereo location of delay 2.                                                                                  |
| Dly3 Pan   | L64–63R                       | Adjusts the stereo location of delay 3.                                                                                  |
| Dly1 Level | 0–127                         | Adjusts the volume of delay 1.                                                                                           |
| Dly2 Level | 0–127                         | Adjusts the volume of delay 2.                                                                                           |
| Dly3 Level | 0–127                         | Adjusts the volume of delay 3.                                                                                           |

#### NOTE

(\*1) 1/64T, 1/64, 1/32T, 1/32, 1/16T, 1/32, 1/16, 1/8T, 1/16, 1/8, 1/4T, 1/8, 1/4, 1/2T, 1/4, 1/2, 1T, 1/2, 1, 2T, 1, 2

(\*2) 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz], BYPASS

## SCENE EFFECT: Rev (Reverb)

If the "SYSTEM EFFECT: Rev" (p. 34) Source parameter is set to "SCENE," the following parameters are shown, allowing you to edit the reverb type and other parameters.

If the parameter is set to "SYS," the screen indicates "Source is System.," and the following parameters are not shown.

| Parameter         | Value                                                                                                                                                                     | Explanation                                                  |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Switch            | OFF, ON                                                                                                                                                                   | Switches the reverb on/off.                                  |
| RevType           | ➔ "Reverb Parameters" (p. 8)                                                                                                                                              |                                                              |
| Level             | 0–127                                                                                                                                                                     | Specifies the output level of the sound with reverb applied. |
| Reverb Parameters | Edit the parameters of the selected reverb type. The available parameters differ depending on the type of reverb you selected in RevType.<br>➔ "Reverb Parameters" (p. 8) |                                                              |

## Reverb Parameters

### 00 OFF

### 01 INTEGRA7Rev (INTEGRA 7 Reverb)

| Parameter | Value                             | Explanation                                                                                                                                                                                      |
|-----------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Char      | ROOM1, ROOM2, HALL1, HALL2, PLATE | Selects the type of reverb.                                                                                                                                                                      |
| PreDelay  | 0–100                             | Adjusts the delay time from the direct sound until the reverb sound is heard.                                                                                                                    |
| Time      | 0.1–10.0 [sec]                    | Adjusts the decay length of the reverb sound.                                                                                                                                                    |
| Density   | 0–127                             | Adjusts the density of the reverb sound.                                                                                                                                                         |
| Diffusion | 0–127                             | Adjusts the change in the density of the reverb over time. The higher the value, the more the density increases with time. The effect of this setting is most pronounced with long reverb times. |
| LF Damp   | 0–100                             | Adjusts the low-frequency portion of the reverb.                                                                                                                                                 |
| HF Damp   | 0–100                             | Adjusts the high-frequency portion of the reverb.                                                                                                                                                |
| Spread    | 0–127                             | Adjusts the reverb spread.                                                                                                                                                                       |
| Tone      | 0–127                             | Adjusts the tonal character of the reverb.                                                                                                                                                       |

### 02 Warm Hall

| Parameter | Value                      | Explanation                                                                                           |
|-----------|----------------------------|-------------------------------------------------------------------------------------------------------|
| PreDelay  | 0.0–100.0                  | Adjusts the delay time from the direct sound until the reverb sound is heard.                         |
| Time      | 0.3–30.0 [sec]             | Adjusts the decay length of the reverb sound.                                                         |
| Pre LPF   | 20–15000 [Hz], BYPASS (*1) | Adjusts the frequency above which to cut the high-frequency portion of the sound entering the reverb. |
| Pre HPF   | BYPASS, 16–15000 [Hz] (*2) | Adjusts the frequency below which to cut the low-frequency portion of the sound entering the reverb.  |
| PreLpLPF  | 20–15000 [Hz], BYPASS (*1) | Adjusts the frequency above which to cut the high-frequency portion of the extended reverberation.    |
| Diffusion | 0–127                      | Adjusts the change in the density of the reverb over time.                                            |
| HF Damp F | 1000–8000 [Hz] (*3)        | Adjusts the frequency above which to cut the high-frequency portion of the reverb.                    |
| HF Damp R | 0.1–1.0                    | Adjusts the amount by which to attenuate the high-frequency portion of the reverb.                    |

### 03 Hall

| Parameter | Value                       | Explanation                                                                                                    |
|-----------|-----------------------------|----------------------------------------------------------------------------------------------------------------|
| PreDelay  | 0.0–100.0                   | Adjusts the delay time from the direct sound until the reverb sound is heard.                                  |
| Time      | 0–127                       | Adjusts the decay length of the reverb sound.                                                                  |
| Size      | 1–8                         | Adjusts the size of room/hall.                                                                                 |
| High Cut  | 160–12500 [Hz], BYPASS (*4) | Adjusts the frequency above which the high-frequency portion of the final output sound is cut (BYPASS: no cut) |
| Density   | 0–127                       | Adjusts the density of the reverb sound.                                                                       |
| Diffusion | 0–127                       | Adjusts how reverb density increases over time. This effect is especially noticeable with long reverb times.   |
| LF Damp F | 50–4000 [Hz] (*5)           | Adjusts the frequency below which the low-frequency portion of the reverb sound is cut.                        |
| LF Damp G | –36–0 [dB]                  | Adjusts the LF damp attenuation amount (0: no effect).                                                         |
| HF DampF  | 4000–12500 [Hz] (*6)        | Adjusts the frequency above which the high-frequency portion of the reverb sound is cut.                       |
| HF Damp G | –36–0 [dB]                  | Adjusts the HF damp attenuation amount (0: no effect).                                                         |

### 04 GS Reverb

| Parameter | Value                                                      | Explanation                                                                               |
|-----------|------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Char      | ROOM1, ROOM2, ROOM3, HALL1, HALL2, PLATE, DELAY, PAN-DELAY | Selects the type of reverb.                                                               |
| Pre LPF   | 0–7                                                        | Adjusts the amount of high-frequency attenuation for the sound being input to the reverb. |
| Time      | 0–127                                                      | Adjusts the decay length of the reverb sound.                                             |
| Feedback  | 0–127                                                      | Adjusts the level at which the reverb sound is returned to the input.                     |

### 05 SRV-2000

| Parameter   | Value                                                                        | Explanation                                                                                             |
|-------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Selection   | R0.3, R1.0, R7.0, R15, R22, R26, R32, R37, H15, H22, H26, H32, H37, P-B, P-A | Selects the type of reverb offered by the Roland SRV-2000 digital reverb.                               |
| PreDelay    | 0–160                                                                        | Adjusts the delay time from the direct sound until the reverb sound is heard.                           |
| Time        | 0.1–99.0 [sec]                                                               | Adjusts the decay length of the reverb sound.                                                           |
| HF Damp     | 0.05–1.00                                                                    | Adjusts the high-frequency portion of the reverb.                                                       |
| Density     | 0–9                                                                          | Adjusts the density of the late reverberation.                                                          |
| Attack Gain | 0–9                                                                          | Adjusts the gain of the early reflections.                                                              |
| Attack Time | 0–9                                                                          | Adjusts the time of the early reflections.                                                              |
| ER Density  | 0–9                                                                          | Adjusts the density of the early reflections.                                                           |
| ER Level    | 0–99                                                                         | Adjusts the volume of the early reflections.                                                            |
| Low Freq    | 0.04–1.00 [kHz]                                                              | Specifies the frequency of the low range.                                                               |
| Low Gain    | –24–+12 [dB]                                                                 | Adjusts the gain of the low range.                                                                      |
| Mid Freq    | 0.25–9.99 [kHz]                                                              | Specifies the frequency of the middle range.                                                            |
| Mid Gain    | –24–+12 [dB]                                                                 | Adjusts the gain of the middle range.                                                                   |
| Mid Q       | 0.2–9.0                                                                      | Adjusts the width of the middle range. Set a higher value to narrow the range to be affected.           |
| HighFreq    | 0.80–9.99 [kHz]                                                              | Specifies the frequency of the high range.                                                              |
| HighGain    | –24–+12 [dB]                                                                 | Adjusts the gain of the high range.                                                                     |
| High Q      | 0.2–9.0                                                                      | Specifies the width of the high-frequency range. Set a higher value to narrow the range to be affected. |

## 06 SRV-2000NL (NON-LINEAR)

| Parameter  | Value            | Explanation                                                                                             |
|------------|------------------|---------------------------------------------------------------------------------------------------------|
| PreDelay   | 0–120            | Adjusts the delay time from the direct sound until the reverb sound is heard.                           |
| ReverbTime | -0.9–+99.0 [sec] | Adjusts the decay length of the reverb sound.                                                           |
| GateTime   | 10–450           | Adjusts the time from when the reverb starts being heard until the reverb sound is cut off.             |
| Low Freq   | 0.04–1.00 [kHz]  | Specifies the frequency of the low range.                                                               |
| Low Gain   | -24–+12 [dB]     | Adjusts the gain of the low range.                                                                      |
| Mid Freq   | 0.25–9.99 [kHz]  | Specifies the frequency of the middle range.                                                            |
| Mid Gain   | -24–+12 [dB]     | Adjusts the gain of the middle range.                                                                   |
| Mid Q      | 0.2–9.0          | Specifies the width of the middle range. Set a higher value to narrow the range to be affected.         |
| HighFreq   | 0.80–9.99 [kHz]  | Specifies the frequency of the high range.                                                              |
| HighGain   | -24–+12 [dB]     | Gain of the high range.                                                                                 |
| Hi Q       | 0.2–9.0          | Specifies the width of the high-frequency range. Set a higher value to narrow the range to be affected. |

## 07 GM2 Reverb

| Parameter | Value                                                                              | Explanation                                   |
|-----------|------------------------------------------------------------------------------------|-----------------------------------------------|
| Char      | SMALL ROOM,<br>MEDIUM ROOM,<br>LARGE ROOM,<br>MEDIUM HALL,<br>LARGE HALL,<br>PLATE | Selects the type of reverb.                   |
| Time      | 0–127                                                                              | Adjusts the decay length of the reverb sound. |

## 08 GatedReverb

| Parameter | Value          | Explanation                                                                              |
|-----------|----------------|------------------------------------------------------------------------------------------|
| Type      | NORMAL         | This is a standard gate reverb.                                                          |
|           | REVERSE        | This is a reverb for which the sound ramps up in volume.                                 |
|           | SWEEP1         | The reverb sound moves from right to left.                                               |
|           | SWEEP2         | The reverb sound moves from left to right.                                               |
| Pre Delay | 0.0–100.0 [ms] | Adjusts the delay time from when the direct sound plays until the reverb sound is heard. |
| Gate Time | 5–500 [ms]     | Adjusts the decay length of the reverb sound.                                            |

### NOTE

- (\*1) 16, 20, 25, 32, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12500, 15000 [Hz], BYPASS
- (\*2) BYPASS, 16, 20, 25, 32, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12500, 15000 [Hz]
- (\*3) 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz]
- (\*4) 160, 200, 250, 320, 400, 500, 640, 800, 1000, 1250, 1600, 2000, 2500, 3200, 4000, 5000, 6400, 8000, 10000, 12500 [Hz], BYPASS
- (\*5) 50, 64, 80, 100, 125, 160, 200, 250, 320, 400, 500, 640, 800, 1000, 1250, 1600, 2000, 2500, 3200, 4000 [Hz]
- (\*6) 4000, 5000, 6400, 8000, 10000, 12500 [Hz]

## SCENE EFFECT: Cho (Chorus)

If the “SYSTEM EFFECT: Cho” (p. 34) Source parameter is set to “SCENE,” the following parameters are shown, allowing you to edit the chorus type and other parameters.

If the parameter is set to “SYS,” the screen indicates “Source is System,” and the following parameters are not shown.

| Parameter         | Value                                                                                                                                                                | Explanation                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Switch            | OFF, ON                                                                                                                                                              | Switches chorus on/off.                                      |
| ChoType           | → “Chorus Parameters” (p. 9)                                                                                                                                         |                                                              |
| Level             | 0–127                                                                                                                                                                | Specifies the output level of the sound with chorus applied. |
| Rev Send          | 0–127                                                                                                                                                                | Specifies the send level to reverb.                          |
| Chorus Parameters | Edit the parameters of the selected chorus. The available parameters differ depending on the type of chorus you selected in ChoType.<br>→ “Chorus Parameters” (p. 9) |                                                              |

## Chorus Parameters

### 00 OFF

### 01 Chorus

This is a stereo chorus.

| Parameter | Value | Explanation                                                       |
|-----------|-------|-------------------------------------------------------------------|
| Rate      | 0–127 | Adjusts the frequency of modulation.                              |
| Depth     | 0–127 | Adjusts the depth of modulation.                                  |
| Feedback  | 0–127 | Adjusts the level at which chorus sound is returned to the input. |

### 02 CE-1 (Chorus)

This models the classic BOSS CE-1 chorus effect unit. It provides a chorus sound with a distinctively analog warmth.

| Parameter | Value | Explanation               |
|-----------|-------|---------------------------|
| Intensity | 0–127 | Adjusts the chorus depth. |

### 03 SDD-320 (Dimension D)

This models Roland’s DIMENSION D (SDD-320). It provides a clear chorus sound.

| Parameter | Value                     | Explanation        |
|-----------|---------------------------|--------------------|
| Mode      | 1, 2, 3, 4, 1+4, 2+4, 3+4 | Switches the mode. |

### 04 Delay

This is a stereo delay.

| Parameter | Value                      | Explanation                                                                                                              |
|-----------|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly Sync  | OFF, ON                    | If this is “ON,” the delay synchronizes with the tempo.                                                                  |
| Dly Msec  | 1–1300                     | Adjusts the delay time from the direct sound until the delay sound is heard.                                             |
| Dly Note  | Note (*1)                  |                                                                                                                          |
| Feedback  | -98–+98 [%]                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp   | 200–8000 [Hz], BYPASS (*2) | Adjusts the frequency above which the delay sound fed back to the effect is filtered out (BYPASS: no cut).               |

## 05 T-Ctrl Dly (Time Control Delay)

A stereo delay in which the delay time can be varied smoothly.

| Parameter    | Value                      | Explanation                                                                                                                                                                                      |
|--------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dly Sync     | OFF, ON                    | If this is "ON," the delay synchronizes with the tempo.                                                                                                                                          |
| Dly Msec     | 1–1300                     | Adjusts the delay time from the direct sound until the delay sound is heard.                                                                                                                     |
| Dly Note     | Note (*1)                  |                                                                                                                                                                                                  |
| Acceleration | 0–15                       | When you change the delay time, this specifies the time over which the current delay time changes to the specified delay time. This affects the speed of pitch change as well as the delay time. |
| Feedback     | -98–+98 [%]                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.                                                                         |
| HF Damp      | 200–8000 [Hz], BYPASS (*2) | Adjusts the proportion of the delay sound that is fed back into the effect (BYPASS: no cut).                                                                                                     |

## 06 Delay → Trem (Delay → Tremolo)

Tremolo is applied to the delay sound.

| Parameter   | Value                      | Explanation                                                                                                              |
|-------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Input       | MONAURAL                   | The input is mono-mixed.                                                                                                 |
|             | STEREO                     | The sound is input in stereo.                                                                                            |
| Dly Sync    | OFF, ON                    | If this is "ON," the delay synchronizes with the tempo.                                                                  |
| Dly Msec    | 1–1300                     | Adjusts the delay time from the direct sound until the delay sound is heard.                                             |
| Dly Note    | Note (*1)                  |                                                                                                                          |
| Feedback    | -98–+98 [%]                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp     | 200–8000 [Hz], BYPASS (*2) | Adjusts the frequency above which the filtered out (BYPASS: no cut).                                                     |
| Trm Switch  | OFF, ON                    | Switches the tremolo effect on/off                                                                                       |
| Trm ModWave | Modulation Wave of panning |                                                                                                                          |
|             | TRI                        | Triangle wave                                                                                                            |
|             | SQR                        | Square wave                                                                                                              |
|             | SIN                        | Sine wave                                                                                                                |
|             | SAW1                       | Sawtooth wave                                                                                                            |
|             | SAW2                       |                                                                                                                          |
|             | TRP                        | Trapezoidal wave                                                                                                         |
| Trm Sync    | OFF, ON                    | If this is "ON," the tremolo synchronizes with the tempo.                                                                |
| Trm Hz      | 0.05–10.00 [Hz]            | Tremolo rate                                                                                                             |
| Trm Note    | Note (*1)                  |                                                                                                                          |
| Trm Depth   | 0–127                      | Tremolo depth                                                                                                            |

## 07 2Tap PanDly (2 Tap Pan Delay)

Delayed sound is heard from the two locations you specify.

| Parameter  | Value                      | Explanation                                                                                                              |
|------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly Sync   | OFF, ON                    | If this is "ON," the delay synchronizes with the tempo.                                                                  |
| Dly Msec   | 1–1300                     | Adjusts the delay time from the direct sound until the second delay sound is heard.                                      |
| Dly Note   | Note (*1)                  |                                                                                                                          |
| Feedback   | -98–+98 [%]                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200–8000 [Hz], BYPASS (*2) | Adjusts the frequency above which the delay sound fed back to the effect is filtered out (BYPASS: no cut).               |
| Dly1 Pan   | L64–63R                    | Stereo location of Delay 1                                                                                               |
| Dly2 Pan   | L64–63R                    | Stereo location of Delay 2                                                                                               |
| Dly1 Level | 0–127                      | Volume of delay 1                                                                                                        |
| Dly2 Level | 0–127                      | Volume of delay 2                                                                                                        |

## 08 3Tap PanDly (3 Tap Pan Delay)

Delayed sound is heard from the three locations you specify.

| Parameter  | Value                      | Explanation                                                                                                              |
|------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly Sync   | OFF, ON                    | If this is "ON," the delay synchronizes with the tempo.                                                                  |
| Dly Msec   | 1–2600                     | Adjusts the delay time from the direct sound until the third delay sound is heard.                                       |
| Dly Note   | Note (*1)                  |                                                                                                                          |
| Feedback   | -98–+98 [%]                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200–8000 [Hz], BYPASS (*2) | Adjusts the frequency above which the delay sound fed back to the effect is filtered out (BYPASS: no cut).               |
| Dly1 Pan   | L64–63R                    | Adjusts the stereo location of Delay 1.                                                                                  |
| Dly2 Pan   | L64–63R                    | Adjusts the stereo location of Delay 2.                                                                                  |
| Dly3 Pan   | L64–63R                    | Adjusts the stereo location of Delay 3.                                                                                  |
| Dly1 Level | 0–127                      | Adjusts the volume of delay 1.                                                                                           |
| Dly2 Level | 0–127                      | Adjusts the volume of delay 2.                                                                                           |
| Dly3 Level | 0–127                      | Adjusts the volume of delay 3.                                                                                           |

## 09 JUNO Chorus (JUNO-106 Chorus)

This models the chorus effects of the Roland JUNO-106.

| Parameter   | Value                                                  | Explanation                                                                                                                            |
|-------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Mode        | I, II, I+II, JX I, JX II, III, I+III, II+III, I+II+III | Selects the type of Chorus.<br><b>I+II, I+III, II+III, I+II+III:</b> The state in which multiple buttons are pressed at the same time. |
| Noise Level | 0–127                                                  | Amount of noise produced by the chorus                                                                                                 |

## 10 JV Chorus

| Parameter   | Value           | Explanation                                                                              |
|-------------|-----------------|------------------------------------------------------------------------------------------|
| Filter Type | OFF             | The filter is not used.                                                                  |
|             | LPF             | This filter cuts off the high frequencies.                                               |
|             | HPF             | This filter cuts off the low frequencies.                                                |
| Cutoff Freq | 200–8000 [Hz]   | Adjusts the center frequency used when the filter cuts a specific frequency region.      |
| Pre Delay   | 0.0–100.0 [ms]  | Adjusts the delay time from when the direct sound plays until the reverb sound is heard. |
| Rate Sync   | OFF, ON         | When this is ON, the delay synchronizes with the tempo.                                  |
| Rate Hz     | 0.05–10.00 [Hz] | Adjusts the frequency of modulation.                                                     |
| Rate Note   | Note (*1)       |                                                                                          |
| Depth       | 0–127           | Adjusts the depth of modulation.                                                         |
| Phase       | 0–180 [deg]     | Adjusts the depth of the chorus sound.                                                   |
| Feedback    | 0–127           | Adjusts how much of the sound that is fed into the chorus is returned to the input.      |

### NOTE

(\*1) 1/64T, 1/64, 1/32T, 1/32, 1/16T, 1/32, 1/16, 1/8T, 1/16, 1/8, 1/4T, 1/8, 1/4, 1/2T, 1/4, 1/2, 1T, 1/2, 1, 2T, 1, 2

(\*2) 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz], BYPASS



## SCENE EFFECT: OD (Overdrive)

| Parameter    | Value   | Explanation                                                                                 |
|--------------|---------|---------------------------------------------------------------------------------------------|
| Drive Sw     | OFF, ON | Turns overdrive on/off.                                                                     |
| Drive        | 0–127   | Adjusts the degree of distortion.                                                           |
| Cho Send Lev | 0–127   | Adjusts the amount of chorus.<br>If you don't want to add the chorus effect, set it to "0." |
| Rev Send Lev | 0–127   | Adjusts the amount of reverb.<br>If you don't want to add the reverb effect, set it to "0." |
| Dly Send Lev | 0–127   | Adjusts the amount of delay.<br>If you don't want to add the delay effect, set it to "0."   |

## ARP COMMON

| Parameter                   | Value                                          | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type                        | 00–65<br>* Refer to "Type/Rhythm list" (p. 12) | Specifies the arpeggio type.<br><b>What is changed by TYPE</b><br>(1) Arpeggio pattern and part arpeggio parameters of parts 1–4<br>(2) Sound (tone) and part level of parts 1–4<br>* For a part whose SCENE ZONE EDIT > Keyboard Sw is ON, the current sound (tone) and part level are maintained even if you change the TYPE.<br>* You can use the SYSTEM > ARPEGGIO > Set Tone setting to turn on/off the function of (2).                                                                                                                                                                              |
| Rytm                        | 00–65<br>* Refer to "Type/Rhythm list" (p. 12) | Specifies the arpeggio's rhythm type.<br><b>What is changed by RHYTHM</b><br>(1) Arpeggio pattern and part arpeggio parameters of part R<br>(2) Sound (drum kit) and part level of part R<br>* If the SCENE ZONE EDIT > Keyboard Sw is ON, the current sound (tone) and part level are maintained even if you change the RHYTHM.<br>* You can use the SYSTEM > ARPEGGIO > Set Drum Kit setting to specify whether this changes (ON) or does not change (OFF).<br>(3) Tempo (SCENE or SYSTEM)<br>* You can use SYSTEM > ARPEGGIO > Set Tempo to specify whether this changes (ON) or does not change (OFF). |
| G-Duration                  | -50–50                                         | Specifies a global duration value that applies a relative adjustment to the duration values of each part.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| G-Shuffle                   | 0–100                                          | Specifies the global shuffle value that applies a relative adjustment to the shuffle values of each part.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Switch                      | OFF, ON                                        | Arpeggio switch. This is linked with the panel button.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Hold Sw                     | OFF, ON                                        | Arpeggio hold switch. This is linked with the panel button.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Keys Sw                     | OFF, ON                                        | KEYS switch for the arpeggio PLAY DETECTOR. If this is "ON," the arpeggio pitch changes according to the key you play. This is linked with the panel button.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Beat Sw                     | OFF, ON                                        | BEAT switch for the arpeggio PLAY DETECTOR. If this is "ON," the arpeggio pattern changes according to the timing of your keyboard performance. This is linked with the panel button.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Detect Sens                 | 1–10                                           | Specifies the BEAT sensitivity of the arpeggio PLAY DETECTOR. Higher values make the pattern change more sensitively. If you're unable to reproduce the same pattern even when you're trying to play the keyboard in the same rhythm, lowering this value might help.                                                                                                                                                                                                                                                                                                                                      |
| In Range Low<br>In Range Up | C- –G9                                         | Specifies the range of keys detected for arpeggio performance.<br>Pressing a key outside the specified range will not affect the arpeggio function.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Type/Rhythm list

|    | Type        | Rytm        |
|----|-------------|-------------|
| 00 | OFF         | OFF         |
| 01 | UP 16th     | KICK        |
| 02 | DOWN 16S    | KICK+HH     |
| 03 | UP&DOWN TRI | KICK+CLAP   |
| 04 | RANDOM 8th  | LOOP        |
| 05 | POLY SYNCNP | LOOP-S      |
| 06 | I-UP 8-16   | I-BEAT[2]   |
| 07 | I-NO 4-8-16 | I-BEAT[3]-1 |
| 08 | I-P 4-8-16R | I-BEAT[3]-2 |
| 09 | I-NO SYNCNP | I-BEAT[0+3] |
| 10 | I-NO FREE   | I-BEAT[3]-3 |
| 11 | I-ENS P3-1  | I-CR78 120  |
| 12 | I-ENS P3-2  | I-CR78 116  |
| 13 | I-ENS P3-3S | I-CR78 112S |
| 14 | I-ENS P34-1 | I-90's 70   |
| 15 | I-ENS P34-2 | I-T808 120  |
| 16 | I-ENS P34-3 | I-T909 120  |
| 17 | I-ENS P34-4 | I-T808 120  |
| 18 | I-ENS P34-5 | I-T909 135  |
| 19 | I-ENS P34-6 | I-POP 80    |
| 20 | I-ENS P34-7 | I-T808 80   |
| 21 | I-ENS P34-8 | I-T808 131  |
| 22 | I-ENS P34-9 | I-CR78 112S |
| 23 | I-ENSP34-10 | I-T707 124  |
| 24 | I-ENSP34-11 | I-ANA 122   |
| 25 | I-ENSP34-12 | I-CR78 109  |
| 26 | I-ENSP34-13 | I-T909 135S |
| 27 | I-ENSP34-14 | I-CR78 118  |
| 28 | I-ENSP34-15 | I-ANA 118   |
| 29 | I-ENSP34-16 | I-T606 112  |
| 30 | I-ENSP34-17 | I-T808 149  |
| 31 | I-ENSP34-18 | I-CR78 124  |
| 32 | I-ENSP34-19 | I-T909 126  |
| 33 | I-ENSP34-20 | I-CR78 98S  |
| 34 | I-ENSP34-21 | I-STD 116S  |
| 35 | I-ENSP34-22 | I-T808 130  |
| 36 | I-ENSP34-23 | I-T606 135  |
| 37 | I-ENSP34-24 | I-CR78 160  |
| 38 | I-ENSP34-25 | I-ANA 130   |
| 39 | I-ENSP34-26 | I-CR78 152  |
| 40 | I-ENSP34-27 | I-T808 138  |
| 41 | I-ENSP234-1 | I-T808 108  |
| 42 | I-ENSP234-2 | I-CR78 98   |
| 43 | I-ENSP234-3 | I-T808 114S |
| 44 | I-ENSP234-4 | I-T606 126S |
| 45 | I-ENSP34-28 | I-HOUSE 120 |
| 46 | I-ENSP34-29 | I-HIPHOP109 |
| 47 | I-ENSP234-5 | I-T808 105  |
| 48 | I-ENSP234-6 | I-JAZZ 110  |
| 49 | I-ENSP234-7 | I-POWER 112 |
| 50 | I-ENSP34-30 | I-T909 125  |
| 51 | I-DLY SYNTH | I-HIPHOP142 |
| 52 | I-DLY PIANO | I-T707 123  |
| 53 | I-DLY PLUCK | I-T808 120  |
| 54 | I-BASS AUTO | I-T909 118  |
| 55 | I-BASS DOWN | I-D&B 140   |
| 56 | I-ENSP34-31 | I-T626 98   |
| 57 | I-ENSP34-32 | I-80'S 95   |
| 58 | I-ENSP34-33 | I-T909 110  |
| 59 | I-ENSP34-34 | I-T808 172  |

|    | Type        | Rytm       |
|----|-------------|------------|
| 60 | I-ENSP34-35 | I-T7&7 175 |
| 61 | I-ENSP34-36 | I-EDM 95   |
| 62 | I-ENSP34-37 | I-CR78 88  |
| 63 | I-ENSP34-38 | I-T7&7 96  |
| 64 | I-ENSP34-39 | I-T909 108 |
| 65 | I-ENSP34-40 | I-T626 105 |



## ARP PART

| Parameter                                           | I-ARP | ARP | STEP | Value             | Explanation                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|-------|-----|------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Switch                                              | ✓     | ✓   | ✓    | OFF, ON           | Specifies whether each part's arpeggio is "ON" or "OFF" when the panel arpeggio switch is ON.                                                                                                                                                                                                                                                             |
| Arp Mode                                            | ✓     | ✓   | ✓    | I-ARP, ARP, STEP  | Sets how the arpeggio operates for each part.<br><b>I-ARP:</b> Analyzes your keyboard performance and plays the arpeggio pattern that best matches it.<br><b>ARP:</b> Plays a simple arpeggio pattern.<br><b>STEP:</b> Plays the pattern data you created using the STEP EDIT function. Use this setting when you want to play original arpeggio phrases. |
| I-ARP Style (*1)<br>* Refer to "Style List" (p. 14) | ✓     | -   | -    | 000–111           | Sets the style for the I-ARPEGGIO. You can select the ARP COMMON type for each part.                                                                                                                                                                                                                                                                      |
| ARP Style (*2)<br>* Refer to "Style List" (p. 14)   | -     | ✓   | -    | 1–128             | Sets the arpeggio style.                                                                                                                                                                                                                                                                                                                                  |
| Variation (*2)                                      | -     | ✓   | -    | 1–11              | Sets the arpeggio style variation. The number of available variations depends on the style.                                                                                                                                                                                                                                                               |
| Arp Step Sw (*2)                                    | -     | ✓   | -    | OFF, ON           | Plays the arpeggio using the pattern data you created with Step Edit.                                                                                                                                                                                                                                                                                     |
| Step Key Shift (*3)                                 | -     | -   | ✓    | OFF, ON           | The pattern data created in Step Edit is played, transposed according to the notes played on the keyboard.                                                                                                                                                                                                                                                |
| Motif (*4)                                          | ✓     | ✓   | -    |                   | Specifies the order in which the notes of the chord you play are sounded as an arpeggio.                                                                                                                                                                                                                                                                  |
|                                                     |       |     |      | UP                | Sounded consecutively starting at the lowest key you press.                                                                                                                                                                                                                                                                                               |
|                                                     |       |     |      | DOWN              | Sounded consecutively starting at the highest key you press.                                                                                                                                                                                                                                                                                              |
|                                                     |       |     |      | UP&DOWN           | Sounded consecutively from the lowest to the highest key, and then back down to the lowest.                                                                                                                                                                                                                                                               |
|                                                     |       |     |      | RANDOM            | The keys you press are sounded in random order.                                                                                                                                                                                                                                                                                                           |
|                                                     |       |     |      | NOTE ORDER        | Sounded in the order in which you press the keys.                                                                                                                                                                                                                                                                                                         |
|                                                     |       |     |      | RHYTHM            | Notes are sounded as specified by the arpeggio pattern, regardless of the pitches that you play on the keyboard. This is useful when playing a rhythm pattern.                                                                                                                                                                                            |
|                                                     |       |     |      | PHRASE            | The pitches specified by the arpeggio pattern are played, but shifted according to the pitches that you play. This is useful when you want to transpose the melody while the arpeggio plays.                                                                                                                                                              |
| Oct Range (*4)                                      | ✓     | ✓   | -    | -3+3              | When you play a chord, priority is given to starting with the lowest pitch. This is effective for a bass part.                                                                                                                                                                                                                                            |
|                                                     |       |     |      |                   | Specifies the range of octaves in which the arpeggio is sounded. You can specify whether the arpeggio is sounded in the octave(s) above (+) or below (-) the notes you play.                                                                                                                                                                              |
| Duration                                            | ✓     | ✓   | ✓    | 0–100 [%]         | Specifies the duration that the notes of the arpeggio pattern are sounded, as a proportion of the note length. You can set this to make the arpeggiated notes sound briefly for a staccato feel, or at their full duration for a tenuto feel.                                                                                                             |
| Grid                                                |       |     |      |                   |                                                                                                                                                                                                                                                                                                                                                           |
| Grid Length (*5)                                    | ✓     | -   | ✓    | 2–64              | Specifies the grid length for the arpeggio pattern.                                                                                                                                                                                                                                                                                                       |
| Grid Note                                           | ✓     | ✓   | ✓    |                   | Specifies the note value represented by one grid of the arpeggio.                                                                                                                                                                                                                                                                                         |
|                                                     |       |     |      | 4th               | quarter note (1 grid = 1 beat)                                                                                                                                                                                                                                                                                                                            |
|                                                     |       |     |      | 8th               | eighth note (2 grids = 1 beat)                                                                                                                                                                                                                                                                                                                            |
|                                                     |       |     |      | 8th_3             | eighth note triplet (3 grids = 1 beat)                                                                                                                                                                                                                                                                                                                    |
|                                                     |       |     |      | 16th              | sixteenth note (4 grids = 1 beat)                                                                                                                                                                                                                                                                                                                         |
|                                                     |       |     |      | 16th_3            | sixteenth note triplets (6 grids = 1 beat)                                                                                                                                                                                                                                                                                                                |
| Grid Note                                           | ✓     | ✓   | ✓    | 32nd              | thirty-second note (8 grids = 1 beat)                                                                                                                                                                                                                                                                                                                     |
|                                                     |       |     |      |                   |                                                                                                                                                                                                                                                                                                                                                           |
| Grid Offset (*1)                                    | ✓     | -   | -    | -63+64            | Specifies the grid position at which the arpeggio pattern starts as an amount of shift from the first grid position.                                                                                                                                                                                                                                      |
| Grid Sync (*1)                                      | ✓     | -   | -    | OFF, ON           | Turn this ON if you want the arpeggio sounded for each part to be synchronized with the grid.                                                                                                                                                                                                                                                             |
| Probability                                         |       |     |      |                   |                                                                                                                                                                                                                                                                                                                                                           |
| Style<br>* Refer to "Style List" (p. 14)            | ✓     | ✓   | ✓    | OFF, 1–64         | Sets an arpeggio pattern with a specified probability of playing back note messages.                                                                                                                                                                                                                                                                      |
| Amount Auto                                         | ✓     | ✓   | ✓    | OFF, ON           | When this is set to "ON," your keyboard performance is analyzed and the degree of playback probability is changed automatically.                                                                                                                                                                                                                          |
| Amount                                              | ✓     | ✓   | ✓    | 0–100%            | Sets the degree of playback probability.                                                                                                                                                                                                                                                                                                                  |
| Amount Dir                                          | ✓     | ✓   | ✓    | Standard, Reverse | Reverses the degree of change for the Probability Amount.                                                                                                                                                                                                                                                                                                 |
| Velo Amount                                         | ✓     | ✓   | ✓    | 0–100%            | Sets the volume of notes that play back based on the Probability Style. Smaller values produce lower playback volumes, and the maximum value produces the normal volume.                                                                                                                                                                                  |
| Grid Offset                                         | ✓     | ✓   | ✓    | -63+64            | Sets the start position of the pattern that's set in Probability Style, by using the shift amount from the start grid.                                                                                                                                                                                                                                    |
| Setting                                             |       |     |      |                   |                                                                                                                                                                                                                                                                                                                                                           |
| Key Sw Sync                                         | ✓     | ✓   | ✓    | OFF, ON           | When this is "ON," you can turn the arpeggio for the relevant parts on/off in part select mode by using the panel button [9]–[13] keyboard switches.                                                                                                                                                                                                      |
| Hold Sw                                             | ✓     | ✓   | ✓    | OFF, ON           | When the panel's arpeggio hold switch is ON, this setting specifies whether the arpeggio performance of each part is held when you release the keyboard (ON) or stops when you release the keyboard (OFF).                                                                                                                                                |
| Transpose                                           | ✓     | ✓   | ✓    | -36+36            | Shifts the arpeggio notes in semitone steps.                                                                                                                                                                                                                                                                                                              |
| Velocity (*)                                        | ✓     | ✓   | -    | REAL, 1–127       | Specifies the velocity of the arpeggiated notes. If you want the velocity to vary according to the strength at which you actually press the key, choose (REAL). If you want the velocity to be a fixed value regardless of your actual playing dynamics, specify that value (1–127).                                                                      |
| Offset Velo                                         | ✓     | ✓   | ✓    | -127+127          | Shifts the velocity values. Use this if the velocity values are not an appropriate match with other parts.                                                                                                                                                                                                                                                |
| Shuffle Rate                                        | ✓     | ✓   | ✓    | 0–100 [%]         | Varies the timing of even-numbered beats, creating a shuffle rhythm. A setting of "50%" sounds the notes at equal timing, and increasing this value produces more of a dotted shuffle feel.                                                                                                                                                               |

## Scene Parameter

| Parameter        | I-ARP | ARP | STEP | Value                            | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------|-----|------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shuffle Reso     | ✓     | ✓   | ✓    |                                  | Specifies the note resolution that is the reference for the shuffle setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                  |       |     |      | 16TH                             | Sixteen note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                  |       |     |      | 8TH                              | Eighth note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Timing           | ✓     | ✓   | ✓    |                                  | Specifies the timing at which the arpeggio pattern changes when the PLAY DETECTOR setting BEAT is ON.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                  |       |     |      | IMMEDIATE                        | Change immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                  |       |     |      | BEAT                             | Change at the beginning of the beat.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  |       |     |      | MEAS                             | Change at the beginning of the measure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Note Off (*5)    | ✓     | -   | ✓    |                                  | Specifies when previously-sounded notes are turned off if the arpeggio pattern changes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                  |       |     |      | NORMAL                           | Sound the note length specified by the arpeggio pattern, and then turn the note off.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                  |       |     |      | IMMEDIATE                        | Turn off immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Poly Remain (*4) | ✓     | ✓   | -    | OFF,<br>1-127                    | If this is other than "OFF," the following two behaviors will be different than normal. Even if the arpeggio pattern is mono, chords played on the keyboard are limited to the specified number of notes. Even if you do not play legato, the individual notes that you play up to the specified number are remembered, and reflected by the arpeggio performance. This produces a result that feels more like keyboard playing than a conventional arpeggio performance.                                                                                                                                                                                                                                   |
| K-Range Lo       | ✓     | ✓   | ✓    | C--G 9                           | Specifies the lower pitch limit that is sounded by the arpeggio. If the arpeggio attempts to play a note that is lower than this, the octave is raised.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| K-Range Oct      | ✓     | ✓   | ✓    | OFF, 0-12                        | Specifies the number of higher octaves in which the arpeggio is sounded, relative to K-Range Lo. If the arpeggio attempts to play a note that is higher than this range, the octave is lowered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Reset Oct (*1)   | ✓     | -   | -    | OFF, ON                          | If this is "ON," when the arpeggio returns to the start grid, it plays from the octave that you pressed, regardless of the Oct Range setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Duck Part        | ✓     | ✓   | ✓    | OFF,<br>PART 1-PART 4,<br>PART R | Velocity duck<br>This temporarily lowers the velocity of a specific note of a specific part when an arpeggio note coincides at the same timing. You can use this to prevent the volume from being excessive when notes overlap, or in a way similar to how a side-chain compressor effect can lower the volume of other instruments at the timing of the kick drum, so that a sense of musical groove is created.                                                                                                                                                                                                                                                                                           |
| Duck Note        | ✓     | ✓   | ✓    | Any,<br>C--G 9                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Duck Rate        | ✓     | ✓   | ✓    | 0-100                            | <b>Duck Part:</b> Enter the part to which the effect applies. For example, to specify the rhythm part, set this to "PART R." If this is "OFF," velocity duck does not occur.<br><b>Duck Note:</b> Specifies the note of the Duck Part sound that is the target. For example, if you want to target the kick drum, specify "C2." If you specify "Any," all notes of that part are the target.<br><b>Duck Rate:</b> Specifies the proportion by which Velocity Duck lowers the velocity value. Higher values produce a greater ducking effect, so that with a setting of "100" there will be no sound at that timing (maximum velocity duck effect). With a setting of "0" there will be no velocity ducking. |

(\*1) This is shown only if Arp Mode is set to "I-ARP."

(\*2) This is shown only if Arp Mode is set to "ARP."

(\*3) This is shown only if Arp Mode is set to "STEP."

(\*4) This is shown only if Arp Mode is set to "I-ARP" and "ARP."

(\*5) This is shown only if Arp Mode is set to "I-ARP" and "STEP."

## Style List

|     | I-ARP Style  | ARP Style | Probability Style |     | I-ARP Style  | ARP Style | Probability Style |
|-----|--------------|-----------|-------------------|-----|--------------|-----------|-------------------|
| 000 | CURRENT TYPE | -         | -                 | 023 | TYPE18 PART4 | 1/8VARI2  | 1MEAS10           |
| 001 | TYPE1 PART1  | SIMPLE    | EIGHT1            | 024 | TYPE19 PART3 | 1/8VARI3  | 1MEAS11           |
| 002 | TYPE5 PART1  | 1/8BASIC1 | EIGHT2            | 025 | TYPE19 PART4 | 1/8VARI4  | 1MEAS12           |
| 003 | TYPE6 PART1  | 1/8BASIC2 | QUARTER1          | 026 | TYPE20 PART3 | 1/8VARI5  | 1MEAS13           |
| 004 | TYPE7 PART1  | 1/8BASIC3 | QUARTER2          | 027 | TYPE20 PART4 | 1/16BASC1 | 1MEAS14           |
| 005 | TYPE8 PART1  | 1/8BASIC4 | HALF1             | 028 | TYPE21 PART3 | 1/16BASC2 | 1MEAS15           |
| 006 | TYPE9 PART1  | 1/8BASIC5 | HALF2             | 029 | TYPE22 PART3 | 1/16BASC3 | 1MEAS16           |
| 007 | TYPE10 PART1 | 1/8SYNC11 | HALF3             | 030 | TYPE22 PART4 | 1/16BASC4 | 1MEAS17           |
| 008 | TYPE11 PART3 | 1/8SYNC12 | HALF4             | 031 | TYPE23 PART3 | 1/16BASC5 | 1MEAS18           |
| 009 | TYPE11 PART4 | 1/8SYNC13 | HALF5             | 032 | TYPE23 PART4 | 1/16SYN11 | 1MEAS19           |
| 010 | TYPE12 PART3 | 1/8SYNC14 | SHORT1            | 033 | TYPE24 PART3 | 1/16SYN12 | 1MEAS20           |
| 011 | TYPE12 PART4 | 1/8SYNC15 | SHORT2            | 034 | TYPE24 PART4 | 1/16SYN13 | 2MEAS1            |
| 012 | TYPE13 PART3 | 1/8SYNC21 | SHORT3            | 035 | TYPE25 PART3 | 1/16SYN14 | 2MEAS2            |
| 013 | TYPE13 PART4 | 1/8SYNC22 | DOUBLE            | 036 | TYPE25 PART4 | 1/16SYN15 | 2MEAS3            |
| 014 | TYPE14 PART3 | 1/8SYNC23 | 1MEAS1            | 037 | TYPE26 PART3 | 1/16SYN21 | 2MEAS4            |
| 015 | TYPE14 PART4 | 1/8SYNC24 | 1MEAS2            | 038 | TYPE26 PART4 | 1/16SYN22 | 2MEAS5            |
| 016 | TYPE15 PART3 | 1/8SYNC25 | 1MEAS3            | 039 | TYPE27 PART3 | 1/16SYN23 | 2MEAS6            |
| 017 | TYPE15 PART4 | 1/8DRIVE1 | 1MEAS4            | 040 | TYPE27 PART4 | 1/16SYN24 | 2MEAS7            |
| 018 | TYPE16 PART3 | 1/8DRIVE2 | 1MEAS5            | 041 | TYPE28 PART3 | 1/16SYN25 | 2MEAS8            |
| 019 | TYPE16 PART4 | 1/8DRIVE3 | 1MEAS6            | 042 | TYPE28 PART4 | 1/16DRVE1 | 2MEAS9            |
| 020 | TYPE17 PART3 | 1/8DRIVE4 | 1MEAS7            | 043 | TYPE29 PART3 | 1/16DRVE2 | 2MEAS10           |
| 021 | TYPE17 PART4 | 1/8DRIVE5 | 1MEAS8            | 044 | TYPE29 PART4 | 1/16DRVE3 | 2MEAS11           |
| 022 | TYPE18 PART3 | 1/8VARI1  | 1MEAS9            | 045 | TYPE30 PART3 | 1/16DRVE4 | 2MEAS12           |

|     | I-ARP Style  | ARP Style | Probability Style |
|-----|--------------|-----------|-------------------|
| 046 | TYPE30 PART4 | 1/16DRVE5 | 2MEAS13           |
| 047 | TYPE31 PART3 | RHYTHM X1 | 2MEAS14           |
| 048 | TYPE31 PART4 | RHYTHM X2 | 2MEAS15           |
| 049 | TYPE32 PART3 | RHYTHM X3 | 2MEAS16           |
| 050 | TYPE32 PART4 | RHYTHM X4 | 2MEAS17           |
| 051 | TYPE33 PART3 | RHYTHM X5 | 2MEAS18           |
| 052 | TYPE33 PART4 | RHYTHM X6 | 2MEAS19           |
| 053 | TYPE34 PART3 | RHYTHM X7 | FEW1              |
| 054 | TYPE34 PART4 | RHYTHM X8 | FEW2              |
| 055 | TYPE35 PART3 | CYCLES3RD | FEW3              |
| 056 | TYPE35 PART4 | CYCLES4TH | FEW4              |
| 057 | TYPE36 PART3 | CYCLES5TH | FEW5              |
| 058 | TYPE36 PART4 | CYCLESMAJ | FEW6              |
| 059 | TYPE37 PART3 | CYCLESMIN | VARI1             |
| 060 | TYPE37 PART4 | CYCMAJ/MN | VARI2             |
| 061 | TYPE38 PART3 | AG PROGR1 | VARI3             |
| 062 | TYPE38 PART4 | AG PROGR2 | VARI4             |
| 063 | TYPE39 PART3 | AG CUTTIN | VARI5             |
| 064 | TYPE39 PART4 | AG 3FINGR | VARI6             |
| 065 | TYPE40 PART3 | AG ARPEGG |                   |
| 066 | TYPE40 PART4 | AG SPANS1 |                   |
| 067 | TYPE41 PART2 | AG SPANS2 |                   |
| 068 | TYPE41 PART3 | AG RIFFS  |                   |
| 069 | TYPE41 PART4 | EG CUTTIN |                   |
| 070 | TYPE42 PART2 | EG RIFFS  |                   |
| 071 | TYPE42 PART3 | EG ODRIF1 |                   |
| 072 | TYPE42 PART4 | EG ODRIF2 |                   |
| 073 | TYPE43 PART2 | EG ARPEGG |                   |
| 074 | TYPE43 PART3 | BLUES GTR |                   |
| 075 | TYPE43 PART4 | GTR TRILL |                   |
| 076 | TYPE44 PART4 | BASS PHR  |                   |
| 077 | TYPE45 PART3 | BS SHUFFL |                   |
| 078 | TYPE45 PART4 | FRETLESBS |                   |
| 079 | TYPE46 PART3 | WALKINGBS |                   |
| 080 | TYPE46 PART4 | BALLADBAS |                   |
| 081 | TYPE47 PART2 | EP PROGR1 |                   |
| 082 | TYPE47 PART3 | EP PROGR2 |                   |
| 083 | TYPE47 PART4 | LTN PIANO |                   |
| 084 | TYPE48 PART2 | FUNKCLAV1 |                   |
| 085 | TYPE48 PART3 | FUNKCLAV2 |                   |
| 086 | TYPE48 PART4 | SYNTHLEAD |                   |
| 087 | TYPE49 PART2 | DANCE SYN |                   |
| 088 | TYPE49 PART3 | HARP      |                   |
| 089 | TYPE49 PART4 | SYN BASS1 |                   |
| 090 | TYPE50 PART3 | SYN BASS2 |                   |
| 091 | TYPE50 PART4 | SYN BASS3 |                   |
| 092 | TYPE56 PART3 | SYN LINE1 |                   |
| 093 | TYPE56 PART4 | SYN LINE2 |                   |
| 094 | TYPE57 PART3 | SYN LINE3 |                   |
| 095 | TYPE57 PART4 | LEADLINE1 |                   |
| 096 | TYPE58 PART3 | LEADLINE2 |                   |
| 097 | TYPE58 PART4 | LEADLINE3 |                   |
| 098 | TYPE59 PART3 | SEQUENCE1 |                   |
| 099 | TYPE59 PART4 | SEQUENCE2 |                   |
| 100 | TYPE60 PART3 | SEQUENCE3 |                   |
| 101 | TYPE60 PART4 | CHORDS 1  |                   |
| 102 | TYPE61 PART3 | CHORDS 2  |                   |
| 103 | TYPE61 PART4 | CHORDS 3  |                   |
| 104 | TYPE62 PART3 | SHORTIES1 |                   |
| 105 | TYPE62 PART4 | SHORTIES2 |                   |
| 106 | TYPE63 PART3 | SHORTIES3 |                   |
| 107 | TYPE63 PART4 | FATTIES 1 |                   |

|     | I-ARP Style  | ARP Style | Probability Style |
|-----|--------------|-----------|-------------------|
| 108 | TYPE64 PART3 | FATTIES 2 |                   |
| 109 | TYPE64 PART4 | FATTIES 3 |                   |
| 110 | TYPE65 PART3 | SHRT&FAT1 |                   |
| 111 | TYPE65 PART4 | SHRT&FAT2 |                   |
| 112 |              | FAT&SHRT1 |                   |
| 113 |              | FAT&SHRT2 |                   |
| 114 |              | MIXTURE 1 |                   |
| 115 |              | MIXTURE 2 |                   |
| 116 |              | MIXTURE 3 |                   |
| 117 |              | COMBINAT1 |                   |
| 118 |              | COMBINAT2 |                   |
| 119 |              | COMBINAT3 |                   |
| 120 |              | COMBINAT4 |                   |
| 121 |              | COMBINAT5 |                   |
| 122 |              | COMBINAT6 |                   |
| 123 |              | COMBINAT7 |                   |
| 124 |              | COMBINAT8 |                   |
| 125 |              | PLEXI 1   |                   |
| 126 |              | PLEXI 2   |                   |
| 127 |              | PLEXI 3   |                   |
| 128 |              | PLEXI 4   |                   |

# Tone Parameters

## TONE COMMON Parameters

\* You can set whether control change messages (CC) are transmitted/received using the "Tone CC Map" system parameter (p. 32).

### TONE COMMON

JUNO-X, JUNO-106, JUNO-106

| Parameter | Value     | Explanation                  |
|-----------|-----------|------------------------------|
| (name)    | Tone name |                              |
| Catg      | CATEGORY  | Selects the tone's category. |

### TONE COMMON

VOCODER

There are no displayed parameters.

### TONE COMMON

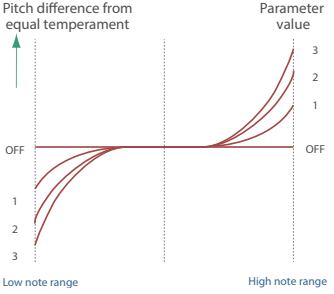
PR-A DRUM, CMN DRUM

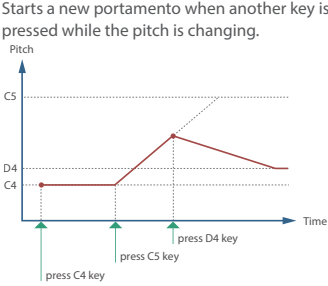
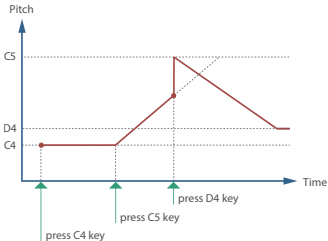
There are no displayed parameters.

### TONE COMMON

XV-5080, RD-PIANO, PR-A, PR-B, PD-C, PR-D, COMMON, PR-X

| Parameter   | Value                                                                             | Explanation                                                                                                                                         | CC# |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| (name)      | Tone name                                                                         |                                                                                                                                                     |     |
| Catg        | CATEGORY                                                                          | Selects the tone's category.                                                                                                                        |     |
| Level       | 0–127                                                                             | Adjusts the overall volume of the tone.                                                                                                             | 110 |
| Pan         | L64–63R                                                                           | Specifies the pan of the tone. "L64" is far left, "0" is center, and "63R" is far right.                                                            |     |
| Priority    | This determines how notes will be managed when the maximum polyphony is exceeded. |                                                                                                                                                     |     |
|             | LAST                                                                              | The last-played voices will be given priority, and currently sounding notes will be turned off in order, beginning with the first-played note.      |     |
|             | LOUDEST                                                                           | The voices with the loudest volume will be given priority, and currently sounding notes will be turned off, beginning with the lowest-volume voice. |     |
| Coarse Tune | -48–+48                                                                           | Adjusts the pitch of the sound up or down in semitone steps (+/-4 octaves).                                                                         |     |
| Fine Tune   | -50–+50                                                                           | Adjusts the pitch of the sound up or down in 1-cent steps (+/-50 cents).                                                                            |     |
| Octave      | -3–+3                                                                             | Adjusts the pitch of the tone's sound up or down in units of an octave (+/-3 octaves).                                                              |     |

| Parameter                                | Value     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CC# |
|------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Stretch                                  | OFF, 1–3  | <p>This setting allows you to apply "stretched tuning" to the tone. (Stretched tuning is a system by which acoustic pianos are normally tuned, causing the lower range to be lower and the higher range to be higher than the mathematical tuning ratios would otherwise dictate.) With a setting of "OFF," the tone's tuning will be equal temperament. A setting of "3" will produce the greatest difference in the pitch of the low and high ranges.</p> <p>The diagram shows the pitch change relative to equal temperament that will occur in the low and high ranges. This setting will have a subtle effect on the way in which chords resonate.</p>                                                                                                                                                                                                                                                                            |     |
| Analog Feel                              | 0–127     | Applies time-varying change to the pitch and volume of the tone that is producing sound, adding a sense of variability. As you increase this value toward the maximum, the variability becomes greater, producing instability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| Mono Poly                                |           | Specifies whether the tone will play polyphonically (POLY) or monophonically (MONO).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 115 |
|                                          | MONO      | Sound only the last-played key one at a time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
|                                          | POLY      | Two or more notes can be played simultaneously.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| Legato Sw                                | OFF, ON   | <p>This is effective when MONO/POLY is set to MONO and Legato Switch is turned "ON."</p> <p>When you press the next key while still holding down the previous key (legato performance), the pitch changes smoothly.</p> <p>The way in which the change occurs depends on the Legato Retrigger Interval.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 116 |
| Retrig Intvl (Legato Retrigger Interval) | 0–12, OFF | <p>When Legato Switch is enabled and you play legato, this specifies whether retriggering occurs (0–12) or does not occur (OFF).</p> <p>If this is "OFF," only the pitch of the currently-sounding tones changes according to the pitch of the key.</p> <p>If this is set to 1–12, retriggering occurs smoothly when the pitch difference during legato performance exceeds the specified value.</p> <p><b>If this is set to "4"</b></p> <p>Using C4 as the reference pitch, playing notes Db4–E4 legato will change only the pitch without retriggering, but playing the F4 note (which is five semitones away from C4) legato will retrigger F4. When F4 is retriggered at this time, F4 now becomes the reference pitch.</p> <p><b>If this is set to "0"</b></p> <p>Each note is retriggered every time regardless of the pitch difference.</p> <p>For acoustic-type sounds in particular, an unnatural impression can occur if only the pitch is changed, so you'll need to adjust the Legato Retrigger Interval.</p> |     |
| Porta Sw                                 | OFF, ON   | <p>Specifies whether the portamento effect will be applied (ON) or not applied (OFF).</p> <p>* Portamento is an effect which smoothly changes the pitch from the first-played key to the next-played key. By applying portamento when the MONO/POLY parameter is "MONO," you can simulate slide performance techniques on a violin or similar instrument.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
| Porta Mode                               |           | Specifies the performance conditions for which portamento will be applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|                                          | NORMAL    | Portamento will always be applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
|                                          | LEGATO    | Applies portamento only when you play legato (i.e., when you press the next key before releasing the previous key).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |

| Parameter    | Value                                                                                                                                                                     | Explanation                                                                                                                                                                                                                                           | CC# |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Porta Type   | RATE                                                                                                                                                                      | Specifies the type of portamento effect.<br>The time it takes will depend on the distance between the two pitches.                                                                                                                                    |     |
|              | TIME                                                                                                                                                                      | The time it takes will be constant.                                                                                                                                                                                                                   |     |
| Porta Start  | When another key is pressed during a pitch change produced by portamento, a new pitch change will begin. This setting specifies the pitch at which the change will begin. |                                                                                                                                                                                                                                                       |     |
|              | PITCH                                                                                                                                                                     | Starts a new portamento when another key is pressed while the pitch is changing.<br>                                                                                 |     |
| Porta Start  | NOTE                                                                                                                                                                      | Portamento will begin from the pitch where the current change would end.<br>                                                                                        |     |
| Porta Time   | 0-1023                                                                                                                                                                    | When portamento is used, this specifies the time over which the pitch will change. Higher settings will cause the pitch change to the next note to take more time.                                                                                    | 5   |
| Porta Crv    | Specifies the pitch change curve for portamento.                                                                                                                          |                                                                                                                                                                                                                                                       |     |
|              | LIN                                                                                                                                                                       | Change on a linear curve.                                                                                                                                                                                                                             |     |
|              | EXP-L                                                                                                                                                                     | Change on a non-linear curve (gentle slope).                                                                                                                                                                                                          |     |
| BendRange Up | EXP-H                                                                                                                                                                     | Change on a non-linear curve (steep slope).                                                                                                                                                                                                           |     |
|              | 0-48                                                                                                                                                                      | Specifies the amount of change in semitone units when the pitch bend wheel is turned all the way upward. For example, if this is "48," turning the pitch bend wheel all the way upward raises the pitch by four octaves.                              | 41  |
| BendRange Dw | 0-48                                                                                                                                                                      | Specifies the amount of change in semitone units when the pitch bend wheel is turned all the way downward. For example, if this is "48," turning the pitch bend wheel all the way downward lowers the pitch by four octaves.                          | 49  |
| BendMode     | NORMAL                                                                                                                                                                    | The pitch bend wheel produces the usual effect.                                                                                                                                                                                                       |     |
|              | CATCH+LAST                                                                                                                                                                | The pitch bend effect applies only to the last-played note. If a note-on occurs while pitch bend is already applied, the new note sounds at the center pitch. The pitch starts changing only after the controller passes through the center position. |     |
| Soft Lv Sens | 0-100                                                                                                                                                                     | Specifies the amount of volume change that occurs when you operate the soft pedal (CC#67). This is effective when specified for piano sounds.                                                                                                         |     |

| Parameter      | Value                  | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CC# |
|----------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Tone PMT       |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| Vel Ctrl       | OFF, ON, RANDOM, CYCLE | Specifies how partials are played according to your keyboard playing dynamics (velocity). If this is "ON," different partials are sounded according to the playing velocity and the Velo Rng Low/Upp and Velo FadeLow/Upp settings. <ul style="list-style-type: none"> <li>If this is "RANDOM" or "CYCLE," each partial is sounded randomly or cyclically.</li> <li>In the case of "RANDOM" or "CYCLE" when Structure 1-2 (3-4) has a setting other than OFF, partials 1 and 2 (3 and 4) are sounded as a pair, either randomly or in alternation.</li> <li>In the case of "RANDOM" or "CYCLE," velocity has no effect, but you'll need to make settings for each partial so that the Velocity Range does not conflict.</li> </ul> |     |
| Level Crv      | EXP                    | When using Vel Ctrl to switch between partials, the crossfade level changes in a non-linear curve.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |
|                | LINEAR                 | When using Vel Ctrl to switch between partials, the crossfade level changes in a linear curve.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| Synth          |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
| Unison Sw      | OFF, ON                | This layers a single sound. If the Unison Switch is on, the number of notes layered on one key will change according to the number of keys you play. <ul style="list-style-type: none"> <li>If the OSC Type is PCM, this is limited to mono playing.</li> <li>If the Legato Sw is on, the Delay Time is ignored while playing legato.</li> <li>Even if Retrig Intvl (Legato Retrig Interval) is specified, it operates as OFF.</li> </ul>                                                                                                                                                                                                                                                                                          | 119 |
| Unison Size    | 2-8                    | If unison is on, this specifies the number of notes that are assigned to each key that is pressed. Increasing the Unison Size increases the polyphony, making it more likely that notes will be cut off.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| Unison Detn    | 0-100                  | Detunes each of the notes that are allocated by the Unison Size number, producing a detuned effect. As you increase this value, each note is detuned more greatly, producing a thicker sound.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |
| BendRngFine Up | 0-100                  | Specifies a fine adjustment in one-cent units to the amount of change when the pitch bend wheel is turned upward.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
| BendRngFine Dw | 0-100                  | Specifies a fine adjustment in one-cent units to the amount of change when the pitch bend wheel is turned downward.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |

| Parameter        | Value   | Explanation                                                                                                                                                                                                                  | CC# |
|------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Synth PMT</b> |         |                                                                                                                                                                                                                              |     |
| Struct12         |         | The sound of partial 1 is modulated by partial 2.                                                                                                                                                                            |     |
|                  | OFF     | OFF                                                                                                                                                                                                                          |     |
|                  | SYNC    | Implements the oscillator sync function that is provided by an analog synthesizer. The partial 1 oscillator is reset at intervals of partial 2's pitch cycle.<br>* This is effective only if OSC Type is "VA" or "PCM-Sync." |     |
|                  | RING    | Implements the ring modulator function that is provided by an analog synthesizer. The output sound of partial 2 is multiplied with partial 1.                                                                                |     |
| Struct34         |         | Implements the cross modulation function that is provided by an analog synthesizer. The output sound of partial 2 is applied as the pitch of partial 1.<br>* XMOD2 is available only when Partial 1 and 3 are OSC Type "VA." |     |
|                  | OFF     | OFF                                                                                                                                                                                                                          |     |
|                  | SYNC    | Implements the oscillator sync function that is provided by an analog synthesizer. The partial 3 oscillator is reset at intervals of partial 4's pitch cycle.<br>* This is effective only if OSC Type is "VA" or "PCM-Sync." |     |
|                  | RING    | Implements the ring modulator function that is provided by an analog synthesizer. The output sound of partial 4 is multiplied with partial 3.                                                                                |     |
| Struct34         |         | Implements the cross modulation function that is provided by an analog synthesizer. The output sound of partial 4 is applied as the pitch of partial 3.<br>* XMOD2 is available only when Partial 1 and 3 are OSC Type "VA." |     |
|                  | OFF     | OFF                                                                                                                                                                                                                          |     |
|                  | SYNC    | Implements the oscillator sync function that is provided by an analog synthesizer. The partial 3 oscillator is reset at intervals of partial 4's pitch cycle.<br>* This is effective only if OSC Type is "VA" or "PCM-Sync." |     |
|                  | RING    | Implements the ring modulator function that is provided by an analog synthesizer. The output sound of partial 4 is multiplied with partial 3.                                                                                |     |
| Ring12 Level     | 0–127   | RING level when Structure1-2 is RING.                                                                                                                                                                                        |     |
| Ring34 Level     | 0–127   | RING level when Structure3-4 is RING.                                                                                                                                                                                        |     |
| Ring OSC1 Lv     | 0–127   | Sets the partial 1 OSC level.<br>* It is effective when Structure1-2 is RING.                                                                                                                                                |     |
| Ring OSC2 Lv     | 0–127   | Sets the partial 2 OSC level.<br>* It is effective when Structure1-2 is RING.                                                                                                                                                |     |
| Ring OSC3 Lv     | 0–127   | Sets the partial 3 OSC level.<br>* It is effective when Structure3-4 is RING.                                                                                                                                                |     |
| Ring OSC4 Lv     | 0–127   | Sets the partial 4 OSC level.<br>* It is effective when Structure3-4 is RING.                                                                                                                                                |     |
| XMd12 Dpth       | 0–10800 | Cross Modulation Depth when Structure1-2 is XMOD.                                                                                                                                                                            |     |
| XMd34 Dpth       | 0–10800 | Cross Modulation Depth when Structure3-4 is XMOD.                                                                                                                                                                            |     |
| XMd OSC1 Lv      | 0–127   | Sets the partial 1 OSC level.<br>* It is effective when Structure1-2 is XMOD/XMOD2.                                                                                                                                          |     |
| XMd OSC2 Lv      | 0–127   | Sets the partial 2 OSC level.<br>* It is effective when Structure1-2 is XMOD/XMOD2.                                                                                                                                          |     |
| XMd OSC3 Lv      | 0–127   | Sets the partial 3 OSC level.<br>* It is effective when Structure3-4 is XMOD/XMOD2.                                                                                                                                          |     |
| XMd OSC4 Lv      | 0–127   | Sets the partial 4 OSC level.<br>* It is effective when Structure3-4 is XMOD/XMOD2.                                                                                                                                          |     |
| Ptl Phs Lock     | OFF, ON | This is available if OSC Type is "VA"; it locks the waveform phase between partials.<br>* It is effective to use this with XMOD2.                                                                                            |     |
| XMd2 12 Dpth     | 0–127   | Cross Modulation Depth when Structure1-2 is XMOD2.                                                                                                                                                                           |     |
| XMd2 34 Dpth     | 0–127   | Cross Modulation Depth when Structure3-4 is XMOD2.                                                                                                                                                                           |     |

## TONE Parameters

TONE  
JUNO-X

| Parameter                     | Value                                                                                                                 | Explanation                                                                                                                                               | CC# |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| LFO WAVEFORM                  | SIN, SAW-DW, SQR, S&H                                                                                                 | Selects the waveform of the LFO.                                                                                                                          | 35  |
| LFO NOTE                      | 1/64T, 1/64, 1/32T, 1/32, 1/16T, 1/32, 1/16, 1/8T, 1/16, 1/8, 1/4T, 1/8, 1/4, 1/2T, 1/4, 1/2, 1T, 1/2, 1, 2T, 1, 2, 4 | Sets the LFO RATE as a note length, when LFO SYNC is "ON."                                                                                                | 31  |
| LFO RATE                      | 0–1023                                                                                                                | Specifies the rate of the LFO cycle.                                                                                                                      | 29  |
| LFO DELAY TIME                | 0–1023                                                                                                                | Adjusts the time from when a key is pressed until LFO modulation starts being applied.                                                                    | 27  |
| LFO SYNC                      | OFF, ON                                                                                                               | Turn this ON to sync the speed of the LFO cycle with the tempo.                                                                                           | 117 |
| OSC LFO MOD                   | 0–100                                                                                                                 | Uses the LFO to vary the pitch (vibrato).                                                                                                                 | 26  |
| FILTER MOD                    | 0–100                                                                                                                 | Adjusts the amount by which the LFO modulates the cutoff frequency.                                                                                       | 28  |
| AMP MOD                       | 0–100                                                                                                                 | Sets how much the LFO changes the AMP volume.                                                                                                             | 30  |
| SUPER SAW                     | OFF, ON                                                                                                               | Set this to "ON" to use the Super SAW wave instead of the pulse wave.                                                                                     | 46  |
| PW MODE                       |                                                                                                                       | Specifies the pulse width mode.                                                                                                                           |     |
|                               |                                                                                                                       | <b>If SUPER SAW = OFF:</b>                                                                                                                                |     |
|                               | LFO                                                                                                                   | The pulse width is affected by the LFO.                                                                                                                   |     |
|                               | MANUAL                                                                                                                | The pulse width is affected by PULSE WIDTH MOD.                                                                                                           |     |
|                               | ENV                                                                                                                   | The pulse width is affected by ENV.                                                                                                                       |     |
|                               |                                                                                                                       | <b>If SUPER SAW = ON:</b>                                                                                                                                 |     |
|                               | LFO                                                                                                                   | Changes the Super SAW Detune depth based on the LFO.                                                                                                      |     |
|                               | MANUAL                                                                                                                | Sets the Super SAW Detune depth based on PULSE WIDTH MOD.                                                                                                 |     |
| PULSE WIDTH MOD / SSAW DETUNE |                                                                                                                       |                                                                                                                                                           |     |
|                               |                                                                                                                       | <b>If SUPER SAW = OFF:</b>                                                                                                                                |     |
|                               | PW MODE = LFO:                                                                                                        | Adjusts the degree of pulse width change based on the LFO.                                                                                                |     |
|                               | PW MODE = MANUAL:                                                                                                     | Adjusts the pulse width value.                                                                                                                            |     |
|                               | PW MODE = ENV:                                                                                                        | Adjusts the degree of pulse width change based on the ENV.                                                                                                |     |
|                               |                                                                                                                       | <b>If SUPER SAW = ON:</b>                                                                                                                                 |     |
|                               | PW MODE = LFO:                                                                                                        | Adjusts the degree of Super SAW Detune change based on the LFO.                                                                                           |     |
|                               | PW MODE = MANUAL:                                                                                                     | Adjusts the Super SAW Detune depth.                                                                                                                       |     |
| SUB OSC PSHIFT                |                                                                                                                       |                                                                                                                                                           |     |
|                               | OFF, ON                                                                                                               | Sets whether to use the settings of the JUNO-106 (OFF) or the JUNO-60 (ON) for the phase setting between the sub-oscillator pulse wave and sawtooth wave. | 25  |
| OSC PITCH                     | -12–+12                                                                                                               | Adjusts the pitch of the oscillator in semitones.                                                                                                         | 20  |

| Parameter             | Value          | Explanation                                                                                                                                                                                                                                                                                                                                                                                             | CC# |
|-----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| OSC DETUNE            | -50→+50        | Shifts the tuning of the sawtooth wave and sub-oscillator to create a detuned effect. Larger values detune the oscillators more, making the sound thicker. Setting this to a positive value raises the pitch of the sawtooth wave while lowering the pitch of the sub-oscillator. Setting this to a negative value lowers the pitch of the sawtooth wave while raising the pitch of the sub-oscillator. | 21  |
| OSC PAN SPLIT         | -50→+50        | Sets the degree of separation for the panning of the pulse wave/Super SAW, sawtooth wave, sub-oscillator and noise.                                                                                                                                                                                                                                                                                     | 55  |
| PW LEVEL / SSAW LEVEL | 0–255          | Adjusts the pulse wave/Super SAW volume.                                                                                                                                                                                                                                                                                                                                                                | 16  |
| SAW LEVEL             | 0–255          | Adjusts the volume of the sawtooth wave.                                                                                                                                                                                                                                                                                                                                                                | 17  |
| SUB LEVEL             | 0–255          | Adjusts the volume of the sub oscillator.                                                                                                                                                                                                                                                                                                                                                               | 18  |
| NOISE LEVEL           | 0–255          | Adjusts the noise volume.                                                                                                                                                                                                                                                                                                                                                                               | 19  |
| HPF-STEP              | 0–3            | Sets the high-pass filter's cutoff frequency in four steps.                                                                                                                                                                                                                                                                                                                                             | 79  |
| VINTAGE FLT TYPE      | R, M, S        | Selects one of three response curves, each modeling the LPF of an analog synthesizer of the past.                                                                                                                                                                                                                                                                                                       | 108 |
| CUTOFF                | 0–1023         | Specifies the cutoff frequency of the low-pass filter. The frequency region above the cutoff frequency is cut, producing a more mellow tonal character.                                                                                                                                                                                                                                                 | 3   |
| RESONANCE             | 0–1023         | Boosts the region of the filter's cutoff frequency. Higher values produce a stronger result, giving the sound a distinctively synthesizer-like character.                                                                                                                                                                                                                                               | 9   |
| FLT ENV DEPTH         | -1023→+1023    | Adjusts the amount by which the cutoff frequency is controlled by the envelope.                                                                                                                                                                                                                                                                                                                         | 81  |
| FLT KEY FOLLOW        | -200–200       | Adjusts the amount by which the keyboard pitch affects the cutoff frequency (key follow). With smaller values, the cutoff frequency becomes lower as you play higher notes.                                                                                                                                                                                                                             | 122 |
| FLT VSENS             | -100→+100      | Adjusts how much the cutoff frequency changes according to how hard you play the keys.                                                                                                                                                                                                                                                                                                                  | 53  |
| AMP LEVEL             | 0–127          | Adjusts the volume of the tone.                                                                                                                                                                                                                                                                                                                                                                         | 110 |
| AMP ENV SEL           | ENV F&A, G-AMP | Specifies whether the volume is controlled by the ENV (ENV F&A) or by the gate signal (G-AMP).                                                                                                                                                                                                                                                                                                          |     |
| AMP VSENS             | -100→+100      | Adjusts how much the AMP volume changes according to how hard you play the keys.                                                                                                                                                                                                                                                                                                                        | 54  |
| PENV ATTACK           | 0–1023         | Specifies the attack time of the pitch envelope.                                                                                                                                                                                                                                                                                                                                                        | 83  |
| PENV DECAY            | 0–1023         | Specifies the decay time of the pitch envelope.                                                                                                                                                                                                                                                                                                                                                         | 80  |
| PENV SUSTAIN          | 0–1023         | Specifies the sustain level of the pitch envelope.                                                                                                                                                                                                                                                                                                                                                      | 85  |
| PENV RELEASE          | 0–1023         | Specifies the release time of the pitch envelope.                                                                                                                                                                                                                                                                                                                                                       | 86  |
| PENV VSENS            | -100→+100      | Adjusts the pitch envelope depth according to how hard you play the keys.                                                                                                                                                                                                                                                                                                                               | 52  |
| PENV DEPTH            | -100→+100      | Specifies the depth of the pitch envelope.                                                                                                                                                                                                                                                                                                                                                              | 22  |
| ENV ATTACK            | 0–1023         | Specifies the ENV Attack time.                                                                                                                                                                                                                                                                                                                                                                          | 89  |
| ENV DECAY             | 0–1023         | Specifies the ENV Decay time.                                                                                                                                                                                                                                                                                                                                                                           | 90  |
| ENV SUSTAIN           | 0–1023         | Specifies the ENV Sustain level.                                                                                                                                                                                                                                                                                                                                                                        | 102 |
| ENV RELEASE           | 0–1023         | Specifies the ENV Release time.                                                                                                                                                                                                                                                                                                                                                                         | 103 |
| BEND FILTER           | -63→+63        | Specifies the range of filter change produced by pitch bend.                                                                                                                                                                                                                                                                                                                                            | 14  |
| BEND PITCH            | 0–1200         | Specifies the range of pitch change produced by pitch bend.                                                                                                                                                                                                                                                                                                                                             | 41  |
| MODULATION LFO        | -63→+63        | Specifies the amount of LFO applied by modulation.                                                                                                                                                                                                                                                                                                                                                      |     |
| PORTA MODE            | OFF, ON        | Turns portamento on/off. If this is "ON," the pitch will change smoothly from one note to the next-played note.                                                                                                                                                                                                                                                                                         | 118 |
| PORTA TIME            | 0–1023         | Adjusts the time over which the portamento pitch change occurs.                                                                                                                                                                                                                                                                                                                                         | 5   |

| Parameter       | Value     | Explanation                                                                                                              | CC# |
|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------|-----|
| PORTA CRV       |           | Specifies the pitch change curve for portamento.                                                                         |     |
|                 | ORIGINAL  | Change according to the original curve of the model.                                                                     |     |
|                 | LINEAR    | Change in a linear curve.                                                                                                |     |
|                 | EXP1      | Change in a non-linear curve (gentle slope).                                                                             |     |
| KEY MODE        | EXP2      | Change in a non-linear curve (steep slope).                                                                              |     |
|                 |           | Specifies how notes are sounded.                                                                                         |     |
|                 | POLY      | Polyphonic                                                                                                               | 119 |
|                 | SOLO      | Monophonic                                                                                                               |     |
| AFT LFO         | UNISON    | Unison                                                                                                                   |     |
|                 | SL-UNISON | Monophonic unison                                                                                                        |     |
| AFT LFO         | -63→+63   | Sets how much aftertouch changes the LFO intensity.<br>* This is only enabled for products with aftertouch.              |     |
| AFT FREQ        | -63→+63   | Sets how much aftertouch changes the low-pass filter intensity.<br>* This is only enabled for keyboards with aftertouch. |     |
| AFT LEVEL       | -63→+63   | Sets how much aftertouch changes the tone volume.<br>* This is only enabled for keyboards with aftertouch.               |     |
| PITCH DRIFT     | 0–255     | Adjusts the slight pitch drift that occurs when notes are played on an analog synthesizer.                               |     |
| CONDITION       | 0–100     | Simulates the changes that occur as a unit ages.                                                                         |     |
| PARAM EXPANSION | OFF, ON   | When this is set to "ON," the LFO RATE, CUTOFF and RESONANCE produce greater change.                                     |     |



## TONE

### JUNO-106

| Parameter        | Value                                            | Explanation                                                                                                                                                                 | CC# |
|------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| LFO RATE         | 0–1023                                           | Specifies the speed of the LFO cycle.                                                                                                                                       | 29  |
| LFO DELAY TIME   | 0–1023                                           | Adjusts the time from when the key is pressed until LFO modulation starts to apply.                                                                                         | 27  |
| OSC RANGE        | 16', 8', 4'                                      | Specifies the octave of the oscillator.                                                                                                                                     |     |
| OSC LFO MOD      | 0–100                                            | Uses the LFO to vary the pitch (vibrato).                                                                                                                                   | 26  |
| PULSE WIDTH MOD  | 0–127                                            | PM MODE = LFO: Adjusts the modulation depth.<br>PM MODE = MANUAL: Adjusts the pulse width.                                                                                  | 50  |
| PW MODE          | LFO, MANUAL                                      | Selects whether the pulse width is modulated by the LFO (LFO) or kept at the fixed value specified by PULSE WIDTH MOD (MANUAL).                                             |     |
| PW SWITCH        | OFF, ON                                          | Turns the pulse wave on/off.                                                                                                                                                |     |
| SAW SWITCH       | OFF, ON                                          | Turns the sawtooth wave on/off.                                                                                                                                             |     |
| SUB LEVEL        | 0–255                                            | Adjusts the volume of the sub oscillator.                                                                                                                                   | 18  |
| NOISE LEVEL      | 0–255                                            | Adjusts the volume of the noise.                                                                                                                                            | 19  |
| HPF-STEP         | 0–3                                              | Sets the high-pass filter's cutoff frequency in four steps.                                                                                                                 |     |
| VINTAGE FLT TYPE | R, M, S                                          | Selects one of three response curves, each modeling the LPF of an analog synthesizer of the past.                                                                           | 108 |
| CUTOFF           | 0–1023                                           | Specifies the cutoff frequency of the low-pass filter. The frequency region above the cutoff frequency is cut, producing a more mellow tonal character.                     | 3   |
| RESONANCE        | 0–1023                                           | Boosts the region of the filter's cutoff frequency. Higher values produce a stronger result, giving the sound a distinctively synthesizer-like character.                   | 9   |
| FLT ENV DEPTH    | -1023–+1023                                      | Adjusts the amount by which the cutoff frequency is controlled by the envelope.                                                                                             | 81  |
| FILTER MOD       | 0–100                                            | Adjusts the amount by which the LFO modulates the cutoff frequency.                                                                                                         | 28  |
| FLT KEY FOLLOW   | 0–200                                            | Adjusts the amount by which the keyboard pitch affects the cutoff frequency (key follow). With smaller values, the cutoff frequency becomes lower as you play higher notes. | 82  |
| AMP ENV SEL      | ENV F&A, G-AMP                                   | Specifies whether the volume is controlled by the ENV (ENV F&A) or by the gate signal (G-AMP).                                                                              |     |
| AMP LEVEL        | 0–127                                            | Adjusts the volume of the tone.                                                                                                                                             | 110 |
| ENV ATTACK       | 0–1023                                           | Specifies the ENV Attack time.                                                                                                                                              | 89  |
| ENV DECAY        | 0–1023                                           | Specifies the ENV Decay time.                                                                                                                                               | 90  |
| ENV SUSTAIN      | 0–1023                                           | Specifies the ENV Sustain level.                                                                                                                                            | 102 |
| ENV RELEASE      | 0–1023                                           | Specifies the ENV Release time.                                                                                                                                             | 103 |
| BEND PITCH       | 0–1200                                           | Specifies the range of pitch change produced by pitch bend.                                                                                                                 | 41  |
| BEND FILTER      | -63–+63                                          | Specifies the range of filter change produced by pitch bend.                                                                                                                | 14  |
| MODULATION LFO   | -63–+63                                          | Specifies the amount of LFO applied by modulation.                                                                                                                          |     |
| PORTA MODE       | OFF, ON                                          | Turns portamento on/off. If this is "ON," the pitch will change smoothly from one note to the next-played note.                                                             | 118 |
| PORTA TIME       | 0–1023                                           | Adjusts the time over which the portamento pitch change occurs.                                                                                                             | 5   |
| PORTA CRV        | Specifies the pitch change curve for portamento. |                                                                                                                                                                             |     |
|                  | ORIGINAL                                         | Change according to the original curve of the model.                                                                                                                        |     |
|                  | LINEAR                                           | Change in a linear curve.                                                                                                                                                   |     |
|                  | EXP1                                             | Change in a non-linear curve (gentle slope).                                                                                                                                |     |
|                  | EXP2                                             | Change in a non-linear curve (steep slope).                                                                                                                                 |     |

| Parameter       | Value                            | Explanation                                                                                                              | CC# |
|-----------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----|
| KEY MODE        | Specifies how notes are sounded. |                                                                                                                          | 119 |
|                 | POLY                             | Polyphonic                                                                                                               |     |
|                 | SOLO                             | Monophonic                                                                                                               |     |
|                 | UNISON                           | Unison                                                                                                                   |     |
|                 | SL-UNISON                        | Monophonic unison                                                                                                        |     |
| AFT LFO         | -63–+63                          | Sets how much aftertouch changes the LFO intensity.<br>* This is only enabled for products with aftertouch.              |     |
| AFT FREQ        | -63–+63                          | Sets how much aftertouch changes the low-pass filter intensity.<br>* This is only enabled for keyboards with aftertouch. |     |
| AFT LEVEL       | -63–+63                          | Sets how much aftertouch changes the tone volume.<br>* This is only enabled for keyboards with aftertouch.               |     |
| PITCH DRIFT     | 0–255                            | Adjusts the slight pitch drift that occurs when notes are played on an analog synthesizer.                               |     |
| PARAM EXPANSION | OFF, ON                          | When this is set to "ON," the LFO RATE, CUTOFF and RESONANCE produce greater change.                                     |     |
| CONDITION       | 0–100                            | Simulates the changes that occur as a unit ages.                                                                         |     |

## TONE

### JUNO-60

| Parameter        | Value                           | Explanation                                                                                                                                               | CC# |
|------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| LFO RATE         | 0–1023                          | Specifies the speed of the LFO cycle.                                                                                                                     |     |
| LFO DELAY TIME   | 0–1023                          | Adjusts the time from when the key is pressed until LFO modulation starts to apply.                                                                       |     |
| OCTAVE           | DOWN, NORMAL, UP                | Specifies the octave of the oscillator.                                                                                                                   |     |
| OSC LFO MOD      | 0–100                           | Uses the LFO to vary the pitch (vibrato).                                                                                                                 |     |
| PULSE WIDTH MOD  | 0–127                           | PM MODE = LFO: Adjusts the degree of pulse width change based on the LFO.                                                                                 |     |
|                  |                                 | PM MODE = MANUAL: Adjusts the pulse width value.                                                                                                          |     |
|                  |                                 | PM MODE = ENV: Adjusts the degree of pulse width change based on the ENV.                                                                                 |     |
| PW MODE          | Specifies the pulse width mode. |                                                                                                                                                           |     |
|                  | LFO                             | The pulse width is affected by the LFO.                                                                                                                   |     |
|                  | MANUAL                          | The pulse width is affected by PULSE WIDTH MOD.                                                                                                           |     |
|                  | ENV                             | The pulse width is affected by ENV.                                                                                                                       |     |
| PW SWITCH        | OFF, ON                         | Turns the pulse wave on/off.                                                                                                                              |     |
| SAW SWITCH       | OFF, ON                         | Turns the sawtooth wave on/off.                                                                                                                           |     |
| SUB LEVEL        | 0–255                           | Adjusts the volume of the sub oscillator.                                                                                                                 | 18  |
| NOISE LEVEL      | 0–255                           | Adjusts the volume of the noise.                                                                                                                          | 19  |
| HPF-STEP         | 0–3                             | Sets the high-pass filter's cutoff frequency in four steps.                                                                                               |     |
| VINTAGE FLT TYPE | R, M, S                         | Selects one of three response curves, each modeling the LPF of an analog synthesizer of the past.                                                         | 108 |
| CUTOFF           | 0–1023                          | Specifies the cutoff frequency of the low-pass filter. The frequency region above the cutoff frequency is cut, producing a more mellow tonal character.   | 3   |
| RESONANCE        | 0–1023                          | Boosts the region of the filter's cutoff frequency. Higher values produce a stronger result, giving the sound a distinctively synthesizer-like character. | 9   |
| FLT ENV DEPTH    | -1023–+1023                     | Adjusts the amount by which the cutoff frequency is controlled by the envelope.                                                                           | 81  |
| FILTER MOD       | 0–100                           | Adjusts the amount by which the LFO modulates the cutoff frequency.                                                                                       | 28  |

| Parameter       | Value                                            | Explanation                                                                                                                                                                 | CC# |
|-----------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FLT KEY FOLLOW  | 0–200                                            | Adjusts the amount by which the keyboard pitch affects the cutoff frequency (key follow). With smaller values, the cutoff frequency becomes lower as you play higher notes. | 82  |
| AMP ENV SEL     | ENV F&A, G-AMP                                   | Specifies whether the volume is controlled by the ENV (ENV F&A) or by the gate signal (G-AMP).                                                                              |     |
| AMP LEVEL       | 0–127                                            | Adjusts the volume of the tone.                                                                                                                                             | 110 |
| ENV ATTACK      | 0–1023                                           | Specifies the ENV Attack time.                                                                                                                                              | 89  |
| ENV DECAY       | 0–1023                                           | Specifies the ENV Decay time.                                                                                                                                               | 90  |
| ENV SUSTAIN     | 0–1023                                           | Specifies the ENV Sustain level.                                                                                                                                            | 102 |
| ENV RELEASE     | 0–1023                                           | Specifies the ENV Release time.                                                                                                                                             | 103 |
| BEND PITCH      | 0–700                                            | Specifies the range of pitch change produced by pitch bend.                                                                                                                 | 41  |
| BEND FILTER     | -63–+63                                          | Specifies the range of filter change produced by pitch bend.                                                                                                                | 14  |
| MODULATION LFO  | -63–+63                                          | Specifies the amount of LFO applied by modulation.                                                                                                                          |     |
| PORTA MODE      | OFF, ON                                          | Turns portamento on/off. If this is "ON," the pitch will change smoothly from one note to the next-played note.                                                             | 118 |
| PORTA TIME      | 0–1023                                           | Adjusts the time over which the portamento pitch change occurs.                                                                                                             | 5   |
| PORTA CRV       | Specifies the pitch change curve for portamento. |                                                                                                                                                                             |     |
|                 | ORIGINAL                                         | Change according to the original curve of the model.                                                                                                                        |     |
|                 | LINEAR                                           | Change in a linear curve.                                                                                                                                                   |     |
|                 | EXP1                                             | Change in a non-linear curve (gentle slope).                                                                                                                                |     |
|                 | EXP2                                             | Change in a non-linear curve (steep slope).                                                                                                                                 |     |
| KEY MODE        | Specifies how notes are sounded.                 |                                                                                                                                                                             | 119 |
|                 | POLY                                             | Polyphonic                                                                                                                                                                  |     |
|                 | SOLO                                             | Monophonic                                                                                                                                                                  |     |
|                 | UNISON                                           | Unison                                                                                                                                                                      |     |
|                 | SL-UNISON                                        | Monophonic unison                                                                                                                                                           |     |
| AFT LFO         | -63–+63                                          | Sets how much aftertouch changes the LFO intensity.<br>* This is only enabled for products with aftertouch.                                                                 |     |
| AFT FREQ        | -63–+63                                          | Sets how much aftertouch changes the low-pass filter intensity.<br>* This is only enabled for keyboards with aftertouch.                                                    |     |
| AFT LEVEL       | -63–+63                                          | Sets how much aftertouch changes the tone volume.<br>* This is only enabled for keyboards with aftertouch.                                                                  |     |
| PITCH DRIFT     | 0–255                                            | Adjusts the slight pitch drift that occurs when notes are played on an analog synthesizer.                                                                                  |     |
| PARAM EXPANSION | OFF, ON                                          | If this is "ON," the range of change for LFO RATE is wider than on the original model.                                                                                      |     |
| CONDITION       | 0–100                                            | Simulates the changes that occur as a unit ages.                                                                                                                            |     |

## TONE

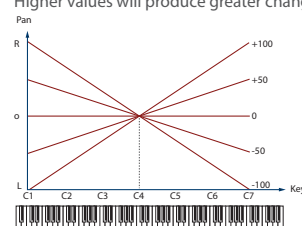
VOCODER, PR-A DRUM, CMN DRUM

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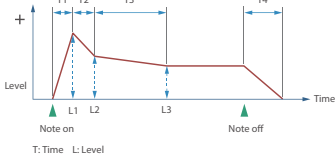
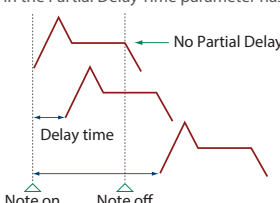
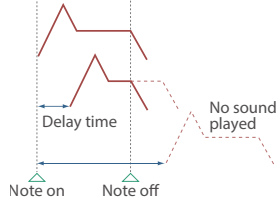
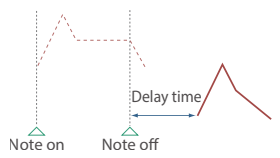
## TONE

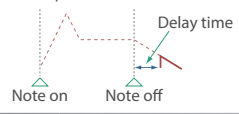
XV-5080, RD-PIANO, PR-A, PR-B, PD-C, PR-D, COMMON, PR-X

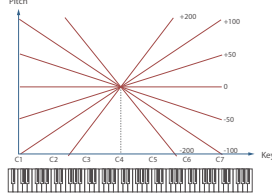
\* The CCs that are supported differ with each partial. (p. 30)

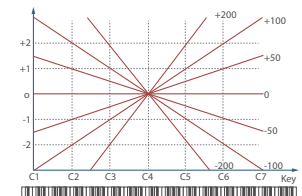
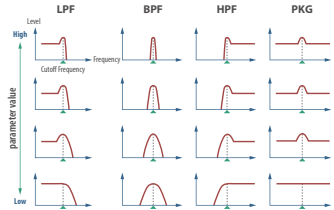
| Parameter       | Value     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tone PMT</b> |           |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Ptl Sw          | OFF, ON   | Use these switch to turn the partials on/off.                                                                                                                                                                                                                                                                                                                                                                                |
| Key Rng Low     | C–G9      | Specify the key range for each partial. Make these settings when you want different key ranges to play different tones.                                                                                                                                                                                                                                                                                                      |
| Key Rng Upp     | C–G9      | Specify the lower limit (Low) and upper limit (Upp) of the key range.                                                                                                                                                                                                                                                                                                                                                        |
| Key Fade Low    | 0–127     | Specifies the degree to which the partial is sounded by notes played below the Key Rng Low. If you don't want the tone to sound at all, set this parameter to "0."                                                                                                                                                                                                                                                           |
| Key Fade Upp    | 0–127     | Specifies the degree to which the partial is sounded by notes played above the Key Rng Upp. If you don't want the tone to sound at all, set this parameter to "0."                                                                                                                                                                                                                                                           |
| Velo Rng Low    | 1–127     | Specify the lower limit (Low) and upper limit (Upp) of the velocities that will sound the partial. Make these settings when you want different partials to sound depending on keyboard playing dynamics.                                                                                                                                                                                                                     |
| Velo Rng Upp    | 1–127     |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Velo FadeLow    | 0–127     | Specifies the degree to which the partial is sounded by notes played more strongly than Velo Rng Low. If you don't want the tone to sound at all, set this parameter to "0."                                                                                                                                                                                                                                                 |
| Velo FadeUpp    | 0–127     | Specifies the degree to which the partial is sounded by notes played more strongly than Velo Rng Upp. If you don't want the tone to sound at all, set this parameter to "0."                                                                                                                                                                                                                                                 |
| <b>Tone PTL</b> |           |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Level           | 0–127     | Sets the volume of the partial. This setting is useful primarily for adjusting the volume balance between partials.                                                                                                                                                                                                                                                                                                          |
| Coarse Tune     | -48–+48   | Adjusts the pitch of the sound up or down in semitone steps (+/-4 octaves).                                                                                                                                                                                                                                                                                                                                                  |
| Fine Tune       | -50–+50   | Adjusts the pitch of the sound up or down in 1-cent steps (+/-50 cents).                                                                                                                                                                                                                                                                                                                                                     |
| Pit Rnd         | 0–1200    | This specifies the width of random pitch deviation that will occur each time a key is pressed. If you do not want the pitch to change randomly, set this to "0."<br>* These values are in units of cents (1/100th of a semitone).                                                                                                                                                                                            |
| Pan             | L64–63R   | Sets the pan of the partial. "L64" is far left, "0" is center, and "63R" is far right.                                                                                                                                                                                                                                                                                                                                       |
| Pan Keyf        | -100–+100 | Use this parameter if you want key position to affect panning. Positive (+) value will cause notes higher than C4 key (center C) to be panned increasingly further toward the right, and negative (-) value will cause notes higher than C4 key (center C) to be panned toward the left. Higher values will produce greater change.<br> |
| Pan Rnd         | 0–63      | Use this parameter when you want the stereo location to change randomly each time you press a key. Higher values will produce a greater amount of change.                                                                                                                                                                                                                                                                    |

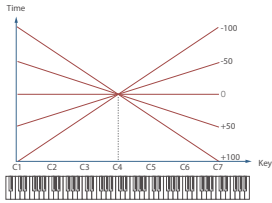
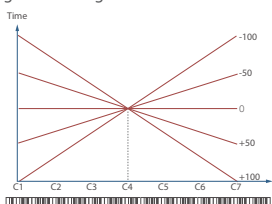
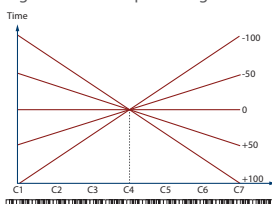
## Tone Parameters

| Parameter | Value           | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pan Alt   | L64–63R         | This setting causes panning to be alternated between left and right each time a key is pressed. Higher values will produce a greater amount of change. “L” or “R” settings will reverse the order in which the pan will alternate between left and right.<br>For example if two partials are set to “L” and “R” respectively, the panning of the two tones will alternate each time they are played.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Env Mode  | NO-SUS, SUSTAIN | If this is set to SUSTAIN, the Envelope Level 3 is held from when the envelope Time 3 has elapsed until note-off.<br>When note-off occurs, the envelope transitions from the current value to the Time 4 segment (release segment).<br>If this is set to NO-SUS, the envelope transitions to the release segment after passing Time 3 regardless of the note-off timing, operating according to the times specified by the envelope.<br>                                                                                                                                                                                                                                                                                                            |
| DlyMod    |                 | <b>Partial Delay</b><br>This produces a time delay between the moment a key is pressed (or released), and the moment the partial actually begins to sound. You can also make settings that shift the timing at which each partial is sounded.<br>This differs from the Delay in the internal effects, in that by changing the sound qualities of the delayed partials and changing the pitch for each partial, you can also perform arpeggio-like passages just by pressing one key.<br>You can also synchronize the partial delay time to the tempo of the external MIDI sequencer.<br>If Retrigger Intvl (Legato Retrigger Interval) is other than OFF, legato operation occurs only when Delay Mode is NORMAL.<br>Also in this case, Retrigger Intvl (Legato Retrigger Interval) operates as “0” (retriggers at each Delay Time). |
| DlyMod    | NORMAL          | The partial begins to play after the time specified in the Partial Delay Time parameter has elapsed.<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|           | HOLD            | Although the partial begins to play after the time specified in the Partial Delay Time parameter has elapsed, if the key is released before the time specified in the Partial Delay Time parameter has elapsed, the partial is not played.<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|           | KEYOFF-NORMAL   | Rather than being played while the key is pressed, the partial begins to play once the period of time specified in the Partial Delay Time parameter has elapsed after release of the key.<br>This is effective in situations such as when simulating noises from guitars and other instruments.<br>                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Parameter             | Value                                                                                                              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DlyMod                | KEYOFF DECAY                                                                                                       | Rather than being played while the key is pressed, the partial begins to play once the period of time specified in the Partial Delay Time parameter has elapsed after release of the key. Here, however, changes in the TVA Envelope begin while the key is pressed, which in many cases means that only the sound from the release portion of the envelope is heard.<br> |
| Dly Time Sync         | OFF, ON                                                                                                            | Set this ON if you want the partial delay time to synchronize with the tempo.                                                                                                                                                                                                                                                                                                                                                                                |
| Dly Time Note         | 1/64T, 1/64, 1/32T, 1/32, 1/16T, 1/32, 1/16, 1/8T, 1/16, 1/8, 1/4T, 1/8, 1/4, 1/2T, 1/4, 1/2, 1T, 1/2, 1, 2T, 1, 2 | This is available when Dly Time Sync is “ON.” It specifies the delay time in terms of a note value.                                                                                                                                                                                                                                                                                                                                                          |
| Dly Time              | 0–1023                                                                                                             | This is available when Dly Time Sync is OFF. It specifies the delay time without regard to the tempo.                                                                                                                                                                                                                                                                                                                                                        |
| Cho Send              | 0–127                                                                                                              | Specifies the level of the signal sent to the chorus for each partial.                                                                                                                                                                                                                                                                                                                                                                                       |
| Rev Send              | 0–127                                                                                                              | Specifies the level of the signal sent to the reverb for each partial.                                                                                                                                                                                                                                                                                                                                                                                       |
| Rx Bend               | OFF, ON                                                                                                            | Specifies for each partial whether MIDI pitch bend messages are received (ON) or not received (OFF).                                                                                                                                                                                                                                                                                                                                                         |
| Rx Expr               | OFF, ON                                                                                                            | Specifies for each partial whether MIDI expression messages are received (ON) or not received (OFF).                                                                                                                                                                                                                                                                                                                                                         |
| Rx Hold               | OFF, ON                                                                                                            | Specifies for each partial whether MIDI hold 1 messages are received (ON) or not received (OFF).                                                                                                                                                                                                                                                                                                                                                             |
| Redamp Sw             | OFF, ON                                                                                                            | If Redamp Sw is ON, you can perform the Half Damper operations used for piano sounds. However, the following conditions must be satisfied in order to use this operation.<br>• Env Mode is NO-SUS<br>• Amp Env’s Level 1 and 2 are 1 or greater<br>• Amp Env’s Times are Time 3 > Time 4                                                                                                                                                                     |
| Out Assign            | DRY, MFX                                                                                                           | Specifies how the sound of each partial will be output.                                                                                                                                                                                                                                                                                                                                                                                                      |
| Wav Gr Type           | INT, EXT                                                                                                           | Specifies the type of wave group.<br>* If an expansion is installed, “EXT” is shown.                                                                                                                                                                                                                                                                                                                                                                         |
| Wav Gr ID             | A - E                                                                                                              | Sets the wave group ID within the specified wave group type.<br>* This is enabled when Wav Gr Type is “INT.”<br>* If an expansion is installed, the expansion name “EXZ***” is shown.                                                                                                                                                                                                                                                                        |
| Wav L No. [Wave Name] | (Wave number)<br>(Wave name)                                                                                       | Specifies the Wave within the wave group ID. For mono use, specify only the left side (L).<br>For stereo use, specify the right side (R) as well.                                                                                                                                                                                                                                                                                                            |
| Wav R No. [Wave Name] | (Wave number)<br>(Wave name)                                                                                       | If the sound will be played in mono, specify only Wav L No., and leave Wav R No. set as “0” (OFF).<br>Sound is not produced if you specify only Wav R No.                                                                                                                                                                                                                                                                                                    |
| Wav Gain              | -18, -12, -6, 0, +6, +12 [dB]                                                                                      | Specifies the gain (amplitude) of the waveform. The value will change in 6 dB (decibel) steps. Each 6 dB increase doubles the gain.                                                                                                                                                                                                                                                                                                                          |
| FXM Sw                | OFF, ON                                                                                                            | This sets whether FXM will be used (ON) or not (OFF).<br>* FXM (Frequency Cross Modulation) uses a specified waveform to apply frequency modulation to the currently selected waveform, creating complex overtones. This is useful for creating dramatic sounds or sound effects.                                                                                                                                                                            |
| FXM Color             | 1–4                                                                                                                | Specifies how FXM will perform frequency modulation. Higher settings result in a grainier sound, while lower settings result in a more metallic sound.                                                                                                                                                                                                                                                                                                       |
| FXM Depth             | 0–16                                                                                                               | Specifies the depth of the modulation produced by FXM.                                                                                                                                                                                                                                                                                                                                                                                                       |

| Parameter    | Value                                                                                           | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pit Keyf     | -200—+200                                                                                       | <p>This specifies the amount of pitch change that will occur when you play a key one octave higher (i.e., 12 keys upward on the keyboard). If you want the pitch to rise one octave as on a conventional keyboard, set this to "+100." If you want the pitch to rise two octaves, set this to "+200." Conversely, set this to a negative (-) value if you want the pitch to fall. With a setting of "0," all keys will produce the same pitch.</p>                                                                                                                                                                                                                                                                                                            |
| Soft EQ Sens | 0–100                                                                                           | Increases the proportion by which the EQ's HighGain is lowered by the amount of pedal. With a setting of "0," this has no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| TVF Type     | <p>Selects the type of TVF filter.</p> <p>* If Filter Type is set to VCF, this will be LPF.</p> | OFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|              | LPF                                                                                             | No filter is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|              | LPF                                                                                             | Low Pass Filter. This cuts the frequencies in the region above the cutoff frequency. Since this cuts the high-frequency region, the sound becomes more mellow. This is the most common filter used in synthesizers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|              | BPF                                                                                             | Band Pass Filter. This leaves only the frequencies in the region of the cutoff frequency, and cuts the rest. This can be useful when creating distinctive sounds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|              | HPF                                                                                             | High Pass Filter. This cuts the frequencies in the region below the cutoff frequency. This is suitable for creating percussive sounds emphasizing their higher tones.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|              | PKG                                                                                             | Peaking Filter. This emphasizes the frequencies in the region of the cutoff frequency. You can use this to create wah-wah effects by employing an LFO to change the cutoff frequency cyclically.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|              | LPF2                                                                                            | Low Pass Filter 2. Although frequency components above the cutoff frequency are cut, the sensitivity of this filter is half that of the LPF. This makes it a comparatively warmer low pass filter. This filter is good for use with simulated instrument sounds such as the acoustic piano.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|              | LPF3                                                                                            | <p>* If you set "LPF2," the setting for the Resonance parameter will be ignored (p. 23).</p> <p>Low Pass Filter 3. Although frequency components above the cutoff frequency are cut, the sensitivity of this filter changes according to the cutoff frequency. While this filter is also good for use with simulated acoustic instrument sounds, the nuance it exhibits differs from that of the LPF2, even with the same TVF Envelope settings.</p> <p>* If you set "LPF3," the setting for the Resonance parameter will be ignored (p. 23).</p>                                                                                                                                                                                                                                                                                              |
| Cutoff       | 0–1023                                                                                          | <p>Selects the frequency at which the filter begins to have an effect on the waveform's frequency components.</p> <p><b>If "LPF/LPF2/LPF3" is selected for the TVF Type</b><br/>Lower cutoff frequency settings reduce a tone's upper harmonics for a more rounded, warmer sound. Higher settings make it sound brighter.</p> <p><b>If "BPF" is selected for the TVF Type</b><br/>Harmonic components will change depending on the TVF cutoff frequency setting. This can be useful when creating distinctive sounds.</p> <p><b>If "HPF" is selected for the TVF Type</b><br/>Higher cutoff frequency settings will reduce lower harmonics to emphasize just the brighter components of the sound.</p> <p><b>If "PKG" is selected for the TVF Type</b><br/>The harmonics to be emphasized will vary depending on cutoff frequency setting.</p> |

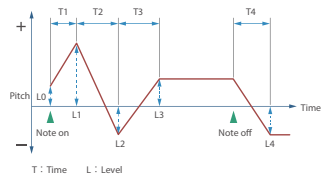
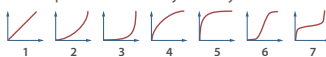
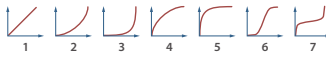
| Parameter    | Value                                                                                                                                                                                                                          | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cutoff Keyf  | -200—+200                                                                                                                                                                                                                      | <p>Use this parameter if you want the cutoff frequency to change according to the key that is pressed. Relative to the cutoff frequency at the key specified by Cutoff Keyf BP (Cutoff Keyfollow Base Point), positive "+" values cause the cutoff frequency to become higher as you play above the reference key, and negative "-" values cause the cutoff frequency to become lower. Higher values will produce greater change.</p>  |
| Cutoff VCrv  | FIXED, 1–7                                                                                                                                                                                                                     | <p>Selects one of the following seven curves that determine how keyboard playing dynamics (velocity) influence the cutoff frequency. Set this to "FIXED" if you don't want the cutoff frequency to be affected by the keyboard velocity.</p>                                                                                                                                                                                           |
| Cutoff VSens | -100—+100                                                                                                                                                                                                                      | Use this parameter when changing the cutoff frequency to be applied as a result of changes in playing velocity. Specify a positive "+" value if you want the cutoff frequency to raise when you play strongly, or a negative "-" value if you want it to lower.                                                                                                                                                                                                                                                           |
| Resonance    | 0–1023                                                                                                                                                                                                                         | <p>Emphasizes the portion of the sound in the region of the cutoff frequency, adding character to the sound. Excessively high settings can produce oscillation, causing the sound to distort.</p>                                                                                                                                                                                                                                    |
| Reso VSens   | -100—+100                                                                                                                                                                                                                      | Use this parameter when changing the resonance to be applied as a result of changes in playing velocity. Specify a positive "+" value if you want resonance to increase when you play strongly, or a negative "-" value if you want it to decrease.                                                                                                                                                                                                                                                                       |
| Bias Lv      | -100—+100                                                                                                                                                                                                                      | Adjusts the angle of the volume change that will occur in the selected Bias Direction. Higher values will produce greater change. Negative (-) values will invert the change direction.                                                                                                                                                                                                                                                                                                                                   |
| Bias Pos     | 0–127                                                                                                                                                                                                                          | Specifies the key relative to which the volume will be modified. A setting of 64 is the C4 key (middle C).                                                                                                                                                                                                                                                                                                                                                                                                                |
| Bias Dir     | Selects the direction in which change will occur starting from the Bias Position.                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | LOWER                                                                                                                                                                                                                          | The volume will be modified for the keyboard area below the Bias Point.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | UPPER                                                                                                                                                                                                                          | The volume will be modified for the keyboard area above the Bias Point.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|              | LOWER&UPPER                                                                                                                                                                                                                    | The volume will be modified symmetrically toward the left and right of the Bias Point.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|              | ALL                                                                                                                                                                                                                            | The volume changes linearly with the bias point at the center.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|              | Selects one of the following seven curves that determine how keyboard dynamics will affect the volume of the partial. Set this to "FIXED" if you don't want the volume of the partial to be affected by the keyboard velocity. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|              | FIXED, 1–7                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Parameter    | Value     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level VSens  | -100–+100 | Set this when you want the volume of the partial to change depending on the force with which you press the keys.<br>Set this to a positive (+) value to have the changes in partial volume increase the more forcefully the keys are played; to make the partial play more softly as you play harder, set this to a negative (-) value.                                                                                                                                 |
| PEnv TKeyf   | -100–+100 | Use this setting if you want the pitch envelope times (Time 2–Time 4) to be affected by the keyboard location.<br>Based on the pitch envelope times for the C4 key, positive (+) value will cause notes higher than C4 to have increasingly shorter times, and negative (-) value will cause them to have increasingly longer times.<br>Higher values will produce greater change.<br> |
| FEnv TKeyf   | -100–+100 | Specify this if you want the filter envelope times (Time 2–Time 4) to vary depending on the keyboard position you play. Relative to the filter envelope times at the C4 key (middle C), positive "+" values shorten the times for notes played in the region above C4, and negative "-" values lengthen the times. Higher values will produce greater change.<br>                     |
| AEnv TKeyf   | -100–+100 | Specify this if you want keyboard position to affect the AMP envelope's times (Time 2–Time 4). Relative to the AMP envelope times at the C4 key (middle C), positive (+) values cause the times to shorten as you play higher on the keyboard, and negative (-) values cause the times to lengthen. Higher values will produce greater change.<br>                                   |
| Vib Pit Sens | -100–+100 | Specifies the amount by which the Pitch Depth of LFO1 is changed by the SCENE PART, MODIFY: Vib Depth.                                                                                                                                                                                                                                                                                                                                                                  |
| Vib Cut Sens | -100–+100 | Specifies how the TVF Depth of LFO1 is affected by the SCENE PART, MODIFY: Vib Depth.                                                                                                                                                                                                                                                                                                                                                                                   |
| Vib Amp Sens | -100–+100 | Specifies how the SCENE PART, MODIFY: Vib Depth affects the Amp Depth of LFO1.                                                                                                                                                                                                                                                                                                                                                                                          |

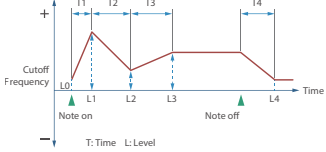
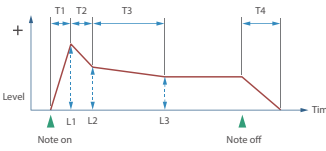
| Parameter                                                | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Explanation                                                                                                                                                                |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mct1 Src<br>Mct2 Src<br>Mct3 Src<br>Mct4 Src             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sets the MIDI message used to change the partial parameter with the Matrix Control.                                                                                        |
|                                                          | OFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Matrix control will not be used.                                                                                                                                           |
|                                                          | CC01–CC31,<br>CC33–CC95                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Controller number 1–31, 33–95                                                                                                                                              |
|                                                          | BEND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Pitch bend                                                                                                                                                                 |
|                                                          | AFT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Aftertouch                                                                                                                                                                 |
|                                                          | SYS-CTRL1–4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MIDI messages assigned by the SYSTEM parameters, MIDI: SysCtrlSrc1–4                                                                                                       |
|                                                          | VELOCITY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Velocity (pressure you press a key with)                                                                                                                                   |
|                                                          | KEYFOLLOW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Keyfollow (keyboard position with C4 as 0)                                                                                                                                 |
|                                                          | TEMPO,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Tempo specified by the tempo assign source                                                                                                                                 |
|                                                          | LFO1,<br>LFO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LFO 1<br>LFO 2                                                                                                                                                             |
|                                                          | PIT-ENV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Pitch envelope                                                                                                                                                             |
|                                                          | TVF-ENV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Filter envelope                                                                                                                                                            |
|                                                          | TVA-ENV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Amp envelope                                                                                                                                                               |
|                                                          | <p>* Velocity and Keyfollow correspond to Note messages.</p> <p>* Although there are no MIDI messages for LFO 1 through AMP Envelope, they can be used as Matrix Control. In this case, you can change the partial settings in realtime by playing tones.</p> <p>* If you want to use common controllers for the entire JUNO-X, select "SYS-CTRL1"–"SYS-CTRL4." MIDI messages used as System Control 1–4 are set with the SysCtrlSrc1–4. For details, refer to "System Parameters" (p. 31).</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                            |
| Mct1 Dst1–4<br>Mct2 Dst1–4<br>Mct3 Dst1–4<br>Mct4 Dst1–4 | <b>NOTE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                            |
|                                                          | <ul style="list-style-type: none"> <li>There are parameters that determine whether or not Pitch Bend, Controller Number 11 (Expression) and Controller Number 64 (Hold 1) are received (p. 22). When these settings are "ON," and the MIDI messages are received, then when any change is made in the settings of the desired parameter, the Pitch Bend, Expression, and Hold 1 settings also change simultaneously. If you want to change the targeted parameters only, then set these to "OFF."</li> <li>There are parameters that let you specify whether specific MIDI messages will be received for each zone in a scene (p. 5). When a tone with Matrix Control settings is assigned to a zone, confirm that any MIDI messages used for the Matrix Control will be received. If the JUNO-X is set up such that reception of MIDI messages is disabled, then the Matrix Control will not function.</li> </ul> |                                                                                                                                                                            |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Selects the partial parameter that is to be controlled when using the Matrix Control.                                                                                      |
|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | When not controlling parameters with the Matrix Control, set this to "OFF." Up to four parameters can be specified for each Matrix Control, and controlled simultaneously. |
|                                                          | OFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Matrix control will not be used.                                                                                                                                           |
|                                                          | PCH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Changes the pitch.                                                                                                                                                         |
|                                                          | CUT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Changes the cutoff frequency.                                                                                                                                              |
|                                                          | RES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Emphasizes the overtones in the region of the cutoff frequency, adding character to the sound.                                                                             |
|                                                          | LEV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Changes the volume level.                                                                                                                                                  |
|                                                          | PAN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Changes the pan.                                                                                                                                                           |
|                                                          | CHO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Changes the amount of chorus.                                                                                                                                              |
|                                                          | REV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Changes the amount of reverb.                                                                                                                                              |
|                                                          | PIT-LFO1, PIT-LFO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Changes the vibrato depth.                                                                                                                                                 |
|                                                          | TVF-LFO1,<br>TVF-LFO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Changes the wah depth.                                                                                                                                                     |
|                                                          | TVA-LFO1,<br>TVA-LFO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Changes the tremolo depth.                                                                                                                                                 |
|                                                          | PAN-LFO1,<br>PAN-LFO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Changes the effect that the LFO will have on pan.                                                                                                                          |
|                                                          | LFO1-RATE<br>LFO2-RATE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Changes the speed of the LFO cycles. The speed will not change if LFO Rate is set to "note."                                                                               |
|                                                          | PIT-ATK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 1 of the pitch envelope.                                                                                                                                  |
|                                                          | PIT-DCY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 2 and Env Time 3 of the pitch envelope.                                                                                                                   |
|                                                          | PIT-REL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 4 of the pitch envelope.                                                                                                                                  |
|                                                          | TVF-ATK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 1 of the FLT envelope.                                                                                                                                    |
|                                                          | TVF-DCY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 2 and Env Time 3 of the FLT envelope.                                                                                                                     |
|                                                          | TVF-REL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 4 of the FLT envelope.                                                                                                                                    |
|                                                          | TVA-ATK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 1 of the AMP envelope.                                                                                                                                    |
|                                                          | TVA-DCY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 2 and Env Time 3 of the AMP envelope.                                                                                                                     |
|                                                          | TVA-REL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Changes the Time 4 of the AMP envelope.                                                                                                                                    |

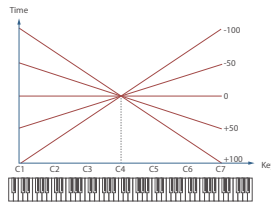


| Parameter                                                | Value                | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mct1 Dst1-4<br>Mct2 Dst1-4<br>Mct3 Dst1-4<br>Mct4 Dst1-4 | PMT                  | <p>If the Matrix Control is used to split partials, set the PMT Velocity Control (p. 17) to "OFF"</p> <ul style="list-style-type: none"> <li>If the Matrix Control is used to split partials, we recommend setting the Mct1 Sns (p. 25) to "+63." Selecting a lower value may prevent switching of the partials. Furthermore, if you want to reverse the effect, set the value to "-63."</li> <li>If you want to use matrix control to switch smoothly between partials, use the Velocity Fade Lower and Velocity Fade Upper (p. 21). The higher the values set, the smoother the switch is between the partials.</li> </ul> |
|                                                          | FXM                  | Changing the depth of frequency modulation produced by FXM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                          | MFx-CTL1-4           | Applies a change to MFx Src1-4. If this is specified for more than one partial, the result will be the summed values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                          | PW                   | Applies change to PW.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                          | PWM                  | Applies change to PWM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | FAT                  | Applies change to FAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | XMOD                 | This setting is valid only for the carrier partial (Partial 1 or 3), and applies change to the XMD12 Dpth or XMD34 Dpth.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                          | LFO1-STP<br>LFO2-STP | This is valid if the LFO1/LFO2 Waveform is STEP; it specifies the step position. In this case, the Sns value is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                          | SSAW-DTN             | This is effective if OSC Type is SuperSAW; it applies change to Super-SAW Detune.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                          | PIT-DPTH             | Changes the depth of the Pitch envelope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                          | TVF-DPTH             | Changes the depth of the Filter envelope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                          | TVA-DPTH             | Changes the depth of the AMP envelope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | XMOD2                | This is effective when Structure 1-2 (3-4) is XMOD2; it applies change to XMD2 12 (34) Dpth.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                          | ATT                  | Changes the OSC level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | R-OSC1-LV            | This is valid only for Partial 1 and 3; when Structure 12 (or Structure 34 in the case of Partial 3) is RING, this changes the OSC level of Partial 1 (or 3 in the case of Partial 3).                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | R-OSC2-LV            | This is valid only for Partial 1 and 3; when Structure 12 (or Structure 34 in the case of Partial 3) is RING, this changes the OSC level of Partial 2 (or 4 in the case of Partial 3).                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | X-OSC1-LV            | This is valid only for Partial 1 and 3; when Structure 12 (or Structure 34 in the case of Partial 3) is XMOD/XMOD2, this changes the OSC level of Partial 1 (or 3 in the case of Partial 3).                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                          | X-OSC2-LV            | This is valid only for Partial 1 and 3; when Structure 12 (or Structure 34 in the case of Partial 3) is XMOD/XMOD2, this changes the OSC level of Partial 2 (or 4 in the case of Partial 3).                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Mct1 Sns1-4<br>Mct2 Sns1-4<br>Mct3 Sns1-4<br>Mct4 Sns1-4 | -63--+63             | <p>Specify the effective depth of the matrix controls. To make an increase in the currently selected value (to get higher values, move to the right, increase rates, and so on), select a positive (+) value; to make a decrease in the currently selected value (to get lower values, move to the left, decrease rates, and so on), select a negative (-) value.</p> <p>For either positive or negative value, greater absolute values will allow greater amounts of change.</p> <p>Set this to "0" if you don't want to apply the effect.</p>                                                                              |
| Damp Free                                                | OFF, 1-127           | For notes above the specified note number, the Env Mode operates as NO-SUS. Use this to simulate the undamped region of a piano sound.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Stereo Width                                             | 0-100                | Adjusts the amount of width when outputting in stereo. This has no effect when outputting in mono.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Stereo Detn                                              | -50--+50             | Specifies the detune between L↔R when outputting in stereo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DF Dcy Offset                                            | -100--+100           | Specifies a fine adjustment to the time over which the sound decays when the Damper Free effect is applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Tempo Sync                                               | OFF, ON              | Set this ON if you want the LFO rate to synchronize with the tempo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Parameter                                           | Value         | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pitch Env</b>                                    |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Pit Depth                                           | -100--+100    | Adjusts the effect of the Pitch Envelope. Higher settings will cause the pitch envelope to produce greater change. Negative (-) value will invert the shape of the envelope. If OSC Type is other than VA, this is limited to $\pm 63$ .                                                                                                                                                                                                                                                           |
| Pit VSens                                           | -100--+100    | Keyboard playing dynamics can be used to control the depth of the pitch envelope. If you want the pitch envelope to have more effect for strongly played notes, set this parameter to a positive (+) value. If you want the pitch envelope to have less effect for strongly played notes, set this to a negative (-) value.                                                                                                                                                                        |
| Pit T1 VSens                                        | -100--+100    | This allows keyboard dynamics to affect the Time 1 of the Pitch envelope. If you want Time 1 to be speeded up for strongly played notes, set this parameter to a positive "+" value. If you want it to be slowed down, set this to a negative "-" value.                                                                                                                                                                                                                                           |
| Pit T4 VSens                                        | -100--+100    | Use this parameter when you want key release speed to affect the Time 4 value of the pitch envelope. If you want Time 4 to be speeded up for quickly released notes, set this parameter to a positive (+) value. If you want it to be slowed down, set this to a negative (-) value.                                                                                                                                                                                                               |
| Pit Time1<br>Pit Time2<br>Pit Time3<br>Pit Time4    | 0-1023        | <p>Specify the pitch envelope times (Time 1–Time 4). Higher settings will result in a longer time until the next pitch is reached. (For example, Time 2 is the time over which the pitch changes from Level 1 to Level 2.)</p> <p>* If ADSR Envelope Switch is ON, the Time 2 has no effect.</p>  <p>T : Time L : Level</p>                                                                                   |
| Pit Lv0<br>Pit Lv1<br>Pit Lv2<br>Pit Lv3<br>Pit Lv4 | -511--+511    | <p>Specify the pitch envelope levels (Level 0–Level 4). It determines how much the pitch changes from the reference pitch (the value set with Coarse Tune or Fine Tune on the Pitch screen) at each point.</p> <p>Positive (+) value will cause the pitch to be higher than the standard pitch, and negative (-) value will cause it to be lower.</p> <p>* If ADSR Envelope Switch is ON, only Level 3 (Sustain) has an effect. Also in this case, settings with a negative value are ignored.</p> |
| Pit VCrv                                            | FIXED,<br>1-7 | <p>Selects one of the following 7 curves that will determine how keyboard playing dynamics will affect the pitch envelope.</p> <p>Set this to "FIXED" if you don't want the pitch envelope to be affected by the keyboard velocity.</p>                                                                                                                                                                       |
| <b>Filter Env</b>                                   |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Filtr Depth                                         | -63--+63      | Specifies the depth of the Filter envelope. Higher settings increase the change produced by the Filter envelope. Negative (-) value will invert the shape of the envelope.                                                                                                                                                                                                                                                                                                                         |
| Filtr VCrv                                          | FIXED,<br>1-7 | <p>Selects one of the following seven types of curve by which keyboard playing dynamics affect the depth of the filter envelope.</p> <p>If you don't want keyboard playing dynamics to affect the filter envelope depth, specify "FIXED."</p>                                                                                                                                                                 |
| Filtr VSens                                         | -100--+100    | Specify this if you want keyboard playing dynamics to affect the filter envelope depth. Specify a positive "+" value if you want the filter envelope to apply more deeply as you play more strongly, or a negative "-" value if you want it to apply less deeply.                                                                                                                                                                                                                                  |

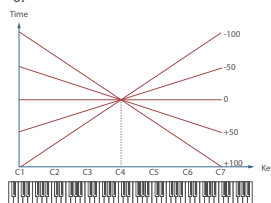
## Tone Parameters

| Parameter                                                     | Value     | Explanation                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filtr T1 VSens                                                | -100~+100 | Specify this if you want keyboard playing dynamics to affect Time 1 of the filter envelope. If you want Time 1 to be speeded up for strongly played notes, set this parameter to a positive "+" value. If you want it to be slowed down, set this to a negative "-" value.                                                                                                  |
| Filtr T4 VSens                                                | -100~+100 | Specify this if you want key release velocity to affect Time 4 of the filter envelope. If you want Time 4 to be speeded up for quickly released notes, set this parameter to a positive (+) value. If you want it to be slowed down, set this to a negative (-) value.                                                                                                      |
| Filtr Time1<br>Filtr Time2<br>Filtr Time3<br>Filtr Time4      | 0~1023    | Specify the filter envelope times (Time 1~Time 4). Higher settings will lengthen the time until the next cutoff frequency level is reached. (For example, Time 2 is the time over which Level 1 will change to Level 2.)<br>* If ADSR Envelope Switch is ON, the Time 2 has no effect.<br> |
| Filtr Lv0<br>Filtr Lv1<br>Filtr Lv2<br>Filtr Lv3<br>Filtr Lv4 | 0~1023    | Specify the filter envelope levels (Level 0~Level 4). Specify the amount of cutoff frequency change at each point relative to the reference cutoff frequency (the cutoff frequency value specified in the Filter screen).<br>* If ADSR Envelope Switch is ON, only Level 3 (Sustain) has an effect.                                                                         |
| <b>Amp Env</b>                                                |           |                                                                                                                                                                                                                                                                                                                                                                             |
| Amp T1 VSens                                                  | -100~+100 | Specify this if you want keyboard dynamics to affect the AMP envelope's Time 1. If you want Time 1 to be speeded up for strongly played notes, set this parameter to a positive (+) value. If you want it to be slowed down, set this to a negative (-) value.                                                                                                              |
| Amp T4 VSens                                                  | -100~+100 | Specify this if you want key release velocity to affect the AMP envelope's Time 4. If you want Time 4 to be speeded up for quickly released notes, set this parameter to a positive (+) value. If you want it to be slowed down, set this to a negative (-) value.                                                                                                          |
| Amp Time1<br>Amp Time2<br>Amp Time3<br>Amp Time4              | 0~1023    | Specify the AMP envelope times (Time 1~Time 4). Higher settings lengthen the time until the next volume level is reached. (For example, Time 2 is the time over which Level 1 will change to Level 2.)<br>* If ADSR Envelope Switch is ON, the Time 2 has no effect.                                                                                                        |
| Amp Lv1<br>Amp Lv2<br>Amp Lv3                                 | 0~1023    | Specify the AMP envelope levels (Level 1~Level 3). These specify the amount of change at each point relative to the reference volume (the partial level value specified in the Amp screen).<br><br>* If ADSR Envelope Switch is ON, only Level 3 (Sustain) has an effect.                |

| Parameter    | Value                                                                                               | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LFO1</b>  |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| L1 Waveform  |                                                                                                     | Selects the waveform of the LFO.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|              | SIN                                                                                                 | Sine wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|              | TRI                                                                                                 | Triangle wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|              | SAW-UP                                                                                              | Sawtooth wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|              | SAW-DW                                                                                              | Sawtooth wave (negative polarity)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|              | SQR                                                                                                 | Square wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|              | RND                                                                                                 | Random wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|              | TRP                                                                                                 | Trapezoidal wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|              | S&H                                                                                                 | Sample & Hold wave (one time per cycle, LFO value is changed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|              | CHS                                                                                                 | Chaos wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| L1 Rate Sync | VSIN                                                                                                | Modified sine wave. The amplitude of a sine wave is randomly varied once each cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|              | STEP                                                                                                | A waveform generated by the data specified by LFO Step 1~16. This produces stepped change with a fixed pattern similar to a step modulator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L1 Rate Sync | OFF, ON                                                                                             | Set this ON if you want the LFO rate to synchronize with the tempo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| L1 Rate Note | 1/64T, 1/64, 1/32T, 1/32, 1/16T, 1/16, 1/8, 1/4T, 1/8, 1/4, 1/2T, 1/4, 1/2, 1T, 1/2, 1, 2T, 1, 2, 4 | Specifies the LFO rate in terms of a note value.<br>* This is effective if Rate Sync is "ON."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| L1 Step Len  | 1~16                                                                                                | Specifies the step size that is looped.<br>* This is effective if Waveform is STEP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| L1 Rate      | 0~1023                                                                                              | Specifies the LFO rate without regard to the tempo. Higher values produce a faster LFO rate (a shorter cycle).<br>* This is effective if Rate Sync is OFF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| L1 Offset    | -100~+100                                                                                           | Raises or lowers the LFO waveform relative to the central value (pitch or cutoff frequency). Positive (+) value will move the waveform so that modulation will occur from the central value upward. Negative (-) value will move the waveform so that modulation will occur from the central value downward.                                                                                                                                                                                                                                                                                                                                                     |
| L1 Rate Detn | 0~127                                                                                               | Subtly changes the LFO cycle speed (Rate parameter) each time you press a key. Higher values produce greater change. This parameter is invalid when Rate is set to "note."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| L1 Dly Time  | 0~1023                                                                                              | Specifies the time elapsed before the LFO effect is applied (the effect continues) after the key is pressed (or released).<br>* After referring to "How to Apply the LFO" (p. 29), change the setting until the desired effect is achieved.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L1 Dly Keyf  | -100~+100                                                                                           | Adjusts the value for the Delay Time parameter depending on the key position, relative to the C4 key (center C). To decrease the time that elapses before the LFO effect is applied (the effect is continuous) with each higher key that is pressed in the upper registers, select a positive (+) value; to increase the elapsed time, select a negative (-) value. Higher values will produce greater change. If you do not want the elapsed time before the LFO effect is applied (the effect is continuous) to change according to the key pressed, set this to "0."<br> |
| L1 Fade Mod  | ON-IN, ON-OUT, OFF-IN, OFF-OUT                                                                      | Specifies how the LFO will be applied.<br>* After referring to "How to Apply the LFO" (p. 29), change the setting until the desired effect is achieved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| L1 Fade Time | 0~1023                                                                                              | Specifies the time over which the LFO amplitude will reach the maximum (minimum).<br>* After referring to "How to Apply the LFO" (p. 29), change the setting until the desired effect is achieved.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



| Parameter        | Value                                                                                                                 | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1 Key Trig      | OFF, ON                                                                                                               | Specifies whether the LFO cycle will be synchronized to begin when the key is pressed (ON) or not (OFF).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| L1 Pit Depth     | -100+100                                                                                                              | Specifies how deeply the LFO will affect pitch.<br>* If OSC Type is other than VA, the range is limited to -63+63.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| L1 Flt Depth     | -100+100                                                                                                              | Specifies how deeply the LFO will affect the cutoff frequency.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| L1 Amp Depth     | -100+100                                                                                                              | Specifies how deeply the LFO will affect the volume.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| L1 Pan Depth     | -63+63                                                                                                                | Specifies how deeply the LFO will affect the pan.<br><b>MEMO</b><br>Positive (+) and negative (-) value for the Depth parameter result in differing kinds of change in pitch and volume. For example, if you set the Depth parameter to a positive (+) value for one partial, and set another partial to the same numerical value, but make it negative (-), the modulation phase for the two partials will be the reverse of each other. This allows you to shift back and forth between two different partials, or combine it with the Pan setting to cyclically change the location of the sound image. |
| L1 Phase Pos     | 0, 1, 2, 3                                                                                                            | Specifies the LFO's starting phase value when Key Trigger is "ON."<br>* This has no effect if Waveform is RND, S&H, or CHS.<br>0 1 cycle<br>1 1/4 cycle<br>2 1/2 cycle<br>3 3/4 cycle                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L1 Stp1-16 Depth | -72+72                                                                                                                | This is effective if Waveform is STEP.<br>Specify the Depth value of each step.<br>If you want to specify this in pitch scale degrees (100 cents), the settings are as follows.<br>1 Pitch Depth: 51, Step: multiples of 6 ... up to one octave of change<br>2 Pitch Depth: 74, Step: multiples of 3 ... up to two octaves of change<br>3 Pitch Depth: 89, Step: multiples of 2 ... up to three octaves of change<br>* If OSC Type is not VA, the Pitch Depth setting range is limited to -63+63, so only "1" above is possible.                                                                           |
| L1 Stp1-16 Curve | 0-36                                                                                                                  | Specifies the type of curve at each step.<br>→ "Step curve types" (p. 29)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>LFO2</b>      |                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| L2 Waveform      | SIN, TRI, SAW-UP, SAW-DW, SQR, RND, TRP, S&H, CHS, VSIN, STEP                                                         | Selects the waveform of the LFO.<br>SIN Sine wave<br>TRI Triangle wave<br>SAW-UP Sawtooth wave<br>SAW-DW Sawtooth wave (negative polarity)<br>SQR Square wave<br>RND Random wave<br>TRP Trapezoidal wave<br>S&H Sample & Hold wave (one time per cycle, LFO value is changed)<br>CHS Chaos wave<br>VSIN Modified sine wave. The amplitude of a sine wave is randomly varied once each cycle.<br>STEP A waveform generated by the data specified by LFO Step 1-16. This produces stepped change with a fixed pattern similar to a step modulator.                                                           |
| L2 Rate Sync     | OFF, ON                                                                                                               | Set this ON if you want the LFO rate to synchronize with the tempo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| L2 Rate Note     | 1/64T, 1/64, 1/32T, 1/32, 1/16T, 1/32, 1/16, 1/8T, 1/16, 1/8, 1/4T, 1/8, 1/4, 1/2T, 1/4, 1/2, 1T, 1/2, 1, 2T, 1, 2, 4 | This is effective if Rate Sync is "ON."<br>Specifies the LFO rate in terms of a note value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| L2 Step Length   | 1-16                                                                                                                  | This is effective if Waveform is STEP.<br>Specifies the step size that is looped.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Parameter    | Value                          | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L2 Rate      | 0-1023                         | This is effective if Rate Sync is OFF.<br>Specifies the LFO rate without regard to the tempo. Higher values produce a faster LFO rate (a shorter cycle).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| L2 Offset    | -100+100                       | Raises or lowers the LFO waveform relative to the central value (pitch or cutoff frequency). Positive (+) value will move the waveform so that modulation will occur from the central value upward. Negative (-) value will move the waveform so that modulation will occur from the central value downward.                                                                                                                                                                                                                                                                                                                                                    |
| L2 Rate Detn | 0-127                          | Subtly changes the LFO cycle speed (Rate parameter) each time you press a key. Higher values produce greater change.<br>This parameter is invalid when Rate is set to "note."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| L2 Dly Time  | 0-1023                         | Specifies the time elapsed before the LFO effect is applied (the effect continues) after the key is pressed (or released).<br>* After referring to "How to Apply the LFO" (p. 29), change the setting until the desired effect is achieved.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| L2 Dly Keyf  | -100+100                       | Adjusts the value for the Delay Time parameter depending on the key position, relative to the C4 key (center C). To decrease the time that elapses before the LFO effect is applied (the effect is continuous) with each higher key that is pressed in the upper registers, select a positive (+) value; to increase the elapsed time, select a negative (-) value. Higher values will produce greater change. If you do not want the elapsed time before the LFO effect is applied (the effect is continuous) to change according to the key pressed, set this to "0."<br> |
| L2 Fade Mod  | ON-IN, ON-OUT, OFF-IN, OFF-OUT | Specifies how the LFO will be applied.<br>* After referring to "How to Apply the LFO" (p. 29), change the setting until the desired effect is achieved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| L2 Fade Time | 0-1023                         | Specifies the time over which the LFO amplitude will reach the maximum (minimum).<br>* After referring to "How to Apply the LFO" (p. 29), change the setting until the desired effect is achieved.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| L2 Key Trig  | OFF, ON                        | Specifies whether the LFO cycle will be synchronized to begin when the key is pressed (ON) or not (OFF).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| L2 Pit Depth | -100+100                       | Specifies how deeply the LFO will affect pitch.<br>* If OSC Type is other than VA, the range is limited to -63+63.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| L2 Flt Depth | -100+100                       | Specifies how deeply the LFO will affect the cutoff frequency.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| L2 Amp Depth | -100+100                       | Specifies how deeply the LFO will affect the volume.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| L2 Pan Depth | -63+63                         | Specifies how deeply the LFO will affect the pan.<br><b>MEMO</b><br>Positive (+) and negative (-) value for the Depth parameter result in differing kinds of change in pitch and volume. For example, if you set the Depth parameter to a positive (+) value for one partial, and set another partial to the same numerical value, but make it negative (-), the modulation phase for the two partials will be the reverse of each other. This allows you to shift back and forth between two different partials, or combine it with the Pan setting to cyclically change the location of the sound image.                                                      |
| L2 Phase Pos | 0, 1, 2, 3                     | Specifies the LFO's starting phase value when Key Trigger is "ON."<br>* This has no effect if Waveform is RND, S&H, or CHS.<br>0 1 cycle<br>1 1/4 cycle<br>2 1/2 cycle<br>3 3/4 cycle                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

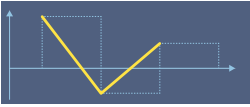
## Tone Parameters

| Parameter        | Value                                                | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L2 Stp1–16 Depth | -72–+72                                              | <p>This is effective if Waveform is STEP. Specify the Depth value of each step. If you want to specify this in pitch scale degrees (100 cents), the settings are as follows.</p> <ol style="list-style-type: none"> <li>Pitch Depth: 51, Step: multiples of 6 ... up to one octave of change</li> <li>Pitch Depth: 74, Step: multiples of 3 ... up to two octaves of change</li> <li>Pitch Depth: 89, Step: multiples of 2 ... up to three octaves of change</li> </ol> <p>* If OSC Type is not VA, the Pitch Depth setting range is limited to -63–+63, so only "1" above is possible.</p> |
| L2 Stp1–16 Curve | 0–36                                                 | Specifies the type of curve at each step.<br>→ "Step curve types" (p. 29)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>EQ</b>        |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Gain L           | -24.0–+24.0 [dB]                                     | Gain of the low range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Gain M           | -24.0–+24.0 [dB]                                     | Gain of the middle range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Gain H           | -24.0–+24.0 [dB]                                     | Gain of the high range                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Freq L           | 20–16000 [Hz]                                        | Frequency of the low range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Freq M           | 20–16000 [Hz]                                        | Frequency of the middle range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Freq H           | 20–16000 [Hz]                                        | Frequency of the high range.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mid Q            | 0.5–16.0                                             | Width of the middle range.<br>Set a higher value to narrow the range to be affected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| EQ Sw            | OFF, ON                                              | Turns the equalizer on/off for each partial.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Synth PTL</b> |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| OSC Type         | Specifies the oscillator type.                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                  | PCM                                                  | PCM is used.<br>The wave of the number specified by the Wav Gr ID and Wav L/R No. is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                  | VA                                                   | A numerically calculated analog-modeled wave is generated.<br>The wave of the number specified by Waveform is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                  | PCM-Sync                                             | The wave of the number specified by Sync Wav No. is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                  | SuperSAW                                             | SuperSAW is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                  | Noise                                                | White noise is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Wav Form         | Specifies the wave that is used when OSC Type is VA. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                  | SAW                                                  | Sawtooth wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | SQR                                                  | Square wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | TRI                                                  | Triangle wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                  | SIN                                                  | sine wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                  | RAMP                                                 | Ramp wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                  | JUNO                                                 | Modulated sawtooth wave                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                  | TRI2                                                 | Triangle wave variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                  | TRI3                                                 | Triangle wave variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Sync Wav No.     | (Wave number)                                        | When "PCM-Sync" is selected for the OSC Type, set the wave to use here.<br>PCM-Sync is an effective oscillator type when "SYNC" is selected for the structure, when partial 1 is set for Structure 1–2 and when partial 3 is set for Structure 3–4.                                                                                                                                                                                                                                                                                                                                         |
|                  | (Wave name)                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Pulse Width      | 0–127                                                | <p>This effect is produced when the waveform is deformed by varying the duty cycle of the pulse width.<br/>It is effective when OSC Type is VA, and is also effective with waveforms other than SQR (square wave).<br/>* If the value is 64, the pulse width has a 50%:50% duty cycle.</p>                                                                                                                                                                                                                                                                                                  |
| PWM Depth        | -63–+63                                              | Specifies the amount (depth) of LFO applied to PW (Pulse Width).<br>PW is modulated according to the LFO2 setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SSaw Detune      | 0–127                                                | <p>Adjusts the Detune depth for SuperSAW. Higher values produce a deeper Detune effect.<br/>* This is effective only when SuperSAW is selected as the OSC Type.</p>                                                                                                                                                                                                                                                                                                                                                                                                                         |

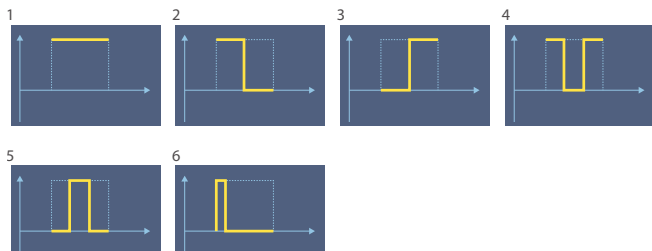
| Parameter       | Value                    | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Click Type      | SOFT, HARD, NATURAL, OFF | Changes the sense of attack by varying the position at which the sound starts.<br>This is available if OSC Type is VA. However, HARD is effective only when Waveform is TRI, TRI2, SIN, or SIN2.                                                                                                                                                                                                                                                                                                 |
| HPF Cutoff      | 0–1023                   | Specifies the cutoff frequency of the -6 dB high-pass filter.<br>* This parameter is effective when Filter Type is VCF.                                                                                                                                                                                                                                                                                                                                                                          |
| Filter Type     | TVF, VCF                 | <p>Selects the type of filter.<br/>* TVF stands for Time Variant Filter, a filter that lets you specify in detail how the frequency components of the sound change over time.<br/>* If you select VCF, the polyphony will be lower than if you select TVF.</p>                                                                                                                                                                                                                                   |
| Flt Slope       | -12, -18, -24 [dB/Oct]   | <p>This button selects the slope (steepness) of the filter.<br/>For VCF, you can choose -12, -18, or -24.<br/>For TVF, only -12 or -24 can be selected.<br/>If Filter Type is TVF, the following limitations apply.</p> <ul style="list-style-type: none"> <li>You can specify only -12 dB or -24 dB. If you specify -18 dB, the sound generator operates internally with the -12 dB setting.</li> <li>If you specify -24 dB, the polyphony will be lower than if you specify -12 dB.</li> </ul> |
| ADSR Env Sw     | OFF, ON                  | This imitates the operation of the ADSR envelope that is provided on an analog synthesizer.<br>If ADSR Env Sw is ON, the "Time 2" parameters of Pitch/Filter/Amp Env Time respectively are ignored, and only the "Level 3" parameters of Pitch/Filter/Amp Env Level are valid.                                                                                                                                                                                                                   |
| Fat             | 0–127                    | Boosts the low-frequency region.<br>This is effective if OSC Type is VA.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| VCF Type        | VCF1, JP, MG, P5         | This parameter is effective when Filter Type is VCF.<br>Each setting simulates the operation of an analog synthesizer's LPF. In particular, MG, JP, and P5 are types that are suitable for reproducing synthesizer sounds of the past.                                                                                                                                                                                                                                                           |
| OSC Attenuator  | 0–255                    | Specifies the OSC level.<br>255 is the reference value. If you want only the self-oscillation of the filter to be heard, set this to "0."                                                                                                                                                                                                                                                                                                                                                        |
| Cutoff Keyf BP  | 0–127                    | Specifies the reference key when using Keyfollow to modify the cutoff frequency. If this is 60, the C4 key (middle C) is the reference key.                                                                                                                                                                                                                                                                                                                                                      |
| VA Invert Sw    | OFF, ON                  | If this is "ON," the phase of the VA waveform is inverted.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| FEnv Fine Depth | -63–+63                  | Finely adjusts the depth of the filter envelope.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PEnvLFOTrig Sw  | OFF, ON                  | <p>If this is "ON," the pitch envelope is cyclically retriggered by LFO1.<br/>* This is effective when Env Mode is SUSTAIN.</p>                                                                                                                                                                                                                                                                                                                                                                  |
| FEnvLFOTrig Sw  | OFF, ON                  | <p>If this is "ON," the filter envelope is cyclically triggered by LFO1.<br/>* This is effective when Env Mode is SUSTAIN.</p>                                                                                                                                                                                                                                                                                                                                                                   |
| AEnvLFOTrig Sw  | OFF, ON                  | <p>If this is "ON," the amp envelope is cyclically triggered by LFO1.<br/>* This is effective when Env Mode is SUSTAIN.</p>                                                                                                                                                                                                                                                                                                                                                                      |

## Step curve types

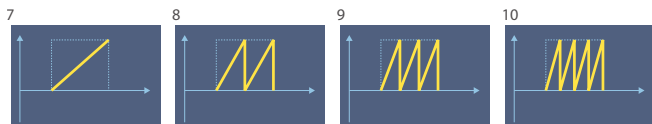
### Step Curve 0



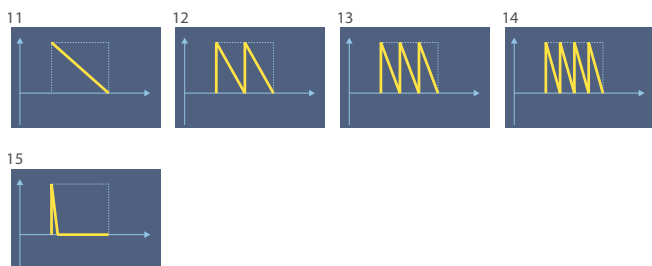
### Step Curve 1–6 (variations of square wave)



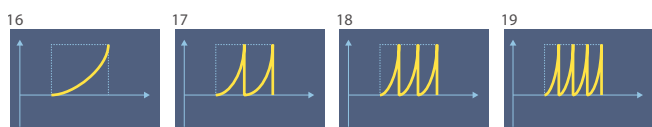
### Step Curve 7–10 (variations of ascending saw)



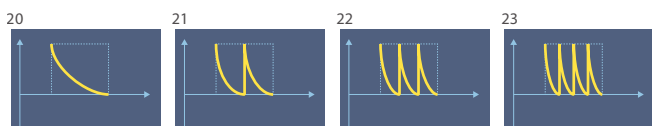
### Step Curve 11–15 (variations of descending saw)



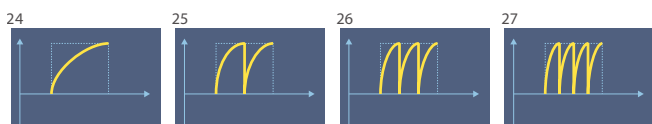
### Step Curve 16–19 (variations of ascending exponential)



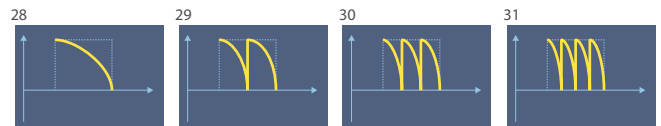
### Step Curve 20–23 (variations of descending exponential)



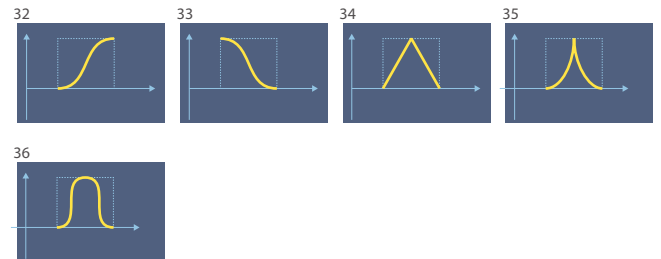
### Step Curve 24–27 (variations of ascending charging curve)



### Step Curve 28–31 (variations of descending charging curve)



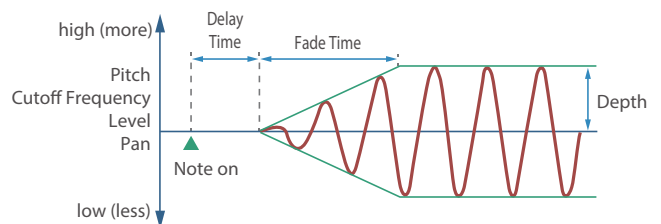
### Step Curve 32–36 (other variations)



## How to Apply the LFO

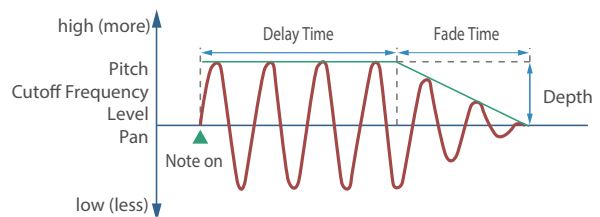
### Apply the LFO gradually after the key is pressed

**Fade Mod (Fade Mode): ON-IN**



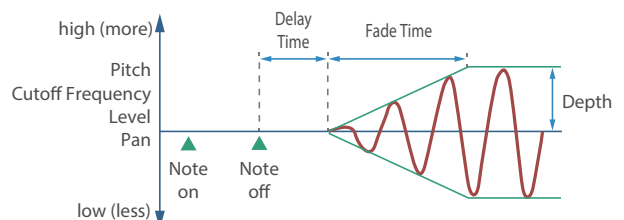
### Apply the LFO immediately when the key is pressed, and then gradually begin to decrease the effect

**Fade Mod (Fade Mode): ON-OUT**



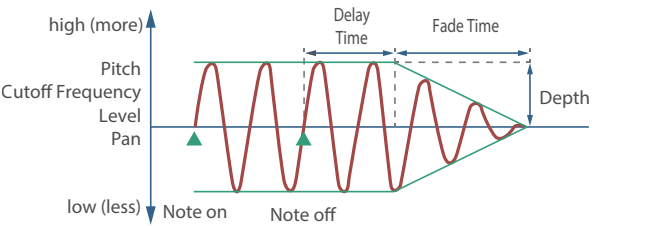
### Apply the LFO gradually after the key is released

**Fade Mod (Fade Mode): OFF-IN**



Apply the LFO from when the key is pressed until it is released, and gradually begin to decrease the effect when the key is released

Fade Mod (Fade Mode): OFF-OUT



List of supported CCs

| Parameter    | PARTIAL 1 | PARTIAL 2 | PARTIAL 3 | PARTIAL 4 |
|--------------|-----------|-----------|-----------|-----------|
| Tone PTL     |           |           |           |           |
| Level        | 16        | 17        | 18        | 19        |
| Fine Tune    | 21        | 31        | 35        | 46        |
| Cutoff       | 3         | 54        | 55        | 56        |
| Resonance    | 9         | 57        | 58        | 59        |
| Filter Env   |           |           |           |           |
| Filtr Depth  | 81        | 63        | 79        | 80        |
| Filtr Time1  | 83        | 82        | 85        | 87        |
| Filtr Time4  | 86        | 102       | 103       | 104       |
| Amp Env      |           |           |           |           |
| Amp Time1    | 89        | 108       | 109       | 111       |
| Amp Time4    | 90        | 112       | 114       | 117       |
| LFO1         |           |           |           |           |
| L1 Rate      | 29        | 20        | 22        | 23        |
| L1 Pit Depth | 26        | 47        | 48        | 50        |
| L1 Amp Depth | 30        | 105       | 106       | 107       |
| L1 Flt Depth | 28        | 60        | 61        | 62        |
| LFO2         |           |           |           |           |
| L2 Rate      | 14        | 24        | 25        | 27        |
| L2 Pit Depth | 15        | 51        | 52        | 53        |

TONE RD SYMPATHETIC RESO Parameter

These parameters adjust the resonance (sympathetic resonance) that occurs when you hold down the damper pedal. On an acoustic piano, holding down the damper pedal allows strings other than those that you play to resonate sympathetically with the played strings, creating a richly expansive resonance. This effect simulates that behavior.

\* MODEL RD can be selected for any of the parts. However, when the RD models are selected only for part 1, the MFX uses the sympathetic resonance effect. The normal MFX is applied when parts 2-4 are selected.

| Parameter        | Value   | Explanation                                                  |
|------------------|---------|--------------------------------------------------------------|
| SYMPATHETIC RESO |         |                                                              |
| SymReso Switch   | OFF, ON | With the ON setting, the effect is applied.                  |
| SymReso Depth    | 0-127   | Effect depth                                                 |
| Cabinet Reso     | 0-127   | Depth of the resonance when the damper pedal is not pressed. |

# System Parameters

## System Parameters

| Parameter       | Value                 | Explanation                                                                                                                                                                                                                                                                                                                       |
|-----------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AGING</b>    |                       |                                                                                                                                                                                                                                                                                                                                   |
| Warm-Up         |                       | Specifies whether the character of the sound (pitch variance) changes according to the varying internal temperature of this unit. This has an effect only on analog synthesizer models. The internal temperature value is shown in the upper right of the top screen of the scene.                                                |
|                 | OFF                   | The character of the sound is not affected by the temperature. Nor is there a temperature indication in the SCENE screen.                                                                                                                                                                                                         |
|                 | ON                    | The internal temperature starts at the value specified by Aging Init Temp, and changes to the actual temperature (REAL). Although it depends on the temperature difference, the REAL temperature is reached in approximately ten minutes. After reaching the REAL temperature, the setting follows the actual temperature change. |
|                 | FAST                  | The internal temperature starts at the value specified by Aging Init Temp, and then quickly (in approximately ten seconds) changes to the actual temperature (REAL) when you play the keyboard. After reaching the REAL temperature, the setting follows the actual temperature change.                                           |
| Init Temp       | FIXED                 | The internal temperature is fixed at the value specified by Aging Init Temp, and the character of the sound is maintained at that point.                                                                                                                                                                                          |
|                 |                       | Specifies the initial internal temperature for Warm-Up.                                                                                                                                                                                                                                                                           |
|                 | REAL                  | The internal temperature that is actually measured will be the initial temperature.                                                                                                                                                                                                                                               |
| Age             | 0–60 [°C]/32–140 [°F] | Virtually specifies the initial internal temperature (Centigrade/Fahrenheit).                                                                                                                                                                                                                                                     |
|                 | OFF, 1–100 [years]    | Simulates the aging of an analog synthesizer's internal components. Increasing this value changes the sound as it would be on a unit that is a corresponding number of years old. This is valid only for sounds of an analog synthesizer model.                                                                                   |
| <b>GENERAL</b>  |                       |                                                                                                                                                                                                                                                                                                                                   |
| MasterTune      | 415.3–466.2           | Adjusts the overall tuning. The displayed value is the frequency of the A4 key (middle A).                                                                                                                                                                                                                                        |
| MasKeyShift     | -24–+24               | Shifts the overall pitch range in semitone steps.                                                                                                                                                                                                                                                                                 |
| ScaleTuneSw     | OFF, ON               | Specifies whether the scene's SCALE TUNE setting is enabled (ON) or disabled (OFF).                                                                                                                                                                                                                                               |
| USB In Lev      | 0–127                 | Adjusts the audio input level of the USB COMPUTER port.                                                                                                                                                                                                                                                                           |
| USB Out Lev     | 0–127                 | Adjusts the audio output level to the USB COMPUTER port.                                                                                                                                                                                                                                                                          |
| AuxIn/BT In Lev | 0–127                 | Adjusts the input level of the AUX IN jack and Bluetooth input.                                                                                                                                                                                                                                                                   |
| USB Audio Thru  | OFF, ON               | Specifies whether the audio input of the USB COMPUTER port is mixed into the audio output of the USB COMPUTER port.<br>If you don't want to output this audio, turn this "OFF."<br>* If you turn this "ON," make settings on the connected equipment so that the audio does not loop back.                                        |
| AUXIN USB Thru  | OFF, ON               | Specifies whether the input of the AUX IN jack is mixed into the audio output of the USB COMPUTER port. If you don't want to output this audio, turn this "OFF."                                                                                                                                                                  |
| LineOut Gain    | -12–+12 [dB]          | Adjusts the output gain of OUTPUT/PHONES.                                                                                                                                                                                                                                                                                         |
| Speaker Sw      | OFF, ON, AUTO         | Specifies whether sound is output from the speakers. The AUTO setting operates as "OFF" when headphones are connected and as "ON" when headphones are not connected.                                                                                                                                                              |
| SPOut Gain      | -12–0 [dB]            | Adjusts the output gain of the speakers.                                                                                                                                                                                                                                                                                          |
| Auto Off        | OFF, 30, 240 (min)    | Specifies whether the unit will turn off automatically after a certain time has elapsed. If you don't want the unit to turn off automatically, choose "OFF" setting.                                                                                                                                                              |
| LED On Bright   | 0–31                  | Adjusts the brightness when the LEDs are lit.                                                                                                                                                                                                                                                                                     |
| LED OffBright   | 0–30                  | When the JUNO-X is operating, you can make the LEDs remain dimly lit when off instead of going completely dark. This setting adjusts the LED brightness when dimly lit.                                                                                                                                                           |

| Parameter         | Value                                        | Explanation                                                                                                                                                                                                                                             |
|-------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LCD Contrast      | 1–10                                         | Adjusts the contrast of the display.                                                                                                                                                                                                                    |
| Scene Lock        |                                              | Specifies whether a confirmation screen appears when you recall a scene.                                                                                                                                                                                |
|                   | OFF                                          | The scene is recalled immediately.                                                                                                                                                                                                                      |
| Scene Lock        | ON                                           | Before the scene is recalled, a confirmation screen appears. To recall the scene, use the [◀][▶] buttons to select "Yes," and then press the [ENTER] (INIT) button.                                                                                     |
| Startup Scene     | 01-01–16-16                                  | Specifies the scene that is recalled when the unit starts up.                                                                                                                                                                                           |
| <b>ARPEGGIO</b>   |                                              |                                                                                                                                                                                                                                                         |
| Set Tone          | OFF, ON                                      | Specifies whether the current sound settings are kept while only the phrase is switched (OFF) or both the phrase and the sound settings are switched (ON).                                                                                              |
| Set DrumKit       | OFF, ON                                      | Specifies whether the current sound settings are kept while only the rhythm is switched (OFF) or both the rhythm and the sound settings are switched (ON).                                                                                              |
| Set Tempo         | OFF, ON                                      | Specifies whether the current tempo setting is kept while only the rhythm is switched (OFF) or both the rhythm and the tempo settings are switched (ON).                                                                                                |
| Arp Sync          |                                              | Specifies the synchronization setting for arpeggio performance when connected to an external device and playing in synchronization.<br>* This is only enabled when Sync Mode is set to "MIDI," "USB COM" or "USB MEM."                                  |
|                   | OFF                                          | Does not synchronize to measures or beats. Arpeggio performance starts at the moment that MIDI messages are received.                                                                                                                                   |
|                   | BEAT                                         | Synchronizes to beats. Arpeggio performance starts at the next beat after MIDI messages are received.                                                                                                                                                   |
| Arp Sync          | MEASURE                                      | Synchronizes to measures. Arpeggio performance starts at the first beat of the next measure after MIDI messages are received.                                                                                                                           |
| <b>TEMPO/SYNC</b> |                                              |                                                                                                                                                                                                                                                         |
| Tempo             | 20.00–300.00                                 | Specifies the system tempo.<br>Hold down the [SHIFT] button and turn the [- VALUE +] knob to change the value in steps of 0.01.                                                                                                                         |
| Tempo Src         | SCENE, SYS                                   | When you switch scenes, this setting specifies whether to use the system tempo (SYS) or the tempo stored in the scene (SCENE).                                                                                                                          |
| Sync Mode         | AUTO, INT, MIDI, USB COM, USB MEM            | Specifies the synchronization signal according to which the JUNO-X operates.                                                                                                                                                                            |
| Sync Out          | OFF, MIDI, USB COM, MIDI/USBCM, USB MEM, ALL | Specifies the connector from which MIDI clock messages etc. are output.                                                                                                                                                                                 |
| <b>Bluetooth</b>  |                                              |                                                                                                                                                                                                                                                         |
| Bluetooth Sw      | OFF, ON                                      | Enables (ON) or disables (OFF) Bluetooth communication.                                                                                                                                                                                                 |
| Pairing           | -                                            | Executing pairing for Bluetooth audio.                                                                                                                                                                                                                  |
| Bluetooth ID      | OFF, 1–9                                     | Specifies the number added to the end of this unit's device name shown in a Bluetooth-connected app.                                                                                                                                                    |
| <b>MIDI</b>       |                                              |                                                                                                                                                                                                                                                         |
| Ctrl Ch           | 1–16, OFF                                    | Specifies the MIDI receive channel on which MIDI messages (program change and bank select) from an external MIDI device can be received to switch programs.<br>If you don't want programs to be switched from a connected MIDI device, turn this "OFF." |
|                   |                                              |                                                                                                                                                                                                                                                         |
| Ctrl Src Sel      | SYS                                          | SysCtrlSrc1–4 are used for tone control.                                                                                                                                                                                                                |
|                   | SCENE                                        | The scene's CtrlSrc1–4 settings are used for tone control.                                                                                                                                                                                              |
| SysCtrlSrc1–4     | OFF, CC01–CC31, CC33–CC95, BEND, AFT         | Specify the MIDI messages that will be used as system controls.                                                                                                                                                                                         |
| Soft Thru         | OFF, ON                                      | If this is "ON," MIDI messages that are input from the MIDI IN connector are re-transmitted without change from the MIDI OUT connector.                                                                                                                 |
| USB-MIDIThru      | OFF, ON                                      | Specifies whether MIDI messages received at the USB COMPUTER port/MIDI IN connector are retransmitted without change from the MIDI OUT connector/USB COMPUTER port (ON) or are not retransmitted (OFF).                                                 |
| USB Driver        | GENERIC, VENDOR                              | Specifies the USB driver setting.                                                                                                                                                                                                                       |

## System Parameters

| Parameter      | Value                                              | Explanation                                                                                                                                                                                                                                     |
|----------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Remote Kbd     | OFF, MIDI IN, USB COM, USB MEM                     | Sets which connector is used for input when you use an external MIDI keyboard instead of the keyboard of the JUNO-X. In this case, the MIDI transmit channel of the external MIDI keyboard does not matter. Normally you will leave this "OFF." |
| Local Sw       | OFF, ON                                            | Turns on/off the connection between the controllers section (keyboard, panel buttons and sliders, pedals, etc.) and the internal sound engine.                                                                                                  |
| Device ID      | 17–32                                              | When transmitting and receiving system exclusive messages, the device ID numbers of both devices must match.                                                                                                                                    |
| Tone CC Map    | OFF, ON                                            | Sets whether control change messages (CC) for the tone parameters are transmitted/received (ON) or not (OFF).                                                                                                                                   |
| <b>MIDI Tx</b> |                                                    |                                                                                                                                                                                                                                                 |
| Tx PC          | OFF, ON                                            | Specifies whether program change messages will be transmitted (ON) or not be transmitted (OFF).                                                                                                                                                 |
| Tx Bank        | OFF, ON                                            | Specifies whether bank select messages will be transmitted (ON) or not be transmitted (OFF).                                                                                                                                                    |
| Tx Edit        | OFF, ON                                            | Specify whether changes you make in the settings of a program will be transmitted as system exclusive messages (ON), or will not be transmitted (OFF).                                                                                          |
| <b>MIDI Rx</b> |                                                    |                                                                                                                                                                                                                                                 |
| Rx PC          | OFF, ON                                            | Specifies whether program change messages will be received (ON) or not be received (OFF).                                                                                                                                                       |
| Rx Bank        | OFF, ON                                            | Specifies whether bank select messages will be received (ON) or not be received (OFF).                                                                                                                                                          |
| Rx Exclusive   | OFF, ON                                            | Specifies whether system exclusive messages will be received (ON) or not be received (OFF).                                                                                                                                                     |
| <b>MIC IN</b>  |                                                    |                                                                                                                                                                                                                                                 |
| Mic In Gain    | -24.0–+24.0 [dB]                                   | Adjusts the input level of the MIC IN connector.                                                                                                                                                                                                |
| Mic Power      | OFF, ON                                            | If this is "ON," plug-in power (5 V) is supplied to the MIC IN connector.                                                                                                                                                                       |
| NS Switch      | OFF, ON                                            | Switches the noise suppressor on/off. The noise suppressor is a function that suppresses noise during periods of silence.                                                                                                                       |
| NS Threshold   | -96–0 [dB]                                         | Adjusts the volume at which noise suppression starts to be applied.                                                                                                                                                                             |
| NS Release     | 0–127                                              | Adjusts the time from when noise suppression starts until the volume reaches "0."                                                                                                                                                               |
| CompSwitch     | OFF, ON                                            | Specifies whether the mic compressor (a compressor applied to the mic input) is used (ON) or not used (OFF).                                                                                                                                    |
| CompAttack     | 0.1–100 [ms]                                       | Specifies the time from when the input to the mic compressor exceeds the CompThreshold level until the volume is compressed.                                                                                                                    |
| CompRelease    | 10–1000 [ms]                                       | Specifies the time from when the input to the mic compressor falls below the CompThreshold level until compression is no longer applied.                                                                                                        |
| CompThreshold  | -60–0 [dB]                                         | Specifies the level at which the mic compressor starts applying compression.                                                                                                                                                                    |
| CompRatio      | 1: 1, 2: 1, 3: 1, 4: 1, 8: 1, 16: 1, 32: 1, INF: 1 | Specifies the compression ratio for the mic compressor.                                                                                                                                                                                         |
| CompKnee       | 0–30 [dB]                                          | Smooths the transition until the mic compressor starts to be applied. Higher values produce a smoother transition.                                                                                                                              |
| CompOutGain    | -24.0–+24.0 [dB]                                   | Specifies the output volume of the mic compressor.                                                                                                                                                                                              |
| Dly Send Lev   | 0–127                                              | Specifies the amount of delay that is applied to the mic input.                                                                                                                                                                                 |
| Rev Send Lev   | 0–127                                              | Specifies the amount of reverb that is applied to the mic input.                                                                                                                                                                                |
| Cho Send Lev   | 0–127                                              | Specifies the amount of chorus that is applied to the mic input.                                                                                                                                                                                |
| Mic Thru       | OFF, ON                                            | If you want the mic to be cut when the vocoder is off, turn this "OFF."                                                                                                                                                                         |

| Parameter          | Value                                             | Explanation                                                                                                                                                                                                                           |
|--------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CONTROLLER</b>  |                                                   |                                                                                                                                                                                                                                       |
| Velocity           | REAL, 1–127                                       | Specifies the velocity value that is transmitted when you play the keyboard.                                                                                                                                                          |
| Velo Crv           | LIGHT, MEDIUM, HEAVY                              | Specifies the keyboard touch.                                                                                                                                                                                                         |
| Velo Offset        | -10–+9                                            | Adjusts the keyboard velocity curve.                                                                                                                                                                                                  |
| Knob Mode          | DIRECT, CATCH                                     | Specifies whether the parameter value corresponding to a controller is immediately updated when you operate that controller (DIRECT) or only after the controller reaches the same position as the parameter's current value (CATCH). |
| Aft Sens           | 0–100                                             | Specifies the sensitivity of aftertouch.                                                                                                                                                                                              |
| <b>BUTTON Func</b> |                                                   |                                                                                                                                                                                                                                       |
| Source             | SCENE, SYS                                        | Specifies whether the functions assigned to these buttons follows the settings of the currently selected scene (SCENE) or the system settings (SYS).                                                                                  |
| S1 Func            | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the S1 button.                                                                                                                                                                                     |
| S1 Mode            | LATCH, MOMENTARY                                  | Specifies how the button operates.                                                                                                                                                                                                    |
| S2 Func            | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the S2 button.                                                                                                                                                                                     |
| S2 Mode            | LATCH, MOMENTARY                                  | Specifies how the button operates.                                                                                                                                                                                                    |
| S3 Func            | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the S3 button.                                                                                                                                                                                     |
| S3 Mode            | LATCH, MOMENTARY                                  | Specifies how the button operates.                                                                                                                                                                                                    |
| <b>SLIDER Func</b> |                                                   |                                                                                                                                                                                                                                       |
| SL1 Source         | SCENE, SYS                                        | Specifies whether the function assigned to the SL1 slider follows the setting of the currently selected scene (SCENE) or the system setting (SYS).                                                                                    |
| SL1                | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the SL1 slider.                                                                                                                                                                                    |
| SL2 Source         | SCENE, SYS                                        | Specifies whether the function assigned to the SL2 slider follows the setting of the currently selected scene (SCENE) or the system setting (SYS).                                                                                    |
| SL2                | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the SL2 slider.                                                                                                                                                                                    |
| <b>PEDAL Func</b>  |                                                   |                                                                                                                                                                                                                                       |
| Hold Source        | SCENE, SYS                                        | Specifies whether the function assigned to the pedal connected to the HOLD jack follows the setting of the currently selected scene (SCENE) or the system setting (SYS).                                                              |
| Hold               | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the pedal connected to the HOLD jack.                                                                                                                                                              |
| Hold Pole          | STANDARD, REVERSE                                 | Specifies the polarity of the pedal connected to the HOLD jack.                                                                                                                                                                       |
| Ctrl Source        | SCENE, SYS                                        | Specifies whether the function assigned to the pedal connected to the CTRL jack follows the setting of the currently selected scene (SCENE) or the system setting (SYS).                                                              |
| Ctrl               | For the values, refer to assignable list (p. 33). | Specifies the function assigned to the pedal connected to the CTRL jack.                                                                                                                                                              |

| Parameter                          | Value      | Explanation                                                                                                                                                                                                         |
|------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PART Btn Asgn</b>               |            |                                                                                                                                                                                                                     |
| 4-8<br>4-8+(S)<br>9-13<br>9-13+(S) |            | Assign the functions of the [4]–[8], [9]–[13] buttons and their functions when pressed while holding down the [SHIFT] button.                                                                                       |
|                                    | No Assign  | No assignment.                                                                                                                                                                                                      |
|                                    | PartSel    | Specifies the current part operated from the panel and in the screen. The part played from the keyboard does not change.                                                                                            |
|                                    | Part+KeySw | Simultaneously operates the current part and the Keyboard SW, so that the selected part can be played from the keyboard. By pressing multiple parts simultaneously, you can turn Keyboard SW on for multiple parts. |
|                                    | KeySw      | Functions as the keyboard switch to change the part played from the keyboard.                                                                                                                                       |
|                                    | PartSw     | Turns on/off the sound of each part. This is useful for DJ-like performances in which you turn each part's sound on/off while you perform.                                                                          |
|                                    | ArpSw      | Switches whether each part is played by the arpeggio.                                                                                                                                                               |
|                                    | EfxSw      | From the left button, turns MFX, DRIVE, DLY, REV, and CHO respectively on/off for all parameters simultaneously.                                                                                                    |

### List of functions that can be assigned to the controllers

| Function                    | S1 Func<br>S2 Func<br>S3 Func | SL1<br>SL2 | HOLD | Ctrl |
|-----------------------------|-------------------------------|------------|------|------|
| OFF                         | ✓                             | ✓          | ✓    | ✓    |
| CC01–31, 32 (OFF),<br>33–95 | ✓                             | ✓          | ✓    | ✓    |
| AFTERTOUCH                  | ✓                             | ✓          | ✓    | ✓    |
| MONO/POLY                   | ✓                             |            | ✓    |      |
| SCENE DOWN                  | ✓                             |            | ✓    |      |
| SCENE UP                    | ✓                             |            | ✓    |      |
| tone DOWN                   | ✓                             |            | ✓    |      |
| tone UP                     | ✓                             |            | ✓    |      |
| PANEL DEC                   | ✓                             |            | ✓    |      |
| PANEL INC                   | ✓                             |            | ✓    |      |
| CHO SW                      | ✓                             |            | ✓    |      |
| REV SW                      | ✓                             |            | ✓    |      |
| DLY SW                      | ✓                             |            | ✓    |      |
| ARP SW                      | ✓                             |            | ✓    |      |
| ARP HOLD                    | ✓                             |            | ✓    |      |
| DETECT KEYS                 | ✓                             |            | ✓    |      |
| DETECT BEAT                 | ✓                             |            | ✓    |      |
| UNISON SW                   | ✓                             |            | ✓    |      |
| BEND MODE                   | ✓                             |            | ✓    |      |
| AUTO TUNING                 | ✓                             |            | ✓    |      |
| TAP TEMPO                   | ✓                             |            | ✓    |      |
| START/STOP                  | ✓                             |            | ✓    |      |
| DRV SW                      | ✓                             |            | ✓    |      |
| VOC/MIC                     | ✓                             |            | ✓    |      |
| BEND DOWN                   |                               | ✓          |      | ✓    |
| BEND UP                     |                               | ✓          |      | ✓    |
| CHO LEVEL                   |                               | ✓          |      | ✓    |
| REV LEVEL                   |                               | ✓          |      | ✓    |
| DLY LEVEL                   |                               | ✓          |      | ✓    |
| ARP SHUFFLE                 |                               | ✓          |      | ✓    |
| ARP DURATION                |                               | ✓          |      | ✓    |
| PART FADE1                  |                               | ✓          |      | ✓    |
| PART FADE2                  |                               | ✓          |      | ✓    |
| LEVEL P-1                   |                               | ✓          |      | ✓    |
| LEVEL P-2                   |                               | ✓          |      | ✓    |
| LEVEL P-3                   |                               | ✓          |      | ✓    |
| LEVEL P-4                   |                               | ✓          |      | ✓    |
| LEVEL P-R                   |                               | ✓          |      | ✓    |
| AGE                         |                               | ✓          |      | ✓    |



## SYSTEM EFFECT: Dly

| Parameter        | Value                                                                                                                                                                 | Explanation                                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source           | SCENE                                                                                                                                                                 | Choose "SCENE" if you want to use the delay settings that are assigned to the scene.<br>➔ "SCENE EFFECT: Dly (Delay)" (p. 7)                                            |
|                  | SYS                                                                                                                                                                   | Choose "SYS" if you want to use the delay settings that are assigned to the system.<br>If this is set to "SYS," you can edit the delay type and the various parameters. |
| Switch           | OFF, ON                                                                                                                                                               | Switches delay on/off.                                                                                                                                                  |
| DlyType          | ➔ "Delay Parameters" (p. 7)                                                                                                                                           |                                                                                                                                                                         |
| Level            | 0–127                                                                                                                                                                 | Specifies the output level of the sound with delay applied.                                                                                                             |
| Rev Send         | 0–127                                                                                                                                                                 | Specifies the send level to reverb.                                                                                                                                     |
| Delay Parameters | Edit the parameters of the selected delay.<br>The available parameters differ depending on the type of chorus you selected in DlyType.<br>➔ "Delay Parameters" (p. 7) |                                                                                                                                                                         |

## SYSTEM EFFECT: Rev

| Parameter         | Value                                                                                                                                                                        | Explanation                                                                                                                                                               |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source            | SCENE                                                                                                                                                                        | Choose "SCENE" if you want to use the Reverb settings that are assigned to the scene.<br>➔ "SCENE EFFECT: Rev (Reverb)" (p. 8)                                            |
|                   | SYS                                                                                                                                                                          | Choose "SYS" if you want to use the Reverb settings that are assigned to the system.<br>If this is set to "SYS," you can edit the Reverb type and the various parameters. |
| Switch            | OFF, ON                                                                                                                                                                      | Switches the reverb on/off.                                                                                                                                               |
| RevType           | ➔ "Reverb Parameters" (p. 8)                                                                                                                                                 |                                                                                                                                                                           |
| Level             | 0–127                                                                                                                                                                        | Specifies the output level of the sound with reverb applied.                                                                                                              |
| Reverb Parameters | Edit the parameters of the selected reverb type.<br>The available parameters differ depending on the type of reverb you selected in RevType.<br>➔ "Reverb Parameters" (p. 8) |                                                                                                                                                                           |

## SYSTEM EFFECT: Cho

| Parameter         | Value                                                                                                                                                                   | Explanation                                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source            | SCENE                                                                                                                                                                   | Choose "SCENE" if you want to use the chorus settings that are assigned to the scene.<br>➔ "SCENE EFFECT: Cho (Chorus)" (p. 9)                                            |
|                   | SYS                                                                                                                                                                     | Choose "SYS" if you want to use the chorus settings that are assigned to the system.<br>If this is set to "SYS," you can edit the chorus type and the various parameters. |
| Switch            | OFF, ON                                                                                                                                                                 | Switches chorus on/off.                                                                                                                                                   |
| ChoType           | ➔ "Chorus Parameters" (p. 9)                                                                                                                                            |                                                                                                                                                                           |
| Level             | 0–127                                                                                                                                                                   | Specifies the output level of the sound with chorus applied.                                                                                                              |
| Rev Send          | 0–127                                                                                                                                                                   | Specifies the send level to reverb.                                                                                                                                       |
| Chorus Parameters | Edit the parameters of the selected chorus.<br>The available parameters differ depending on the type of chorus you selected in ChoType.<br>➔ "Chorus Parameters" (p. 9) |                                                                                                                                                                           |

## SYSTEM EQ/COMP

| Parameter          | Value                                              | Explanation                                                                                                                                                                                          |
|--------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MASTER EQ</b>   |                                                    |                                                                                                                                                                                                      |
| Switch             | OFF, ON                                            | Specifies whether the mastering EQ (an equalizer applied to the entire sound generator of the JUNO-X) is used (ON) or not used (OFF).                                                                |
| In Gain            | -24–+24 [dB]                                       | Adjusts the amount of boost/cut for the input to the EQ.                                                                                                                                             |
| Low Gain           | -24–+24 [dB]                                       | Gain of the low range.                                                                                                                                                                               |
| Low Freq           | 20–16000 [Hz]                                      | Frequency of the low range.                                                                                                                                                                          |
| Mid1 Gain          | -24–+24 [dB]                                       | Gain of the middle frequency range 1.                                                                                                                                                                |
| Mid1 Freq          | 20–16000 [Hz]                                      | Frequency of the middle range 1.                                                                                                                                                                     |
| Mid1 Q             | 0.5, 1.0, 2.0, 4.0, 8.0, 16.0                      | Width of the middle frequency range 1. Set a higher value to narrow the range to be affected.                                                                                                        |
| Mid2 Gain          | -24–+24 [dB]                                       | Gain of the middle frequency range 2.                                                                                                                                                                |
| Mid2 Freq          | 20–16000 [Hz]                                      | Frequency of the middle range 2.                                                                                                                                                                     |
| Mid2 Q             | 0.5, 1.0, 2.0, 4.0, 8.0, 16.0                      | Width of the middle frequency range 2. Set a higher value to narrow the range to be affected.                                                                                                        |
| Mid3 Gain          | -24–+24 [dB]                                       | Gain of the middle frequency range 3.                                                                                                                                                                |
| Mid3 Freq          | 20–16000 [Hz]                                      | Frequency of the middle range 3.                                                                                                                                                                     |
| Mid3 Q             | 0.5, 1.0, 2.0, 4.0, 8.0, 16.0                      | Width of the middle frequency range 3. Set a higher value to narrow the range to be affected.                                                                                                        |
| High Gain          | -24–+24 [dB]                                       | Gain of the high range.                                                                                                                                                                              |
| High Freq          | 20–16000 [Hz]                                      | Frequency of the high range.                                                                                                                                                                         |
| <b>MASTER COMP</b> |                                                    |                                                                                                                                                                                                      |
| Switch             | OFF, ON                                            | Specifies whether the mastering COMP (a compressor applied to the entire sound generator of the JUNO-X) is used (ON) or not used (OFF).                                                              |
| Low Attack         | 0.1–100 [ms]                                       | Specifies the time from when the input exceeds Low Thres until compression is applied to the volume of the low-frequency band.                                                                       |
| Low Rels           | 10–1000 [ms]                                       | In a state when compression is already being applied, this specifies the time from when the input decreases below Low Thres until the low-frequency band stops being compressed.                     |
| Low Thres          | -60–0 [dB]                                         | Specifies the volume level at which compression starts for the low-frequency band.                                                                                                                   |
| Low Ratio          | 1: 1, 2: 1, 3: 1, 4: 1, 8: 1, 16: 1, 32: 1, INF: 1 | Specifies the compression ratio for the low-frequency band.                                                                                                                                          |
| Low Knee           | 0–30 [dB]                                          | This is a function that smooths the onset of compression from the uncompressed state; it gradually applies compression starting earlier than Low Thres. Higher values produce a smoother transition. |
| Low Gain           | -24–+24 [dB]                                       | Specifies the output volume of the low-frequency band.                                                                                                                                               |
| Mid Attack         | 0.1–100 [ms]                                       | Specifies the time from when the input exceeds Mid Thres until compression is applied to the volume of the mid-frequency band.                                                                       |
| Mid Rels           | 10–1000 [ms]                                       | In a state when compression is already being applied, this specifies the time from when the input decreases below Mid Thres until the mid-frequency band stops being compressed.                     |
| Mid Thres          | -60–0 [dB]                                         | Specifies the volume level at which compression starts for the mid-frequency band.                                                                                                                   |
| Mid Ratio          | 1: 1, 2: 1, 3: 1, 4: 1, 8: 1, 16: 1, 32: 1, INF: 1 | Specifies the compression ratio for the mid-frequency band.                                                                                                                                          |
| Mid Knee           | 0–30 [dB]                                          | This is a function that smooths the onset of compression from the uncompressed state; it gradually applies compression starting earlier than Mid Thres. Higher values produce a smoother transition. |
| Mid Gain           | -24–+24 [dB]                                       | Specifies the output volume of the mid-frequency band.                                                                                                                                               |
| High Attack        | 0.1–100 [ms]                                       | Specifies the time from when the input exceeds High Thres until compression is applied to the volume of the high-frequency band.                                                                     |

| Parameter  | Value                                              | Explanation                                                                                                                                                                                           |
|------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High Rels  | 10–1000 [ms]                                       | In a state when compression is already being applied, this specifies the time from when the input decreases below High Thres until the high-frequency band stops being compressed.                    |
| High Thres | -60–0 [dB]                                         | Specifies the volume level at which compression starts for the high-frequency band.                                                                                                                   |
| High Ratio | 1: 1, 2: 1, 3: 1, 4: 1, 8: 1, 16: 1, 32: 1, INF: 1 | Specifies the compression ratio for the high-frequency band.                                                                                                                                          |
| High Knee  | 0–30 [dB]                                          | This is a function that smooths the onset of compression from the uncompressed state; it gradually applies compression starting earlier than High Thres. Higher values produce a smoother transition. |
| High Gain  | -24–+24 [dB]                                       | Specifies the output volume of the high-frequency band.                                                                                                                                               |
| Splt Low   | 16–16000 [Hz]                                      | Specifies the frequency at which the low-frequency band (Low) and mid-frequency band (Mid) are divided.                                                                                               |
| Splt High  | 16–16000 [Hz]                                      | Specifies the frequency at which the high-frequency band (High) and mid-frequency band (Mid) are divided.                                                                                             |

## SYSTEM COLOR SET

| Parameter | Value | Explanation                                            |
|-----------|-------|--------------------------------------------------------|
| Color Set | 1–10  | Saves and switches color settings for each set number. |

### OFF COLOR

|                |       |  |
|----------------|-------|--|
| Arp Off        |       |  |
| Scene Off      |       |  |
| Part/Func Off  |       |  |
| ModelBank Off  |       |  |
| Sc1-8 Off      |       |  |
| Sc9-16 Off     |       |  |
| KeyCtrl Off    |       |  |
| StepEditOff    |       |  |
| Model Off      |       |  |
| Categ Off      |       |  |
| User Off       |       |  |
| Part Off       |       |  |
| Part+KeySw Off |       |  |
| KeySw Off      |       |  |
| PartSw Off     | Off,  |  |
| ArpSw Off      | O,    |  |
| EfxSw Off      | O(b), |  |
| No Assign      | Y,    |  |
| St1-4 Off      | Y(b), |  |
| St5-8 Off      | W,    |  |
| St9-12 Off     | W(b), |  |
| St13-16 Off    | G,    |  |
|                | G(b), |  |
|                | B,    |  |
| Arp On         | B(b), |  |
| Scene On       | R,    |  |
| Part/Func On   | R(b), |  |
| ModelBank On   | V,    |  |
| Sc1-8 On       | V(b), |  |
| Sc9-16 On      | B2,   |  |
| KeyCtrl On     | B2(b) |  |
| StepEdit On    |       |  |
| Model On       |       |  |
| Categ On       |       |  |
| User On        |       |  |
| Part On        |       |  |
| Part+KeySw On  |       |  |
| KeySw On       |       |  |
| PartSw On      |       |  |
| ArpSw On       |       |  |
| EfxSw On       |       |  |
| St1-4 On       |       |  |
| St5-8 On       |       |  |
| St9-12 On      |       |  |
| St13-16 On     |       |  |

Specify the illumination color of the button for <parameter name> Off (when the corresponding button is off), <parameter name> On (when the corresponding button is on).

**Off:** Unlit  
**O:** Orange,  
**Y:** Yellow,  
**W:** White,  
**G:** Green,  
**B:** Blue,  
**R:** Red,  
**V:** Violet,  
**B2:** Bright blue  
 (b) indicates blinking.

## MODEL ASSIGN

Hold down the [MODEL BANK] button and press a button from [1]–[16] to display the settings screen.

| Parameter | Value         | Explanation                                                                                                                                        |
|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Attr      |               | Specifies the assignment attributes.                                                                                                               |
|           | MODEL         | Select models by the Mdl 1–8 parameters.                                                                                                           |
|           | CATEGORY      | Select categories by the Catg 1–8 parameters.                                                                                                      |
|           | USER          | No assignments are possible.<br>* If an expansion is installed, "EXZ****" is shown after "USER." In this case, the expansion name is shown in 1–8. |
| Mdl1–8    | model (*1)    | This is shown only when Attr = MODEL.<br>Specify the model that you want to assign.                                                                |
| Catg1–8   | category (*2) | This is shown only when Attr = CATEGORY.<br>Specify the category name that you want to assign.                                                     |

(\*1) OFF, Z-Core, COMMON, PR-A, PR-B, PR-C, PR-D, XV-5080, JUNO-X, JUNO-106, JUNO-60, VOCODER, RD-PIANO, PR-X

\* If an expansion is installed, "EXZ\*\*\*\*" is shown at the end.

(\*2) OFF, No Assign, Ac.Piano, Pop Piano, E.Grand Piano, E.Piano1, E.Piano2, E.Organ, Pipe Organ, Reed Organ, Harpsichord, Clav, Celesta, Accordion, Harmonica, Bell, Mallet, Ac.Guitar, E.Guitar, Dist.Guitar, Ac.Bass, E.Bass, Synth Bass, Plucked/Stroke, Solo Strings, Ensemble Strs, Orchestral, Solo Brass, Ensemble Brass, Wind, Flute, Sax, Recorder, Vox/Choir, Scat, Synth Lead, Synth Brass, Synth Pad/Str, Synth Bellpad, Synth PolyKey, Synth FX, Synth Seq/Pop, Phrase, Pulsating, Beat&Groove, Hit, Sound FX, Drums, Percussion, Stack, Zone, Vocoder

\* If an expansion is installed, "EXZ On" is shown after "User On."

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## MFX Common Parameters

| Parameter     | Value                                                                            | Explanation                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type          | Selects the MFX type.                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| Switch        | OFF, ON                                                                          | Switches the MFX on/off.                                                                                                                                                                                                                                                                                                                                                               |
| Cho Send      | 0–127                                                                            | Adjusts the amount of chorus. If you don't want to add the chorus effect, set it to "0."                                                                                                                                                                                                                                                                                               |
| Rev Send      | 0–127                                                                            | Adjusts the amount of reverb. If you don't want to add the reverb effect, set it to "0."                                                                                                                                                                                                                                                                                               |
| MFX parameter | Differs depending on the MFX type.                                               | ➔ For details, refer to the parameters for each MFX.                                                                                                                                                                                                                                                                                                                                   |
| Src1–4        | OFF, CC01–CC31, CC33–CC95, BEND, AFT, SYS-CTRL1, SYS-CTRL2, SYS-CTRL3, SYS-CTRL4 | Specifies the MIDI message that will control the corresponding MFX CONTROL parameter.                                                                                                                                                                                                                                                                                                  |
| Sens1–4       | -63–+63                                                                          | Specifies the depth of MFX CONTROL. Specify a positive "+" value if you want to change the value of the assigned destination in a positive direction (larger, toward the right, faster, etc.), or specify a negative value "-" if you want to change the value in a negative direction (smaller, toward the left, slower, etc.). Larger values will allow a greater amount of control. |
| Asgn1–4       | Differs depending on the MFX type.                                               | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                                                                                                                                                                              |

## 00 Thru

## Filter

## 01 Equalizer

This is a four-band stereo equalizer (low, mid x 2, high).



| Parameter | Value                                                                                              | Explanation                                                                                 |
|-----------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Low Freq  | 20, 25, 31, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400 [Hz]                                 | Frequency of the low range                                                                  |
| Low Gain  | -15–+15 [dB]                                                                                       | Amount of boost/cut for the low-frequency range                                             |
| Mid1 Freq | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Frequency of the middle range 1                                                             |
| Mid1 Gain | -15–+15 [dB]                                                                                       | Gain of the middle range 1                                                                  |
| Mid1 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                            | Width of the middle range 1<br>Set a higher value for Q to narrow the range to be affected. |
| Mid2 Freq | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Frequency of the middle range 2                                                             |
| Mid2 Gain | -15–+15 [dB]                                                                                       | Gain of the middle range 2                                                                  |

| Parameter | Value                                                              | Explanation                                                                                 |
|-----------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Mid2 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                            | Width of the middle range 2<br>Set a higher value for Q to narrow the range to be affected. |
| HighFreq  | 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12500, 16000 [Hz] | Frequency of the high range                                                                 |
| High Gain | -15–+15 [dB]                                                       | Amount of boost/cut for the high-frequency range                                            |
| Level     | 0–127                                                              | Output Level                                                                                |
| Asgn1–4   | OFF, Low Gain, High Gain, Level                                    | Specifies the parameters that are assigned to MFX CONTROL assign 1–4.                       |

## 02 Mid-Side EQ (Mid-Side Equalizer)

This effect allows the left/right signals that have similar phase to be tonally adjusted in a different way than the left/right signals that have different phase.

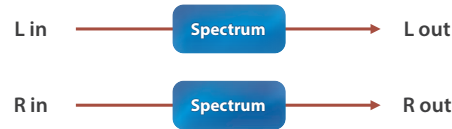


| Parameter   | Value                                                                                              | Explanation                                                                                               |
|-------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| M EQ Switch | OFF, ON                                                                                            | Switches whether to apply tonal adjustment to left/right input signals whose phase is similar (in phase). |
| M In G      | -12.00–+12.00 [dB]                                                                                 | Volume of left/right input signals whose phase is similar (in phase)                                      |
| M Low F     | 20, 25, 31, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400 [Hz]                                 | Frequency of the low range                                                                                |
| M Low G     | -12.00–+12.00 [dB]                                                                                 | Amount of boost/cut for the low-frequency range                                                           |
| M Mid1 F    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Frequency of the middle range 1                                                                           |
| M Mid1G     | -12.00–+12.00 [dB]                                                                                 | Gain of the middle range 1                                                                                |
| M Mid1 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                            | Width of the middle range 1<br>Set a higher value for Q to narrow the range to be affected.               |
| M Mid2 F    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Frequency of the middle range 2                                                                           |
| M Mid2G     | -12.00–+12.00 [dB]                                                                                 | Gain of the middle range 2                                                                                |
| M Mid2 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                            | Width of the middle range 2<br>Set a higher value for Q to narrow the range to be affected.               |
| M Mid3 F    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Frequency of the middle range 3                                                                           |
| M Mid3G     | -12.00–+12.00 [dB]                                                                                 | Gain of the middle range 3                                                                                |
| M Mid3 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                            | Width of the middle range 3<br>Set a higher value for Q to narrow the range to be affected.               |
| M High F    | 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12500, 16000 [Hz]                                 | Frequency of the high range                                                                               |
| M HighG     | -12.00–+12.00 [dB]                                                                                 | Amount of boost/cut for the high-frequency range                                                          |

| Parameter   | Value                                                                                                                               | Explanation                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| S EQ Switch | OFF, ON                                                                                                                             | Switches whether to apply tonal adjustment to left/right input signals whose phase is distant (opposite phase). |
| S In G      | -12.00—+12.00 [dB]                                                                                                                  | Volume of left/right input signals whose phase is distant (opposite phase)                                      |
| S Low F     | 20, 25, 31, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400 [Hz]                                                                  | Frequency of the low range                                                                                      |
| S Low G     | -12.00—+12.00 [dB]                                                                                                                  | Amount of boost/cut for the low-frequency range                                                                 |
| S Mid1 F    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz]                                  | Frequency of the middle range 1                                                                                 |
| S Mid1 G    | -12.00—+12.00 [dB]                                                                                                                  | Gain of the middle range 1                                                                                      |
| S Mid1 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                                                             | Width of the middle range 1<br>Set a higher value for Q to narrow the range to be affected.                     |
| S Mid2 F    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz]                                  | Frequency of the middle range 2                                                                                 |
| S Mid2 G    | -12.00—+12.00 [dB]                                                                                                                  | Gain of the middle range 2                                                                                      |
| S Mid2 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                                                             | Width of the middle range 2<br>Set a higher value for Q to narrow the range to be affected.                     |
| S Mid3 F    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz]                                  | Frequency of the middle range 3                                                                                 |
| S Mid3 G    | -12.00—+12.00 [dB]                                                                                                                  | Gain of the middle range 3                                                                                      |
| S Mid3 Q    | 0.5, 1.0, 2.0, 4.0, 8.0                                                                                                             | Width of the middle range 3<br>Set a higher value for Q to narrow the range to be affected.                     |
| S High F    | 2000, 2500, 3150, 4000, 5000, 6300, 8000, 10000, 12500, 16000 [Hz]                                                                  | Frequency of the high range                                                                                     |
| S High G    | -12.00—+12.00 [dB]                                                                                                                  | Amount of boost/cut for the high-frequency range                                                                |
| Level       | 0–127                                                                                                                               | Output Level                                                                                                    |
| Asgn1–4     | OFF, M Low Gain, M Mid1 Gain, M Mid2 Gain, M Mid3 Gain, M High Gain, S Low Gain, S Mid1 Gain, S Mid2 Gain, S Mid3 Gain, S High Gain | Specifies the parameters that are assigned to assign 1–4.                                                       |

## 03 Spectrum

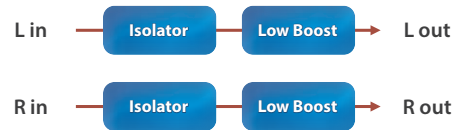
This is a stereo spectrum. Spectrum is a type of filter which modifies the timbre by boosting or cutting the level at specific frequencies.



| Parameter | Value                   | Explanation                                                                          |
|-----------|-------------------------|--------------------------------------------------------------------------------------|
| Band1     | -15—+15 [dB]            | Gain of each frequency band                                                          |
| Band2     |                         |                                                                                      |
| Band3     |                         |                                                                                      |
| Band4     |                         |                                                                                      |
| Band5     |                         |                                                                                      |
| Band6     |                         |                                                                                      |
| Band7     |                         |                                                                                      |
| Band8     |                         |                                                                                      |
| Q         | 0.5, 1.0, 2.0, 4.0, 8.0 | Simultaneously adjusts the width of the adjusted ranges for all the frequency bands. |
| Level     | 0–127                   | Output Level                                                                         |
| Asgn1–4   | OFF, Level              | Specifies the parameters that are assigned to assign 1–4.                            |

## 04 Isolator

This is an equalizer which cuts the volume greatly, allowing you to add a special effect to the sound by cutting the volume in varying ranges.



| Parameter  | Value                                 | Explanation                                                                                                                                                                                         |
|------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low Level  | -60—+4 [dB]                           | These boost and cut each of the High, Middle, and Low frequency ranges.<br>At -60 dB, the sound becomes inaudible. 0 dB is equivalent to the input level of the sound.                              |
| Mid Level  | -60—+4 [dB]                           |                                                                                                                                                                                                     |
| High Level | -60—+4 [dB]                           |                                                                                                                                                                                                     |
| Low AP Sw  | OFF, ON                               | Turns the Anti-Phase function on and off for the Low frequency ranges.<br>When turned on, the counter-channel of stereo sound is inverted and added to the signal.                                  |
| Low AP Lv  | 0–127                                 | Adjusts the level settings for the Low frequency ranges.<br>Adjusting this level for certain frequencies allows you to lend emphasis to specific parts (This is effective only for stereo source.). |
| Mid AP Sw  | OFF, ON                               | Settings of the Anti-Phase function for the Middle frequency ranges.<br>The parameters are the same as for the Low frequency ranges.                                                                |
| Mid AP Lv  | 0–127                                 |                                                                                                                                                                                                     |
| Boost Sw   | OFF, ON                               | Turns Low Booster on/off.<br>This emphasizes the bottom to create a heavy bass sound.                                                                                                               |
| Boost Lv   | 0–127                                 | Increasing this value gives you a heavier low end.<br>Depending on the Isolator and filter settings this effect may be hard to distinguish.                                                         |
| Level      | 0–127                                 | Output Level                                                                                                                                                                                        |
| Asgn1–4    | OFF, Low Level, Mid Level, High Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                           |

## 05 Low Boost

Boosts the volume of the lower range, creating powerful lows.



| Parameter  | Value                                      | Explanation                                               |
|------------|--------------------------------------------|-----------------------------------------------------------|
| Boost Freq | 50, 56, 63, 71, 80, 90, 100, 112, 125 [Hz] | Center frequency at which the lower range will be boosted |
| Boost Gain | 0–+12 [dB]                                 | Center frequency at which the lower range will be boosted |
| Boost Wid  | WIDE, MID, NARROW                          | Width of the lower range that will be boosted             |
| Low Gain   | -15–+15 [dB]                               | Amount of boost/cut for the low-frequency range           |
| High Gain  | -15–+15 [dB]                               | Amount of boost/cut for the high-frequency range          |
| Level      | 0–127                                      | Output Level                                              |
| Asgn1–4    | OFF, Boost Freq, Boost Gain                | Specifies the parameters that are assigned to assign 1–4. |

## 06 SuperFilter

This is a filter with an extremely sharp slope. The cutoff frequency can be varied cyclically.

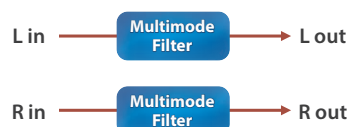


| Parameter | Value                                         | Explanation                                                                                                                                                                                                                                                                           |
|-----------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type      | LPF, BPF, HPF, NOTCH                          | Filter type<br>Frequency range that will pass through each filter<br><b>LPF</b> : Frequencies below the cutoff<br><b>BPF</b> : Frequencies in the region of the cutoff<br><b>HPF</b> : Frequencies above the cutoff<br><b>NOTCH</b> : Frequencies other than the region of the cutoff |
| Slope     | -12, -24, -36 [dB]                            | Amount of attenuation per octave<br><b>-12 dB</b> : Gentle<br><b>-24 dB</b> : Steep<br><b>-36 dB</b> : Extremely steep                                                                                                                                                                |
| Cutoff    | 0–127                                         | Cutoff frequency of the filter<br>Increasing this value will raise the cutoff frequency.                                                                                                                                                                                              |
| Resonance | 0–100                                         | Filter resonance level<br>Increasing this value will emphasize the region near the cutoff frequency.                                                                                                                                                                                  |
| Gain      | 0–+12 [dB]                                    | Amount of boost for the filter output                                                                                                                                                                                                                                                 |
| Mod Sw    | OFF, ON                                       | On/off switch for cyclic change                                                                                                                                                                                                                                                       |
| Mod Wave  | TRI, SQR, SIN, SAW1, SAW2<br><br>SAW1<br>SAW2 | How the cutoff frequency will be modulated<br><b>TRI</b> : Triangle wave<br><b>SQR</b> : Square wave<br><b>SIN</b> : Sine wave<br><b>SAW1</b> : Sawtooth wave (upward)<br><b>SAW2</b> : Sawtooth wave (downward)                                                                      |
| Rate Sync | OFF, ON                                       | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➡ "Tempo" (p. 3, p. 31)                                                                                                                                                                                       |
| Rate      | 0.05–10.00 [Hz]                               | Frequency of modulation                                                                                                                                                                                                                                                               |
| Rate Note | Note<br>➡ "Note" (p. 72)                      | Frequency of modulation                                                                                                                                                                                                                                                               |

| Parameter | Value                                | Explanation                                                                                          |
|-----------|--------------------------------------|------------------------------------------------------------------------------------------------------|
| Depth     | 0–127                                | Depth of modulation                                                                                  |
| Attack    | 0–127                                | Speed at which the cutoff frequency will change This is effective if Mod Wave is SQR, SAW1, or SAW2. |
| Level     | 0–127                                | Output Level                                                                                         |
| Asgn1–4   | OFF, Cutoff, Resonance, Rate, Attack | Specifies the parameters that are assigned to assign 1–4.                                            |

## 07 MM Filter (Multi-mode Filter)

This is a filter that is adjusted for effective use in a DJ performance.



| Parameter | Value                                                     | Explanation                                                                                                                |
|-----------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Type      | LPF/HPF, LPF, HPF, BPF                                    | Type of filter<br><b>LPF/HPF</b> : The filter type is automatically switched according to the Filter Tone parameter value. |
| Tone      | 0–255                                                     | Frequency at which the filter operates                                                                                     |
| Color     | 0–255                                                     | Filter resonance level<br>Higher values more strongly emphasize the region of the operating frequency.                     |
| Slope     | -12, -24, -36 [dB]                                        | Amount of attenuation per octave<br><b>-12 dB</b> : gentle<br><b>-24 dB</b> : steep<br><b>-36 dB</b> : extremely steep     |
| Gain      | 0–+12 [dB]                                                | Amount of boost for the filter output                                                                                      |
| Level     | 0–127                                                     | Output Level                                                                                                               |
| Asgn1–4   | OFF, Filter Type, Filter Tone, Filter Color, Filter Slope | Specifies the parameters that are assigned to assign 1–4.                                                                  |

## 08 Step Filter

This is a filter whose cutoff frequency can be modulated in steps. You can specify the pattern by which the cutoff frequency will change.



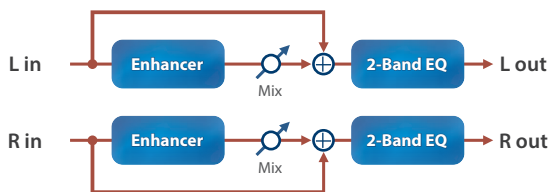
| Parameter | Value                    | Explanation                                                                                                                                                                                                                                                                           |
|-----------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1–16 | 0–127                    | Cutoff frequency at each step                                                                                                                                                                                                                                                         |
| Rate Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➡ "Tempo" (p. 3, p. 31)                                                                                                                                                                                       |
| Rate      | 0.05–10.00 [Hz]          | Frequency of modulation                                                                                                                                                                                                                                                               |
| Rate Note | Note<br>➡ "Note" (p. 72) | Frequency of modulation                                                                                                                                                                                                                                                               |
| Attack    | 0–127                    | Speed at which the cutoff frequency changes between steps                                                                                                                                                                                                                             |
| Type      | LPF, BPF, HPF, NOTCH     | Filter type<br>Frequency range that will pass through each filter<br><b>LPF</b> : Frequencies below the cutoff<br><b>BPF</b> : Frequencies in the region of the cutoff<br><b>HPF</b> : Frequencies above the cutoff<br><b>NOTCH</b> : Frequencies other than the region of the cutoff |



| Parameter | Value                        | Explanation                                                                                                         |
|-----------|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Slope     | -12, -24, -36 [dB]           | Amount of attenuation per octave<br><b>-12 dB:</b> Gentle<br><b>-24 dB:</b> Steep<br><b>-36 dB:</b> Extremely steep |
| Reso      | 0–127                        | Filter resonance level<br>Increasing this value will emphasize the region near the cutoff frequency.                |
| Gain      | 0–+12 [dB]                   | Amount of boost for the filter output                                                                               |
| Level     | 0–127                        | Output Level                                                                                                        |
| Asgn1–4   | OFF, Rate, Attack, Resonance | Specifies the parameters that are assigned to assign 1–4.                                                           |

## 09 Enhancer

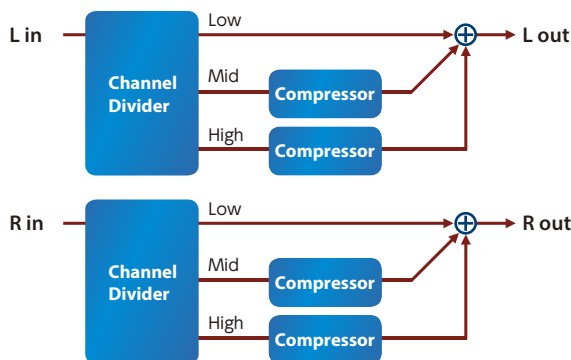
Controls the overtone structure of the high frequencies, adding sparkle and tightness to the sound.



| Parameter | Value          | Explanation                                               |
|-----------|----------------|-----------------------------------------------------------|
| Sens      | 0–127          | Sensitivity of the enhancer                               |
| Mix       | 0–127          | Level of the overtones generated by the enhancer          |
| Low Gain  | -15–+15 [dB]   | Amount of boost/cut for the low-frequency range           |
| High Gain | -15–+15 [dB]   | Amount of boost/cut for the high-frequency range          |
| Level     | 0–127          | Output Level                                              |
| Asgn1–4   | OFF, Sens, Mix | Specifies the parameters that are assigned to assign 1–4. |

## 10 Exciter

This adds dynamics to the sound, by dynamically bringing up the high end using a split-band compressor.



| Parameter       | Value          | Explanation                                                                     |
|-----------------|----------------|---------------------------------------------------------------------------------|
| Band2 Threshold | -80.0–0.0 (dB) | Raises the midrange frequency levels when they fall below the specified amount. |
| Band2 Gain      | 0–+24 (dB)     | Sets how much to raise the levels when the midrange volume is low.              |
| Band3 Threshold | -80.0–0.0 (dB) | Raises the high-end frequency levels when they fall below the specified amount. |
| Band3 Gain      | 0–+24 (dB)     | Sets how much to raise the levels when the high-end frequency volume is low.    |
| Split1 Freq     | 2000–5000 (Hz) | Frequency at which the low and midrange frequencies are split                   |

| Parameter   | Value                                               | Explanation                                                        |
|-------------|-----------------------------------------------------|--------------------------------------------------------------------|
| Split2 Freq | 3000–10000 (Hz)                                     | Frequency at which the midrange and high-end frequencies are split |
| Level       | 0–127                                               | Output Level                                                       |
| Asgn1–4     | OFF, Band2 Thrs, Band2 Gain, Band3 Thrs, Band3 Gain | Specifies the parameters that are assigned to assign 1–4.          |

## 11 Auto Wah

Cyclically controls a filter to create cyclic change in timbre.



| Parameter | Value                                      | Explanation                                                                                                                                                              |
|-----------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode      | LPF, BPF                                   | Filter type<br><b>LPF:</b> The wah effect will be applied over a wide frequency range.<br><b>BPF:</b> The wah effect will be applied over a narrow frequency range.      |
| Manual    | 0–127                                      | Center frequency at which the wah effect is applied                                                                                                                      |
| Peak      | 0–127                                      | Width of the frequency region at which the wah effect is applied<br>Increasing this value will make the frequency region narrower.                                       |
| Sens      | 0–127                                      | Sensitivity with which the filter is modified                                                                                                                            |
| Polarity  | UP, DOWN                                   | Direction in which the filter will move<br><b>UP:</b> The filter will change toward a higher frequency.<br><b>DOWN:</b> The filter will change toward a lower frequency. |
| Rate Sync | OFF, ON                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                                                                          |
| Rate      | 0.05–10.00 [Hz]                            | Frequency of modulation of wah effect                                                                                                                                    |
| Rate Note | Note<br>➔ "Note" (p. 72)                   |                                                                                                                                                                          |
| Depth     | 0–127                                      | Depth at which the wah effect is modulated                                                                                                                               |
| Phase     | 0–180 [deg]                                | Adjusts the degree of phase shift of the left and right sounds when the wah effect is applied.                                                                           |
| Low Gain  | -15–+15 [dB]                               | Amount of boost/cut for the low-frequency range                                                                                                                          |
| High Gain | -15–+15 [dB]                               | Amount of boost/cut for the high-frequency range                                                                                                                         |
| Level     | 0–127                                      | Output Level                                                                                                                                                             |
| Asgn1–4   | OFF, Manual, Sens, Rate (Hz), Depth, Phase | Specifies the parameters that are assigned to assign 1–4.                                                                                                                |

## 12 Humanizer

Adds a vowel character to the sound, making it similar to a human voice.



| Parameter | Value   | Explanation                                      |
|-----------|---------|--------------------------------------------------|
| Drive Sw  | OFF, ON | Overdrive on/off                                 |
| Drive     | 0–127   | Degree of distortion<br>Also changes the volume. |

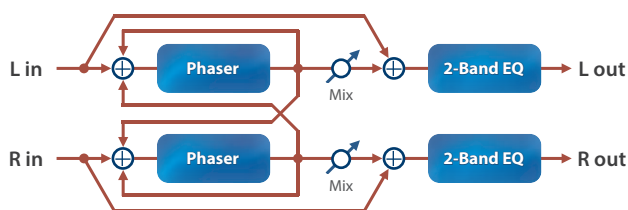


| Parameter   | Value                                     | Explanation                                                                                                                                                                                        |
|-------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vowel1      | a, e, i, o, u                             | Selects the vowel.                                                                                                                                                                                 |
| Vowel2      | a, e, i, o, u                             |                                                                                                                                                                                                    |
| Rate Sync   | OFF, ON                                   | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                                                                                    |
| Rate        | 0.05–10.00 [Hz]                           | Frequency at which the two vowels switch                                                                                                                                                           |
| Rate Note   | Note<br>→ "Note" (p. 72)                  |                                                                                                                                                                                                    |
| Depth       | 0–127                                     | Effect depth                                                                                                                                                                                       |
| In Sync Sw  | OFF, ON                                   | LFO reset on/off<br>Determines whether the LFO for switching the vowels is reset by the input signal (ON) or not (OFF).                                                                            |
| InSyncThres | 0–127                                     | Volume level at which reset is applied                                                                                                                                                             |
| Manual      | 0–100                                     | Point at which Vowel 1/2 switch<br><b>0–49:</b> Vowel 1 will have a longer duration.<br><b>50:</b> Vowel 1 and 2 will be of equal duration.<br><b>51–100:</b> Vowel 2 will have a longer duration. |
| Low Gain    | -15–+15 [dB]                              | Amount of boost/cut for the low-frequency range                                                                                                                                                    |
| High Gain   | -15–+15 [dB]                              | Amount of boost/cut for the high-frequency range                                                                                                                                                   |
| Pan         | L64–63R                                   | Stereo location of the output sound                                                                                                                                                                |
| Level       | 0–127                                     | Output Level                                                                                                                                                                                       |
| Asgn1–4     | OFF, Drive, Rate (Hz), Depth, Manual, Pan | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                          |

## Phaser

### 13 Phaser

This is a stereo phaser. A phase-shifted sound is added to the original sound and modulated.



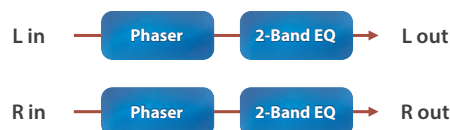
| Parameter | Value                      | Explanation                                                                                                                                                                                                                                                                                                                  |
|-----------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode      | 4-STAGE, 8-STAGE, 12-STAGE | Number of stages in the phaser                                                                                                                                                                                                                                                                                               |
| Manual    | 0–127                      | Adjusts the basic frequency from which the sound will be modulated.                                                                                                                                                                                                                                                          |
| Rate Sync | OFF, ON                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                                                                                                                                                                                                              |
| Rate      | 0.05–10.00 [Hz]            | Frequency of modulation                                                                                                                                                                                                                                                                                                      |
| Rate Note | Note<br>→ "Note" (p. 72)   |                                                                                                                                                                                                                                                                                                                              |
| Depth     | 0–127                      | Depth of modulation                                                                                                                                                                                                                                                                                                          |
| Polarity  | INVERSE, SYNCHRO           | Selects whether the left and right phase of the modulation will be the same or the opposite.<br><b>INVERSE:</b> The left and right phase will be opposite.<br>When using a mono source, this spreads the sound.<br><b>SYNCHRO:</b> The left and right phase will be the same.<br>Select this when inputting a stereo source. |
| Resonance | 0–127                      | Amount of feedback                                                                                                                                                                                                                                                                                                           |

| Parameter | Value                             | Explanation                                                                                                                  |
|-----------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Feedback  | -98–+98 [%]                       | Adjusts the proportion of the phaser sound that is fed back into the effect.<br>Negative (-) settings will invert the phase. |
| Mix       | 0–127                             | Level of the phase-shifted sound                                                                                             |
| Low Gain  | -15–+15 [dB]                      | Amount of boost/cut for the low-frequency range                                                                              |
| High Gain | -15–+15 [dB]                      | Amount of boost/cut for the high-frequency range                                                                             |
| Level     | 0–127                             | Output Level                                                                                                                 |
| Asgn1–4   | OFF, Manual, Rate (Hz), Resonance | Specifies the parameters that are assigned to assign 1–4.                                                                    |

### 14 Small Phaser

This simulates an analog phaser of the past.

It is particularly suitable for electric piano.

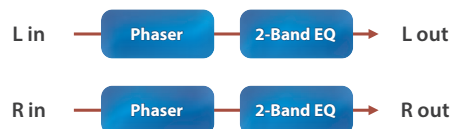


| Parameter | Value        | Explanation                                               |
|-----------|--------------|-----------------------------------------------------------|
| Rate      | 0–100        | Frequency of modulation                                   |
| Color     | 1, 2         | Modulation character                                      |
| Low Gain  | -15–+15 [dB] | Amount of boost/cut for the low-frequency range           |
| High Gain | -15–+15 [dB] | Amount of boost/cut for the high-frequency range          |
| Level     | 0–127        | Output Level                                              |
| Asgn1–4   | OFF, Rate    | Specifies the parameters that are assigned to assign 1–4. |

### 15 Script 90

This simulates a different analog phaser than Small Phaser.

It is particularly suitable for electric piano.



| Parameter | Value        | Explanation                                               |
|-----------|--------------|-----------------------------------------------------------|
| Speed     | 0–100        | Speed of modulation                                       |
| Depth     | 0–127        | Depth of modulation                                       |
| Low Gain  | -15–+15 [dB] | Amount of boost/cut for the low-frequency range           |
| High Gain | -15–+15 [dB] | Amount of boost/cut for the high-frequency range          |
| Level     | 0–127        | Output Level                                              |
| Asgn1–4   | OFF, Speed   | Specifies the parameters that are assigned to assign 1–4. |

16 Script 100

This simulates an analog phaser of the past.

L in

Phaser

L out

R in

Phaser

R out

| Parameter | Value                                            | Explanation                                                                                              |
|-----------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Rate Sync | OFF, ON                                          | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)          |
| Rate      | 0.05–10.00 [Hz]                                  | Frequency of modulation                                                                                  |
| Rate Note | Note<br>➔ "Note" (p. 72)                         |                                                                                                          |
| Duty      | –50–50                                           | Adjusts the ratio of speeds at which the modulation rises or falls.                                      |
| Min       | 0–100                                            | Lower limit reached by modulation                                                                        |
| Max       | 0–100                                            | Upper limit reached by modulation                                                                        |
| Manual Sw | OFF, ON                                          | Applies modulation according to the value of the Manual parameter, rather than modulating automatically. |
| Manual    | 0–100                                            | Adjusts the basic frequency from which the sound will be modulated.                                      |
| Resonance | 0–66                                             | Amount of feedback                                                                                       |
| Mix       | 0–127                                            | Level of the phase-shifted sound                                                                         |
| Level     | 0–127                                            | Output Level                                                                                             |
| Asgn1–4   | OFF, Rate (Hz), Min, Max, Manual, Resonance, Mix | Specifies the parameters that are assigned to assign 1–4.                                                |

17 Step Phaser

This is a stereo phaser. The phaser effect will be varied gradually.

L in

Step Phaser

Mix

2-Band EQ

L out

R in

Step Phaser

Mix

2-Band EQ

R out

| Parameter | Value                      | Explanation                                                                                                                                                                                                                                                                                                                  |
|-----------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode      | 4-STAGE, 8-STAGE, 12-STAGE | Number of stages in the phaser                                                                                                                                                                                                                                                                                               |
| Manual    | 0–127                      | Adjusts the basic frequency from which the sound will be modulated.                                                                                                                                                                                                                                                          |
| Rate Sync | OFF, ON                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                                                                                                                                                                                                                              |
| Rate      | 0.05–10.00 [Hz]            | Frequency of modulation                                                                                                                                                                                                                                                                                                      |
| Rate Note | Note<br>➔ "Note" (p. 72)   |                                                                                                                                                                                                                                                                                                                              |
| Depth     | 0–127                      | Depth of modulation                                                                                                                                                                                                                                                                                                          |
| Polarity  | INVERSE, SYNCHRO           | Selects whether the left and right phase of the modulation will be the same or the opposite.<br><b>INVERSE:</b> The left and right phase will be opposite.<br>When using a mono source, this spreads the sound.<br><b>SYNCHRO:</b> The left and right phase will be the same.<br>Select this when inputting a stereo source. |
| Resonance | 0–127                      | Amount of feedback                                                                                                                                                                                                                                                                                                           |
| Feedback  | –98–+98 [%]                | Adjusts the proportion of the phaser sound that is fed back into the effect.<br>Negative (–) settings will invert the phase.                                                                                                                                                                                                 |

| Parameter   | Value                                      | Explanation                                                                                     |
|-------------|--------------------------------------------|-------------------------------------------------------------------------------------------------|
| S Rate Sync | OFF, ON                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31) |
| S. Rate     | 0.10–20.00 [Hz]                            | Rate of the step-wise change in the phaser effect                                               |
| S. Rate Nt  | Note<br>➔ "Note" (p. 72)                   |                                                                                                 |
| Mix         | 0–127                                      | Level of the phase-shifted sound                                                                |
| Low Gain    | –15–+15 [dB]                               | Amount of boost/cut for the low-frequency range                                                 |
| High Gain   | –15–+15 [dB]                               | Amount of boost/cut for the high-frequency range                                                |
| Level       | 0–127                                      | Output Level                                                                                    |
| Asgn1–4     | OFF, Manual, Rate, Resonance, S. Rate, Mix | Specifies the parameters that are assigned to assign 1–4.                                       |

18 M StagePhsr (Multi Stage Phaser)

Extremely high settings of the phase difference produce a deep phaser effect.

L in

Mix

Multi Stage Phaser

Mix

2-Band EQ

Pan L

Pan R

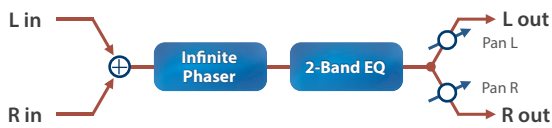
L out

R out

| Parameter | Value                                                    | Explanation                                                                                     |
|-----------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Mode      | 4-STAGE, 8-STAGE, 12-STAGE, 16-STAGE, 20-STAGE, 24-STAGE | Number of stages in the phaser                                                                  |
| Manual    | 0–127                                                    | Adjusts the basic frequency from which the sound will be modulated.                             |
| Rate Sync | OFF, ON                                                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31) |
| Rate      | 0.05–10.00 [Hz]                                          | Frequency of modulation                                                                         |
| Rate Note | Note<br>➔ "Note" (p. 72)                                 |                                                                                                 |
| Depth     | 0–127                                                    | Depth of modulation                                                                             |
| Resonance | 0–127                                                    | Amount of feedback                                                                              |
| Mix       | 0–127                                                    | Level of the phase-shifted sound                                                                |
| Pan       | L64–63R                                                  | Stereo location of the output sound                                                             |
| Low Gain  | –15–+15 [dB]                                             | Amount of boost/cut for the low-frequency range                                                 |
| High Gain | –15–+15 [dB]                                             | Amount of boost/cut for the high-frequency range                                                |
| Level     | 0–127                                                    | Output Level                                                                                    |
| Asgn1–4   | OFF, Manual, Rate (Hz), Resonance, Mix, Pan              | Specifies the parameters that are assigned to assign 1–4.                                       |

## 19 Inf Phaser (Infinite Phaser)

A phaser that continues raising/lowering the frequency at which the sound is modulated.



| Parameter | Value                           | Explanation                                                                                              |
|-----------|---------------------------------|----------------------------------------------------------------------------------------------------------|
| Mode      | 1–4                             | Higher values will produce a deeper phaser effect.                                                       |
| Speed     | -100–+100                       | Speed at which to raise or lower the frequency at which the sound is modulated (+: upward / -: downward) |
| Resonance | 0–127                           | Amount of feedback                                                                                       |
| Mix       | 0–127                           | Level of the phase-shifted sound                                                                         |
| Pan       | L64–63R                         | Stereo location of the output sound                                                                      |
| Low Gain  | -15–+15 [dB]                    | Amount of boost/cut for the low-frequency range                                                          |
| High Gain | -15–+15 [dB]                    | Amount of boost/cut for the high-frequency range                                                         |
| Level     | 0–127                           | Output Level                                                                                             |
| Asgn1–4   | OFF, Speed, Resonance, Mix, Pan | Specifies the parameters that are assigned to assign 1–4.                                                |

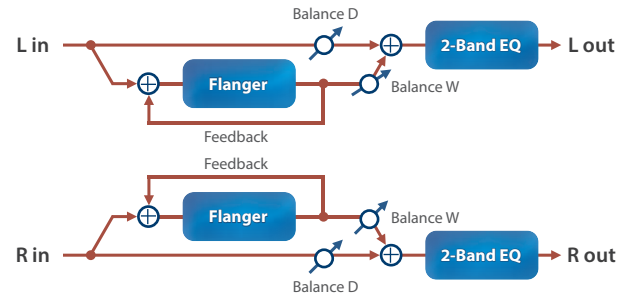
## Flanger

### 20 Flanger

This is a stereo flanger (The LFO has the same phase for left and right.).

It produces a metallic resonance that rises and falls like a jet airplane taking off or landing.

A filter is provided so that you can adjust the timbre of the flanged sound.



| Parameter | Value                                                                                              | Explanation                                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type      | OFF, LPF, HPF                                                                                      | Filter type<br><b>OFF:</b> No filter is used<br><b>LPF:</b> Cuts the frequency range above the Cutoff Freq<br><b>HPF:</b> Cuts the frequency range below the Cutoff Freq |
| Cutoff    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Basic frequency of the filter                                                                                                                                            |
| Pre Delay | 0.0–100 [ms]                                                                                       | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                                                           |
| Rate Sync | OFF, ON                                                                                            | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➡ "Tempo" (p. 3, p. 31)                                                                          |
| Rate      | 0.05–10.00 [Hz]                                                                                    | Frequency of modulation                                                                                                                                                  |
| Rate Note | Note<br>➡ "Note" (p. 72)                                                                           |                                                                                                                                                                          |
| Depth     | 0–127                                                                                              | Depth of modulation                                                                                                                                                      |
| Phase     | 0–180 [deg]                                                                                        | Spatial spread of the sound                                                                                                                                              |
| Feedback  | -98–+98 [%]                                                                                        | Adjusts the proportion of the flanger sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.                                            |
| Low Gain  | -15–+15 [dB]                                                                                       | Amount of boost/cut for the low-frequency range                                                                                                                          |
| High Gain | -15–+15 [dB]                                                                                       | Amount of boost/cut for the high-frequency range                                                                                                                         |
| Balance   | D100: 0W–D0: 100W                                                                                  | Volume balance between the direct sound (D) and the flanger sound (W)                                                                                                    |
| Level     | 0–127                                                                                              | Output Level                                                                                                                                                             |
| Asgn1–4   | OFF, Rate (Hz), Feedback, Balance                                                                  | Specifies the parameters that are assigned to assign 1–4.                                                                                                                |

## 21 SBF-325 (Flanger)

This effect reproduces Roland's SBF-325 analog flanger.

It provides three types of flanging effect (which adds a metallic resonance to the original sound) and a chorus-type effect.

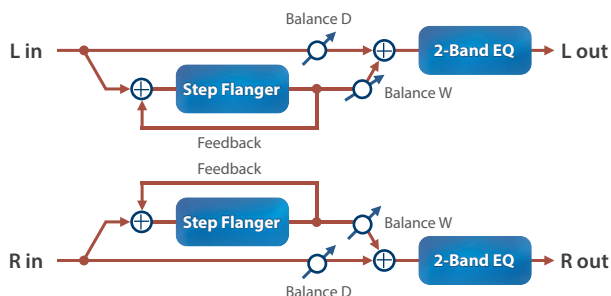


| Parameter  | Value                         | Explanation                                                                                                                                                                                             |
|------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode       |                               | Types of flanging effect                                                                                                                                                                                |
|            | FL1                           | A typical mono flanger                                                                                                                                                                                  |
|            | FL2                           | A stereo flanger that preserves the stereo positioning of the original sound                                                                                                                            |
|            | FL3                           | A cross-mix flanger that produces a more intense effect                                                                                                                                                 |
| CHO        |                               | A chorus effect                                                                                                                                                                                         |
|            |                               |                                                                                                                                                                                                         |
| Rate Sync  | OFF, ON                       | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                                                                                         |
| Rate       | 0.02–5.00 [Hz]                |                                                                                                                                                                                                         |
| Rate Note  | Note<br>→ "Note" (p. 72)      | Modulation frequency of the flanger effect                                                                                                                                                              |
| Depth      | 0–127                         | Modulation depth of the flanger effect                                                                                                                                                                  |
| Manual     | 0–127                         | Center frequency at which the flanger effect is applied                                                                                                                                                 |
| Feedback   | 0–127                         | Amount by which the flanging effect is boosted<br>If Mode is CHO, this setting is ignored.                                                                                                              |
| RMod Phase | NORM, INV                     | Phase of the right channel modulation:<br>Normally, you will leave this at Normal (NORM).<br>If you specify Inverted (INV), the modulation (upward/downward movement) of the right channel is inverted. |
| L Phase    | NORM, INV                     | Phase when mixing the flanging sound with the original sound                                                                                                                                            |
| R Phase    | NORM, INV                     | <b>NORM:</b> normal phase<br><b>INV:</b> inverse phase                                                                                                                                                  |
| Level      | 0–127                         | Output Level                                                                                                                                                                                            |
| Asgn1–4    | OFF, Rate (Hz), Depth, Manual | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                               |

## 22 StepFlanger

This is a flanger in which the flanger pitch changes in steps.

The speed at which the pitch changes can also be specified in terms of a note-value of a specified tempo.



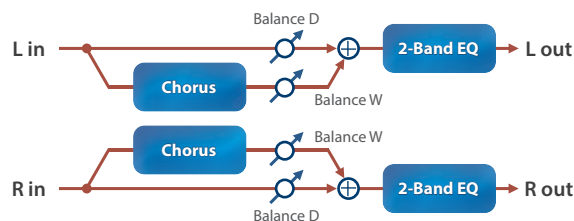
| Parameter | Value         | Explanation                                                                                                                                                              |
|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type      | OFF, LPF, HPF | Filter type<br><b>OFF:</b> No filter is used<br><b>LPF:</b> Cuts the frequency range above the Cutoff Freq<br><b>HPF:</b> Cuts the frequency range below the Cutoff Freq |

| Parameter    | Value                                                                                              | Explanation                                                                                                                   |
|--------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Cutoff       | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Basic frequency of the filter                                                                                                 |
| Pre Delay    | 0.0–100 [ms]                                                                                       | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                |
| Rate Sync    | OFF, ON                                                                                            | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                               |
| Rate         | 0.05–10.00 [Hz]                                                                                    |                                                                                                                               |
| Rate Note    | Note<br>→ "Note" (p. 72)                                                                           | Frequency of modulation                                                                                                       |
| Depth        | 0–127                                                                                              | Depth of modulation                                                                                                           |
| Phase        | 0–180 [deg]                                                                                        | Spatial spread of the sound                                                                                                   |
| Feedback     | -98–+98 [%]                                                                                        | Adjusts the proportion of the flanger sound that is fed back into the effect.<br>Negative (-) settings will invert the phase. |
| S. Rate Sync | OFF, ON                                                                                            | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                               |
| S. Rate      | 0.10–20.00 [Hz]                                                                                    |                                                                                                                               |
| S. Rate Nt   | Note<br>→ "Note" (p. 72)                                                                           | Rate (period) of pitch change                                                                                                 |
| Low Gain     | -15–+15 [dB]                                                                                       | Amount of boost/cut for the low-frequency range                                                                               |
| High Gain    | -15–+15 [dB]                                                                                       | Amount of boost/cut for the high-frequency range                                                                              |
| Balance      | D100: 0W–D0: 100W                                                                                  | Volume balance between the direct sound (D) and the flanger sound (W)                                                         |
| Level        | 0–127                                                                                              | Output Level                                                                                                                  |
| Asgn1–4      | OFF, Rate, Feedback, S. Rate, Balance                                                              | Specifies the parameters that are assigned to assign 1–4.                                                                     |

## Chorus

## 23 Chorus

This is a stereo chorus. A filter is provided so that you can adjust the timbre of the chorus sound.

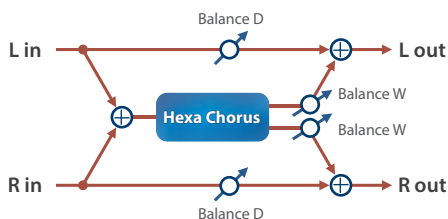


| Parameter | Value                                                                                              | Explanation                                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type      | OFF, LPF, HPF                                                                                      | Filter type<br><b>OFF:</b> No filter is used<br><b>LPF:</b> Cuts the frequency range above the Cutoff Freq<br><b>HPF:</b> Cuts the frequency range below the Cutoff Freq |
| Cutoff    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000 [Hz] | Basic frequency of the filter                                                                                                                                            |
| Pre Delay | 0.0–100 [ms]                                                                                       | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                                                            |

| Parameter | Value                    | Explanation                                                                                     |
|-----------|--------------------------|-------------------------------------------------------------------------------------------------|
| Rate Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| Rate      | 0.05–10.00 [Hz]          | Frequency of modulation                                                                         |
| Rate Note | Note<br>→ "Note" (p. 72) |                                                                                                 |
| Depth     | 0–127                    | Depth of modulation                                                                             |
| Phase     | 0–180 [deg]              | Spatial spread of the sound                                                                     |
| Low Gain  | -15–+15 [dB]             | Amount of boost/cut for the low-frequency range                                                 |
| High Gain | -15–+15 [dB]             | Amount of boost/cut for the high-frequency range                                                |
| Balance   | D100: 0W–D0: 100W        | Volume balance between the direct sound (D) and the chorus sound (W)                            |
| Level     | 0–127                    | Output Level                                                                                    |
| Asgn1–4   | OFF, Rate (Hz), Balance  | Specifies the parameters that are assigned to assign 1–4.                                       |

## 24 Hexa-Chorus

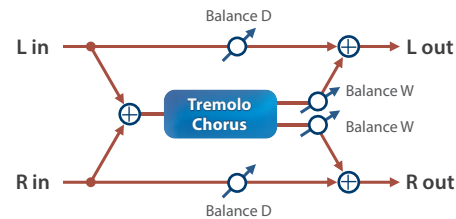
Uses a six-phase chorus (six layers of chorused sound) to give richness and spatial spread to the sound.



| Parameter  | Value                    | Explanation                                                                                                                                                                                                              |
|------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Delay  | 0.0–100 [ms]             | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                                                                                                            |
| Rate Sync  | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                                                                                                          |
| Rate       | 0.05–10.00 [Hz]          | Frequency of modulation                                                                                                                                                                                                  |
| Rate Note  | Note<br>→ "Note" (p. 72) |                                                                                                                                                                                                                          |
| Depth      | 0–127                    | Depth of modulation                                                                                                                                                                                                      |
| PreDly Dev | 0–20                     | Adjusts the differences in Pre Delay between each chorus sound.                                                                                                                                                          |
| Depth Dev  | -20–+20                  | Adjusts the difference in modulation depth between each chorus sound.                                                                                                                                                    |
| Pan Dev    | 0–20                     | Adjusts the difference in stereo location between each chorus sound.<br><b>0:</b> All chorus sounds will be in the center.<br><b>20:</b> Each chorus sound will be spaced at 60 degree intervals relative to the center. |
| Balance    | D100: 0W–D0: 100W        | Volume balance between the direct sound (D) and the chorus sound (W)                                                                                                                                                     |
| Level      | 0–127                    | Output Level                                                                                                                                                                                                             |
| Asgn1–4    | OFF, Rate (Hz), Balance  | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                |

## 25 Trem Chorus (Tremolo Chorus)

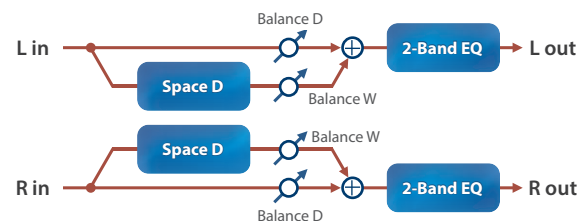
This is a chorus effect with added Tremolo (cyclic modulation of volume).



| Parameter    | Value                          | Explanation                                                                                     |
|--------------|--------------------------------|-------------------------------------------------------------------------------------------------|
| Pre Delay    | 0.0–100 [ms]                   | Adjusts the delay time from the direct sound until the chorus sound is heard.                   |
| Cho Sync     | OFF, ON                        | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| C. Rate      | 0.05–10.00 [Hz]                | Modulation frequency of the chorus effect                                                       |
| C. Rate Nt   | Note<br>→ "Note" (p. 72)       |                                                                                                 |
| Cho Depth    | 0–127                          | Modulation depth of the chorus effect                                                           |
| Trm Sync     | OFF, ON                        | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| T. Rate      | 0.05–10.00 [Hz]                | Modulation frequency of the tremolo effect                                                      |
| T. Rate Nt   | Note<br>→ "Note" (p. 72)       |                                                                                                 |
| Trm Separate | 0–127                          | Depth of the tremolo effect                                                                     |
| Trm Phase    | 0–180 [deg]                    | Spread of the tremolo effect                                                                    |
| Balance      | D100: 0W–D0: 100W              | Volume balance between the direct sound (D) and the tremolo chorus sound (W)                    |
| Level        | 0–127                          | Output Level                                                                                    |
| Asgn1–4      | OFF, C. Rate, T. Rate, Balance | Specifies the parameters that are assigned to assign 1–4.                                       |

## 26 Space-D

This is a multiple chorus that applies two-phase modulation in stereo. It gives no impression of modulation, but produces a transparent chorus effect.

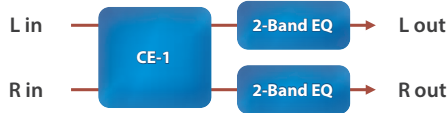


| Parameter | Value                    | Explanation                                                                                     |
|-----------|--------------------------|-------------------------------------------------------------------------------------------------|
| Pre Delay | 0.0–100 [ms]             | Adjusts the delay time from the direct sound until the chorus sound is heard.                   |
| Rate Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| Rate      | 0.05–10.00 [Hz]          | Frequency of modulation                                                                         |
| Rate Note | Note<br>→ "Note" (p. 72) |                                                                                                 |
| Depth     | 0–127                    | Depth of modulation                                                                             |
| Phase     | 0–180 [deg]              | Spatial spread of the sound                                                                     |
| Low Gain  | -15–+15 [dB]             | Amount of boost/cut for the low-frequency range                                                 |
| High Gain | -15–+15 [dB]             | Amount of boost/cut for the high-frequency range                                                |
| Balance   | D100: 0W–D0: 100W        | Volume balance between the direct sound (D) and the chorus sound (W)                            |

| Parameter | Value                   | Explanation                                               |
|-----------|-------------------------|-----------------------------------------------------------|
| Level     | 0–127                   | Output Level                                              |
| Asgn1–4   | OFF, Rate (Hz), Balance | Specifies the parameters that are assigned to assign 1–4. |

27 CE-1 (Chorus)

This models the classic BOSS CE-1 chorus effect unit. It provides a chorus sound with a distinctively analog warmth.



| Parameter | Value          | Explanation                                               |
|-----------|----------------|-----------------------------------------------------------|
| Intensity | 0–127          | Chorus depth                                              |
| Low Gain  | -15–+15 [dB]   | Amount of boost/cut for the low-frequency range           |
| High Gain | -15–+15 [dB]   | Amount of boost/cut for the high-frequency range          |
| Level     | 0–127          | Output Level                                              |
| Asgn1–4   | OFF, Intensity | Specifies the parameters that are assigned to assign 1–4. |

28 SDD-320 (DIMENSION D)

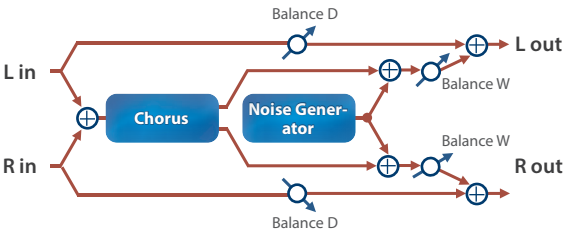
This models Roland’s DIMENSION D (SDD-320). It provides a clear chorus sound.



| Parameter | Value                     | Explanation                                               |
|-----------|---------------------------|-----------------------------------------------------------|
| Mode      | 1, 2, 3, 4, 1+4, 2+4, 3+4 | Switches the mode.                                        |
| Low Gain  | -15–+15 [dB]              | Amount of boost/cut for the low-frequency range           |
| High Gain | -15–+15 [dB]              | Amount of boost/cut for the high-frequency range          |
| Level     | 0–127                     | Output Level                                              |
| Asgn1–4   | OFF, Mode                 | Specifies the parameters that are assigned to assign 1–4. |

29 JUNO Chorus (JUNO-106 Chorus)

This models the chorus effects of the Roland JUNO-106.

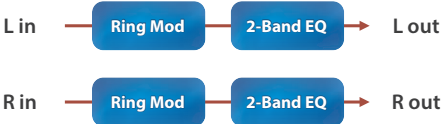


| Parameter | Value                                                  | Explanation                                                                                                                |
|-----------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Mode      | I, II, I+II, JX I, JX II, III, I+III, II+III, I+II+III | Type of Chorus<br><b>I+II, I+III, II+III, I+II+III</b> : The state in which multiple buttons are pressed at the same time. |
| Noise Lv  | 0–127                                                  | Volume of the noise produced by chorus                                                                                     |
| Balance   | D100: 0W–D0: 100W                                      | Volume balance between the dry sound (D) and effect sound (W)                                                              |
| Level     | 0–127                                                  | Output Level                                                                                                               |
| Asgn1–4   | OFF, Noise Level, Balance                              | Specifies the parameters that are assigned to assign 1–4.                                                                  |

Modulation

30 Ring Mod (Ring modulator)

This is an effect that applies amplitude modulation (AM) to the input signal, producing bell-like sounds. You can also change the modulation frequency in response to changes in the volume of the sound sent into the effect.



| Parameter | Value                         | Explanation                                                                                                                                                                                                                           |
|-----------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frequency | 0–127                         | Adjusts the frequency at which modulation is applied.                                                                                                                                                                                 |
| Sens      | 0–127                         | Adjusts the amount of frequency modulation applied.                                                                                                                                                                                   |
| Polarity  | UP, DOWN                      | Determines whether the frequency modulation moves towards higher frequencies or lower frequencies.<br><b>UP</b> : The filter will change toward a higher frequency.<br><b>DOWN</b> : The filter will change toward a lower frequency. |
| Low Gain  | -15–+15 [dB]                  | Amount of boost/cut for the low-frequency range                                                                                                                                                                                       |
| High Gain | -15–+15 [dB]                  | Amount of boost/cut for the high-frequency range                                                                                                                                                                                      |
| Balance   | D100: 0W–D0: 100W             | Volume balance between the direct sound (D) and the effect sound (W)                                                                                                                                                                  |
| Level     | 0–127                         | Output Level                                                                                                                                                                                                                          |
| Asgn1–4   | OFF, Frequency, Sens, Balance | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                             |

## 31 Tremolo

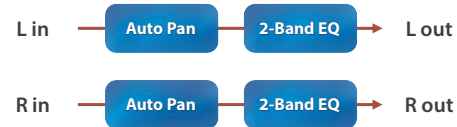
Cyclically changes the volume.



| Parameter | Value                          | Explanation                                                                                                                                                           |
|-----------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mod Wave  | TRI, SQR, SIN, SAW1, SAW2, TRP | Modulation Wave<br><b>TRI</b> : Triangle wave<br><b>SQR</b> : Square wave<br><b>SIN</b> : Sine wave<br><b>SAW1/2</b> : Sawtooth wave<br><b>TRP</b> : Trapezoidal wave |
|           | SAW1 SAW2                      |                                                                                                                                                                       |
| Rate Sync | OFF, ON                        | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                                                                       |
| Rate      | 0.05–10.00 [Hz]                |                                                                                                                                                                       |
| Rate Note | Note<br>➔ "Note" (p. 72)       | Frequency of the change                                                                                                                                               |
| Depth     | 0–127                          | Depth to which the effect is applied                                                                                                                                  |
| Low Gain  | -15–+15 [dB]                   | Amount of boost/cut for the low-frequency range                                                                                                                       |
| High Gain | -15–+15 [dB]                   | Amount of boost/cut for the high-frequency range                                                                                                                      |
| Level     | 0–127                          | Output Level                                                                                                                                                          |
| Asgn1–4   | OFF, Rate (Hz), Depth          | Specifies the parameters that are assigned to assign 1–4.                                                                                                             |

## 32 Auto Pan

Cyclically modulates the stereo location of the sound.



| Parameter | Value                          | Explanation                                                                                                                                                           |
|-----------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mod Wave  | TRI, SQR, SIN, SAW1, SAW2, TRP | Modulation Wave<br><b>TRI</b> : Triangle wave<br><b>SQR</b> : Square wave<br><b>SIN</b> : Sine wave<br><b>SAW1/2</b> : Sawtooth wave<br><b>TRP</b> : Trapezoidal wave |
|           | SAW1 R SAW2 L                  |                                                                                                                                                                       |
| Rate Sync | OFF, ON                        | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                                                                       |
| Rate      | 0.05–10.00 [Hz]                |                                                                                                                                                                       |
| Rate Note | Note<br>➔ "Note" (p. 72)       | Frequency of the change                                                                                                                                               |
| Depth     | 0–127                          | Depth to which the effect is applied                                                                                                                                  |
| Low Gain  | -15–+15 [dB]                   | Amount of boost/cut for the low-frequency range                                                                                                                       |
| High Gain | -15–+15 [dB]                   | Amount of boost/cut for the high-frequency range                                                                                                                      |
| Level     | 0–127                          | Output Level                                                                                                                                                          |
| Asgn1–4   | OFF, Rate (Hz), Depth          | Specifies the parameters that are assigned to assign 1–4.                                                                                                             |

## 33 Slicer

By applying successive cuts to the sound, this effect turns a conventional sound into a sound that appears to be played as a backing phrase. This is especially effective when applied to sustain-type sounds.



| Parameter   | Value                    | Explanation                                                                                                             |
|-------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Step 1–16   | 0–127                    | Level at each step                                                                                                      |
| Rate Sync   | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                         |
| Rate        | 0.05–10.00 [Hz]          |                                                                                                                         |
| Rate Note   | Note<br>➔ "Note" (p. 72) | Rate at which the 16-step sequence will cycle                                                                           |
| Attack      | 0–127                    | Speed at which the level changes between steps                                                                          |
| In Sync Sw  | OFF, ON                  | Specifies whether an input note will cause the sequence to resume from the first step of the sequence (ON) or not (OFF) |
| InSyncThres | 0–127                    | Volume at which an input note will be detected                                                                          |

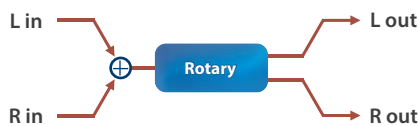


| Parameter | Value                           | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode      | LEGATO, SLASH                   | Sets the manner in which the volume changes as one step progresses to the next.<br><b>LEGATO:</b> The change in volume from one step's level to the next remains unaltered. If the level of a following step is the same as the one preceding it, there is no change in volume.<br><b>SLASH:</b> The level is momentarily set to 0 before progressing to the level of the next step. This change in volume occurs even if the level of the following step is the same as the preceding step. |
| Shuffle   | 0–127                           | Timing of volume changes in levels for even-numbered steps (step 2, step 4, step 6...). The higher the value, the later the beat progresses.                                                                                                                                                                                                                                                                                                                                                 |
| Level     | 0–127                           | Output Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Asgn1–4   | OFF, Rate (Hz), Attack, Shuffle | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## 34 Rotary

This simulates a classic rotary speaker of the past.

Since the operation of the high-frequency and low-frequency rotors can be specified independently, the distinctive modulation can be reproduced realistically. This is most effective on organ patches.

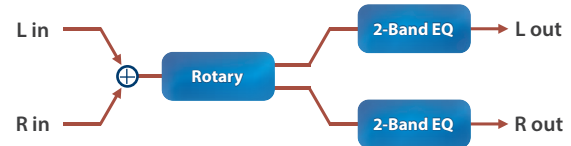


| Parameter  | Value             | Explanation                                                                                                                                                                                                        |
|------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speed      | SLOW, FAST        | Simultaneously switch the rotational speed of the low frequency rotor and high frequency rotor.<br><b>SLOW:</b> Slows down the rotation to the Slow Rate.<br><b>FAST:</b> Speeds up the rotation to the Fast Rate. |
| Wf Slow    | 0.05–10.00 [Hz]   | Slow speed (SLOW) of the low frequency rotor                                                                                                                                                                       |
| Wf Fast    | 0.05–10.00 [Hz]   | Fast speed (FAST) of the low frequency rotor                                                                                                                                                                       |
| Wf Accel   | 0–15              | Adjusts the time it takes the low frequency rotor to reach the newly selected speed when switching from fast to slow (or slow to fast) speed. Lower values will require longer times.                              |
| Wf Level   | 0–127             | Volume of the low frequency rotor                                                                                                                                                                                  |
| Tw Slow    | 0.05–10.00 [Hz]   | Settings of the high frequency rotor<br>The parameters are the same as for the low frequency rotor                                                                                                                 |
| Tw Fast    | 0.05–10.00 [Hz]   |                                                                                                                                                                                                                    |
| Tw Accel   | 0–15              |                                                                                                                                                                                                                    |
| Tw Level   | 0–127             |                                                                                                                                                                                                                    |
| Separation | 0–127             | Spatial dispersion of the sound                                                                                                                                                                                    |
| Level      | 0–127             | Output Level                                                                                                                                                                                                       |
| Asgn1–4    | OFF, Speed, Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                          |

## 35 VK Rotary

This type provides modified response for the rotary speaker, with the low end boosted further.

This effect features the same specifications as the VK-7's built-in rotary speaker.



| Parameter | Value                                          | Explanation                                                                                                                                                       |
|-----------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speed     | SLOW, FAST                                     | Rotational speed of the rotating speaker<br><b>SLOW:</b> Slow<br><b>FAST:</b> Fast                                                                                |
| Brake     | OFF, ON                                        | Switches the rotation of the rotary speaker. When this is turned on, the rotation will gradually stop. When it is turned off, the rotation will gradually resume. |
| Wf Slow   | 0.05–10.00 [Hz]                                | Low-speed rotation speed of the woofer                                                                                                                            |
| Wf Fast   | 0.05–10.00 [Hz]                                | High-speed rotation speed of the woofer                                                                                                                           |
| Wf Trs Up | 0–127                                          | Adjusts the rate at which the woofer rotation speeds up when the rotation is switched from Slow to Fast.                                                          |
| Wf Trs Dw | 0–127                                          | Adjusts the rate at which the woofer rotation speeds up when the rotation is switched from Fast to Slow.                                                          |
| Wf Level  | 0–127                                          | Volume of the woofer                                                                                                                                              |
| Tw Slow   | 0.05–10.00 [Hz]                                | Settings of the tweeter<br>The parameters are the same as for the woofer.                                                                                         |
| Tw Fast   | 0.05–10.00 [Hz]                                |                                                                                                                                                                   |
| Tw Trs Up | 0–127                                          |                                                                                                                                                                   |
| Tw Trs Dw | 0–127                                          |                                                                                                                                                                   |
| Tw Level  | 0–127                                          |                                                                                                                                                                   |
| Spread    | 0–10                                           | Sets the rotary speaker stereo image. The higher the value set, the wider the sound is spread out.                                                                |
| Low Gain  | -15–+15 [dB]                                   | Amount of boost/cut for the low-frequency range                                                                                                                   |
| High Gain | -15–+15 [dB]                                   | Amount of boost/cut for the high-frequency range                                                                                                                  |
| Level     | 0–127                                          | Output Level                                                                                                                                                      |
| OD Switch | OFF, ON                                        | Overdrive on/off                                                                                                                                                  |
| OD Gain   | 0–127                                          | Overdrive input level<br>Higher values will increase the distortion.                                                                                              |
| OD Drive  | 0–127                                          | Degree of distortion                                                                                                                                              |
| OD Level  | 0–127                                          | Volume of the overdrive                                                                                                                                           |
| Asgn1–4   | OFF, Speed, Brake, OD Gain, OD Drive, OD Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                         |

## Drive / Amp

### 36 Overdrive

This is an overdrive that provides heavy distortion.



| Parameter  | Value                             | Explanation                                                                                                                                                                |
|------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drive      | 0–127                             | Degree of distortion<br>Also changes the volume.                                                                                                                           |
| Tone       | 0–127                             | Sound quality of the Overdrive effect                                                                                                                                      |
| Amp Switch | OFF, ON                           | Turns the Amp Simulator on/off.                                                                                                                                            |
| AmpType    | SMALL, BUILT-IN, 2-STACK, 3-STACK | Type of guitar amp<br><b>SMALL:</b> Small amp<br><b>BUILT-IN:</b> Single-unit type amp<br><b>2-STACK:</b> Large double stack amp<br><b>3-STACK:</b> Large triple stack amp |
| Low Gain   | -15–+15 [dB]                      | Amount of boost/cut for the low-frequency range                                                                                                                            |
| High Gain  | -15–+15 [dB]                      | Amount of boost/cut for the high-frequency range                                                                                                                           |
| Pan        | L64–63R                           | Stereo location of the output sound                                                                                                                                        |
| Level      | 0–127                             | Output Level                                                                                                                                                               |
| Asgn1–4    | OFF, Drive, Tone, Pan             | Specifies the parameters that are assigned to assign 1–4.                                                                                                                  |

### 37 Distortion

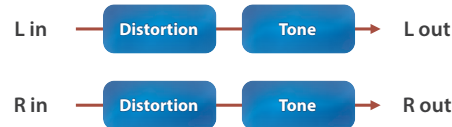
Produces a more intense distortion than Overdrive.



| Parameter  | Value                             | Explanation                                                                                                                                                                |
|------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drive      | 0–127                             | Degree of distortion<br>Also changes the volume.                                                                                                                           |
| Tone       | 0–127                             | Sound quality of the Overdrive effect                                                                                                                                      |
| Amp Switch | OFF, ON                           | Turns the Amp Simulator on/off.                                                                                                                                            |
| AmpType    | SMALL, BUILT-IN, 2-STACK, 3-STACK | Type of guitar amp<br><b>SMALL:</b> Small amp<br><b>BUILT-IN:</b> Single-unit type amp<br><b>2-STACK:</b> Large double stack amp<br><b>3-STACK:</b> Large triple stack amp |
| Low Gain   | -15–+15 [dB]                      | Amount of boost/cut for the low-frequency range                                                                                                                            |
| High Gain  | -15–+15 [dB]                      | Amount of boost/cut for the high-frequency range                                                                                                                           |
| Pan        | L64–63R                           | Stereo location of the output sound                                                                                                                                        |
| Level      | 0–127                             | Output Level                                                                                                                                                               |
| Asgn1–4    | OFF, Drive, Tone, Pan             | Specifies the parameters that are assigned to assign 1–4.                                                                                                                  |

### 38 T-Scream

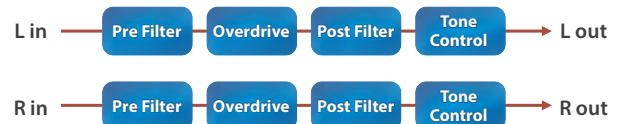
This models a classic analog overdrive. It is distinctive in adding an appropriate amount of overtones without muddying the sound.



| Parameter  | Value                 | Explanation                                               |
|------------|-----------------------|-----------------------------------------------------------|
| Distortion | 0–127                 | Degree of distortion<br>Also changes the volume.          |
| Tone       | 0–127                 | Tonal character of the overdrive                          |
| Level      | 0–127                 | Output Level                                              |
| Asgn1–4    | OFF, Distortion, Tone | Specifies the parameters that are assigned to assign 1–4. |

### 39 Fuzz

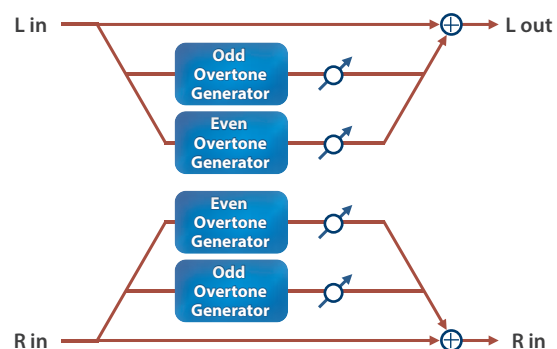
Adds overtones and intensely distorts the sound.



| Parameter | Value            | Explanation                                                       |
|-----------|------------------|-------------------------------------------------------------------|
| Drive     | 0–127            | Adjusts the depth of distortion.<br>This also changes the volume. |
| Tone      | 0–100            | Sound quality of the Overdrive effect                             |
| Level     | 0–127            | Output Level                                                      |
| Asgn1–4   | OFF, Drive, Tone | Specifies the parameters that are assigned to assign 1–4.         |

### 40 Fattener (Tone Fattener)

This effect applies distinctive distortion, adding overtones to give more depth to the sound.



| Parameter  | Value                      | Explanation                                               |
|------------|----------------------------|-----------------------------------------------------------|
| Odd Level  | 0–400 [%]                  | Raising the value adds odd-order overtones.               |
| Even Level | 0–400 [%]                  | Raising the value adds even-order overtones.              |
| Level      | 0–127                      | Output Level                                              |
| Asgn1–4    | OFF, Odd Level, Even Level | Specifies the parameters that are assigned to assign 1–4. |

## 41 HMS Distort (HMS Distortion)

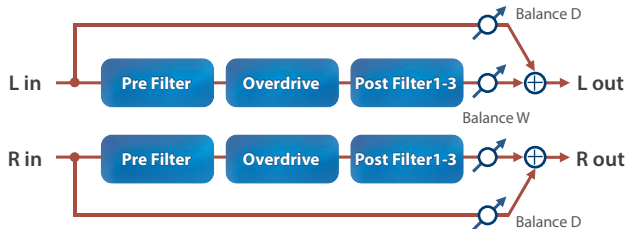
This is a distortion-type effect that models the vacuum tube amp section of a rotary speaker of the past.



| Parameter | Value           | Explanation                                               |
|-----------|-----------------|-----------------------------------------------------------|
| Dist      | 0–127           | Strength of distortion                                    |
| Level     | 0–127           | Output Level                                              |
| Asgn1–4   | OFF, Distortion | Specifies the parameters that are assigned to assign 1–4. |

## 42 Saturator

This effect combines overdrive and filter.

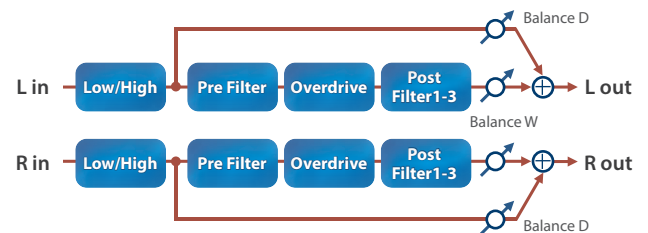


| Parameter  | Value                    | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Type   | THRU, LPF, HPF, LSV, HSV | Type of filter that precedes the distortion processing<br><b>THRU</b> : No filter is applied<br><b>LPF</b> : A filter that passes the sound below the specified frequency<br><b>HPF</b> : A filter that passes the sound above the specified frequency<br><b>LSV</b> : A filter that boosts/cuts the sound below the specified frequency<br><b>HSV</b> : A filter that boosts/cuts the sound above the specified frequency |
| Pre Freq   | 20–16000 [Hz]            | Frequency at which the pre-distortion filter operates                                                                                                                                                                                                                                                                                                                                                                      |
| Pre Gain   | -24.0–+24.0 [dB]         | For the LSV/HSV types, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                             |
| Drive      | 0.0–48.0 [dB]            | Strength of distortion                                                                                                                                                                                                                                                                                                                                                                                                     |
| Post1 Type | THRU, LPF, HPF, LSV, HSV | Type of filter 1 which follows the distortion processing                                                                                                                                                                                                                                                                                                                                                                   |
| Post1Frq   | 20–16000 [Hz]            | Frequency at which post-distortion filter 1 operates                                                                                                                                                                                                                                                                                                                                                                       |
| Post1Gain  | -24.0–+24.0 [dB]         | For the LSV/HSV types, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                             |
| Post2 Type | THRU, LPF, HPF, LSV, HSV | Type of filter 2 which follows the distortion processing                                                                                                                                                                                                                                                                                                                                                                   |
| Post2Frq   | 20–16000 [Hz]            | Frequency at which post-distortion filter 2 operates                                                                                                                                                                                                                                                                                                                                                                       |
| Post2Gain  | -24.0–+24.0 [dB]         | For the LSV/HSV types, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                             |
| Post3 Type | THRU, LPF, HPF, BPF, PKG | Type of filter 3 which follows the distortion processing<br><b>THRU</b> : No filter is applied<br><b>LPF</b> : A filter that passes the sound below the specified frequency<br><b>HPF</b> : A filter that passes the sound above the specified frequency<br><b>BPF</b> : A filter that passes only the specified frequency<br><b>PKG</b> : A filter that boosts/cuts the specified frequency                               |
| Post3Frq   | 20–16000 [Hz]            | Frequency at which post-distortion filter 3 operates                                                                                                                                                                                                                                                                                                                                                                       |
| Post3Gain  | -24.0–+24.0 [dB]         | For the PKG type, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                                  |

| Parameter | Value                            | Explanation                                                                        |
|-----------|----------------------------------|------------------------------------------------------------------------------------|
| Post3 Q   | 0.5–16.0                         | Width of the frequency range affected by the filter                                |
| Sense     | -60.0–0.0 [dB]                   | Adjust this value so that the sound is not made louder when distortion is applied. |
| PostGain  | -48.0 +12.0 [dB]                 | Gain following distortion processing                                               |
| Balance   | D100: 0W–D0: 100W                | Volume balance between the dry sound (D) and effect sound (W)                      |
| Level     | 0–127                            | Output Level                                                                       |
| Asgn1–4   | OFF, Drive, Drive Balance, Level | Specifies the parameters that are assigned to assign 1–4.                          |

## 43 W Saturator (Worm Saturator)

This is a variety of saturator, and is distinctive for its warmer sound.

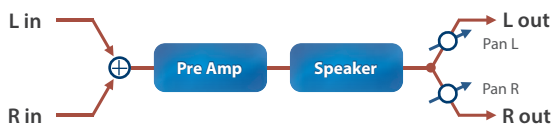


| Parameter  | Value                    | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LowFreq    | 20–16000 [Hz]            | Input filter (low range) Boosts/cuts the sound below the specified frequency.                                                                                                                                                                                                                                                                                                                                               |
| LowGain    | -24.0–+24.0 [dB]         | Input filter (low range) Amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                                |
| Hi Slope   | THRU, -12dB, -24dB       | Amount of attenuation per octave<br><b>THRU</b> : No attenuation<br><b>-12 dB</b> : Gentle<br><b>-24 dB</b> : Steep                                                                                                                                                                                                                                                                                                         |
| Hi Freq    | 20–16000 [Hz]            | Input filter (high range) Boosts/cuts the sound above the specified frequency.                                                                                                                                                                                                                                                                                                                                              |
| Pre1 Type  | THRU, LPF, HPF, LSV, HSV | Types of filter that precedes the distortion processing<br><b>THRU</b> : No filter is applied<br><b>LPF</b> : A filter that passes the sound below the specified frequency<br><b>HPF</b> : A filter that passes the sound above the specified frequency<br><b>LSV</b> : A filter that boosts/cuts the sound below the specified frequency<br><b>HSV</b> : A filter that boosts/cuts the sound above the specified frequency |
| Pre1Frq    | 20–16000 [Hz]            | Frequency at which the pre-distortion filter operates                                                                                                                                                                                                                                                                                                                                                                       |
| Pre1Gain   | -24.0–+24.0 [dB]         | For the LSV/HSV types, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                              |
| Drive      | 0.0–48.0 [dB]            | Strength of distortion                                                                                                                                                                                                                                                                                                                                                                                                      |
| Post1 Type | THRU, LPF, HPF, LSV, HSV | Type of filter 1 which follows the distortion processing                                                                                                                                                                                                                                                                                                                                                                    |
| Post1Frq   | 20–16000 [Hz]            | Frequency at which post-distortion filter 1 operates                                                                                                                                                                                                                                                                                                                                                                        |
| Post1Gain  | -24.0–+24.0 [dB]         | For the LSV/HSV types, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                              |
| Post2 Type | THRU, LPF, HPF, LSV, HSV | Type of filter 2 which follows the distortion processing                                                                                                                                                                                                                                                                                                                                                                    |
| Post2Frq   | 20–16000 [Hz]            | Frequency at which post-distortion filter 2 operates                                                                                                                                                                                                                                                                                                                                                                        |
| Post2Gain  | -24.0–+24.0 [dB]         | For the LSV/HSV types, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                                              |

| Parameter  | Value                                        | Explanation                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Post3 Type | THRU, LPF, HPF, BPF, PKG                     | Type of filter 3 which follows the distortion processing<br><b>THRU</b> : No filter is applied<br><b>LPF</b> : A filter that passes the sound below the specified frequency<br><b>HPF</b> : A filter that passes the sound above the specified frequency<br><b>BPF</b> : A filter that passes only the specified frequency<br><b>PKG</b> : A filter that boosts/cuts the specified frequency |
| Post3Frq   | 20–16000 [Hz]                                | Frequency at which post-distortion filter 3 operates                                                                                                                                                                                                                                                                                                                                         |
| Post3Gain  | -24.0–+24.0 [dB]                             | For the PKG type, the amount of boost/cut                                                                                                                                                                                                                                                                                                                                                    |
| Post3 Q    | 0.5–16.0                                     | Width of the frequency range affected by the filter                                                                                                                                                                                                                                                                                                                                          |
| Sense      | -60.0–0.0 [dB]                               | Adjust this value so that the sound is not made louder when distortion is applied.                                                                                                                                                                                                                                                                                                           |
| PostGain   | -48.0–+12.0 [dB]                             | Gain following distortion processing                                                                                                                                                                                                                                                                                                                                                         |
| Balance    | D100: 0W–D0: 100W                            | Volume balance between the dry sound (D) and effect sound (W)                                                                                                                                                                                                                                                                                                                                |
| Level      | 0–127                                        | Output Level                                                                                                                                                                                                                                                                                                                                                                                 |
| Asgn1–4    | OFF, LowGain, Hi Freq, Drive, Balance, Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                                                                                                                                                                                    |

## 44 Gt Amp Sim (Guitar Amp Simulator)

This is an effect that simulates the sound of a guitar amplifier.

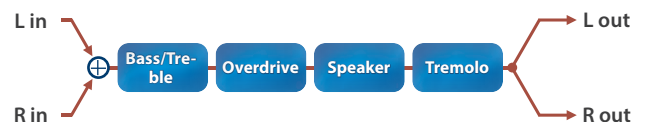


| Parameter  | Value             | Explanation                                                                                                                                            |
|------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Amp Sw | OFF, ON           | Turns the amp switch on/off.                                                                                                                           |
| ATyp       |                   | Type of guitar amp                                                                                                                                     |
|            | JC-120            | This models the sound of the Roland JC-120.                                                                                                            |
|            | CLEAN TWIN        | This models a Fender Twin Reverb.                                                                                                                      |
|            | MATCH DRIVE       | This models the sound input to left input on a Matchless D/C-30. A simulation of the latest tube amp widely used in styles from blues rock and fusion. |
|            | BG LEAD           | This models the lead sound of the MESA/ Boogie combo amp. The sound of a tube amp typical of the late '70s to '80s.                                    |
|            | MS1959I           | This models the sound input to Input I on a Marshall 1959. This is a trebly sound suited to hard rock.                                                 |
|            | MS1959II          | This models the sound input to Input II on a Marshall 1959.                                                                                            |
|            | MS1959I+II        | The sound of connecting inputs I and II of the guitar amp in parallel, creating a sound with a stronger low end than I.                                |
|            | SLDN LEAD         | This models a Soldano SLO-100. This is the typical sound of the eighties.                                                                              |
|            | METAL 5150        | This models the lead channel of a Peavey EVH 5150.                                                                                                     |
|            | METAL LEAD        | This is distortion sound that is ideal for performances of heavy riffs.                                                                                |
|            | OD-1              | This models the sound of the BOSS OD-1. This produces sweet, mild distortion.                                                                          |
|            | OD-2 TURBO        | This is the high-gain overdrive sound of the BOSS OD-2.                                                                                                |
|            | DISTORTION        | This gives a basic, traditional distortion sound.                                                                                                      |
|            | FUZZ              | A fuzz sound with rich harmonic content.                                                                                                               |
| Drive      | 0–127             | Volume and amount of distortion of the amp                                                                                                             |
| Master Lv  | 0–127             | Volume of the entire pre-amp                                                                                                                           |
| Gain       | LOW, MIDDLE, HIGH | Amount of pre-amp distortion                                                                                                                           |

| Parameter    | Value                              | Explanation                                                                                                                                                                                    |
|--------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bass         | 0–127                              | Tone of the bass/mid/treble frequency range                                                                                                                                                    |
| Middle       | 0–127                              |                                                                                                                                                                                                |
| Treble       | 0–127                              |                                                                                                                                                                                                |
| Presence     | 0–127                              | Tone for the ultra-high frequency range                                                                                                                                                        |
| Bright       | OFF, ON                            | Turning this "On" produces a sharper and brighter sound.<br>* This parameter applies to the "JC-120," "CLEAN TWIN," "MATCH DRIVE," and "BG LEAD" Pre Amp Types.                                |
| Speaker Sw   | OFF, ON                            | Selects whether the sound will be sent through the speaker simulation (ON) or not (OFF)                                                                                                        |
| STyp         |                                    | Cabinet      Diameter (in inches) and number of the speaker      Microphone                                                                                                                    |
|              | SMALL 1                            | small open-back enclosure      10      dynamic                                                                                                                                                 |
|              | SMALL 2                            | small open-back enclosure      10      dynamic                                                                                                                                                 |
|              | MIDDLE                             | open back enclosure      12 x 1      dynamic                                                                                                                                                   |
|              | JC-120                             | open back enclosure      12 x 2      dynamic                                                                                                                                                   |
|              | BUILT-IN 1                         | open back enclosure      12 x 2      dynamic                                                                                                                                                   |
|              | BUILT-IN 2                         | open back enclosure      12 x 2      condenser                                                                                                                                                 |
|              | BUILT-IN 3                         | open back enclosure      12 x 2      condenser                                                                                                                                                 |
|              | BUILT-IN 4                         | open back enclosure      12 x 2      condenser                                                                                                                                                 |
|              | BUILT-IN 5                         | open back enclosure      12 x 2      condenser                                                                                                                                                 |
|              | BG STACK 1                         | sealed enclosure      12 x 2      condenser                                                                                                                                                    |
|              | BG STACK 2                         | large sealed enclosure      12 x 2      condenser                                                                                                                                              |
|              | MS STACK 1                         | large sealed enclosure      12 x 4      condenser                                                                                                                                              |
|              | MS STACK 2                         | large sealed enclosure      12 x 4      condenser                                                                                                                                              |
|              | METAL STACK                        | large double stack      12 x 4      condenser                                                                                                                                                  |
|              | 2-STACK                            | large double stack      12 x 4      condenser                                                                                                                                                  |
|              | 3-STACK                            | large triple stack      12 x 4      condenser                                                                                                                                                  |
| Mic Setting  | 1–3                                | Adjusts the location of the microphone that is recording the sound of the speaker. This can be adjusted in three steps, with the microphone becoming more distant in the order of 1, 2, and 3. |
| Mic Level    | 0–127                              | Volume of the microphone                                                                                                                                                                       |
| Direct Level | 0–127                              | Volume of the direct sound                                                                                                                                                                     |
| Pan          | L64–63R                            | Stereo location of the output sound                                                                                                                                                            |
| Level        | 0–127                              | Output Level                                                                                                                                                                                   |
| Asgn1–4      | OFF, Volume, Master Lv, Pan, Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                      |

## 45 EP Amp Sim (RD EP Amp Simulator)

This is an effect that was developed for the RD series SuperNatural E.Piano.



| Parameter  | Value      | Explanation                                                     |
|------------|------------|-----------------------------------------------------------------|
| Bass       | -50–+50    | Amount of low-frequency boost/cut                               |
| Treble     | -50–+50    | Amount of high-frequency boost/cut                              |
| Tremolo Sw | OFF, ON    | Tremolo on/off                                                  |
| Type       |            | Type of tremolo effect                                          |
|            | OLDCASE MO | A standard electric piano sound of the early '70s (mono)        |
|            | OLDCASE ST | A standard electric piano sound of the early '70s (stereo)      |
|            | NEWCASE    | A standard electric piano sound of the late '70s and early '80s |
|            | DYNO       | A classic modified electric piano                               |
|            | WURLY      | A classic electric piano of the '60s                            |

| Parameter  | Value                                       | Explanation                                                                                     |
|------------|---------------------------------------------|-------------------------------------------------------------------------------------------------|
| Speed Sync | OFF, ON                                     | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| Speed      | 0.05–10.00 [Hz]                             | Rate of the tremolo effect                                                                      |
| Speed Nt   | Note<br>→ "Note" (p. 72)                    |                                                                                                 |
| Depth      | 0–127                                       | Depth of the tremolo effect                                                                     |
| Shape      | 0–20                                        | Adjusts the waveform of the tremolo.                                                            |
| AMP        | OFF, ON                                     | Turns the speaker and distortion on/off                                                         |
| Speaker    | LINE, OLD, NEW, WURLY, TWIN                 | Type of speaker If LINE is selected, the sound will not be sent through the speaker simulation. |
| Drive      | 0–127                                       | Degree of distortion<br>Also changes the volume.                                                |
| Level      | 0–127                                       | Output Level                                                                                    |
| Asgn1–4    | OFF, Bass, Treble, Tremolo Sw, Speed, Depth | Specifies the parameters that are assigned to assign 1–4.                                       |

## 46 Speaker Sim (Speaker Simulator)

Simulates the speaker type and mic settings used to record the speaker sound.

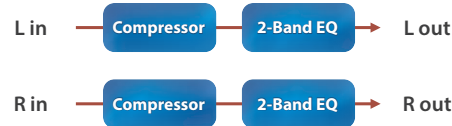


| Parameter   | Value                               | Explanation                                                                                                                                                                                       |
|-------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type        | Cabinet                             | Diameter (in inches) and number of the speaker                                                                                                                                                    |
|             | Microphone                          |                                                                                                                                                                                                   |
|             | SMALL 1                             | small open-back enclosure 10 dynamic                                                                                                                                                              |
|             | SMALL 2                             | small open-back enclosure 10 dynamic                                                                                                                                                              |
|             | MIDDLE                              | open back enclosure 12 x 1 dynamic                                                                                                                                                                |
|             | JC-120                              | open back enclosure 12 x 2 dynamic                                                                                                                                                                |
|             | BUILT-IN 1                          | open back enclosure 12 x 2 dynamic                                                                                                                                                                |
|             | BUILT-IN 2                          | open back enclosure 12 x 2 condenser                                                                                                                                                              |
|             | BUILT-IN 3                          | open back enclosure 12 x 2 condenser                                                                                                                                                              |
|             | BUILT-IN 4                          | open back enclosure 12 x 2 condenser                                                                                                                                                              |
|             | BUILT-IN 5                          | open back enclosure 12 x 2 condenser                                                                                                                                                              |
|             | BG STACK 1                          | sealed enclosure 12 x 2 condenser                                                                                                                                                                 |
|             | BG STACK 2                          | large sealed enclosure 12 x 2 condenser                                                                                                                                                           |
|             | MS STACK 1                          | large sealed enclosure 12 x 4 condenser                                                                                                                                                           |
|             | MS STACK 2                          | large sealed enclosure 12 x 4 condenser                                                                                                                                                           |
| Mic Setting | Metal Stack                         | large double stack 12 x 4 condenser                                                                                                                                                               |
|             | 2-STACK                             | large double stack 12 x 4 condenser                                                                                                                                                               |
|             | 3-STACK                             | large triple stack 12 x 4 condenser                                                                                                                                                               |
|             | Mic Setting                         | Adjusts the location of the microphone that is recording the sound of the speaker.<br>This can be adjusted in three steps, with the microphone becoming more distant in the order of 1, 2, and 3. |
|             | Mic Level                           | Volume of the microphone                                                                                                                                                                          |
| Direct Lv   | 0–127                               | Volume of the direct sound                                                                                                                                                                        |
| Level       | 0–127                               | Output Level                                                                                                                                                                                      |
| Asgn1–4     | OFF, Mic Level, Direct Level, Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                         |

## Comp / Limiter

### 47 Compressor

Flattens out high levels and boosts low levels, smoothing out fluctuations in volume.



| Parameter | Value                                   | Explanation                                                                                                                                                                                          |
|-----------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attack    | 0–124                                   | Sets the speed at which compression starts.                                                                                                                                                          |
| Release   | 0–124                                   | Adjusts the time after the signal volume falls below the Threshold Level until compression is no longer applied.                                                                                     |
| Threshold | –60–0 [dB]                              | Adjusts the volume at which compression begins.                                                                                                                                                      |
| Knee      | 0–30 [dB]                               | This is a function that smooths the onset of compression from the uncompressed state; it gradually applies compression starting earlier than Threshold. Higher values produce a smoother transition. |
| Ratio     | 1: 1, 1.5: 1, 2: 1, 4: 1, 16: 1, INF: 1 | Compression ratio                                                                                                                                                                                    |
| Post Gain | 0–+18 [dB]                              | Level of the output sound                                                                                                                                                                            |
| Level     | 0–127                                   | Output Level                                                                                                                                                                                         |
| Asgn1–4   | OFF, Attack, Threshold, Level           | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                            |

### 48 M/S Comp (Mid-Side Compressor)

This effect allows the left/right signals that have similar phase to be adjusted to a different sense of volume than the left/right signals that have different phase.



| Parameter | Value                                   | Explanation                                                                                                                                                                                        |
|-----------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M Comp Sw | OFF, ON                                 | Switches whether to adjust the sense of volume for left/right input signals whose phase is similar (in phase).                                                                                     |
| M Attack  | 0–124                                   | Sets the speed at which compression starts                                                                                                                                                         |
| M Release | 0–124                                   | Adjusts the time after the signal volume falls below the M Thres Level until compression is no longer applied.                                                                                     |
| M Thres   | –60–0 [dB]                              | Adjusts the volume at which compression begins.                                                                                                                                                    |
| M Knee    | 0–30 [dB]                               | This is a function that smooths the onset of compression from the uncompressed state; it gradually applies compression starting earlier than M Thres. Higher values produce a smoother transition. |
| M Ratio   | 1: 1, 1.5: 1, 2: 1, 4: 1, 16: 1, INF: 1 | Compression ratio                                                                                                                                                                                  |
| M Gain    | 0–+18 [dB]                              | Level of the output sound                                                                                                                                                                          |
| S Comp Sw | OFF, ON                                 | Switches whether to adjust the sense of volume for left/right input signals whose phase is distant (opposite phase).                                                                               |
| S Attack  | 0–124                                   | Sets the speed at which compression starts                                                                                                                                                         |
| S Release | 0–124                                   | Adjusts the time after the signal volume falls below the S Thres Level until compression is no longer applied.                                                                                     |
| S Thres   | –60–0 [dB]                              | Adjusts the volume at which compression begins.                                                                                                                                                    |

| Parameter | Value                                                                                             | Explanation                                                                                                                                                                                        |
|-----------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S Knee    | 0–30 [dB]                                                                                         | This is a function that smooths the onset of compression from the uncompressed state; it gradually applies compression starting earlier than S Thres. Higher values produce a smoother transition. |
| S Ratio   | 1: 1, 1.5: 1, 2: 1, 4: 1, 16: 1, INF: 1                                                           | Compression ratio                                                                                                                                                                                  |
| S Gain    | 0–+18 [dB]                                                                                        | Level of the output sound                                                                                                                                                                          |
| Level     | 0–127                                                                                             | Output Level                                                                                                                                                                                       |
| Asgn1–4   | OFF, M Attack, M Release, M Threshold, M Post Gain, S Attack, S Release, S Threshold, S Post Gain | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                          |

## 49 Limiter

Compresses signals that exceed a specified volume level, preventing distortion from occurring.



| Parameter | Value                          | Explanation                                                                                                      |
|-----------|--------------------------------|------------------------------------------------------------------------------------------------------------------|
| Release   | 0–127                          | Adjusts the time after the signal volume falls below the Threshold Level until compression is no longer applied. |
| Threshold | 0–127                          | Adjusts the volume at which compression begins.                                                                  |
| Ratio     | 1.5: 1, 2: 1, 4: 1, 100: 1     | Compression ratio                                                                                                |
| Post Gain | 0–+18 [dB]                     | Level of the output sound                                                                                        |
| Low Gain  | -15–+15 [dB]                   | Amount of boost/cut for the low-frequency range                                                                  |
| High Gain | -15–+15 [dB]                   | Amount of boost/cut for the high-frequency range                                                                 |
| Level     | 0–127                          | Output Level                                                                                                     |
| Asgn1–4   | OFF, Release, Threshold, Level | Specifies the parameters that are assigned to assign 1–4.                                                        |

## 50 Sustainer

By compressing loud input and boosting low input, this effect keeps the volume consistent to produce a sustain effect without distortion.

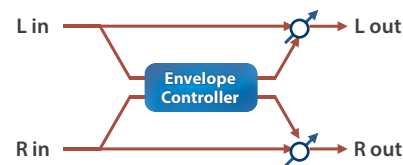


| Parameter | Value        | Explanation                                                                                                            |
|-----------|--------------|------------------------------------------------------------------------------------------------------------------------|
| Sustain   | 0–127        | Adjusts the range in which a low input signal is boosted to a consistent volume. Higher values produce longer sustain. |
| Attack    | 0–127        | Time until the volume is compressed                                                                                    |
| Release   | 0–127        | Time until compression is removed                                                                                      |
| Post Gain | -15–+15 [dB] | Level of the output sound                                                                                              |
| Low Gain  | -15–+15 [dB] | Amount of boost/cut for the low-frequency range                                                                        |
| High Gain | -15–+15 [dB] | Amount of boost/cut for the high-frequency range                                                                       |
| Level     | 0–127        | Output Level                                                                                                           |

| Parameter | Value                         | Explanation                                               |
|-----------|-------------------------------|-----------------------------------------------------------|
| Asgn1–4   | OFF, Sustain, Attack, Release | Specifies the parameters that are assigned to assign 1–4. |

## 51 Transient

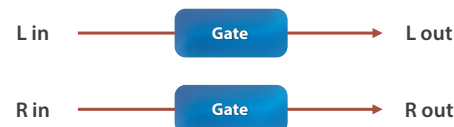
This effect lets you control the way in which the sound attacks and decays.



| Parameter | Value                | Explanation                                                                                                  |
|-----------|----------------------|--------------------------------------------------------------------------------------------------------------|
| Attack    | -50–+50              | Character of the attack. Higher values make the attack more aggressive; lower values make the attack milder. |
| Release   | -50–+50              | Character of the decay. Higher values make the sound linger; lower values make the sound cutoff quickly.     |
| Out Gain  | -24–+12 [dB]         | Output gain                                                                                                  |
| Sens      | LOW, MID, HIGH       | Quickness with which the attack is detected                                                                  |
| Level     | 0–127                | Output Level                                                                                                 |
| Asgn1–4   | OFF, Attack, Release | Specifies the parameters that are assigned to assign 1–4.                                                    |

## 52 Gate

Cuts the reverb's delay according to the volume of the sound sent into the effect. Use this when you want to create an artificial-sounding decrease in the reverb's decay.



| Parameter | Value                   | Explanation                                                                                                                                                                                                                                             |
|-----------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Threshold | 0–127                   | Volume level at which the gate begins to close                                                                                                                                                                                                          |
| Mode      | GATE, DUCK              | Type of gate<br><b>GATE:</b> The gate will close when the volume of the original sound decreases, cutting the original sound.<br><b>DUCK (Duking):</b> The gate will close when the volume of the original sound increases, cutting the original sound. |
| Attack    | 0–127                   | Adjusts the time it takes for the gate to fully open after being triggered.                                                                                                                                                                             |
| Hold      | 0–127                   | Adjusts the time it takes for the gate to start closing after the source sound falls beneath the Threshold.                                                                                                                                             |
| Release   | 0–127                   | Adjusts the time it takes the gate to fully close after the hold time.                                                                                                                                                                                  |
| Balance   | D100: 0W–D0: 100W       | Volume balance between the direct sound (D) and the effect sound (W)                                                                                                                                                                                    |
| Level     | 0–127                   | Output Level                                                                                                                                                                                                                                            |
| Asgn1–4   | OFF, Threshold, Balance | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                                               |

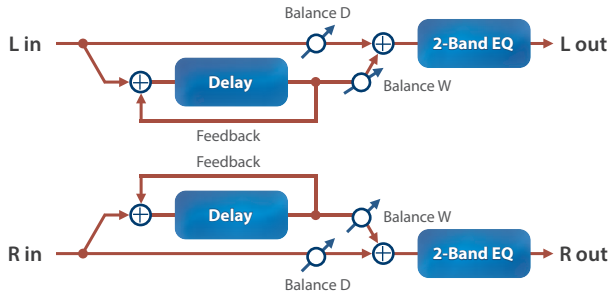


## Delay

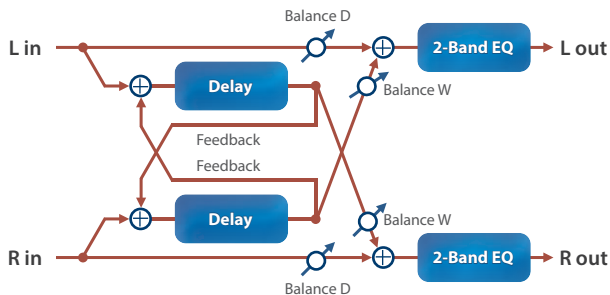
## 53 Delay

This is a stereo delay.

**When Feedback Mode is NORMAL:**



**When Feedback Mode is CROSS:**

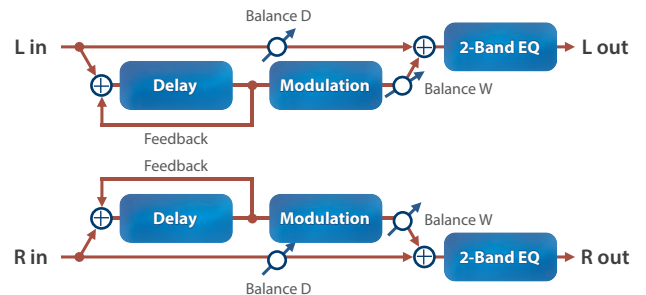


| Parameter  | Value                                                                                                      | Explanation                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly L Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                          |
| DL Time    | 1–1300                                                                                                     | Adjusts the time until the left delay sound is heard.                                                                    |
| DLTime Nt  | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                          |
| Dly R Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                          |
| DR Time    | 1–1300                                                                                                     | Adjusts the time until the right delay sound is heard.                                                                   |
| DRTIME Nt  | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                          |
| Phase L    | NORMAL, INVERSE                                                                                            | Phase of left and right delay sound                                                                                      |
| Phase R    | NORMAL, INVERSE                                                                                            | <b>NORMAL:</b> Non-inverted<br><b>INVERT:</b> Inverted                                                                   |
| Fbk Mode   | NORMAL, CROSS                                                                                              | Selects the way in which delay sound is fed back into the effect. (See the figures above.)                               |
| Feedback   | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                         |
| Low Gain   | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                          |
| High Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                         |
| Balance    | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the delay sound (W)                                                      |
| Level      | 0–127                                                                                                      | Output Level                                                                                                             |
| Asgn1–4    | OFF, Feedback, Balance                                                                                     | Specifies the parameters that are assigned to assign 1–4.                                                                |

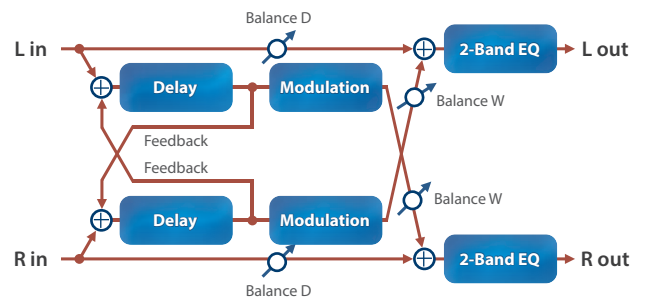
## 54 Mod Delay (Modulation Delay)

Adds modulation to the delayed sound.

**When Feedback Mode is NORMAL:**



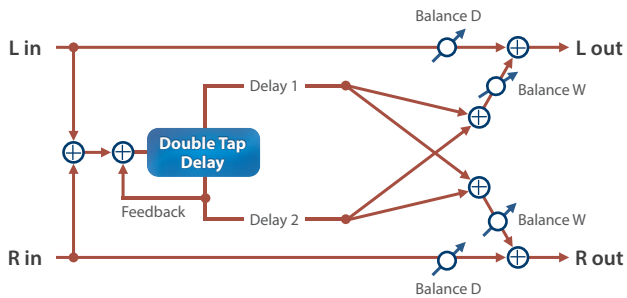
**When Feedback Mode is CROSS:**



| Parameter  | Value                                                                                                      | Explanation                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly L Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                          |
| DL Time    | 1–1300                                                                                                     | Adjusts the time until the left delay sound is heard.                                                                    |
| DLTime Nt  | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                          |
| Dly R Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                          |
| DR Time    | 1–1300                                                                                                     | Adjusts the time until the right delay sound is heard.                                                                   |
| DRTIME Nt  | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                          |
| Fbk Mode   | NORMAL, CROSS                                                                                              | Selects the way in which delay sound is fed back into the effect. (See the figures above.)                               |
| Feedback   | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                         |
| Rate Sync  | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                          |
| Rate       | 0.05–10.00 [Hz]                                                                                            | Frequency of modulation                                                                                                  |
| Rate Note  | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                          |
| Depth      | 0–127                                                                                                      | Depth of modulation                                                                                                      |
| Phase      | 0–180 [deg]                                                                                                | Spatial spread of the sound                                                                                              |
| Low Gain   | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                          |
| High Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                         |
| Balance    | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the delay sound (W)                                                      |
| Level      | 0–127                                                                                                      | Output Level                                                                                                             |
| Asgn1–4    | OFF, Feedback, Rate (Hz), Balance                                                                          | Specifies the parameters that are assigned to assign 1–4.                                                                |



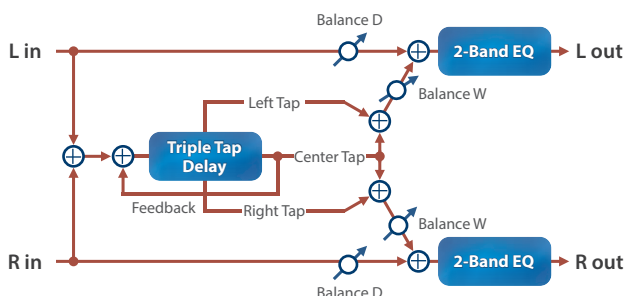
## 55 2Tap PanDly (2 Tap Pan Delay)



| Parameter    | Value                                                                                                      | Explanation                                                                                                                           |
|--------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Delay Sync   | OFF, ON                                                                                                    | If this is "ON," the delay synchronizes with the tempo.                                                                               |
| D. Time (ms) | 1–2600                                                                                                     | Adjusts the time until the second delay sound is heard.                                                                               |
| D. Time (Nt) | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Delay Fbk    | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.              |
| Dly HF       | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                                      |
| Dly1 Pan     | L64–63R                                                                                                    | Stereo location of Delay 1                                                                                                            |
| Dly2 Pan     | L64–63R                                                                                                    | Stereo location of Delay 2                                                                                                            |
| Dly1 Lv      | 0–127                                                                                                      | Volume of delay 1                                                                                                                     |
| Dly2 Lv      | 0–127                                                                                                      | Volume of delay 2                                                                                                                     |
| Low Gain     | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                                       |
| High Gain    | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                                      |
| Balance      | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level        | 0–127                                                                                                      | Output Level                                                                                                                          |
| Asgn1–4      | OFF, D. Time (ms), Delay Fbk, Balance                                                                      | Specifies the parameters that are assigned to assign 1–4.                                                                             |

## 56 3Tap PanDly (3 Tap Pan Delay)

Produces three delay sounds; center, left and right.

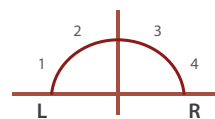
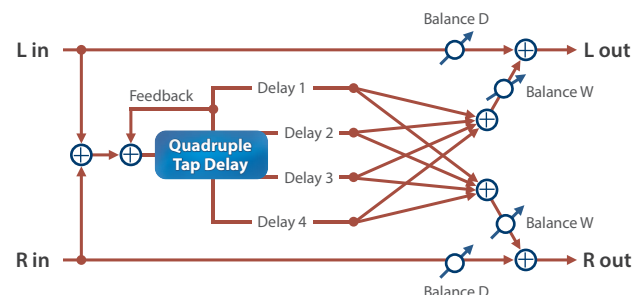


| Parameter  | Value                    | Explanation                                                                                  |
|------------|--------------------------|----------------------------------------------------------------------------------------------|
| Dly L Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm. → "Tempo" (p. 3, p. 31) |
| DL. Time   | 1–2600                   | Adjusts the time until the left delay sound is heard.                                        |
| DLTime Nt  | Note<br>→ "Note" (p. 72) |                                                                                              |

| Parameter  | Value                                                                                                      | Explanation                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Dly R Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm. → "Tempo" (p. 3, p. 31)                             |
| DR. Time   | 1–2600                                                                                                     | Adjusts the time until the right delay sound is heard.                                                                   |
| DRTIME Nt  | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                          |
| Dly C Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm. → "Tempo" (p. 3, p. 31)                             |
| DC. Time   | 1–2600                                                                                                     | Adjusts the time until the center delay sound is heard.                                                                  |
| DCTIME Nt  | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                          |
| C Feedback | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| HF Damp    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                         |
| Left Lv    | 0–127                                                                                                      | Volume of each delay sound                                                                                               |
| Right Lv   | 0–127                                                                                                      |                                                                                                                          |
| Center Lv  | 0–127                                                                                                      |                                                                                                                          |
| Low Gain   | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                          |
| High Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                         |
| Balance    | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the delay sound (W)                                                      |
| Level      | 0–127                                                                                                      | Output Level                                                                                                             |
| Asgn1–4    | OFF, C Feedback, Balance                                                                                   | Specifies the parameters that are assigned to assign 1–4.                                                                |

## 57 4Tap PanDly (4 Tap Pan Delay)

This effect has four delays.



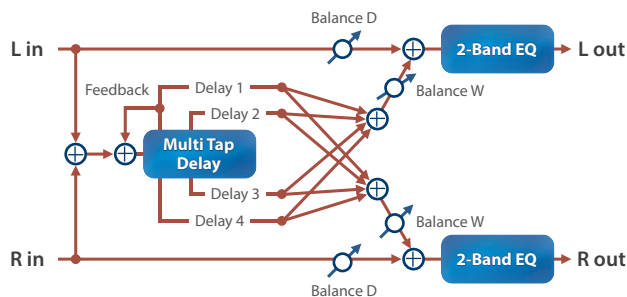
| Parameter | Value                    | Explanation                                                                                  |
|-----------|--------------------------|----------------------------------------------------------------------------------------------|
| Dly1 Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm. → "Tempo" (p. 3, p. 31) |
| D1. Time  | 1–2600                   | Adjusts the time from the original sound until delay 1 sounds is heard.                      |
| D1Time Nt | Note<br>→ "Note" (p. 72) |                                                                                              |
| Dly2 Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm. → "Tempo" (p. 3, p. 31) |
| D2. Time  | 1–2600                   | Adjusts the time from the original sound until delay 2 sounds is heard.                      |
| D2Time Nt | Note<br>→ "Note" (p. 72) |                                                                                              |

| Parameter | Value                                                                                                      | Explanation                                                                                                                 |
|-----------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Dly3 Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                             |
| D3. Time  | 1–2600                                                                                                     | Adjusts the time from the original sound until delay 3 sounds is heard.                                                     |
| D3Time Nt | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                             |
| Dly4 Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                             |
| D4. Time  | 1–2600                                                                                                     | Adjusts the time from the original sound until delay 4 sounds is heard.                                                     |
| D4Time Nt | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                             |
| Dly1 Fbk  | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect.<br>Negative (-) settings will invert the phase. |
| HF Damp   | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                            |
| Dly1 Lv   | 0–127                                                                                                      | Volume of each delay                                                                                                        |
| Dly2 Lv   | 0–127                                                                                                      |                                                                                                                             |
| Dly3 Lv   | 0–127                                                                                                      |                                                                                                                             |
| Dly4 Lv   | 0–127                                                                                                      |                                                                                                                             |
| Low Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                             |
| High Gain | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                            |
| Balance   | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the delay sound (W)                                                         |
| Level     | 0–127                                                                                                      | Output Level                                                                                                                |
| Asgn1–4   | OFF, Dly1 Fbk, Balance                                                                                     | Specifies the parameters that are assigned to assign 1–4.                                                                   |

| Parameter | Value                                                                                                      | Explanation                                                                                                                 |
|-----------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| D3. Time  | 1–2600                                                                                                     | Adjusts the time from the original sound until delay 3 sounds is heard.                                                     |
| D3Time Nt | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                             |
| Dly4 Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                             |
| D4. Time  | 1–2600                                                                                                     | Adjusts the time from the original sound until delay 4 sounds is heard.                                                     |
| D4Time Nt | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                             |
| Dly1 Fbk  | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect.<br>Negative (-) settings will invert the phase. |
| HF Damp   | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                            |
| Dly1 Pan  | L64–63R                                                                                                    | Stereo location of Delays 1–4                                                                                               |
| Dly2 Pan  | L64–63R                                                                                                    |                                                                                                                             |
| Dly3 Pan  | L64–63R                                                                                                    |                                                                                                                             |
| Dly4 Pan  | L64–63R                                                                                                    |                                                                                                                             |
| Dly1 Lv   | 0–127                                                                                                      | Volume of each delay                                                                                                        |
| Dly2 Lv   | 0–127                                                                                                      |                                                                                                                             |
| Dly3 Lv   | 0–127                                                                                                      |                                                                                                                             |
| Dly4 Lv   | 0–127                                                                                                      |                                                                                                                             |
| Low Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                             |
| High Gain | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                            |
| Balance   | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the effect sound (W)                                                        |
| Level     | 0–127                                                                                                      | Output Level                                                                                                                |
| Asgn1–4   | OFF, Dly1 Fbk, Balance                                                                                     | Specifies the parameters that are assigned to assign 1–4.                                                                   |

## 58 MultiTapDly (Multi Tap Delay)

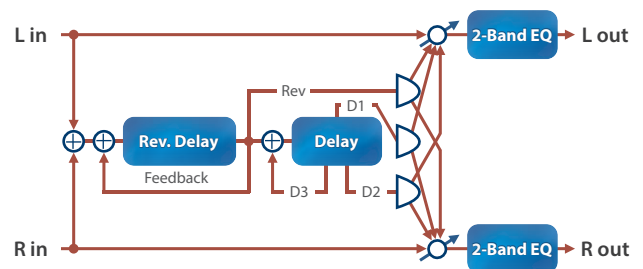
This effect has four delays. Each of the Delay Time parameters can be set to a note length based on the selected tempo. You can also set the panning and level of each delay sound.



| Parameter | Value                    | Explanation                                                                                     |
|-----------|--------------------------|-------------------------------------------------------------------------------------------------|
| Dly1 Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31) |
| D1. Time  | 1–2600                   | Adjusts the time from the original sound until delay 1 sounds is heard.                         |
| D1Time Nt | Note<br>➔ "Note" (p. 72) |                                                                                                 |
| Dly2 Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31) |
| D2. Time  | 1–2600                   | Adjusts the time from the original sound until delay 2 sounds is heard.                         |
| D2Time Nt | Note<br>➔ "Note" (p. 72) |                                                                                                 |
| Dly3 Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31) |

## 59 Reverse Dly (Reverse Delay)

This is a reverse delay that adds a reversed and delayed sound to the input sound. A tap delay is connected immediately after the reverse delay.

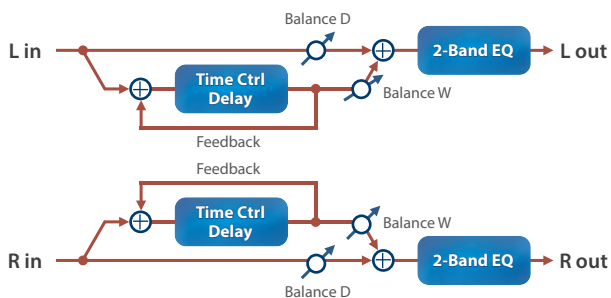


| Parameter   | Value                                                                                                      | Explanation                                                                                                                   |
|-------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Threshold   | 0–127                                                                                                      | Volume at which the reverse delay will begin to be applied                                                                    |
| RDly Sync   | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>➔ "Tempo" (p. 3, p. 31)                               |
| RD. Time    | 1–1300                                                                                                     | Delay time from when sound is input into the reverse delay until the delay sound is heard                                     |
| RD. Time Nt | Note<br>➔ "Note" (p. 72)                                                                                   |                                                                                                                               |
| RDly Fbk    | -98–+98 [%]                                                                                                | Proportion of the delay sound that is to be returned to the input of the reverse delay (negative (-) values invert the phase) |
| RDly HF     | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Frequency at which the high-frequency content of the reverse-delayed sound will be cut (BYPASS: no cut)                       |
| RDly Pan    | L64–63R                                                                                                    | Panning of the reverse delay sound                                                                                            |
| RDly Level  | 0–127                                                                                                      | Volume of the reverse delay sound                                                                                             |

| Parameter | Value                                                                                                      | Explanation                                                                                                               |
|-----------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Dly1 Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                           |
| D1. Time  | 1–1300                                                                                                     | Delay time from when sound is input into the tap delay until the delay sound is heard                                     |
| D1Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                           |
| Dly2 Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                           |
| D2. Time  | 1–1300                                                                                                     | Delay time from when sound is input into the tap delay until the delay sound is heard                                     |
| D2Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                           |
| Dly3 Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                           |
| D3. Time  | 1–1300                                                                                                     | Delay time from when sound is input into the tap delay until the delay sound is heard                                     |
| D3Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                           |
| Dly3 Fbk  | -98–+98 [%]                                                                                                | Proportion of the delay sound that is to be returned to the input of the tap delay (negative (-) values invert the phase) |
| Dly HF    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Frequency at which the hi-frequency content of the tap delay sound will be cut (BYPASS: no cut).                          |
| Dly1 Pan  | L64–63R                                                                                                    | Panning of the tap delay sounds                                                                                           |
| Dly2 Pan  | L64–63R                                                                                                    |                                                                                                                           |
| Dly1 Lv   | 0–127                                                                                                      | Volume of the tap delay sounds                                                                                            |
| Dly2 Lv   | 0–127                                                                                                      |                                                                                                                           |
| Low Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                                           |
| High Gain | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                                          |
| Balance   | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the delay sound (W)                                                       |
| Level     | 0–127                                                                                                      | Output Level                                                                                                              |
| Asgn1–4   | OFF, RDly Fbk, Dly3 Fbk, Balance                                                                           | Specifies the parameters that are assigned to assign 1–4.                                                                 |

## 60 TimeCtrlDly (Time Control Delay)

A stereo delay in which the delay time can be varied smoothly.

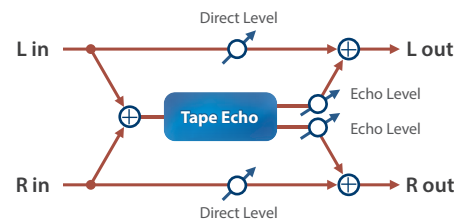


| Parameter    | Value                    | Explanation                                                                                                                                                                          |
|--------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delay Sync   | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                                                                      |
| D. Time      | 1–1300                   | Delay time from when the original sound is heard to when the delay sound is heard                                                                                                    |
| D. Time Nt   | Note<br>→ "Note" (p. 72) |                                                                                                                                                                                      |
| Acceleration | 0–15                     | Adjusts the speed which the Delay Time changes from the current setting to a specified new setting. The rate of change for the Delay Time directly affects the rate of pitch change. |
| Feedback     | -98–+98 [%]              | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.                                                             |

| Parameter | Value                                                                                                      | Explanation                                                                                      |
|-----------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| HF Damp   | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut). |
| Low Gain  | -15–+15 [dB]                                                                                               | Amount of boost/cut for the low-frequency range                                                  |
| High Gain | -15–+15 [dB]                                                                                               | Amount of boost/cut for the high-frequency range                                                 |
| Balance   | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the delay sound (W)                              |
| Level     | 0–127                                                                                                      | Output Level                                                                                     |
| Asgn1–4   | OFF, D. Time, Feedback, Balance                                                                            | Specifies the parameters that are assigned to assign 1–4.                                        |

## 61 Tape Echo

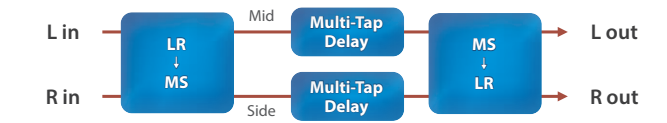
A virtual tape echo that produces a realistic tape delay sound. This simulates the tape echo section of a Roland RE-201 Space Echo.



| Parameter   | Value                             | Explanation                                                                                                                                                                                     |
|-------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode        | S, M, L, S+M, S+L, M+L, S+M+L     | Combination of playback heads to use. Select from three different heads with different delay times.<br><b>S</b> : short<br><b>M</b> : middle<br><b>L</b> : long                                 |
| Repeat Rate | 0–127                             | Tape speed. Increasing this value will shorten the spacing of the delayed sounds.                                                                                                               |
| Intensity   | 0–127                             | Amount of delay repeats                                                                                                                                                                         |
| Bass        | -15–+15 [dB]                      | Boost/cut for the lower range of the echo sound                                                                                                                                                 |
| Treble      | -15–+15 [dB]                      | Boost/cut for the upper range of the echo sound                                                                                                                                                 |
| Head S Pan  | L64–63R                           | Independent panning for the short, middle, and long playback heads                                                                                                                              |
| Head M Pan  | L64–63R                           |                                                                                                                                                                                                 |
| Head L Pan  | L64–63R                           |                                                                                                                                                                                                 |
| Distortion  | 0–5                               | Amount of tape-dependent distortion to be added. This simulates the slight tonal changes that can be detected by signal-analysis equipment. Increasing this value will increase the distortion. |
| Wf Rate     | 0–127                             | Speed of wow/flutter (complex variation in pitch caused by tape wear and rotational irregularity)                                                                                               |
| Wf Depth    | 0–127                             | Depth of wow/flutter                                                                                                                                                                            |
| Echo Level  | 0–127                             | Volume of the echo sound                                                                                                                                                                        |
| Direct Lv   | 0–127                             | Volume of the original sound                                                                                                                                                                    |
| Level       | 0–127                             | Output Level                                                                                                                                                                                    |
| Asgn1–4     | OFF, Mode, Repeat Rate, Intensity | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                       |

62
M/S Delay (Mid-Side Delay)

This effect applies different amounts of delay to left/right signals of similar phase and differing phase.

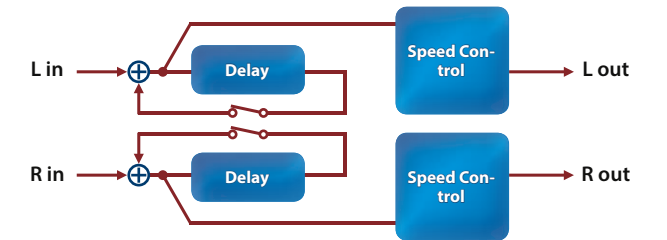


| Parameter   | Value                                                                                                      | Explanation                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| MD Level    | 0–127                                                                                                      | Delay volume of left/right input signals whose phase is similar (in phase)                                               |
| MD Mode     | 2TAP, 3TAP, 4TAP                                                                                           | Delay divisions for the input signals whose left/right phase is similar (identical phase)                                |
| MD Tm Sync  | OFF, ON                                                                                                    | If this is “ON,” the delay synchronizes with the tempo.                                                                  |
| MD. Time    | 1–1300                                                                                                     | Adjusts the time from the original sound until the delay sound is heard.                                                 |
| MDTime Nt   | Note<br>⇒ “Note” (p. 72)                                                                                   |                                                                                                                          |
| MD Feedback | –98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| MD HFDamp   | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                         |
| MD1 Pan     | L64–63R                                                                                                    | Panning of the first delay sound                                                                                         |
| MD2 Pan     | L64–63R                                                                                                    | Panning of the second delay sound                                                                                        |
| MD3 Pan     | L64–63R                                                                                                    | Panning of the third delay sound                                                                                         |
| MD4 Pan     | L64–63R                                                                                                    | Panning of the fourth delay sound                                                                                        |
| SD Level    | 0–127                                                                                                      | Delay volume of left/right input signals whose phase is distant (opposite phase)                                         |
| SD Mode     | 2TAP, 3TAP, 4TAP                                                                                           | Delay divisions for the input signals whose left/right phase is distant (reverse phase)                                  |
| SD Tm Sync  | OFF, ON                                                                                                    | If this is “ON,” the delay synchronizes with the tempo.                                                                  |
| SD Time     | 1–1300                                                                                                     | Adjusts the time from the original sound until the delay sound is heard.                                                 |
| SDTime Nt   | Note<br>⇒ “Note” (p. 72)                                                                                   |                                                                                                                          |
| SD Feedback | –98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| SD HFDamp   | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect is filtered out (BYPASS: no cut).                         |
| SD1 Pan     | L64–63R                                                                                                    | Panning of the first delay sound                                                                                         |
| SD2 Pan     | L64–63R                                                                                                    | Panning of the second delay sound                                                                                        |
| SD3 Pan     | L64–63R                                                                                                    | Panning of the third delay sound                                                                                         |
| SD4 Pan     | L64–63R                                                                                                    | Panning of the fourth delay sound                                                                                        |
| Level       | 0–127                                                                                                      | Output Level                                                                                                             |
| Asgn1–4     | OFF, MD Level, MD Feedback, SD Level, SD Feedback                                                          | Specifies the parameters that are assigned to assign 1–4.                                                                |

Looper

63
DJFX Looper

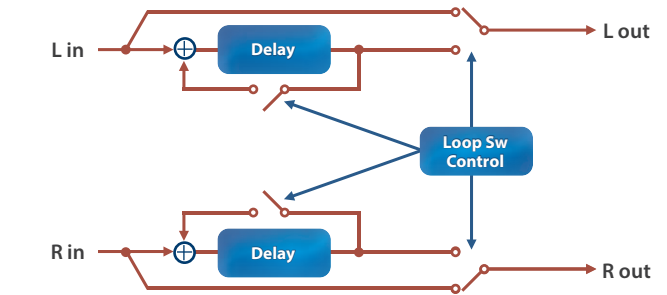
Loops a short portion of the input sound. You can vary the playback direction and playback speed of the input sound to add turntable-type effects.



| Parameter | Value                       | Explanation                                                                                                                                                                                                                                                        |
|-----------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length    | 230–23 (not straight)       | Specifies the length of the loop.                                                                                                                                                                                                                                  |
| Speed     | –1.00–+1.00                 | Specifies the playback direction and playback speed.<br>– <b>direction</b> : Reverse playback<br>+ <b>direction</b> : Normal playback<br>0: Stop playback<br>As the value moves away from 0, the playback speed becomes faster.                                    |
| Loop Sw   | OFF, ON                     | If you turn this on while the sound is heard, the sound at that point will be looped. Turn this off to cancel the loop.<br>* If the effect is recalled with this ON, this parameter must be turned OFF and then turned ON again in order to make the loop operate. |
| Level     | 0–127                       | Output Level                                                                                                                                                                                                                                                       |
| Asgn1–4   | OFF, Length, Speed, Loop Sw | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                                                          |

64
BPM Looper

Loops a short portion of the input sound. This can automatically turn the loop on/off in synchronization with the rhythm.



| Parameter | Value                    | Explanation                                                                                                                                         |
|-----------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Length    | 230–23 (not straight)    | Specifies the length of the loop.                                                                                                                   |
| Rate Sync | OFF, ON                  | If this is “ON,” the rate synchronizes with the tempo of the rhythm.<br>⇒ “Tempo” (p. 3, p. 31)                                                     |
| Rate      | 0.05–10.00 [Hz]          | Cycle at which the loop automatically turns on/off                                                                                                  |
| Rate Note | Note<br>⇒ “Note” (p. 72) |                                                                                                                                                     |
| Timing    | 1–8                      | Specifies the timing within the cycle at which the loop automatically starts (which step of the eight timing divisions at which the sound is heard) |
| Lenth     | 1–8                      | Specifies the length at which the loop automatically ends within the cycle (the number of times that the 1/8-length of sound is heard)              |

| Parameter | Value                  | Explanation                                                                                                                                                                                                                          |
|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Loop Mode | OFF, AUTO, ON          | If this is AUTO, the loop automatically turns on/off in synchronization with the rhythm.<br>* If the effect is recalled with this ON, this parameter must first be set to something other than ON in order to make the loop operate. |
| Level     | 0–127                  | Output Level                                                                                                                                                                                                                         |
| Asgn1–4   | OFF, Length, Rate (Hz) | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                                                            |

## Lo-Fi

### 65 LOFI Comp (Lo-Fi Compressor)

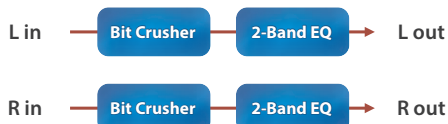
Degrades the sound quality.



| Parameter   | Value               | Explanation                                                                                                                                                                    |
|-------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Filter  | 1–6                 | Selects the type of filter applied to the sound before it passes through the Lo-Fi effect.<br><b>1</b> : Compressor off<br><b>2–6</b> : Compressor on                          |
| LoFi Type   | 1–9                 | Degrades the sound quality. The sound quality grows poorer as this value is increased.                                                                                         |
| Post Filter | OFF, LPF, HPF       | Type of filter<br><b>OFF</b> : No filter is used<br><b>LPF</b> : Cuts the frequency range above the Cutoff Freq<br><b>HPF</b> : Cuts the frequency range below the Cutoff Freq |
| Cutoff      | 200–8000 [Hz]       | Basic frequency of the Post Filter                                                                                                                                             |
| Low Gain    | -15–+15 [dB]        | Amount of boost/cut for the low-frequency range                                                                                                                                |
| High Gain   | -15–+15 [dB]        | Amount of boost/cut for the high-frequency range                                                                                                                               |
| Balance     | D100: 0W–D0: 100W   | Volume balance between the direct sound (D) and the effect sound (W)                                                                                                           |
| Level       | 0–127               | Output Level                                                                                                                                                                   |
| Asgn1–4     | OFF, Balance, Level | Specifies the parameters that are assigned to assign 1–4.                                                                                                                      |

### 66 Bit Crusher

Produces an extreme lo-fi effect.

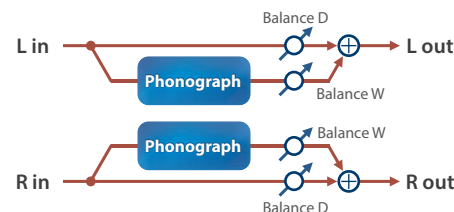


| Parameter   | Value        | Explanation                                      |
|-------------|--------------|--------------------------------------------------|
| Sample Rate | 0–127        | Adjusts the sample rate.                         |
| Bit Down    | 0–20         | Adjusts the bit depth.                           |
| Filter      | 0–127        | Adjusts the filter depth.                        |
| Low Gain    | -15–+15 [dB] | Amount of boost/cut for the low-frequency range  |
| High Gain   | -15–+15 [dB] | Amount of boost/cut for the high-frequency range |
| Level       | 0–127        | Output Level                                     |

| Parameter | Value                    | Explanation                                               |
|-----------|--------------------------|-----------------------------------------------------------|
| Asgn1–4   | OFF, Sample Rate, Filter | Specifies the parameters that are assigned to assign 1–4. |

## 67 Phonograph

Recreates the sound of an analog record being played on a record player. This lets you simulate the unique noises produced when a record is played, as well as the variations that occur when the record spins.

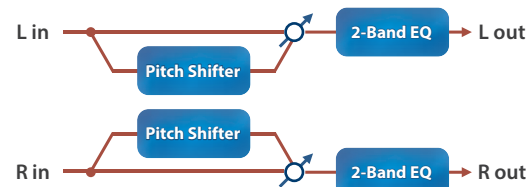


| Parameter       | Value             | Explanation                                                                                                                                  |
|-----------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Signal Dist     | 0–127             | Sets the amount of distortion.                                                                                                               |
| Frequency Range | 0–127             | Sets the frequency characteristics of the playback system. Smaller values create the feeling of an older system with narrow frequency bands. |
| Disc Type       | LP, EP, SP        | Sets the turntable rotation speed. This has an effect on the scratch noise cycle.                                                            |
| Scratch NZ Lev  | 0–127             | Sets the volume of noise created by scratches in the record.                                                                                 |
| Dust NZ Lev     | 0–127             | Sets the volume of noise created by dust on the record.                                                                                      |
| Hiss NZ Lev     | 0–127             | Sets the volume of continuous hiss noise.                                                                                                    |
| Total NZ Lev    | 0–127             | Sets the volume of noise overall.                                                                                                            |
| Wow             | 0–127             | Sets the amount of variation in record spin (long cycle).                                                                                    |
| Flutter         | 0–127             | Sets the amount of variation in record spin (short cycle).                                                                                   |
| Random          | 0–127             | Sets the amount of non-cyclical variation in record spin.                                                                                    |
| Total W/F       | 0–127             | Sets the volume of variation in record spin overall.                                                                                         |
| Balance         | D100: 0W–D0: 100W | Sets the volume balance between the original sound (D) and the effect sound (W).                                                             |
| Level           | 0–127             | Sets the output volume.                                                                                                                      |

## Pitch

### 68 PitchShifr (Pitch Shifter)

A stereo pitch shifter.

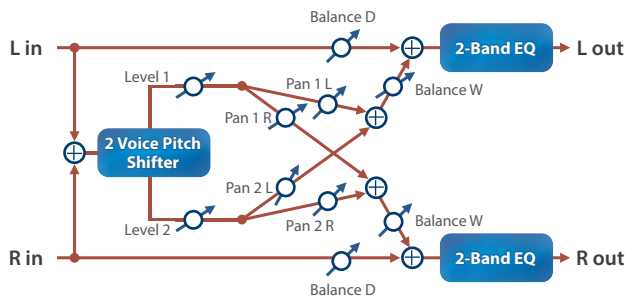


| Parameter  | Value         | Explanation                                                                                     |
|------------|---------------|-------------------------------------------------------------------------------------------------|
| Coarse     | -24–+12 [sem] | Adjusts the pitch of the pitch shifted sound in semitone steps.                                 |
| Fine       | -100–+100     | Adjusts the pitch of the pitch shifted sound in 2-cent steps.                                   |
| Delay Sync | OFF, ON       | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |

| Parameter  | Value                                | Explanation                                                                                                                      |
|------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| D. Time    | 1–1300                               |                                                                                                                                  |
| D. Time Nt | Note<br>→ "Note" (p. 72)             | Adjusts the delay time from the direct sound until the pitch shifted sound is heard.                                             |
| Feedback   | -98–+98 [%]                          | Adjusts the proportion of the pitch shifted sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| Low Gain   | -15–+15 [dB]                         | Amount of boost/cut for the low-frequency range                                                                                  |
| High Gain  | -15–+15 [dB]                         | Amount of boost/cut for the high-frequency range                                                                                 |
| Balance    | D100: 0W–D0: 100W                    | Volume balance between the direct sound (D) and the pitch shifted sound (W)                                                      |
| Level      | 0–127                                | Output Level                                                                                                                     |
| Asgn1–4    | OFF, Coarse, Fine, Feedback, Balance | Specifies the parameters that are assigned to assign 1–4.                                                                        |

## 69 2V PShifter (2 Voice Pitch Shifter)

Shifts the pitch of the original sound. This 2-voice pitch shifter has two pitch shifters, and can add two pitch shifted sounds to the original sound.

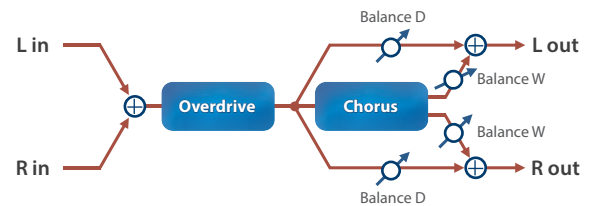


| Parameter   | Value                    | Explanation                                                                                                                      |
|-------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| P1 Coarse   | -24–+12 [sem]            | Adjusts the pitch of Pitch Shift 1 in semitone steps.                                                                            |
| P1 Fine     | -100–+100                | Adjusts the pitch of Pitch Shift 1 in 2-cent steps.                                                                              |
| P1 Dly Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                  |
| P1D.Time    | 1–1300                   |                                                                                                                                  |
| P1DRate Nt  | Note<br>→ "Note" (p. 72) | Adjusts the delay time from the direct sound until the Pitch Shift 1 sound is heard.                                             |
| P1 Feedback | -98–+98 [%]              | Adjusts the proportion of the pitch shifted sound that is fed back into the effect. Negative (-) settings will invert the phase. |
| P1 Pan      | L64–63R                  | Stereo location of the Pitch Shift 1 sound                                                                                       |
| P1 Level    | 0–127                    | Volume of the Pitch Shift 1 sound                                                                                                |
| P2 Coarse   | -24–+12 [sem]            |                                                                                                                                  |
| P2 Fine     | -100–+100                |                                                                                                                                  |
| P2 Dly Sync | OFF, ON                  |                                                                                                                                  |
| P2D.Time    | 1–1300                   | Settings of the Pitch Shift 2 sound. The parameters are the same as for the Pitch Shift 1 sound.                                 |
| P2DRate Nt  | Note                     |                                                                                                                                  |
| P2 Feedback | -98–+98 [%]              |                                                                                                                                  |
| P2 Pan      | L64–63R                  |                                                                                                                                  |
| P2 Level    | 0–127                    |                                                                                                                                  |
| Low Gain    | -15–+15 [dB]             | Amount of boost/cut for the low-frequency range                                                                                  |
| High Gain   | -15–+15 [dB]             | Amount of boost/cut for the high-frequency range                                                                                 |
| Balance     | D100: 0W–D0: 100W        | Volume balance between the direct sound (D) and the pitch shifted sound (W)                                                      |
| Level       | 0–127                    | Output Level                                                                                                                     |

| Parameter | Value                                                                                          | Explanation                                               |
|-----------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Asgn1–4   | OFF, P1 Coarse, P1 Fine, P1 Feedback, P1 Pan, P2 Coarse, P2 Fine, P2 Feedback, P2 Pan, Balance | Specifies the parameters that are assigned to assign 1–4. |

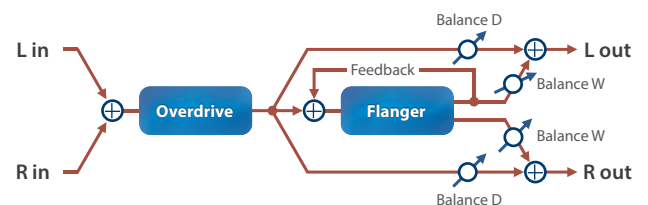
## Combination

### 70 OD → Chorus (Overdrive → Chorus)



| Parameter  | Value                                   | Explanation                                                                                                                             |
|------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| OD Drive   | 0–127                                   | Degree of distortion<br>Also changes the volume.                                                                                        |
| OD Pan     | L64–63R                                 | Stereo location of the overdrive sound                                                                                                  |
| Cho PreDly | 0.0–100 [ms]                            | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                           |
| Cho Sync   | OFF, ON                                 | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                         |
| C. Rate    | 0.05–10.00 [Hz]                         |                                                                                                                                         |
| C. Rate Nt | Note<br>→ "Note" (p. 72)                | Frequency of modulation                                                                                                                 |
| Cho Depth  | 0–127                                   | Depth of modulation                                                                                                                     |
| Cho Bal    | D100: 0W–D0: 100W                       | Adjusts the volume balance between the sound that is sent through the chorus (W) and the sound that is not sent through the chorus (D). |
| Level      | 0–127                                   | Output Level                                                                                                                            |
| Asgn1–4    | OFF, OD Drive, OD Pan, C. Rate, Cho Bal | Specifies the parameters that are assigned to assign 1–4.                                                                               |

### 71 OD → Flanger (Overdrive → Flanger)

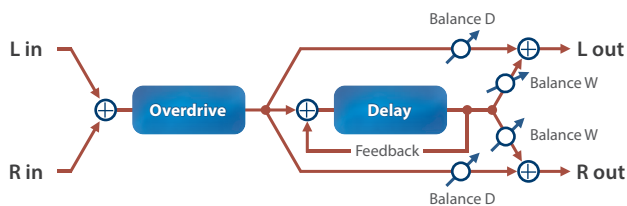


| Parameter  | Value                    | Explanation                                                                                     |
|------------|--------------------------|-------------------------------------------------------------------------------------------------|
| OD Drive   | 0–127                    | Degree of distortion<br>Also changes the volume.                                                |
| OD Pan     | L64–63R                  | Stereo location of the overdrive sound                                                          |
| Flg PreDly | 0.0–100 [ms]             | Adjusts the delay time from the direct sound until the flanger sound is heard.                  |
| Flg Sync   | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| F. Rate    | 0.05–10.00 [Hz]          |                                                                                                 |
| F. Rate Nt | Note<br>→ "Note" (p. 72) | Frequency of modulation                                                                         |



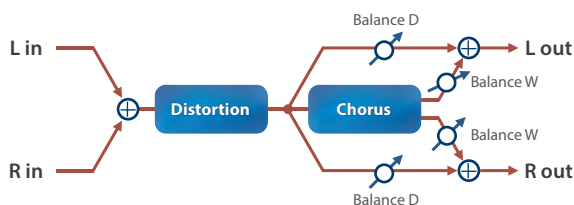
| Parameter | Value                                            | Explanation                                                                                                                               |
|-----------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Flg Depth | 0–127                                            | Depth of modulation                                                                                                                       |
| Flg Fbk   | -98–+98 [%]                                      | Adjusts the proportion of the flanger sound that is fed back into the effect. Negative (-) settings will invert the phase.                |
| Flg Bal   | D100: 0W–D0: 100W                                | Adjusts the volume balance between the sound that is sent through the flanger (W) and the sound that is not sent through the flanger (D). |
| Level     | 0–127                                            | Output Level                                                                                                                              |
| Asgn1–4   | OFF, OD Drive, OD Pan, F. Rate, Flg Fbk, Flg Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                 |

## 72 OD → Delay (Overdrive → Delay)



| Parameter  | Value                                                                                                      | Explanation                                                                                                                           |
|------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| OD Drive   | 0–127                                                                                                      | Degree of distortion<br>Also changes the volume.                                                                                      |
| OD Pan     | L64–63R                                                                                                    | Stereo location of the overdrive sound                                                                                                |
| Delay Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| D. Time    | 1–2600                                                                                                     | Delay time from when the original sound is heard to when the delay sound is heard                                                     |
| D. Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Delay Fbk  | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.              |
| Dly HF     | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect will be cut (BYPASS: no cut).                                          |
| Dly Bal    | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level      | 0–127                                                                                                      | Output Level                                                                                                                          |
| Asgn1–4    | OFF, OD Drive, OD Pan, Delay Fbk, Dly Bal                                                                  | Specifies the parameters that are assigned to assign 1–4.                                                                             |

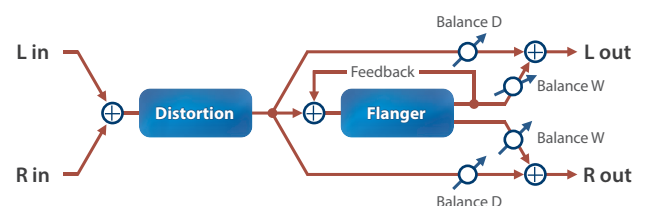
## 73 DS → Chorus (Distortion → Chorus)



| Parameter  | Value        | Explanation                                                                   |
|------------|--------------|-------------------------------------------------------------------------------|
| Dist Drive | 0–127        | Degree of distortion<br>Also changes the volume.                              |
| Dist Pan   | L64–63R      | Stereo location of the overdrive sound                                        |
| Cho PreDly | 0.0–100 [ms] | Adjusts the delay time from the direct sound until the chorus sound is heard. |

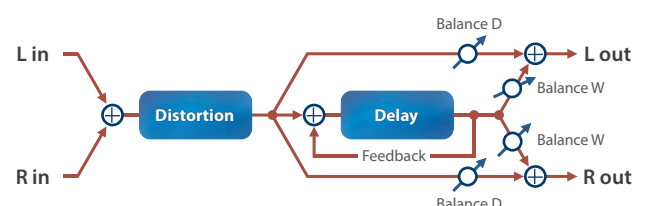
| Parameter  | Value                                       | Explanation                                                                                                                             |
|------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Cho Sync   | OFF, ON                                     | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                         |
| C. Rate    | 0.05–10.00 [Hz]                             | Frequency of modulation                                                                                                                 |
| C. Rate Nt | Note<br>→ "Note" (p. 72)                    |                                                                                                                                         |
| Cho Depth  | 0–127                                       | Depth of modulation                                                                                                                     |
| Cho Bal    | D100: 0W–D0: 100W                           | Adjusts the volume balance between the sound that is sent through the chorus (W) and the sound that is not sent through the chorus (D). |
| Level      | 0–127                                       | Output Level                                                                                                                            |
| Asgn1–4    | OFF, Dist Drive, Dist Pan, C. Rate, Cho Bal | Specifies the parameters that are assigned to assign 1–4.                                                                               |

## 74 DS → Flanger (Distortion → Flanger)



| Parameter  | Value                                                | Explanation                                                                                                                               |
|------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Dist Drive | 0–127                                                | Degree of distortion<br>Also changes the volume.                                                                                          |
| Dist Pan   | L64–63R                                              | Stereo location of the overdrive sound                                                                                                    |
| Flg PreDly | 0.0–100 [ms]                                         | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                            |
| Flg Sync   | OFF, ON                                              | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                           |
| F. Rate    | 0.05–10.00 [Hz]                                      | Frequency of modulation                                                                                                                   |
| F. Rate Nt | Note<br>→ "Note" (p. 72)                             |                                                                                                                                           |
| Flg Depth  | 0–127                                                | Depth of modulation                                                                                                                       |
| Flg Fbk    | -98–+98 [%]                                          | Adjusts the proportion of the flanger sound that is fed back into the effect. Negative (-) settings will invert the phase.                |
| Flg Bal    | D100: 0W–D0: 100W                                    | Adjusts the volume balance between the sound that is sent through the flanger (W) and the sound that is not sent through the flanger (D). |
| Level      | 0–127                                                | Output Level                                                                                                                              |
| Asgn1–4    | OFF, Dist Drive, Dist Pan, F. Rate, Flg Fbk, Flg Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                 |

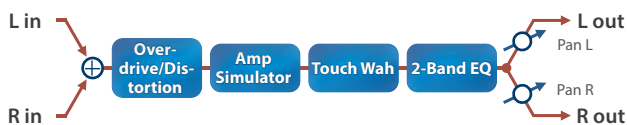
## 75 DS → Delay (Distortion → Delay)



| Parameter  | Value   | Explanation                                      |
|------------|---------|--------------------------------------------------|
| Dist Drive | 0–127   | Degree of distortion<br>Also changes the volume. |
| Dist Pan   | L64–63R | Stereo location of the overdrive sound           |

| Parameter  | Value                                                                                                      | Explanation                                                                                                                           |
|------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Delay Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| D. Time    | 1–2600                                                                                                     | Delay time from when the original sound is heard to when the delay sound is heard                                                     |
| D. Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Delay Fbk  | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.           |
| Dly HF     | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect will be cut (BYPASS: no cut).                                          |
| Dly Bal    | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level      | 0–127                                                                                                      | Output Level                                                                                                                          |
| Asgn1–4    | OFF, Dist Drive, Dist Pan, Delay Fbk, Dly Bal                                                              | Specifies the parameters that are assigned to assign 1–4.                                                                             |

## 76 OD/DS → T. Wah (Overdrive/Distortion → Touch Wah)



| Parameter    | Value                             | Explanation                                                                                                                                                                |
|--------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drive Switch | OFF, ON                           | Turns overdrive/distortion on/off                                                                                                                                          |
| D. Type      | OVERDRIVE, DISTORTION             | Type of distortion                                                                                                                                                         |
| Drive        | 0–127                             | Degree of distortion<br>Also changes the volume.                                                                                                                           |
| Tone         | 0–127                             | Sound quality of the Overdrive effect                                                                                                                                      |
| Amp Switch   | OFF, ON                           | Turns the Amp Simulator on/off.                                                                                                                                            |
| AmpType      | SMALL, BUILT-IN, 2-STACK, 3-STACK | Type of guitar amp<br><b>SMALL:</b> Small amp<br><b>BUILT-IN:</b> Single-unit type amp<br><b>2-STACK:</b> Large double stack amp<br><b>3-STACK:</b> Large triple stack amp |
| TWah Switch  | OFF, ON                           | Wah on/off                                                                                                                                                                 |
| TWah Mode    | LPF, BPF                          | Filter type<br><b>LPF:</b> The wah effect will be applied over a wide frequency range.<br><b>BPF:</b> The wah effect will be applied over a narrow frequency range.        |
| TWah Polar   | DOWN, UP                          | Direction in which the filter will move<br><b>DOWN:</b> The filter will change toward a lower frequency.<br><b>UP:</b> The filter will change toward a higher frequency.   |
| TWah Sens    | 0–127                             | Sensitivity with which the filter is modified                                                                                                                              |
| TWah Manual  | 0–127                             | Center frequency at which the wah effect is applied                                                                                                                        |
| TWah Peak    | 0–127                             | Width of the frequency region at which the wah effect is applied<br>Increasing this value will make the frequency region narrower.                                         |
| TWah Bal     | D100: 0W–D0: 100W                 | Adjusts the volume balance between the sound that is sent through the wah (W) and the sound that is not sent through the wah (D).                                          |
| Low Gain     | -15–+15 [dB]                      | Amount of boost/cut for the low-frequency range                                                                                                                            |
| High Gain    | -15–+15 [dB]                      | Amount of boost/cut for the high-frequency range                                                                                                                           |

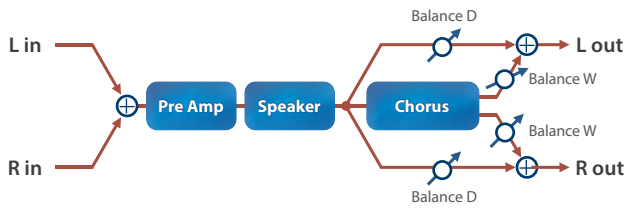
| Parameter | Value                                                             | Explanation                                               |
|-----------|-------------------------------------------------------------------|-----------------------------------------------------------|
| Level     | 0–127                                                             | Output Level                                              |
| Asgn1–4   | OFF, Drive, Tone, TWah Sens, TWah Manual, TWah Peak, TWah Balance | Specifies the parameters that are assigned to assign 1–4. |

## 77 OD/DS → A. Wah (Overdrive/Distortion → Auto Wah)



| Parameter    | Value                                                                  | Explanation                                                                                                                                                                |
|--------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drive Switch | OFF, ON                                                                | Turns overdrive/distortion on/off                                                                                                                                          |
| D. Type      | OVERDRIVE, DISTORTION                                                  | Type of distortion                                                                                                                                                         |
| Drive        | 0–127                                                                  | Degree of distortion<br>Also changes the volume.                                                                                                                           |
| Tone         | 0–127                                                                  | Sound quality of the Overdrive effect                                                                                                                                      |
| Amp Switch   | OFF, ON                                                                | Turns the Amp Simulator on/off.                                                                                                                                            |
| AmpType      | SMALL, BUILT-IN, 2-STACK, 3-STACK                                      | Type of guitar amp<br><b>SMALL:</b> Small amp<br><b>BUILT-IN:</b> Single-unit type amp<br><b>2-STACK:</b> Large double stack amp<br><b>3-STACK:</b> Large triple stack amp |
| AWah Switch  | OFF, ON                                                                | Wah on/off                                                                                                                                                                 |
| AWah Mode    | LPF, BPF                                                               | Filter type<br><b>LPF:</b> The wah effect will be applied over a wide frequency range.<br><b>BPF:</b> The wah effect will be applied over a narrow frequency range.        |
| AWah Manual  | 0–127                                                                  | Center frequency at which the wah effect is applied                                                                                                                        |
| AWah Peak    | 0–127                                                                  | Width of the frequency region at which the wah effect is applied<br>Increasing this value will make the frequency region narrower.                                         |
| AWah Sync    | OFF, ON                                                                | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                                                            |
| AWRate       | 0.05–10.00 [Hz]                                                        | Frequency of modulation                                                                                                                                                    |
| AWRate Nt    | Note<br>→ "Note" (p. 72)                                               |                                                                                                                                                                            |
| AWah Depth   | 0–127                                                                  | Depth at which the wah effect is modulated                                                                                                                                 |
| AWah Bal     | D100: 0W–D0: 100W                                                      | Adjusts the volume balance between the sound that is sent through the wah (W) and the sound that is not sent through the wah (D).                                          |
| Low Gain     | -15–+15 [dB]                                                           | Amount of boost/cut for the low-frequency range                                                                                                                            |
| High Gain    | -15–+15 [dB]                                                           | Amount of boost/cut for the high-frequency range                                                                                                                           |
| Level        | 0–127                                                                  | Output Level                                                                                                                                                               |
| Asgn1–4      | OFF, Drive, Tone, AWah Manual, AWah Peak, AWRate, AWah Depth, AWah Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                                                  |

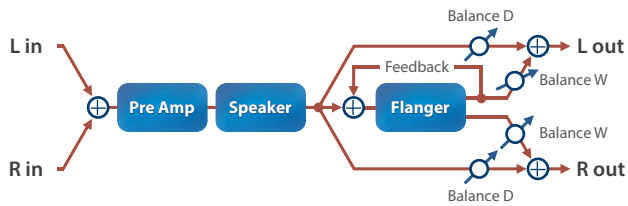
## 78 Gt → Chorus (Guitar Amp Simulator → Chorus)



| Parameter  | Value              | Explanation                                                                                                                                     |
|------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Amp Sw | OFF, ON            | Turns the amp switch on/off.                                                                                                                    |
| ATyp       | Type of guitar amp |                                                                                                                                                 |
|            | JC-120             | This models the sound of the Roland JC-120.                                                                                                     |
|            | CLEAN TWIN         | This models a Fender Twin Reverb.                                                                                                               |
|            | MATCH DRIVE        | This models the sound input to left input on a Matchless D/C-30. A simulation of the latest tube amp widely used in styles from blues and rock. |
|            | BG LEAD            | This models the lead sound of the MESA/ Boogie combo amp. The sound of a tube amp typical of the late '70s to '80s.                             |
|            | MS1959I            | This models the sound input to Input I on a Marshall 1959. This is a treble sound suited to hard rock.                                          |
|            | MS1959II           | This models the sound input to Input II on a Marshall 1959.                                                                                     |
|            | MS1959I+II         | The sound of connecting inputs I and II of the guitar amp in parallel, creating a sound with a stronger low end than I.                         |
|            | SLDN LEAD          | This models a Soldano SLO-100. This is the typical sound of the eighties.                                                                       |
|            | METAL 5150         | This models the lead channel of a Peavey EVH 5150.                                                                                              |
|            | METAL LEAD         | This is distortion sound that is ideal for performances of heavy riffs.                                                                         |
|            | OD-1               | This models the sound of the BOSS OD-1. This produces sweet, mild distortion.                                                                   |
|            | OD-2 TURBO         | This is the high-gain overdrive sound of the BOSS OD-2.                                                                                         |
|            | DISTORTION         | This gives a basic, traditional distortion sound.                                                                                               |
|            | FUZZ               | A fuzz sound with rich harmonic content.                                                                                                        |
| Drive      | 0–127              | Volume and amount of distortion of the amp                                                                                                      |
| Master Lv  | 0–127              | Volume of the entire pre-amp                                                                                                                    |
| Gain       | LOW, MIDDLE, HIGH  | Amount of pre-amp distortion                                                                                                                    |
| Bass       | 0–127              | Tone of the bass/mid/treble frequency range                                                                                                     |
| Middle     | 0–127              |                                                                                                                                                 |
| Treble     | 0–127              |                                                                                                                                                 |
| Speaker Sw | OFF, ON            | Selects whether the sound will be sent through the speaker simulation (ON) or not (OFF).                                                        |

| Parameter  | Value                                                          | Explanation                                                                                                                             |                                                |            |
|------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------|
| STyp       |                                                                | Cabinet                                                                                                                                 | Diameter (in inches) and number of the speaker | Microphone |
|            | SMALL 1                                                        | small open-back enclosure                                                                                                               | 10                                             | dynamic    |
|            | SMALL 2                                                        | small open-back enclosure                                                                                                               | 10                                             | dynamic    |
|            | MIDDLE                                                         | open back enclosure                                                                                                                     | 12 x 1                                         | dynamic    |
|            | JC-120                                                         | open back enclosure                                                                                                                     | 12 x 2                                         | dynamic    |
|            | BUILT-IN 1                                                     | open back enclosure                                                                                                                     | 12 x 2                                         | dynamic    |
|            | BUILT-IN 2                                                     | open back enclosure                                                                                                                     | 12 x 2                                         | condenser  |
|            | BUILT-IN 3                                                     | open back enclosure                                                                                                                     | 12 x 2                                         | condenser  |
|            | BUILT-IN 4                                                     | open back enclosure                                                                                                                     | 12 x 2                                         | condenser  |
|            | BUILT-IN 5                                                     | open back enclosure                                                                                                                     | 12 x 2                                         | condenser  |
|            | BG STACK 1                                                     | sealed enclosure                                                                                                                        | 12 x 2                                         | condenser  |
|            | BG STACK 2                                                     | large sealed enclosure                                                                                                                  | 12 x 2                                         | condenser  |
|            | MS STACK 1                                                     | large sealed enclosure                                                                                                                  | 12 x 4                                         | condenser  |
|            | MS STACK 2                                                     | large sealed enclosure                                                                                                                  | 12 x 4                                         | condenser  |
|            | METAL STACK                                                    | large double stack                                                                                                                      | 12 x 4                                         | condenser  |
|            | 2-STACK                                                        | large double stack                                                                                                                      | 12 x 4                                         | condenser  |
|            | 3-STACK                                                        | large triple stack                                                                                                                      | 12 x 4                                         | condenser  |
| Chorus Sw  | OFF, ON                                                        | Chorus on/off                                                                                                                           |                                                |            |
| Cho PreDly | 0.0–100 [ms]                                                   | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                           |                                                |            |
| C. Rate    | 0.05–10.00 [Hz]                                                | Frequency of modulation                                                                                                                 |                                                |            |
| Cho Depth  | 0–127                                                          | Depth of modulation                                                                                                                     |                                                |            |
| Cho Bal    | D100: 0W–D0: 100W                                              | Adjusts the volume balance between the sound that is sent through the chorus (W) and the sound that is not sent through the chorus (D). |                                                |            |
| Level      | 0–127                                                          | Output Level                                                                                                                            |                                                |            |
| Asgn1–4    | OFF, Volume, Master Lv, Chorus Sw, C. Rate, Cho Depth, Cho Bal | Specifies the parameters that are assigned to assign 1–4.                                                                               |                                                |            |

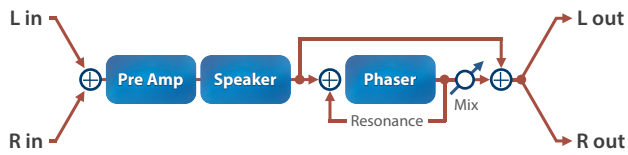
## 79 Gt → Flanger (Guitar Amp Simulator → Flanger)



| Parameter  | Value              | Explanation                                                                                                                                     |
|------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Amp Sw | OFF, ON            | Turns the amp switch on/off.                                                                                                                    |
| ATyp       | Type of guitar amp |                                                                                                                                                 |
|            | JC-120             | This models the sound of the Roland JC-120.                                                                                                     |
|            | CLEAN TWIN         | This models a Fender Twin Reverb.                                                                                                               |
|            | MATCH DRIVE        | This models the sound input to left input on a Matchless D/C-30. A simulation of the latest tube amp widely used in styles from blues and rock. |
|            | BG LEAD            | This models the lead sound of the MESA/ Boogie combo amp. The sound of a tube amp typical of the late '70s to '80s.                             |
|            | MS1959I            | This models the sound input to Input I on a Marshall 1959. This is a trebly sound suited to hard rock.                                          |
|            | MS1959II           | This models the sound input to Input II on a Marshall 1959.                                                                                     |
|            | MS1959I+II         | The sound of connecting inputs I and II of the guitar amp in parallel, creating a sound with a stronger low end than I.                         |
|            | SLDN LEAD          | This models a Soldano SLO-100. This is the typical sound of the eighties.                                                                       |
|            | METAL 5150         | This models the lead channel of a Peavey EVH 5150.                                                                                              |
|            | METAL LEAD         | This is distortion sound that is ideal for performances of heavy riffs.                                                                         |
|            | OD-1               | This models the sound of the BOSS OD-1. This produces sweet, mild distortion.                                                                   |
|            | OD-2 TURBO         | This is the high-gain overdrive sound of the BOSS OD-2.                                                                                         |
|            | DISTORTION         | This gives a basic, traditional distortion sound.                                                                                               |
|            | FUZZ               | A fuzz sound with rich harmonic content.                                                                                                        |
| Drive      | 0–127              | Volume and amount of distortion of the amp                                                                                                      |
| Master Lv  | 0–127              | Volume of the entire pre-amp                                                                                                                    |
| Gain       | LOW, MIDDLE, HIGH  | Amount of pre-amp distortion                                                                                                                    |
| Bass       | 0–127              | Tone of the bass/mid/treble frequency range                                                                                                     |
| Middle     | 0–127              |                                                                                                                                                 |
| Treble     | 0–127              |                                                                                                                                                 |
| Speaker Sw | OFF, ON            | Selects whether the sound will be sent through the speaker simulation (ON) or not (OFF)                                                         |

| Parameter  | Value                                                                    | Explanation                                                                                                                               |                                                |            |
|------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------|
| STyp       |                                                                          | Cabinet                                                                                                                                   | Diameter (in inches) and number of the speaker | Microphone |
|            | SMALL 1                                                                  | small open-back enclosure                                                                                                                 | 10                                             | dynamic    |
|            | SMALL 2                                                                  | small open-back enclosure                                                                                                                 | 10                                             | dynamic    |
|            | MIDDLE                                                                   | open back enclosure                                                                                                                       | 12 x 1                                         | dynamic    |
|            | JC-120                                                                   | open back enclosure                                                                                                                       | 12 x 2                                         | dynamic    |
|            | BUILT-IN 1                                                               | open back enclosure                                                                                                                       | 12 x 2                                         | dynamic    |
|            | BUILT-IN 2                                                               | open back enclosure                                                                                                                       | 12 x 2                                         | condenser  |
|            | BUILT-IN 3                                                               | open back enclosure                                                                                                                       | 12 x 2                                         | condenser  |
|            | BUILT-IN 4                                                               | open back enclosure                                                                                                                       | 12 x 2                                         | condenser  |
|            | BUILT-IN 5                                                               | open back enclosure                                                                                                                       | 12 x 2                                         | condenser  |
|            | BG STACK 1                                                               | sealed enclosure                                                                                                                          | 12 x 2                                         | condenser  |
|            | BG STACK 2                                                               | large sealed enclosure                                                                                                                    | 12 x 2                                         | condenser  |
|            | MS STACK 1                                                               | large sealed enclosure                                                                                                                    | 12 x 4                                         | condenser  |
|            | MS STACK 2                                                               | large sealed enclosure                                                                                                                    | 12 x 4                                         | condenser  |
|            | METAL STACK                                                              | large double stack                                                                                                                        | 12 x 4                                         | condenser  |
|            | 2-STACK                                                                  | large double stack                                                                                                                        | 12 x 4                                         | condenser  |
|            | 3-STACK                                                                  | large triple stack                                                                                                                        | 12 x 4                                         | condenser  |
| Flg Switch | OFF, ON                                                                  | Flanger on/off                                                                                                                            |                                                |            |
| Flg PreDly | 0.0–100 [ms]                                                             | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                            |                                                |            |
| F. Rate    | 0.05–10.00 [Hz]                                                          | Frequency of modulation                                                                                                                   |                                                |            |
| Flg Depth  | 0–127                                                                    | Depth of modulation                                                                                                                       |                                                |            |
| Flg Fbk    | -98–+98 [%]                                                              | Adjusts the proportion of the flanger sound that is fed back into the effect. Negative (-) settings will invert the phase.                |                                                |            |
| Flg Bal    | D100: 0W–D0: 100W                                                        | Adjusts the volume balance between the sound that is sent through the flanger (W) and the sound that is not sent through the flanger (D). |                                                |            |
| Level      | 0–127                                                                    | Output Level                                                                                                                              |                                                |            |
| Asgn1–4    | OFF, Volume, Master Lv, Flg Switch, F. Rate, Flg Depth, Flg Fbk, Flg Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                 |                                                |            |

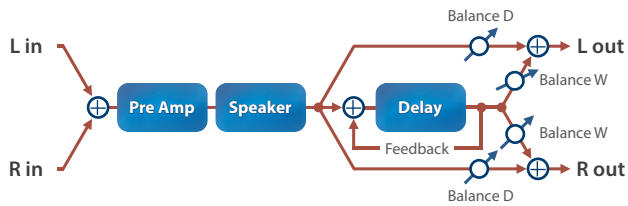
## 80 Gt → Phaser (Guitar Amp Simulator → Phaser)



| Parameter  | Value              | Explanation                                                                                                                                     |
|------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Amp Sw | OFF, ON            | Turns the amp switch on/off.                                                                                                                    |
| ATyp       | Type of guitar amp |                                                                                                                                                 |
|            | JC-120             | This models the sound of the Roland JC-120.                                                                                                     |
|            | CLEAN TWIN         | This models a Fender Twin Reverb.                                                                                                               |
|            | MATCH DRIVE        | This models the sound input to left input on a Matchless D/C-30. A simulation of the latest tube amp widely used in styles from blues and rock. |
|            | BG LEAD            | This models the lead sound of the MESA/ Boogie combo amp. The sound of a tube amp typical of the late '70s to '80s.                             |
|            | MS1959I            | This models the sound input to Input I on a Marshall 1959. This is a trebly sound suited to hard rock.                                          |
|            | MS1959II           | This models the sound input to Input II on a Marshall 1959.                                                                                     |
|            | MS1959I+II         | The sound of connecting inputs I and II of the guitar amp in parallel, creating a sound with a stronger low end than I.                         |
|            | SLDN LEAD          | This models a Soldano SLO-100. This is the typical sound of the eighties.                                                                       |
|            | METAL 5150         | This models the lead channel of a Peavey EVH 5150.                                                                                              |
|            | METAL LEAD         | This is distortion sound that is ideal for performances of heavy riffs.                                                                         |
|            | OD-1               | This models the sound of the BOSS OD-1. This produces sweet, mild distortion.                                                                   |
|            | OD-2 TURBO         | This is the high-gain overdrive sound of the BOSS OD-2.                                                                                         |
|            | DISTORTION         | This gives a basic, traditional distortion sound.                                                                                               |
|            | FUZZ               | A fuzz sound with rich harmonic content.                                                                                                        |
| Drive      | 0–127              | Volume and amount of distortion of the amp                                                                                                      |
| Master Lv  | 0–127              | Volume of the entire pre-amp                                                                                                                    |
| Gain       | LOW, MIDDLE, HIGH  | Amount of pre-amp distortion                                                                                                                    |
| Bass       | 0–127              | Tone of the bass/mid/treble frequency range                                                                                                     |
| Middle     | 0–127              |                                                                                                                                                 |
| Treble     | 0–127              |                                                                                                                                                 |
| Speaker Sw | OFF, ON            | Selects whether the sound will be sent through the speaker simulation (ON) or not (OFF).                                                        |

| Parameter | Value      | Explanation                                                                          |                                                                     |            |
|-----------|------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------|
| STyp      |            | Cabinet                                                                              | Diameter (in inches) and number of the speaker                      | Microphone |
|           | SMALL 1    | small open-back enclosure                                                            | 10                                                                  | dynamic    |
|           | SMALL 2    | small open-back enclosure                                                            | 10                                                                  | dynamic    |
|           | MIDDLE     | open back enclosure                                                                  | 12 x 1                                                              | dynamic    |
|           | JC-120     | open back enclosure                                                                  | 12 x 2                                                              | dynamic    |
|           | BUILT-IN 1 | open back enclosure                                                                  | 12 x 2                                                              | dynamic    |
|           | BUILT-IN 2 | open back enclosure                                                                  | 12 x 2                                                              | condenser  |
|           | BUILT-IN 3 | open back enclosure                                                                  | 12 x 2                                                              | condenser  |
|           | BUILT-IN 4 | open back enclosure                                                                  | 12 x 2                                                              | condenser  |
|           | BUILT-IN 5 | open back enclosure                                                                  | 12 x 2                                                              | condenser  |
|           | BG STACK 1 | sealed enclosure                                                                     | 12 x 2                                                              | condenser  |
|           | BG STACK 2 | large sealed enclosure                                                               | 12 x 2                                                              | condenser  |
|           | MS STACK 1 | large sealed enclosure                                                               | 12 x 4                                                              | condenser  |
|           | MS STACK 2 | large sealed enclosure                                                               | 12 x 4                                                              | condenser  |
| Phaser Sw | OFF, ON    | Phaser on/off                                                                        |                                                                     |            |
|           | P. Rate    | 0.05–10.00 [Hz]                                                                      | Frequency of modulation                                             |            |
|           | Phs Manual | 0–127                                                                                | Adjusts the basic frequency from which the sound will be modulated. |            |
|           | Phs Depth  | 0–127                                                                                | Depth of modulation                                                 |            |
|           | Phs Reso   | 0–127                                                                                | Amount of feedback                                                  |            |
|           | Phs Mix    | 0–127                                                                                | Level of the phase-shifted sound                                    |            |
|           | Level      | 0–127                                                                                | Output Level                                                        |            |
|           | Asgn1–4    | OFF, Volume, Master Lv, Phaser Sw, P. Rate, Phs Manual, Phs Depth, Phs Reso, Phs Mix | Specifies the parameters that are assigned to assign 1–4.           |            |
|           |            |                                                                                      |                                                                     |            |
|           |            |                                                                                      |                                                                     |            |
|           |            |                                                                                      |                                                                     |            |
|           |            |                                                                                      |                                                                     |            |
|           |            |                                                                                      |                                                                     |            |
|           |            |                                                                                      |                                                                     |            |

## 81 Gt → Delay (Guitar Amp Simulator → Delay)

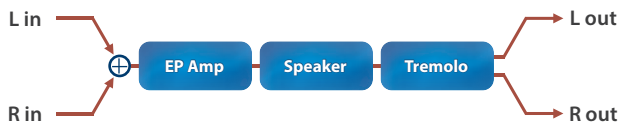


| Parameter  | Value              | Explanation                                                                                                                                     |
|------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre Amp Sw | OFF, ON            | Turns the amp switch on/off.                                                                                                                    |
| ATyp       | Type of guitar amp |                                                                                                                                                 |
|            | JC-120             | This models the sound of the Roland JC-120.                                                                                                     |
|            | CLEAN TWIN         | This models a Fender Twin Reverb.                                                                                                               |
|            | MATCH DRIVE        | This models the sound input to left input on a Matchless D/C-30. A simulation of the latest tube amp widely used in styles from blues and rock. |
|            | BG LEAD            | This models the lead sound of the MESA/ Boogie combo amp. The sound of a tube amp typical of the late '70s to '80s.                             |
|            | MS1959I            | This models the sound input to Input I on a Marshall 1959. This is a trebly sound suited to hard rock.                                          |
|            | MS1959II           | This models the sound input to Input II on a Marshall 1959.                                                                                     |
|            | MS1959I+II         | The sound of connecting inputs I and II of the guitar amp in parallel, creating a sound with a stronger low end than I.                         |
|            | SLDN LEAD          | This models a Soldano SLO-100. This is the typical sound of the eighties.                                                                       |
|            | METAL 5150         | This models the lead channel of a Peavey EVH 5150.                                                                                              |
|            | METAL LEAD         | This is distortion sound that is ideal for performances of heavy riffs.                                                                         |
|            | OD-1               | This models the sound of the BOSS OD-1. This produces sweet, mild distortion.                                                                   |
|            | OD-2 TURBO         | This is the high-gain overdrive sound of the BOSS OD-2.                                                                                         |
|            | DISTORTION         | This gives a basic, traditional distortion sound.                                                                                               |
|            | FUZZ               | A fuzz sound with rich harmonic content.                                                                                                        |
| Drive      | 0–127              | Volume and amount of distortion of the amp                                                                                                      |
| Master Lv  | 0–127              | Volume of the entire pre-amp                                                                                                                    |
| Gain       | LOW, MIDDLE, HIGH  | Amount of pre-amp distortion                                                                                                                    |
| Bass       | 0–127              | Tone of the bass/mid/treble frequency range                                                                                                     |
| Middle     | 0–127              |                                                                                                                                                 |
| Treble     | 0–127              |                                                                                                                                                 |
| Speaker Sw | OFF, ON            | Selects whether the sound will be sent through the speaker simulation (ON) or not (OFF).                                                        |

| Parameter | Value                                                                                                      | Explanation                                                                                                                           |
|-----------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| STyp      |                                                                                                            | Cabinet Diameter (in inches) and number of the speaker Microphone                                                                     |
|           | SMALL 1                                                                                                    | small open-back enclosure 10 dynamic                                                                                                  |
|           | SMALL 2                                                                                                    | small open-back enclosure 10 dynamic                                                                                                  |
|           | MIDDLE                                                                                                     | open back enclosure 12 x 1 dynamic                                                                                                    |
|           | JC-120                                                                                                     | open back enclosure 12 x 2 dynamic                                                                                                    |
|           | BUILT-IN 1                                                                                                 | open back enclosure 12 x 2 dynamic                                                                                                    |
|           | BUILT-IN 2                                                                                                 | open back enclosure 12 x 2 condenser                                                                                                  |
|           | BUILT-IN 3                                                                                                 | open back enclosure 12 x 2 condenser                                                                                                  |
|           | BUILT-IN 4                                                                                                 | open back enclosure 12 x 2 condenser                                                                                                  |
|           | BUILT-IN 5                                                                                                 | open back enclosure 12 x 2 condenser                                                                                                  |
|           | BG STACK 1                                                                                                 | sealed enclosure 12 x 2 condenser                                                                                                     |
|           | BG STACK 2                                                                                                 | large sealed enclosure 12 x 2 condenser                                                                                               |
|           | MS STACK 1                                                                                                 | large sealed enclosure 12 x 4 condenser                                                                                               |
|           | MS STACK 2                                                                                                 | large sealed enclosure 12 x 4 condenser                                                                                               |
|           | METAL STACK                                                                                                | large double stack 12 x 4 condenser                                                                                                   |
|           | 2-STACK                                                                                                    | large double stack 12 x 4 condenser                                                                                                   |
|           | 3-STACK                                                                                                    | large triple stack 12 x 4 condenser                                                                                                   |
| Delay Sw  | OFF, ON                                                                                                    | Delay on/off                                                                                                                          |
| Dly Time  | 1–1300                                                                                                     | Delay time from when the original sound is heard to when the delay sound is heard                                                     |
| Delay Fbk | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.              |
| Dly HF    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Frequency at which the high-frequency portion of the delay sound will be cut (BYPASS: no cut)                                         |
| Dly Bal   | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level     | 0–127                                                                                                      | Output Level                                                                                                                          |
| Asgn1–4   | OFF, Volume, Master Lv, Delay Sw, Dly Time, Delay Fbk, Dly Bal                                             | Specifies the parameters that are assigned to assign 1–4.                                                                             |

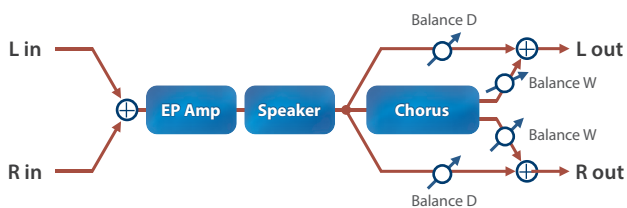


## 82 EP → Tremolo (EP Amp Simulator → Tremolo)



| Parameter    | Value                                              | Explanation                                                                                        |
|--------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Type         |                                                    | Type of amp                                                                                        |
|              | OLDCASE                                            | A standard electric piano sound of the early '70s                                                  |
|              | NEWCASE                                            | A standard electric piano sound of the late '70s and early '80s                                    |
|              | WURLY                                              | A standard electric piano sound of the '60s                                                        |
| Bass         | -50+50                                             | Amount of low-frequency boost/cut                                                                  |
| Treble       | -50+50                                             | Amount of high-frequency boost/cut                                                                 |
| Tremolo Sw   | OFF, ON                                            | Tremolo on/off                                                                                     |
| Tremolo Sync | OFF, ON                                            | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)    |
| T. Speed     | 0.05–10.00 [Hz]                                    |                                                                                                    |
| T. Spd Nt    | Note<br>→ "Note" (p. 72)                           | Rate of the tremolo effect                                                                         |
| Trm Depth    | 0–127                                              | Depth of the tremolo effect                                                                        |
| Trm Duty     | -10+10                                             | Adjusts the duty cycle of the LFO waveform used to apply tremolo.                                  |
| Sp Type      | LINE, OLD, NEW, WURLY, TWIN                        | Type of speaker<br>If LINE is selected, the sound will not be sent through the speaker simulation. |
| OD Switch    | OFF, ON                                            | Overdrive on/off                                                                                   |
| OD Gain      | 0–127                                              | Overdrive input level                                                                              |
| OD Drive     | 0–127                                              | Degree of distortion<br>Also changes the volume.                                                   |
| Level        | 0–127                                              | Output Level                                                                                       |
| Asgn1–4      | OFF, Bass, Treble, Tremolo Sw, T. Speed, Trm Depth | Specifies the parameters that are assigned to assign 1–4.                                          |

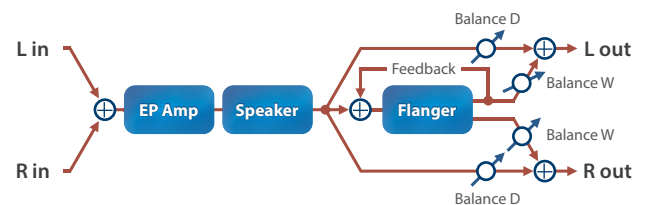
## 83 EP → Chorus (EP Amp Simulator → Chorus)



| Parameter  | Value                    | Explanation                                                                                     |
|------------|--------------------------|-------------------------------------------------------------------------------------------------|
| Type       |                          | Type of amp                                                                                     |
|            | OLDCASE                  | A standard electric piano sound of the early '70s                                               |
|            | NEWCASE                  | A standard electric piano sound of the late '70s and early '80s                                 |
|            | WURLY                    | A standard electric piano sound of the '60s                                                     |
| Bass       | -50+50                   | Amount of low-frequency boost/cut                                                               |
| Treble     | -50+50                   | Amount of high-frequency boost/cut                                                              |
| Cho Switch | OFF, ON                  | Chorus on/off                                                                                   |
| Cho PreDly | 0.0–100 [ms]             | Adjusts the delay time from the direct sound until the chorus sound is heard.                   |
| Cho Sync   | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31) |
| C. Rate    | 0.05–10.00 [Hz]          |                                                                                                 |
| C. Rate Nt | Note<br>→ "Note" (p. 72) | Frequency of modulation                                                                         |
| Cho Depth  | 0–127                    | Depth of modulation                                                                             |

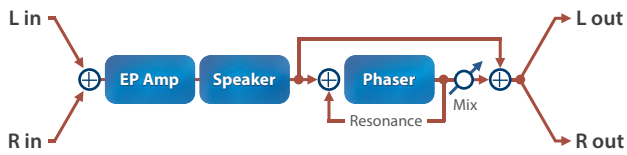
| Parameter | Value                                                      | Explanation                                                                                                                             |
|-----------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Cho Bal   | D100: 0W–D0: 100W                                          | Adjusts the volume balance between the sound that is sent through the chorus (W) and the sound that is not sent through the chorus (D). |
| Sp Type   | LINE, OLD, NEW, WURLY, TWIN                                | Type of speaker<br>If LINE is selected, the sound will not be sent through the speaker simulation.                                      |
| OD Switch | OFF, ON                                                    | Overdrive on/off                                                                                                                        |
| OD Gain   | 0–127                                                      | Overdrive input level                                                                                                                   |
| OD Drive  | 0–127                                                      | Degree of distortion<br>Also changes the volume.                                                                                        |
| Level     | 0–127                                                      | Output Level                                                                                                                            |
| Asgn1–4   | OFF, Bass, Treble, Cho Switch, C. Rate, Cho Depth, Cho Bal | Specifies the parameters that are assigned to assign 1–4.                                                                               |

## 84 EP → Flanger (EP Amp Simulator → Flanger)



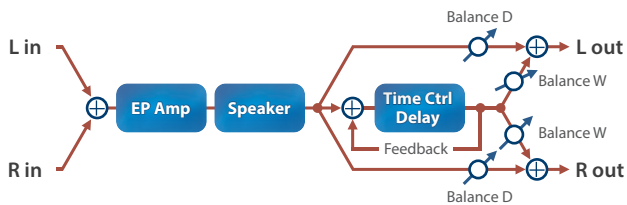
| Parameter  | Value                                                               | Explanation                                                                                                                               |
|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Type       |                                                                     | Type of amp                                                                                                                               |
|            | OLDCASE                                                             | A standard electric piano sound of the early '70s                                                                                         |
|            | NEWCASE                                                             | A standard electric piano sound of the late '70s and early '80s                                                                           |
|            | WURLY                                                               | A standard electric piano sound of the '60s                                                                                               |
| Bass       | -50+50                                                              | Amount of low-frequency boost/cut                                                                                                         |
| Treble     | -50+50                                                              | Amount of high-frequency boost/cut                                                                                                        |
| Flg Switch | OFF, ON                                                             | Flanger on/off                                                                                                                            |
| Flg PreDly | 0.0–100 [ms]                                                        | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                            |
| Flg Sync   | OFF, ON                                                             | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                           |
| F. Rate    | 0.05–10.00 [Hz]                                                     |                                                                                                                                           |
| F. Rate Nt | Note<br>→ "Note" (p. 72)                                            | Frequency of modulation                                                                                                                   |
| Flg Depth  | 0–127                                                               | Depth of modulation                                                                                                                       |
| Flg Fbk    | -98+98 [%]                                                          | Adjusts the proportion of the flanger sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.             |
| Flg Bal    | D100: 0W–D0: 100W                                                   | Adjusts the volume balance between the sound that is sent through the flanger (W) and the sound that is not sent through the flanger (D). |
| Sp Type    | LINE, OLD, NEW, WURLY, TWIN                                         | Type of speaker<br>If LINE is selected, the sound will not be sent through the speaker simulation.                                        |
| OD Switch  | OFF, ON                                                             | Overdrive on/off                                                                                                                          |
| OD Gain    | 0–127                                                               | Overdrive input level                                                                                                                     |
| OD Drive   | 0–127                                                               | Degree of distortion<br>Also changes the volume.                                                                                          |
| Level      | 0–127                                                               | Output Level                                                                                                                              |
| Asgn1–4    | OFF, Bass, Treble, Flg Switch, F. Rate, Flg Depth, Flg Fbk, Flg Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                 |

## 85 EP → Phaser (EP Amp Simulator → Phaser)



| Parameter  | Value                                                                            | Explanation                                                                                        |
|------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Type       |                                                                                  | Type of amp                                                                                        |
|            | OLDCASE                                                                          | A standard electric piano sound of the early '70s                                                  |
|            | NEWCASE                                                                          | A standard electric piano sound of the late '70s and early '80s                                    |
| Bass       | -50→+50                                                                          | Amount of low-frequency boost/cut                                                                  |
| Treble     | -50→+50                                                                          | Amount of high-frequency boost/cut                                                                 |
| Phs Switch | OFF, ON                                                                          | Phaser on/off                                                                                      |
| Phs Sync   | OFF, ON                                                                          | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>⇒ "Tempo" (p. 3, p. 31)    |
| P. Rate    | 0.05–10.00 [Hz]                                                                  |                                                                                                    |
| P. Rate Nt | Note<br>⇒ "Note" (p. 72)                                                         | Frequency of modulation                                                                            |
| Phs Manual | 0–127                                                                            | Adjusts the basic frequency from which the sound will be modulated.                                |
| Phs Depth  | 0–127                                                                            | Depth of modulation                                                                                |
| Phs Reso   | 0–127                                                                            | Amount of feedback                                                                                 |
| Phs Mix    | 0–127                                                                            | Level of the phase-shifted sound                                                                   |
| Sp Type    | LINE, OLD, NEW, WURLY, TWIN                                                      | Type of speaker<br>If LINE is selected, the sound will not be sent through the speaker simulation. |
| OD Switch  | OFF, ON                                                                          | Overdrive on/off                                                                                   |
| OD Gain    | 0–127                                                                            | Overdrive input level                                                                              |
| OD Drive   | 0–127                                                                            | Degree of distortion<br>Also changes the volume.                                                   |
| Level      | 0–127                                                                            | Output Level                                                                                       |
| Asgn1–4    | OFF, Bass, Treble, Phs Switch, P. Rate, Phs Manual, Phs Depth, Phs Reso, Phs Mix | Specifies the parameters that are assigned to assign 1–4.                                          |

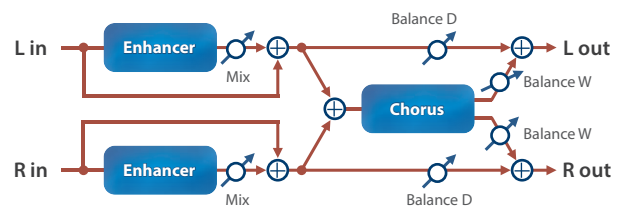
## 86 EP → Delay (EP Amp Simulator → Delay)



| Parameter  | Value                    | Explanation                                                                                     |
|------------|--------------------------|-------------------------------------------------------------------------------------------------|
| Type       |                          | Type of amp                                                                                     |
|            | OLDCASE                  | A standard electric piano sound of the early 70s                                                |
|            | NEWCASE                  | A standard electric piano sound of the late 70s and early 80s                                   |
| Bass       | -50→+50                  | Amount of low-frequency boost/cut                                                               |
| Treble     | -50→+50                  | Amount of high-frequency boost/cut                                                              |
| Dly Switch | OFF, ON                  | Delay on/off                                                                                    |
| Delay Sync | OFF, ON                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>⇒ "Tempo" (p. 3, p. 31) |
| D. Time    | 1–1300                   |                                                                                                 |
| D. Time Nt | Note<br>⇒ "Note" (p. 72) | Delay time from when the original sound is heard to when the delay sound is heard               |

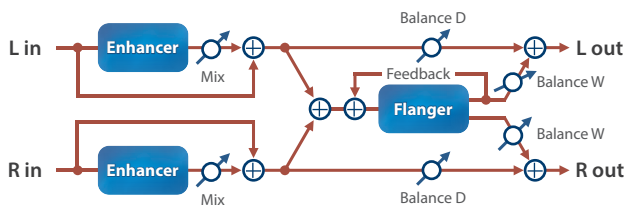
| Parameter | Value                                                                                                      | Explanation                                                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dly Accel | 0–15                                                                                                       | Speed at which the current delay time changes to the specified delay time when you change the delay time.<br>The speed of the pitch change will change simultaneously with the delay time. |
| Delay Fbk | -98→+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.                                                                |
| Dly HF    | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Frequency at which the high-frequency portion of the delay sound will be cut (BYPASS: no cut)                                                                                              |
| Dly Bal   | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D).                                                      |
| Sp Type   | LINE, OLD, NEW, WURLY, TWIN                                                                                | Type of speaker<br>If LINE is selected, the sound will not be sent through the speaker simulation.                                                                                         |
| OD Switch | OFF, ON                                                                                                    | Overdrive on/off                                                                                                                                                                           |
| OD Gain   | 0–127                                                                                                      | Overdrive input level                                                                                                                                                                      |
| OD Drive  | 0–127                                                                                                      | Degree of distortion<br>Also changes the volume.                                                                                                                                           |
| Level     | 0–127                                                                                                      | Output Level                                                                                                                                                                               |
| Asgn1–4   | OFF, Bass, Treble, Dly Switch, D. Time, Delay Fbk                                                          | Specifies the parameters that are assigned to assign 1–4.                                                                                                                                  |

## 87 Enhncr → Cho (Enhancer → Chorus)



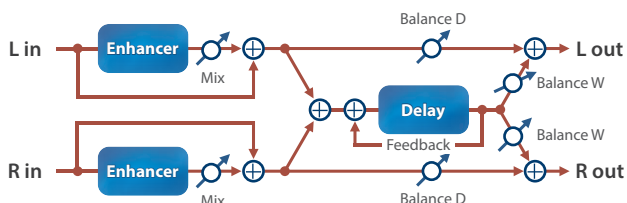
| Parameter  | Value                                    | Explanation                                                                                                                             |
|------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Enh Sens   | 0–127                                    | Sensitivity of the enhancer                                                                                                             |
| Enh Mix    | 0–127                                    | Level of the overtones generated by the enhancer                                                                                        |
| Cho PreDly | 0.0–100 [ms]                             | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                           |
| Cho Sync   | OFF, ON                                  | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>⇒ "Tempo" (p. 3, p. 31)                                         |
| C. Rate    | 0.05–10.00 [Hz]                          |                                                                                                                                         |
| C. Rate Nt | Note<br>⇒ "Note" (p. 72)                 | Frequency of modulation                                                                                                                 |
| Cho Depth  | 0–127                                    | Depth of modulation                                                                                                                     |
| Cho Bal    | D100: 0W–D0: 100W                        | Adjusts the volume balance between the sound that is sent through the chorus (W) and the sound that is not sent through the chorus (D). |
| Level      | 0–127                                    | Output Level                                                                                                                            |
| Asgn1–4    | OFF, Enh Sens, Enh Mix, C. Rate, Cho Bal | Specifies the parameters that are assigned to assign 1–4.                                                                               |

## 88 Enhncr → Fl (Enhancer → Flanger)



| Parameter  | Value                                             | Explanation                                                                                                                               |
|------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Enh Sens   | 0–127                                             | Sensitivity of the enhancer                                                                                                               |
| Enh Mix    | 0–127                                             | Level of the overtones generated by the enhancer                                                                                          |
| Flg PreDly | 0.0–100 [ms]                                      | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                            |
| Flg Sync   | OFF, ON                                           | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                           |
| F. Rate    | 0.05–10.00 [Hz]                                   | Frequency of modulation                                                                                                                   |
| F. Rate Nt | Note<br>→ "Note" (p. 72)                          |                                                                                                                                           |
| Flg Depth  | 0–127                                             | Depth of modulation                                                                                                                       |
| Flg Fbk    | -98–+98 [%]                                       | Adjusts the proportion of the flanger sound that is fed back into the effect. Negative (-) settings will invert the phase.                |
| Flg Bal    | D100: 0W–D0: 100W                                 | Adjusts the volume balance between the sound that is sent through the flanger (W) and the sound that is not sent through the flanger (D). |
| Level      | 0–127                                             | Output Level                                                                                                                              |
| Asgn1–4    | OFF, Enh Sens, Enh Mix, F. Rate, Flg Fbk, Flg Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                 |

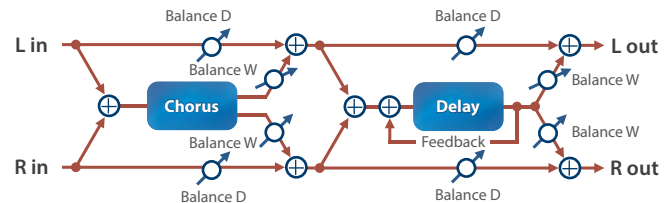
## 89 Enhncr → Dly (Enhancer → Delay)



| Parameter  | Value                                                                                                      | Explanation                                                                                                                           |
|------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Enh Sens   | 0–127                                                                                                      | Sensitivity of the enhancer                                                                                                           |
| Enh Mix    | 0–127                                                                                                      | Level of the overtones generated by the enhancer                                                                                      |
| Delay Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| D. Time    | 1–2600                                                                                                     | Delay time from when the original sound is heard to when the delay sound is heard                                                     |
| D. Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Delay Fbk  | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.              |
| Dly HF     | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect will be cut (BYPASS: no cut).                                          |
| Dly Bal    | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level      | 0–127                                                                                                      | Output Level                                                                                                                          |

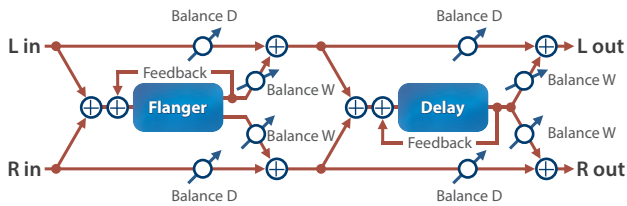
| Parameter | Value                                               | Explanation                                               |
|-----------|-----------------------------------------------------|-----------------------------------------------------------|
| Asgn1–4   | OFF, Enh Sens, Enh Mix, D. Time, Delay Fbk, Dly Bal | Specifies the parameters that are assigned to assign 1–4. |

## 90 Chorus → Dly (Chorus → Delay)



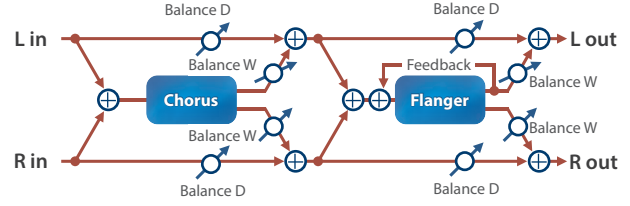
| Parameter  | Value                                                                                                      | Explanation                                                                                                                           |
|------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Cho PreDly | 0.0–100 [ms]                                                                                               | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                         |
| Cho Sync   | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| C. Rate    | 0.05–10.00 [Hz]                                                                                            | Frequency of modulation                                                                                                               |
| C. Rate Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Cho Depth  | 0–127                                                                                                      | Depth of modulation                                                                                                                   |
| Cho Bal    | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the chorus sound (W)                                                                  |
| Delay Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| D. Time    | 1–2600                                                                                                     | Delay time from when the original sound is heard to when the delay sound is heard                                                     |
| D. Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Delay Fbk  | -98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect. Negative (-) settings will invert the phase.              |
| Dly HF     | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect will be cut (BYPASS: no cut).                                          |
| Dly Bal    | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level      | 0–127                                                                                                      | Output Level                                                                                                                          |
| Asgn1–4    | OFF, C. Rate, Cho Bal, Delay Fbk, Dly Bal                                                                  | Specifies the parameters that are assigned to assign 1–4.                                                                             |

## 91 Flanger → Dly (Flanger → Delay)



| Parameter  | Value                                                                                                      | Explanation                                                                                                                           |
|------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Flg PreDly | 0.0–100 [ms]                                                                                               | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                        |
| Flg Sync   | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| F. Rate    | 0.05–10.00 [Hz]                                                                                            | Frequency of modulation                                                                                                               |
| F. Rate Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Flg Depth  | 0–127                                                                                                      | Depth of modulation                                                                                                                   |
| Flg Fbk    | –98–+98 [%]                                                                                                | Adjusts the proportion of the flanger sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.         |
| Flg Bal    | D100: 0W–D0: 100W                                                                                          | Volume balance between the direct sound (D) and the flanger sound (W)                                                                 |
| Delay Sync | OFF, ON                                                                                                    | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                       |
| D. Time    | 1–2600                                                                                                     | Delay time from when the original sound is heard to when the delay sound is heard                                                     |
| D. Time Nt | Note<br>→ "Note" (p. 72)                                                                                   |                                                                                                                                       |
| Delay Fbk  | –98–+98 [%]                                                                                                | Adjusts the proportion of the delay sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.           |
| Dly HF     | 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000, 2500, 3150, 4000, 5000, 6300, 8000, BYPASS [Hz] | Adjusts the frequency above which sound fed back to the effect will be cut (BYPASS: no cut).                                          |
| Dly Bal    | D100: 0W–D0: 100W                                                                                          | Adjusts the volume balance between the sound that is sent through the delay (W) and the sound that is not sent through the delay (D). |
| Level      | 0–127                                                                                                      | Output Level                                                                                                                          |
| Asgn1–4    | OFF, F. Rate, Flg Fbk, Flg Bal, Delay Fbk, Dly Bal                                                         | Specifies the parameters that are assigned to assign 1–4.                                                                             |

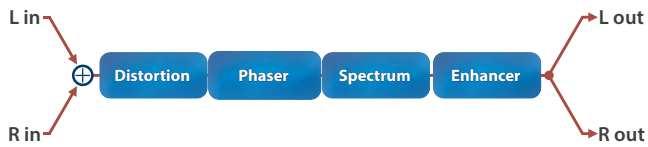
## 92 Chorus → Fl (Chorus → Flanger)



| Parameter  | Value                                            | Explanation                                                                                                                               |
|------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Cho PreDly | 0.0–100 [ms]                                     | Adjusts the delay time from the direct sound until the chorus sound is heard.                                                             |
| Cho Sync   | OFF, ON                                          | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                           |
| C. Rate    | 0.05–10.00 [Hz]                                  | Modulation frequency of the chorus effect                                                                                                 |
| C. Rate Nt | Note<br>→ "Note" (p. 72)                         |                                                                                                                                           |
| Cho Depth  | 0–127                                            | Modulation depth of the chorus effect                                                                                                     |
| Cho Bal    | D100: 0W–D0: 100W                                | Volume balance between the direct sound (D) and the chorus sound (W)                                                                      |
| Flg PreDly | 0.0–100 [ms]                                     | Adjusts the delay time from the direct sound until the flanger sound is heard.                                                            |
| Flg Sync   | OFF, ON                                          | If this is "ON," the rate synchronizes with the tempo of the rhythm.<br>→ "Tempo" (p. 3, p. 31)                                           |
| F. Rate    | 0.05–10.00 [Hz]                                  | Modulation frequency of the flanger effect                                                                                                |
| F. Rate Nt | Note<br>→ "Note" (p. 72)                         |                                                                                                                                           |
| Flg Depth  | 0–127                                            | Modulation depth of the flanger effect                                                                                                    |
| Flg Fbk    | –98–+98 [%]                                      | Adjusts the proportion of the flanger sound that is fed back into the effect.<br>Negative (-) settings will invert the phase.             |
| Flg Bal    | D100: 0W–D0: 100W                                | Adjusts the volume balance between the sound that is sent through the flanger (W) and the sound that is not sent through the flanger (D). |
| Level      | 0–127                                            | Output Level                                                                                                                              |
| Asgn1–4    | OFF, C. Rate, Cho Bal, F. Rate, Flg Fbk, Flg Bal | Specifies the parameters that are assigned to assign 1–4.                                                                                 |

## 93 JD Multi

Recreates the effects included in group A of the JD-800.



| Parameter     | Value                        | Explanation                                                                                                                                  |
|---------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Seq           | DS - PH - SP - EN            | Selects the connection order of the effects.<br><br><b>DS:</b> Distortion<br><b>PH:</b> Phaser<br><b>SP:</b> Spectrum<br><b>EN:</b> Enhancer |
|               | DS - PH - EN - SP            |                                                                                                                                              |
|               | DS - SP - PH - EN            |                                                                                                                                              |
|               | DS - SP - EN - PH            |                                                                                                                                              |
|               | DS - EN - PH - SP            |                                                                                                                                              |
|               | DS - EN - SP - PH            |                                                                                                                                              |
|               | PH - DS - SP - EN            |                                                                                                                                              |
|               | PH - DS - EN - SP            |                                                                                                                                              |
|               | PH - SP - DS - EN            |                                                                                                                                              |
|               | PH - SP - EN - DS            |                                                                                                                                              |
|               | PH - EN - DS - SP            |                                                                                                                                              |
|               | PH - EN - SP - DS            |                                                                                                                                              |
|               | SP - DS - PH - EN            |                                                                                                                                              |
|               | SP - DS - EN - PH            |                                                                                                                                              |
|               | SP - PH - DS - EN            |                                                                                                                                              |
|               | SP - PH - EN - DS            |                                                                                                                                              |
|               | SP - EN - DS - PH            |                                                                                                                                              |
|               | SP - EN - PH - DS            |                                                                                                                                              |
|               | EN - DS - PH - SP            |                                                                                                                                              |
|               | EN - DS - SP - PH            |                                                                                                                                              |
|               | EN - PH - DS - SP            |                                                                                                                                              |
|               | EN - PH - SP - DS            |                                                                                                                                              |
|               | EN - SP - DS - PH            |                                                                                                                                              |
|               | EN - SP - PH - DS            |                                                                                                                                              |
| DS Switch     | OFF, ON                      | Turns the distortion on/off.                                                                                                                 |
| DS Type       | Sets the type of distortion. |                                                                                                                                              |
|               | MELLOW DRV                   | Softer distortion with a slightly darker sound.                                                                                              |
|               | OVERDRIVE                    | Distortion that resembles a vacuum tube amp being driven.                                                                                    |
|               | CRY DRV                      | Distortion that emphasizes the high end.                                                                                                     |
|               | MELLOW DST                   | Gives the feeling of distortion playing through a large amp.                                                                                 |
|               | LIGHT DST                    | Strong distortion with a bright sound.                                                                                                       |
|               | FAT DIST                     | Thick distortion that emphasizes the low and high ends.                                                                                      |
|               | FUZZ DIST                    | Distortion that's even more powerful than FAT DIST.                                                                                          |
| DS Drive      | 0–100                        | Sets the amount of distortion.                                                                                                               |
| DS Level      | 0–100                        | Sets the distortion output level.                                                                                                            |
| PH Switch     | OFF, ON                      | Turns the phaser on/off.                                                                                                                     |
| PH Manual     | 50 [Hz]–15.0 [kHz]           | Sets the basic frequency from which the sound is modulated with the phaser effect.                                                           |
| PH Rate       | 0.1–10.0 [Hz]                | Sets the cycle of the phaser modulation.                                                                                                     |
| PH Depth      | 0–100                        | Sets the depth of the phaser modulation.                                                                                                     |
| PH Resonance  | 0–100                        | Sets the amount of feedback for the phaser. Increasing the value creates a more unusual sound.                                               |
| PH Mix        | 0–100                        | Sets the level of the phase-shifted sound.                                                                                                   |
| SP Switch     | OFF, ON                      | Turns the spectrum on/off.                                                                                                                   |
| SP Band Ctrl1 | –15–+15 [dB]                 | Sets the gain (amount of boost/cut) in the 250 Hz range.                                                                                     |
| SP Band Ctrl2 | –15–+15 [dB]                 | Sets the gain (amount of boost/cut) in the 500 Hz range.                                                                                     |
| SP Band Ctrl3 | –15–+15 [dB]                 | Sets the gain (amount of boost/cut) in the 1000 Hz range.                                                                                    |

| Parameter     | Value                                               | Explanation                                                                                        |
|---------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------|
| SP Band Ctrl4 | –15–+15 [dB]                                        | Sets the gain (amount of boost/cut) in the 2000 Hz range.                                          |
| SP Band Ctrl5 | –15–+15 [dB]                                        | Sets the gain (amount of boost/cut) in the 4000 Hz range.                                          |
| SP Band Ctrl6 | –15–+15 [dB]                                        | Sets the gain (amount of boost/cut) in the 8000 Hz range.                                          |
| SP Width      | 1–5                                                 | Sets the bandwidth for changing the levels, common to all bands.                                   |
| EH Switch     | OFF, ON                                             | Turns the enhancer on/off.                                                                         |
| EH Sens       | 0–100                                               | Sets how easily the enhancer effect is applied.                                                    |
| EH Mix        | 0–100                                               | Sets the ratio at which the harmonics generated by the enhancer are mixed with the original sound. |
| Pan           | L64–63R                                             | Changes the pan.                                                                                   |
| Level         | 0–127                                               | Sets the output volume.                                                                            |
| Asgn1–4       | OFF, Band2 Thrs, Band2 Gain, Band3 Thrs, Band3 Gain | Specifies the parameters that are assigned to assign 1–4.                                          |

| Note  |                           |       |                        |       |                            |
|-------|---------------------------|-------|------------------------|-------|----------------------------|
| 1/64T | Sixty-fourth-note triplet | 1/64  | Sixty-fourth note      | 1/32T | Thirty-second-note triplet |
| 1/32  | Thirty-second note        | 1/16T | Sixteenth-note triplet | 1/32. | Dotted thirty-second note  |
| 1/16  | Sixteenth note            | 1/8T  | Eighth-note triplet    | 1/16. | Dotted sixteenth note      |
| 1/8   | Eighth note               | 1/4T  | Quarter-note triplet   | 1/8.  | Dotted eighth note         |
| 1/4   | Quarter note              | 1/2T  | Half-note triplet      | 1/4.  | Dotted quarter note        |
| 1/2   | Half note                 | 1T    | Whole-note triplet     | 1/2.  | Dotted half note           |
| 1     | Whole note                | 2T    | Double-note triplet    | 1.    | Dotted whole note          |
| 2     | Double note               |       |                        |       |                            |